Asymptotic Transseries and Inversion of Four Combinatorial Sequences

Rooted Trees, the Double Factorial, the Partition Numbers, and the Swing Factorial:
one Lambert-cored reversion calculus and its four specializations

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Abstract

Four combinatorial sequences are studied here to all orders and then inverted: the numbers a_n of unlabelled rooted trees on n vertices (OEIS A000081, the trees underlying Butcher series), the double factorial $n!!$, the partition numbers $p(n)$ (A000041), and the swing factorial $\mathrm{sw}(n) = n!/\lfloor n/2 \rfloor !^2$ (A056040). For each we give the complete forward asymptotic transseries and then the asymptotic inverse: given a value, recover the index, to all orders and with an explicit error theory.

The four subjects look unrelated and their forward analyses genuinely are — a Pólya-tree singularity analysis, a pair of gamma-ratio branches, Rademacher's convergent root-of-unity series, and a second pair of gamma-ratio branches. Their *inversions* are not. Every one of them, after a monomial change of variable, is the inversion of an exponential–power model

$$F(x) = C \exp(ax^{\alpha})\, x^{b} \exp Q(x^{-\delta}),$$

whose dominant phase is inverted exactly by the Lambert W-function and whose remaining corrections are generated by one triangular Bell-polynomial recurrence. Chapter 0 develops that apparatus once, in the generality the four cases require, with complete proofs; Chapters 1–4 supply only the parameter dictionary for each sequence and the mathematics genuinely specific to it. Chapter 5 compares the four and records what the common frame does and does not explain.

This volume is a consolidation of twelve independently written articles, three on each subject, filed in this repository on 3 September 2026. Where the three treatments of a subject agreed, one presentation was kept; where they disagreed, the disagreement is stated in the text and resolved; where a result was asserted without proof, a proof is supplied or the statement is demoted to a remark that says what is actually established. Corrections to the sources are boxed in the text and listed in Appendix B; provenance is in Appendix A.

Contents

Chapter 1

The common apparatus

All twelve filed articles carry a private copy of the material of this chapter. The presentation below follows, in the main, the two sections *A universal coefficient calculus for exponential–power inversion* and *Lambert-normalized inverse transseries* of Swing_Factorial_Transseries_Article, which state the reversion in the greatest generality of the twelve. Four things have been changed. (i) The Lagrange fixed-point lemma is reproved: the swing article's proof appeals to Lagrange–Bürmann inversion in a setting where the required invertibility hypothesis fails, and the gap is closed here by a universality argument (theorem 1.15). (ii) The exponential–power model is separated into a *formal* and an *analytic* reading, and every asymptotic claim is given an explicit remainder hypothesis; the sources routinely pass from a coefficient recurrence to a Poincaré statement without one. (iii) The dominant core is axiomatized (theorem 1.8) so that the factorial-type phase of Part 3 — which does *not* belong to the exponential–power model — is covered by the same reversion theorem. (iv) The discrete-inverse material of Butcher_Tree_Transseries-2, double_factorial_transseries-2 and the three partition articles is merged into one statement with a residue-class lattice.

1.1 Standing conventions

Throughout the volume R denotes a commutative $\mathbb{Q}$-algebra, and $R[[t]]$, $R[[t,u]]$ formal power series rings over it. For a formal series f we write $[t^n]\,f$ for the coefficient of t^n. The symbols $\mathrm{e}, \mathrm{i}, \mathrm{d}$ are the upright constants and differential; $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are the usual number systems, and $\mathbb{N} = \{0, 1, 2, \dots\}$. Asymptotic statements use $\mathcal{O}$ and o in the Landau sense as the stated variable tends to $+\infty$; an implied constant is always allowed to depend on every quantity held fixed in the statement (in particular on the truncation index N) and on nothing else.

The sources disagree on three points of bookkeeping. We fix them once.

Remark 1.1 (Conventions adopted, and where the sources differ).

1. *Ordinary versus exponential Bell polynomials.* The partition articles and Butcher_Tree_Transseries-2 take the *exponential* partial Bell polynomials $B_{n,k}$ as primitive and convert; Swing_Factorial_Transseries_Article and double_factorial_transseries-2 take the *ordinary* ones $\widehat{B}_{n,k}$ as primitive. The volume defines both (theorems 1.9 and 1.10), records the conversion (theorem 1.11), and uses $\widehat{B}$ in every reversion formula and B in every exponentiation or logarithm of a *unit* series. The reason is not taste: reversion multiplies a correction series whose coefficients carry no factorials, and $\widehat{B}$ is the transform that leaves them alone.

2. *Sign of the logarithmic phase.* We write the dominant phase as

$$L = aX + b\log X,$$

with b a *signed* constant. Butcher_Tree_Transseries-2 writes $Z = X - p \log X$ and the partition articles write $\rho - 2\log\rho = L$; both are the case $b = -p < 0$. Every formula below is stated in the signed convention, and the dictionary of table 1.1 gives b for each chapter. One dividend of the signed convention is theorem 1.21: the Lambert branch is W_0 when $b > 0$ and W_{-1} when $b < 0$, with no case analysis beyond the sign.

3. *Truncation index.* A truncation "of order N" retains the terms with index $n = 1, \dots, N$ *inclusive*, together with the leading term. Swing_Factorial_Transseries_Article uses the exclusive convention for its forward table ("the truncation labelled N retains $A_0, \dots, A_{N-1}$") and the inclusive one for its inverse table ("the first N correction terms are retained"); the two tables of that article are therefore not on a common scale. The volume is inclusive throughout.

1.2 The exponential–power model

1.2.1 Formal and analytic readings

Definition 1.2 (The exponential–power model)**.** Fix constants $C > 0$, $a > 0$, $\alpha \in (0, 1]$, $\delta > 0$, $b \in \mathbb{R}$, and a formal series without constant term,

$$Q(t) = \sum_{j \geq 1} q_j t^j \in t\,\mathbb{R}[[t]].$$

The associated *exponential–power model* is the expression

$$F(x) = C \exp\bigl(ax^\alpha\bigr)\, x^b \, \exp Q\bigl(x^{-\delta}\bigr), \tag{1.1}$$

understood through its logarithm

$$\log F(x) = \log C + ax^\alpha + b \log x + Q\bigl(x^{-\delta}\bigr). \tag{1.2}$$

We call $(C, a, \alpha, b, \delta, Q)$ the *model data*.

A positive function $F \in C^1([x_0, \infty))$ is said to be *of exponential–power type with data* $(C, a, \alpha, b, \delta, Q)$ when, for every $M \in \mathbb{N}$, there is a constant K_M with

$$\left| \log F(x) - \log C - ax^\alpha - b \log x - \sum_{j=1}^{M} q_j x^{-j\delta} \right| \leq K_M x^{-(M+1)\delta} \qquad (x \geq x_0), \tag{1.3}$$

and *differentiably* of that type when in addition

$$(\log F)'(x) = a\alpha x^{\alpha - 1} + \frac{b}{x} + \mathcal{O}\bigl(x^{-1-\delta}\bigr) \qquad (x \to \infty). \tag{1.4}$$

Remark 1.3 (Why the two readings are kept apart)*.* The series Q is *not* assumed convergent. In Parts 2, 3 and 5 it is divergent of Gevrey type, which is what makes theorem 1.46 bite there. It is *not* divergent in Part 4: for the partition numbers $Q(t) = \log(1-t)$ has radius of convergence 1, every algebraic expansion of that chapter converges, and theorem 1.46 correspondingly has no content there — the error floor of Part 4 is set by the exponentially small Rademacher remainder and not by a least term. This is precisely why the two readings are kept apart: convergence of Q is a property of the subject, not of the frame. Every purely algebraic assertion below — coefficient recurrences, closed coefficient formulae, degree bounds — is a statement about (1.2) as a formal object and needs no analytic hypothesis at all. Every assertion with an error term uses (1.3), and every assertion that inverts uses (1.4) as well: an asymptotic expansion may not be differentiated without a hypothesis, and the inversion of F is controlled by $(\log F)'$. In each chapter (1.4) is verified directly from a digamma, Laplace or Rademacher representation, never inferred from (1.3).

1.2.2 Reduction to a linear phase

Lemma 1.4 (Monomial α-reduction). *Let F be of exponential–power type with data $(C, a, \alpha, b, \delta, Q)$ on $[x_0, \infty)$, and put*

$$\xi = x^\alpha, \qquad x = \xi^{1/\alpha}, \qquad G(\xi) := F\big(\xi^{1/\alpha}\big). \tag{1.5}$$

Then G is of exponential–power type on $[x_0^\alpha, \infty)$ with data

$$\Big(C,\ a,\ 1,\ \frac{b}{\alpha},\ \frac{\delta}{\alpha},\ Q\Big), \tag{1.6}$$

that is,

$$\log G(\xi) = \log C + a\xi + \frac{b}{\alpha}\log\xi + Q\big(\xi^{-\delta/\alpha}\big). \tag{1.7}$$

The coefficients q_j are unchanged. If moreover F is differentiably of its type then G is differentiably of its type. Finally $x \mapsto x^\alpha$ is an increasing bijection of $[x_0, \infty)$ onto $[x_0^\alpha, \infty)$, so F is strictly increasing if and only if G is, and then

$$F^{-1}(y) = \big(G^{-1}(y)\big)^{1/\alpha}. \tag{1.8}$$

Proof. Substituting $x = \xi^{1/\alpha}$ into (1.2) and using $x^\alpha = \xi$, $\log x = \alpha^{-1}\log\xi$ and $x^{-j\delta} = \xi^{-j\delta/\alpha}$ gives (1.7) term by term; the truncated inequality (1.3) transforms into the corresponding inequality for G with the same constants K_M, because $x^{-(M+1)\delta} = \xi^{-(M+1)\delta/\alpha}$. For the derivative, $\mathrm{d}x/\mathrm{d}\xi = \alpha^{-1}\xi^{1/\alpha-1}$, so

$$(\log G)'(\xi) = (\log F)'\big(\xi^{1/\alpha}\big)\cdot\frac{\xi^{1/\alpha-1}}{\alpha} = \Big(a\alpha\xi^{1-1/\alpha} + \frac{b}{\xi^{1/\alpha}} + \mathcal{O}\big(\xi^{-(1+\delta)/\alpha}\big)\Big)\frac{\xi^{1/\alpha-1}}{\alpha} = a + \frac{b/\alpha}{\xi} + \mathcal{O}\big(\xi^{-1-\delta/\alpha}\big),$$

which is (1.4) for G with α replaced by 1. The last two assertions are immediate from strict monotonicity of $x \mapsto x^\alpha$ on $(0, \infty)$. □

Remark 1.5 (Normalization used from here on). By theorem 1.4 every statement about the model (1.1) reduces to the case $\alpha = 1$, and we assume $\alpha = 1$ from section 1.4 onwards. We assume in addition $\delta = 1$; this is the value of δ/α in every chapter of the volume (table 1.1). Theorem 1.27 states the general-δ reversion, on the two-generator monomial grid it requires.

Remark 1.6 (The affine shift is taken *before* the reduction). Chapters 3 and 4 apply the model not to the combinatorial index n but to a shifted variable $x = n - x_*$ ($x_* = 1/24$ for the partition numbers, $x_* = -1$ for the double factorial). This is not a loss of generality but a genuine choice of normal form, and the order of operations matters. If F has data $(C, a, \alpha, b, \delta, Q)$ in x and one re-expands in $n = x + x_*$, then for $\alpha < 1$

$$a(n - x_*)^\alpha = an^\alpha + a\sum_{k\ge1}\binom{\alpha}{k}(-x_*)^k n^{\alpha-k},$$

so the phase acquires an entire new tower of powers $n^{\alpha-k}$: the re-expanded object no longer has model data with a single exponent grid. The partition articles say this explicitly — their infinite algebraic coefficient arrays in the variable $n^{-1/2}$ "arise entirely from re-expanding the shift $N = n - 1/24$". The volume therefore normalizes the shift first, inverts in x, and returns to the index by $n = x + x_*$, an exact identity that costs nothing.

Remark 1.7 (Which chapters the model covers, and which it does not). Chapters 2, 4 and 5 are governed by theorem 1.2: the rooted-tree counts and the swing factorial with $\alpha = 1$, the partition numbers with $\alpha = 1/2$ (table 1.1). Part 3 is not.

Claim repaired. It is natural, and it is asserted in the merge brief for this volume, that the case $\alpha = 1$ of theorem 1.2 covers the double factorial alongside the swing factorial and the rooted-tree counts. It does not. For the swing factorial $n!/\lfloor n/2\rfloor!^2$ the growth is $2^n n^{\pm1/2}$ and the model applies with

$a = \log 2$. For the double factorial,

$$\log\bigl((x+1)!!\bigr) = \frac{x+1}{2}\bigl(\log(x+1) - 1\bigr) + \mathcal{O}(\log x),$$

so the dominant phase is $\frac{1}{2}s\log s$ with $s = x+1$, not ax: no choice of the data $(C, a, \alpha, b, \delta, Q)$ with $\alpha \le 1$ reproduces it, since $s\log s/s^{\alpha} \to \infty$ for every $\alpha \le 1$. The three double factorial articles are consistent with this; they never claim membership of the exponential–power family, and they invert the phase $\frac{s}{2}(\log s - 1)$ directly. **Correction.** The chapter states the reversion for an axiomatized *dominant core* (theorems 1.8 and 1.29) of which both the linear-plus-logarithmic phase $aX + b\log X$ and the factorial phase $X(\kappa\log X + d)$ are instances, both exactly invertible by Lambert W (theorems 1.20 and 1.23). The theorem named in the Part 0 interface, theorem 1.25, is stated for the exponential–power model as specified, and theorem 1.29 is the version Part 3 cites.

Definition 1.8 (Admissible dominant core)**.** Let $X_* \ge 0$. A function $\Phi_0 \in C^2((X_*, \infty))$ is an *admissible core* when

1. $\Phi_0'(X) > 0$ for $X > X_*$ and $\Phi_0(X) \to +\infty$ as $X \to +\infty$;
2. Φ_0' is eventually monotone and $\Phi_0'(X) \to \Lambda_\infty \in (0, +\infty]$;
3. $X\Phi_0'(X) \to +\infty$.

Its *core slope* is Φ_0'. Given a target L in the range of Φ_0, the *core solution* $X = X(L)$ is the unique large solution of $\Phi_0(X) = L$.

The two cores used in this volume are

$$\Phi_0(X) = aX + b\log X \quad \text{(Part 2, 4, 5)}, \qquad \Phi_0(X) = X\,(\kappa\log X + d) \quad \text{(Part 3)}, \tag{1.9}$$

with $a, \kappa > 0$. Both are admissible on a suitable half-line, and both are inverted exactly by W in section 1.4.

1.3 The coefficient calculus

1.3.1 Two families of partial Bell polynomials

Definition 1.9 (Ordinary partial Bell polynomials)**.** For a sequence $\gamma = (\gamma_1, \gamma_2, \dots)$ in R and $n, k \in \mathbb{N}$ put

$$\widehat{B}_{n,k}(\gamma) := [t^n]\Bigl(\sum_{r\ge1}\gamma_r t^r\Bigr)^k. \tag{1.10}$$

Explicitly,

$$\widehat{B}_{n,k}(\gamma) = \sum_{\substack{m_1, m_2, \dots \ge 0\\ \sum_r m_r = k,\ \sum_r r m_r = n}} \frac{k!}{\prod_{r\ge1} m_r!}\prod_{r\ge1}\gamma_r^{m_r}. \tag{1.11}$$

In particular $\widehat{B}_{0,0} = 1$, $\widehat{B}_{n,0} = 0$ for $n \ge 1$, $\widehat{B}_{n,k} = 0$ for $k > n$, $\widehat{B}_{n,n} = \gamma_1^n$, and $\widehat{B}_{n,k}$ depends only on $\gamma_1, \dots, \gamma_{n-k+1}$.

Definition 1.10 (Exponential partial Bell polynomials)**.** For $x = (x_1, x_2, \dots)$ in R and $n \ge k \ge 0$ put

$$B_{n,k}(x_1, x_2, \dots) := n! \sum_{\substack{m_1, m_2, \dots \ge 0\\ \sum_j m_j = k,\ \sum_j j m_j = n}} \prod_{j\ge1}\frac{1}{m_j!}\Bigl(\frac{x_j}{j!}\Bigr)^{m_j}, \tag{1.12}$$

and let $B_n := \sum_{k=0}^{n} B_{n,k}$ be the complete polynomial. These are characterized by the generating identities

$$\frac{1}{k!}\Big(\sum_{j\ge 1} x_j \frac{t^j}{j!}\Big)^k = \sum_{n\ge k} B_{n,k}(x_1, x_2, \dots)\frac{t^n}{n!}, \tag{1.13}$$

$$\exp\Big(\sum_{j\ge 1} x_j \frac{t^j}{j!}\Big) = \sum_{n\ge 0} B_n(x_1, \dots, x_n)\frac{t^n}{n!}. \tag{1.14}$$

Lemma 1.11 (Conversion). *For all $n \ge k \ge 0$,*

$$\widehat{B}_{n,k}(\gamma_1, \gamma_2, \dots) = \frac{k!}{n!}\, B_{n,k}(1!\,\gamma_1,\, 2!\,\gamma_2, \dots). \tag{1.15}$$

Proof. Set $x_j = j!\,\gamma_j$ in (1.13). The left-hand side becomes $(k!)^{-1}\big(\sum_j \gamma_j t^j\big)^k$, whose coefficient of t^n is $\widehat{B}_{n,k}(\gamma)/k!$; the right-hand side has coefficient $B_{n,k}(1!\gamma_1, \dots)/n!$. □

1.3.2 Powers and logarithms of a unit series

Lemma 1.12 (Powers and logarithms). *Let $w = \sum_{r\ge 1} \gamma_r t^r \in tR[[t]]$ and $n \ge 1$.*

1. *For every β in R (or in any commutative $\mathbb{Q}$-algebra containing R),*

$$[t^n]\,(1+w)^\beta = \sum_{k=1}^{n} \binom{\beta}{k} \widehat{B}_{n,k}(\gamma), \qquad \binom{\beta}{k} = \frac{\beta(\beta-1)\cdots(\beta-k+1)}{k!}. \tag{1.16}$$

2. *For an integer $j \ge 1$,*

$$[t^n]\,(1+w)^{-j} = \sum_{k=0}^{n} (-1)^k \binom{j+k-1}{k} \widehat{B}_{n,k}(\gamma). \tag{1.17}$$

3.
$$[t^n]\,\log(1+w) = \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k} \widehat{B}_{n,k}(\gamma). \tag{1.18}$$

Equivalently, in exponential form, for $U(t) = 1 + \sum_{j\ge 1} u_j t^j$,

$$[t^N]\,U(t)^\beta = \frac{1}{N!}\sum_{k=0}^{N} \beta^{\underline{k}}\, B_{N,k}(1!u_1, 2!u_2, \dots), \qquad \beta^{\underline{k}} = \beta(\beta-1)\cdots(\beta-k+1), \tag{1.19}$$

$$[t^N]\,\log U(t) = \frac{1}{N!}\sum_{k=1}^{N} (-1)^{k-1}(k-1)!\, B_{N,k}(1!u_1, 2!u_2, \dots). \tag{1.20}$$

Proof. All four identities are the substitution of w into a formal series in one variable. Since $w \in tR[[t]]$, the family $(w^k)_{k\ge 0}$ is summable in the t-adic topology and $[t^n]\,w^k = 0$ for $k > n$, so every sum displayed is finite and the substitution is legitimate. Now $(1+w)^\beta = \sum_{k\ge 0}\binom{\beta}{k} w^k$ gives (1.16) on applying theorem 1.9; $\binom{-j}{k} = (-1)^k\binom{j+k-1}{k}$ gives (1.17); and $\log(1+w) = \sum_{k\ge 1}(-1)^{k+1} w^k/k$ gives (1.18). For the exponential forms write $U = 1 + w$ with $\gamma_j = u_j$ and substitute (1.15) into (1.16) and (1.18), using $\binom{\beta}{k} = \beta^{\underline{k}}/k!$ and $(-1)^{k+1}/k = (-1)^{k-1}(k-1)!/k!$. □

Corollary 1.13 (Exponential and logarithmic jets). *Let $D(t) = \sum_{j\ge 1} d_j t^j \in tR[[t]]$ and $A(t) = \exp D(t) = \sum_{n\ge 0} A_n t^n$. Then $A_0 = 1$,*

$$A_n = \frac{1}{n!} B_n(1!d_1, 2!d_2, \dots, n!d_n) = \sum_{\substack{k_1, k_2, \dots \ge 0\\ \sum_j j k_j = n}} \prod_{j\ge 1} \frac{d_j^{k_j}}{k_j!}, \tag{1.21}$$

and the coefficients obey the triangular recurrence

$$nA_n = \sum_{j=1}^{n} j\, d_j\, A_{n-j} \qquad (n \geq 1). \tag{1.22}$$

Conversely, if $A = 1 + \sum_{n\geq 1} A_n t^n$ *is a unit series then* $d_n = [t^n]\,\log A$ *is given by* (1.20).

Proof. The first equality is (1.14) with $x_j = j!d_j$; the second is the expansion of $\exp$ as a product over j. For (1.22) differentiate $A = \exp D$ to get $A' = D'A$ and compare coefficients of t^{n-1}. The converse is (1.20). □

1.3.3 Lagrange–Bürmann inversion and the fixed-point form

Theorem 1.14 (Lagrange–Bürmann)**.** *Let* $\varphi \in R[[w]]$ *with* $\varphi(0)$ *invertible in* R*. There is a unique* $W \in tR[[t]]$ *with*

$$W = t\,\varphi(W), \tag{1.23}$$

and for every $G \in R[[w]]$ *and every* $n \geq 1$*,*

$$[t^n]\,G\big(W(t)\big) = \frac{1}{n}[w^{n-1}]\,\Big(G'(w)\,\varphi(w)^n\Big). \tag{1.24}$$

In particular

$$[t^n]\,W(t) = \frac{1}{n}[w^{n-1}]\,\varphi(w)^n. \tag{1.25}$$

Proof. Existence and uniqueness: (1.23) determines $[t^n]\,W$ from $[t^{<n}]\,W$, because $t\varphi(W)$ modulo t^{n+1} depends only on W modulo t^n.

For (1.24) work in the field of formal Laurent series $R((w))$ and let Res denote the coefficient of w^{-1}. Two facts are needed.

(a) $\operatorname{Res} f' = 0$ *for every* $f \in R((w))$, since w^{-1} has no antiderivative among the monomials: $(w^k)' = kw^{k-1}$ is a multiple of w^{-1} only for $k = 0$, when it vanishes.

(b) Change of variables. Put $\psi(w) := w/\varphi(w)$, a series in $wR[[w]]$ with $\psi'(0) = \varphi(0)^{-1}$ invertible; then W is the compositional inverse of ψ, so $\psi(W(t)) = t$. For $f \in R((w))$,

$$\operatorname{Res}_w\big(f(w)\big) = \operatorname{Res}_t\big(f(W(t))\,W'(t)\big). \tag{1.26}$$

Indeed both sides are R-linear and continuous, so it suffices to check $f = w^k$. For $k \neq -1$, $W^kW' = \big(W^{k+1}/(k+1)\big)'$ and (a) gives $0 = \operatorname{Res}_w w^k$. For $k = -1$, write $W = t\vartheta(t)$ with ϑ a unit; then $W'/W = t^{-1} + \vartheta'/\vartheta$ and the second summand lies in $R[[t]]$, so $\operatorname{Res}_t(W'/W) = 1 = \operatorname{Res}_w w^{-1}$.

Now apply (1.26) to $f(w) := G(w)\psi(w)^{-n-1}\psi'(w)$, noting that $f(W(t))W'(t) = G(W(t))\,t^{-n-1}\,\frac{\mathrm{d}}{\mathrm{d}t}\psi(W(t)) = G(W(t))\,t^{-n-1}$ because $\psi(W(t)) = t$:

$$[t^n]\,G(W(t)) = \operatorname{Res}_t\big(G(W(t))\,t^{-n-1}\big) = \operatorname{Res}_w\big(G(w)\,\psi(w)^{-n-1}\psi'(w)\big).$$

Since $\psi^{-n-1}\psi' = -\frac{1}{n}\big(\psi^{-n}\big)'$, fact (a) and the product rule give

$$\operatorname{Res}_w\big(G\,\psi^{-n-1}\psi'\big) = -\frac{1}{n}\operatorname{Res}_w\big(G\cdot(\psi^{-n})'\big) = \frac{1}{n}\operatorname{Res}_w\big(G'\,\psi^{-n}\big) = \frac{1}{n}\operatorname{Res}_w\big(G'(w)\,\varphi(w)^n w^{-n}\big),$$

which is (1.24). Taking $G(w) = w$ gives (1.25). □

The reversions of this volume all take the form $U = \Psi(t, U)$ in which the rôle of $\varphi(0)$ is played by $\Psi(t, 0)$, an element of $tR[[t]]$, which is *never* invertible. Theorem 1.14 therefore does not apply directly, and the following universality lemma is what makes the passage legitimate.

Lemma 1.15 (Lagrange's formula without an invertibility hypothesis)**.** *Let S be any commutative $\mathbb{Q}$-algebra and $\varphi \in S[[w]]$ arbitrary (in particular $\varphi(0)$ need not be invertible). Let $W \in zS[[z]]$ be the unique solution of $W = z\varphi(W)$. Then for every $m \geq 1$,*

$$[z^m]\,W = \frac{1}{m}[w^{m-1}]\,\varphi(w)^m. \tag{1.27}$$

Proof. Existence and uniqueness of W hold as in theorem 1.14 (the recursion is z-adic and uses no invertibility). Fix m. Iterating $W = z\varphi(W)$ shows that $[z^m]\,W$ is a universal polynomial with integer coefficients in $\varphi_0, \varphi_1, \ldots, \varphi_{m-1}$, where $\varphi = \sum_i \varphi_i w^i$; the right-hand side of (1.27) is a universal polynomial with rational coefficients in the same variables. Both sides are therefore elements P_m, $\widetilde{P}_m$ of the polynomial ring $\mathcal{P} := \mathbb{Q}[\varphi_0, \ldots, \varphi_{m-1}]$, and they are obtained from φ by the unique $\mathbb{Q}$-algebra homomorphism $\mathcal{P} \to S$ sending $\varphi_i \mapsto \varphi_i$. It thus suffices to prove $P_m = \widetilde{P}_m$ in $\mathcal{P}$. The localization map $\mathcal{P} \to \mathcal{P}[\varphi_0^{-1}]$ is injective ($\mathcal{P}$ is an integral domain), and in $\mathcal{P}[\varphi_0^{-1}]$ the element φ_0 is invertible, so theorem 1.14 applies and gives $P_m = \widetilde{P}_m$ there. Injectivity transports the equality back to $\mathcal{P}$, and the homomorphism transports it to S. □

Lemma 1.16 (Lagrange fixed-point formula)**.** *Let $\Psi \in tR[[t,u]]$, that is, a formal series in t and u each of whose monomials has t-degree at least one. Then the equation*

$$U(t) = \Psi\big(t, U(t)\big) \tag{1.28}$$

has a unique solution $U \in tR[[t]]$, and for every $n \geq 1$

$$[t^n]\,U = \sum_{m=1}^{n} \frac{1}{m}[t^n u^{m-1}]\,\Psi(t,u)^m. \tag{1.29}$$

Proof. Uniqueness and existence: $\Psi(t,U)$ modulo t^{n+1} depends only on U modulo t^n because Ψ is divisible by t; so (1.28) determines the coefficients of U recursively, and the resulting series solves the equation by t-adic continuity.

For (1.29) apply theorem 1.15 over the ring $S := R[[t]]$ with $\varphi(w) := \Psi(t,w) \in S[[w]]$ and marker z: the unique $\widetilde{U} \in zS[[z]]$ with $\widetilde{U} = z\varphi(\widetilde{U})$ satisfies

$$[z^m]\,\widetilde{U} = \frac{1}{m}[w^{m-1}]\,\Psi(t,w)^m \in S.$$

Since $\Psi \in tR[[t,u]]$, we have $\Psi^m \in t^m R[[t,u]]$, whence $[t^n]\,[w^{m-1}]\,\Psi(t,w)^m = 0$ for $m > n$. Consequently the family $\big([z^m]\,\widetilde{U}\big)_{m\geq 1}$ is summable in the t-adic topology of S, the specialization $z = 1$ is defined coefficientwise in t, and the resulting series $U := \sum_{m\geq 1}[z^m]\,\widetilde{U}$ satisfies $U = \Psi(t,U)$ by the same summability. Uniqueness identifies it with the solution of (1.28), and extracting $[t^n]$ gives (1.29) with the sum truncated at $m = n$. □

Source claim. Swing_Factorial_Transseries_Article, Lemma *Lagrange fixed-point formula*, proves (1.29) by writing "Ordinary Lagrange–Bürmann inversion in z gives $U = \sum_{m\geq 1} z^m m^{-1}[u^{m-1}]\Psi(t,u)^m$". The same step, with the same silence, appears in Partition_Number_Transseries_and_Inverse ("Introduce $u = z\Phi_\rho(u)$. Lagrange–Bürmann gives …"), in Partition_Numbers_Transseries_and_Inverse, and in double_factorial_transseries-2. **Defect.** The classical Lagrange–Bürmann theorem in the form (1.25) requires $\varphi(0)$ to be invertible in the coefficient ring. Here $\varphi(0) = \Psi(t,0) \in tR[[t]]$, a non-unit, and the residue proof of theorem 1.14 breaks at the change of variables: $\psi = w/\varphi$ is not a well-defined element of $wR[[t]][[w]]$. **Repair.** Theorem 1.15: the identity is a polynomial identity in the coefficients of φ over $\mathbb{Q}$, hence holds in the polynomial ring by injectivity of localization at φ_0, hence in every $\mathbb{Q}$-algebra. double_factorial_transseries-2 comes closest to noticing the issue — "Although A is a divergent Gevrey series, it is a valid formal power series with nonzero linear coefficient" — but that remark addresses convergence, not invertibility, and its own G_Λ does have an invertible linear coefficient, so the gap there is only in the inherited lemma.

Lemma 1.17 (Fixed-point formula with a quadratic self-interaction). *Let $\Psi = h + f$ with $h \in u^2R[[u]]$ and $f \in tR[[t,u]]$. Then $U = \Psi(t,U)$ has a unique solution $U \in tR[[t]]$, and for every $n \geq 1$*

$$[t^n]\, U = \sum_{m=1}^{2n-1} \frac{1}{m} [t^n u^{m-1}]\, \Psi(t,u)^m. \tag{1.30}$$

Proof. Existence and uniqueness. Write $U = U_{<n} + e_n t^n + \cdots$. Then $h(U) = h(U_{<n}) + h'(U_{<n})\, e_n t^n + \cdots$, and $h'(U_{<n}) \in tR[[t]]$ because $h' \in uR[[u]]$ and $U_{<n} \in tR[[t]]$; so e_n does not occur in $[t^n]\, h(U)$. It does not occur in $[t^n]\, f(t,U)$ either, since f is divisible by t. Hence the equation determines e_n, as $[t^n]\, \Psi(t, U_{<n})$, from $U_{<n}$; the resulting series solves the equation by t-adic continuity.

The closed formula. Introduce a marker z and let $\widetilde{U} \in zS[[z]]$, $S := R[[t]]$, solve $\widetilde{U} = z\Psi(t, \widetilde{U})$. Theorem 1.15 gives $[z^m]\, \widetilde{U} = m^{-1}[u^{m-1}]\, \Psi(t,u)^m$. We claim

$$[t^n u^{m-1}]\, \Psi(t,u)^m = 0 \qquad \text{whenever } m > 2n-1. \tag{1.31}$$

Expand $\Psi^m = (h+f)^m$ into 2^m products; a product using h in σ of the m slots and f in the remaining $m - \sigma$ has u-degree at least 2σ and t-degree at least $m - \sigma$. For the u-degree to equal $m-1$ we need $2\sigma \leq m-1$, hence the t-degree is at least $m - \frac{m-1}{2} = \frac{m+1}{2}$; so $[t^n]$ of that product vanishes unless $n \geq (m+1)/2$, which is (1.31). Consequently the family $([z^m]\, \widetilde{U})_{m \geq 1}$ is t-adically summable, the specialization $z = 1$ is defined coefficientwise, and, again by summability, the resulting series solves $U = \Psi(t,U)$. Uniqueness identifies it with U, and extracting $[t^n]$ gives (1.30). ☐

Theorem 1.16 is the case $h = 0$ of theorem 1.17, in which the argument above sharpens the range of m from $2n-1$ to n.

1.3.4 Perturbed inversion around an exactly invertible core

The reversions carried out in Chapters 1–4 all reduce to one step: a core F that can be inverted exactly is perturbed by a correction E, and the inverse of $F + \varepsilon E$ is required to all orders in ε. The answer is a single differential formula, stated here once. It is the tool behind theorem 4.62 in the ordinate of the partition chapter and behind theorem 5.39 in the phase coordinate of the swing chapter, and those two are the same identity in different letters.

Theorem 1.18 (Perturbed inversion). *Let F and E be analytic in a neighbourhood of $\mu_0 \in \mathbb{C}$, let $F'(\mu_0) \neq 0$, and put $y := F(\mu_0)$. Write $\mu_0(\cdot)$ for the local inverse of F, so that $\mu_0(y) = \mu_0$ and $\mu_0'(y) = 1/F'(\mu_0(y))$, and let*

$$\mathcal{D} := \frac{1}{F'(\mu)} \frac{\mathrm{d}}{\mathrm{d}\mu}, \tag{1.32}$$

which is the derivation $\mathrm{d}/\mathrm{d}y$ read in the coordinate $y = F(\mu)$. Then there are a neighbourhood $\mathcal{U}$ of μ_0 and an $\varepsilon_0 > 0$ such that for $|\varepsilon| < \varepsilon_0$ the equation

$$F(\mu) + \varepsilon E(\mu) = y \tag{1.33}$$

has exactly one root $\mu(y;\varepsilon)$ in $\mathcal{U}$; it is analytic in ε there, and

$$\mu(y;\varepsilon) = \mu_0(y) + \sum_{M \geq 1} \frac{(-\varepsilon)^M}{M!}\, \mathcal{D}^{M-1}\left(\frac{E(\mu_0(y))^M}{F'(\mu_0(y))}\right), \tag{1.34}$$

the series converging for $|\varepsilon| < \varepsilon_0$, with Mth coefficient

$$\frac{(-1)^M}{M!}\, \mathcal{D}^{M-1}\left(\frac{E(\mu_0)^M}{F'(\mu_0)}\right) = \frac{(-1)^M}{M} \operatorname*{Res}_{\mu=\mu_0} \frac{E(\mu)^M}{\big(F(\mu) - y\big)^M}. \tag{1.35}$$

If moreover $E = \sum_j \varepsilon_j E_j$ with finitely many E_j analytic near μ_0, then for every multi-index $\boldsymbol{m} \neq \boldsymbol{0}$ with $M := |\boldsymbol{m}|$ the coefficient of $\boldsymbol{\varepsilon}^{\boldsymbol{m}}$ in $\mu - \mu_0$ is

$$V_{\boldsymbol{m}} = \frac{(-1)^M}{\boldsymbol{m}!}\,\mathcal{D}^{M-1}\left(\frac{E^{\boldsymbol{m}}(\mu_0)}{F'(\mu_0)}\right) = \frac{(-1)^M(M-1)!}{\boldsymbol{m}!}\operatorname*{Res}_{\mu=\mu_0}\frac{E^{\boldsymbol{m}}(\mu)}{\big(F(\mu)-F(\mu_0)\big)^M}, \tag{1.36}$$

where $E^{\boldsymbol{m}} := \prod_j E_j^{m_j}$ and $\boldsymbol{m}! := \prod_j m_j!$.

Proof. Let γ be a small positively oriented circle about μ_0, contained with its interior $\mathcal{U}$ in the common domain of analyticity and enclosing no zero of $F(\cdot) - y$ other than μ_0; such a γ exists because $F'(\mu_0) \neq 0$ makes μ_0 a simple zero, hence isolated. Put $m_\gamma := \min_\gamma |F - y| > 0$ and $M_\gamma := \max_\gamma |E|$, and let $\varepsilon_0 := m_\gamma/(M_\gamma + 1)$. For $|\varepsilon| < \varepsilon_0$ we have $|\varepsilon E| < |F - y|$ on γ, so Rouché gives exactly one zero $\mu(y;\varepsilon)$ of $F - y + \varepsilon E$ inside γ, and it is simple. The argument principle, applied to that function, expresses it as

$$\mu(y;\varepsilon) = \frac{1}{2\pi\mathrm{i}}\int_\gamma \zeta\,\mathrm{d}_\zeta \log\big(F(\zeta) - y + \varepsilon E(\zeta)\big), \tag{1.37}$$

which is analytic in ε on $|\varepsilon| < \varepsilon_0$ because the integrand is and γ is compact.

Subtract from (1.37) the same expression at $\varepsilon = 0$, whose value is μ_0, and expand

$$\log\Big(1 + \varepsilon\frac{E}{F-y}\Big) = \sum_{M\geq 1}\frac{(-1)^{M-1}}{M}\,\varepsilon^M\,\frac{E^M}{(F-y)^M},$$

uniformly convergent on γ since $|\varepsilon E/(F-y)| < 1$ there. Integrating by parts once turns $\zeta\,\mathrm{d}_\zeta(\cdot)$ into $-(\cdot)\,\mathrm{d}\zeta$ (the boundary term vanishes on a closed contour), so the coefficient of ε^M is

$$\frac{(-1)^M}{M}\operatorname*{Res}_{\zeta=\mu_0}\frac{E(\zeta)^M}{\big(F(\zeta)-F(\mu_0)\big)^M},$$

which is the right-hand side of (1.35).

It remains to identify that residue with the differential expression. Change coordinate by $w = F(\zeta)$, $\zeta = \mu_0(w)$, $\mathrm{d}\zeta = \mu_0'(w)\,\mathrm{d}w$, legitimate on a neighbourhood of μ_0 because $F'(\mu_0) \neq 0$:

$$\operatorname*{Res}_{\zeta=\mu_0}\frac{E(\zeta)^M}{(F(\zeta)-y)^M}\,\mathrm{d}\zeta = \operatorname*{Res}_{w=y}\frac{E(\mu_0(w))^M\,\mu_0'(w)}{(w-y)^M}\,\mathrm{d}w = \frac{1}{(M-1)!}\frac{\mathrm{d}^{M-1}}{\mathrm{d}y^{M-1}}\left(\frac{E(\mu_0(y))^M}{F'(\mu_0(y))}\right),$$

the last step being the residue of a pole of order M. Multiplying by $(-1)^M/M$ and using $\mathrm{d}/\mathrm{d}y = \mathcal{D}$ gives (1.34) and (1.35).

For (1.36) replace E by $\sum_j \varepsilon_j E_j$ and expand the Mth power by the multinomial theorem: the monomial $\boldsymbol{\varepsilon}^{\boldsymbol{m}}$ occurs with coefficient $M!/\boldsymbol{m}!$, which cancels the $M!$ in (1.34). □

The identity (1.34) is needed in Chapter 4 with a correction E that is a *divergent* formal series, where no contour is available. What survives is the coefficientwise identity, and it survives for the reason the Lagrange fixed-point formula does.

Corollary 1.19 (Formal form of the reversion). *Fix $M \geq 1$ and let $\mathcal{P} := \mathbb{Q}[e_0, \ldots, e_{M-1}, f_2, \ldots, f_M][f_1^{-1}]$. Both of the following are elements of $\mathcal{P}$, obtained by writing out the indicated recursions with e_i and f_i standing for $E^{(i)}(\mu_0)$ and $F^{(i)}(\mu_0)$:*

1. *the coefficient μ_M of ε^M produced by solving* (1.33) *by formal Taylor iteration, i.e. by substituting $\mu = \mu_0 + \sum_{k\geq 1}\mu_k\varepsilon^k$, expanding F and E in their Taylor jets at μ_0, and equating coefficients of ε^M;*

2. *the expression $\frac{(-1)^M}{M!}\mathcal{D}^{M-1}\big(E(\mu_0)^M/F'(\mu_0)\big)$ of* (1.34)*, expanded by the Leibniz rule.*

They are equal in $\mathcal{P}$. Consequently (1.34) *and* (1.36) *hold coefficientwise in any commutative $\mathbb{Q}$-algebra carrying jets e_i, f_i with f_1 invertible — in particular when F and E are formal, possibly divergent, series, and when $\mathcal{D}$ is the derivation* (1.32) *written in any coordinate.*

Proof. Each recursion visibly terminates and involves only $e_0, \dots, e_{M-1}$, $f_2, \dots, f_M$ and divisions by f_1: in (i) the coefficient of ε^M in $F(\mu_0 + \sum_{k\geq 1} \mu_k \varepsilon^k) + \varepsilon E(\mu_0 + \sum_{k\geq 1} \mu_k \varepsilon^k)$ equals $f_1 \mu_M$ plus a polynomial in $f_2, \dots, f_M$, $e_0, \dots, e_{M-1}$ and $\mu_1, \dots, \mu_{M-1}$, so μ_M is determined by one division by f_1; in (ii) each application of $\mathcal{D} = f_1^{-1} \mathrm{d}/\mathrm{d}\mu$ lowers the available jet order by one and contributes one factor f_1^{-1}, and $M-1$ applications to a function of $e_0, \dots$ and $f_1, \dots$ reach at most e_{M-1} and f_M. So both lie in $\mathcal{P}$.

Choose N with $f_1^N \mu_M$ and $f_1^N \cdot$ (ii) both in $\mathcal{A} := \mathbb{Q}[e_0, \dots, e_{M-1}, f_1, \dots, f_M]$, and regard them as polynomial functions on $\mathbb{C}^{2M}$. Every point of $\mathbb{C}^{2M}$ with $f_1 \neq 0$ is realized by a pair of *polynomials*: take $\mu_0 = 0$, $E(\mu) := \sum_{i=0}^{M-1} e_i \mu^i / i!$ and $F(\mu) := \sum_{i=1}^{M} f_i \mu^i / i!$, which are entire with $F'(0) = f_1 \neq 0$ and with the prescribed jets. For such a pair theorem 1.18 applies, and the Taylor coefficients in ε of the analytic root $\mu(y;\varepsilon)$ satisfy the recursion of (i) — substituting the convergent expansion into (1.33) and equating coefficients is legitimate — so (i) and (ii) agree at that point. Two elements of $\mathcal{A}$ agreeing on the Zariski-dense open set $\{f_1 \neq 0\}$ of $\mathbb{C}^{2M}$ are equal in $\mathbb{C}[e_\bullet, f_\bullet]$, hence equal in $\mathcal{A}$ and, after dividing by f_1^N, in $\mathcal{P}$. The final assertion follows by applying the unique $\mathbb{Q}$-algebra homomorphism $\mathcal{P} \to S$ determined by the given jets, exactly as in theorem 1.15. □

1.4 The Lambert core

1.4.1 Exact inversion of the linear–logarithmic phase

Theorem 1.20 (Exact solution of the dominant block)**.** *Let $a > 0$, $b \in \mathbb{R}$, and consider, for $X > 0$,*

$$\Phi_0(X) := aX + b \log X = L. \tag{1.38}$$

1. *If $b = 0$ the unique solution is $X = L/a$.*
2. *If $b \neq 0$, put $W := (a/b)X$. Then* (1.38) *holds if and only if*

$$W \mathrm{e}^W = z(L), \qquad z(L) := \frac{a}{b} \, \mathrm{e}^{L/b}, \tag{1.39}$$

so that the solutions of (1.38) *are exactly*

$$X = \frac{b}{a} \, W_k\Big(\frac{a}{b} \, \mathrm{e}^{L/b}\Big) \tag{1.40}$$

as k ranges over the branches of the Lambert function for which $W_k(z(L))$ is defined.

3. Branch rule. *Suppose $b \neq 0$.*
   - *If $b > 0$ then $z(L) > 0$, equation* (1.38) *has exactly one positive solution for every real L, Φ_0 is strictly increasing on $(0, \infty)$, and $X = (b/a) W_0(z(L))$.*
   - *If $b < 0$ then $z(L) = -(a/|b|)\mathrm{e}^{-L/|b|}$ lies in $(-\mathrm{e}^{-1}, 0)$ precisely when*

$$L > L_c := |b| \Big(1 - \log \frac{|b|}{a}\Big) = \Phi_0\Big(\frac{|b|}{a}\Big), \tag{1.41}$$

and then (1.38) *has exactly two positive solutions, namely $(b/a)W_0(z(L)) < |b|/a < (b/a)W_{-1}(z(L))$. Φ_0 is strictly increasing on $(|b|/a, \infty)$ and the large solution is*

$$X = \frac{b}{a} \, W_{-1}\Big(\frac{a}{b} \, \mathrm{e}^{L/b}\Big) = \frac{|b|}{a} \Big(-W_{-1}\big(z(L)\big)\Big). \tag{1.42}$$

In both cases the large solution satisfies $X \to +\infty$ and $X = L/a + \mathcal{O}(\log L)$ as $L \to +\infty$.

4. Slope identities. *With* $W = (a/b)X$ *and* $b \neq 0$,

$$\Phi_0'(X) = a + \frac{b}{X} = a\,\frac{1+W}{W}, \qquad \frac{\mathrm{d}X}{\mathrm{d}L} = \frac{1}{\Phi_0'(X)} = \frac{1}{a}\,\frac{W}{1+W}. \tag{1.43}$$

Proof. (1) is immediate. For (2), multiply (1.38) by $1/b$ and exponentiate:

$$\mathrm{e}^{L/b} = \mathrm{e}^{(a/b)X}\,X = \mathrm{e}^{W}\cdot\frac{b}{a}\,W,$$

because $X = (b/a)W$. Multiplying by a/b gives (1.39); the step is reversible, and no logarithm of a negative number occurs, which is what makes the argument uniform in the sign of b. By definition, the solutions of $W\mathrm{e}^{W} = z$ are the values $W_k(z)$, whence (1.40).

(3) We have $\Phi_0'(X) = a + b/X$. For $b > 0$ this is positive on $(0,\infty)$ and Φ_0 increases from $-\infty$ to $+\infty$, so there is exactly one solution; correspondingly $W = (a/b)X > 0$ and W_0 is the unique real branch on $(0,\infty)$. For $b < 0$, Φ_0' vanishes only at $X_* = |b|/a$, is negative on $(0, X_*)$ and positive on (X_*,∞); Φ_0 decreases from $+\infty$ to $\Phi_0(X_*) = |b| - |b|\log(|b|/a) = L_c$ and then increases back to $+\infty$. Hence there are two positive solutions exactly when $L > L_c$, one in each monotonicity interval. On the W side $W = (a/b)X < 0$, and the two real branches of W on $(-\mathrm{e}^{-1}, 0)$ are W_0, with values in $(-1,0)$, and W_{-1}, with values in $(-\infty,-1)$; the map $W \mapsto (b/a)W$ reverses order, so W_{-1} produces the larger X, namely the one exceeding $|b|/a$. Also $z(L) > -\mathrm{e}^{-1}$ is equivalent to $(a/|b|)\mathrm{e}^{-L/|b|} < \mathrm{e}^{-1}$, that is to (1.41). Finally $aX = L - b\log X$ with $X \to \infty$ gives $aX = L + \mathcal{O}(\log L)$.

(4) $a + b/X = a + b\cdot\dfrac{a}{bW} = a\big(1 + W^{-1}\big)$, and $\mathrm{d}X/\mathrm{d}L$ is the reciprocal of Φ_0' by the inverse function theorem, applicable because $\Phi_0' \neq 0$ on the relevant interval. □

Corollary 1.21 (Branch rule). *For* $b \neq 0$ *the branch of the Lambert function carrying the large solution of* $L = aX + b\log X$ *is determined by the sign of* b *alone:* W_0 *if* $b > 0$, W_{-1} *if* $b < 0$.

Remark 1.22 (Why the branch is not a detail). Three of the four chapters have $b < 0$ and therefore use W_{-1}; the odd phase of the swing factorial has $b = +1/2$ and uses W_0. Swing_Factorial_Transseries_Article puts the point sharply: the principal branch "would produce the small root of the simplified equation and is irrelevant to the large-index inverse". For $b < 0$ the two roots are the two sides of a genuine fold at $X_* = |b|/a$, and choosing wrongly yields an object all of whose asymptotic corrections are then computed correctly about the wrong point.

1.4.2 The factorial core

Proposition 1.23 (Exact inversion of a log-linear phase). *Let* $\kappa > 0$ *and* $d \in \mathbb{R}$, *and consider*

$$\Phi_0(X) := X\big(\kappa\log X + d\big) = L, \qquad X > 0. \tag{1.44}$$

For $L > 0$ *put*

$$v := W_0\!\left(\frac{L}{\kappa}\,\mathrm{e}^{d/\kappa}\right), \qquad X := \exp\!\Big(v - \frac{d}{\kappa}\Big). \tag{1.45}$$

Then X *is the unique solution of* (1.44) *with* $\kappa\log X + d > 0$, *and*

$$\Phi_0'(X) = \kappa\log X + d + \kappa = \kappa\,(v+1), \qquad \frac{\mathrm{d}X}{\mathrm{d}L} = \frac{1}{\kappa(v+1)}. \tag{1.46}$$

Moreover Φ_0 *is an admissible core in the sense of theorem 1.8 on* $\big(\mathrm{e}^{-(d+\kappa)/\kappa}, \infty\big)$, *with* $\Phi_0'(X) \to +\infty$.

Proof. Put $s = \log X$. Then (1.44) reads $\mathrm{e}^{s}(\kappa s + d) = L$, that is $\big(s + \frac{d}{\kappa}\big)\mathrm{e}^{s+d/\kappa} = (L/\kappa)\mathrm{e}^{d/\kappa}$. The right-hand side is positive, so W_0 is the unique real branch and $s + d/\kappa = v$, which is (1.45). Uniqueness under the stated side condition holds because $\kappa s + d > 0$ forces $s + d/\kappa > 0$, and $\sigma \mapsto \sigma\mathrm{e}^{\sigma}$ is injective on $(0,\infty)$. Differentiating, $\Phi_0'(X) = \kappa\log X + d + \kappa = \kappa(s + d/\kappa) + \kappa = \kappa(v+1)$. Admissibility: Φ_0' is increasing, tends to $+\infty$, is positive exactly for $\log X > -(d+\kappa)/\kappa$, and $X\Phi_0'(X) \to \infty$. □

Remark 1.24 (The double factorial normalization). Part 3 uses $\kappa = \frac{1}{2}$, $d = -\frac{1}{2}$ and the centred variable $s = x+1$, so that $\Phi_0(s) = \frac{s}{2}(\log s - 1)$, $(L/\kappa)\mathrm{e}^{d/\kappa} = 2L/\mathrm{e}$, $X = \mathrm{e}^{v+1}$ with $v = W_0(2L/\mathrm{e})$, and $\Phi_0'(X) = \frac{1}{2}(v+1) = \frac{1}{2}\log X$. This reproduces the core of all three double factorial articles; the quantity they call Λ is $v+1 = \log X$.

1.5 All-orders reversion around the core

1.5.1 The Lambert-centred inverse

Theorem 1.25 (All-orders reversion around the Lambert core). *Let R be a commutative $\mathbb{Q}$-algebra, let $a \in R$ be invertible, let $b \in R$, and let $Q = \sum_{j\geq 1} q_j t^j \in tR[[t]]$. Define*

$$\Psi(t,u) := -\frac{1}{a}\left[b\log(1+tu) + Q\left(\frac{t}{1+tu}\right)\right]. \tag{1.47}$$

1. *$\Psi \in tR[[t,u]]$.*
2. *There is a unique $U = \sum_{n\geq 1} c_n t^n \in tR[[t]]$ with $U = \Psi(t,U)$, and*
$$c_n = \sum_{m=1}^{n} \frac{1}{m}[t^n u^{m-1}]\,\Psi(t,u)^m \qquad (n \geq 1). \tag{1.48}$$
3. Triangular Bell recurrence. *Put $\gamma_1 := 0$ and $\gamma_r := c_{r-1}$ for $r \geq 2$. Then, for $n \geq 1$,*
$$c_n = -\frac{1}{a}\left[b\sum_{k=1}^{n}\frac{(-1)^{k+1}}{k}\widehat{B}_{n,k}(\gamma) + \sum_{j=1}^{n} q_j \sum_{k=0}^{n-j}(-1)^k\binom{j+k-1}{k}\widehat{B}_{n-j,k}(\gamma)\right], \tag{1.49}$$
and only $c_1,\dots,c_{n-1}$ occur on the right-hand side. In particular c_n lies in the $\mathbb{Q}$-subalgebra of R generated by a^{-1}, b and $q_1,\dots,q_n$; no logarithm appears.
4. Formal exactness. *In $t^{-1}R[[t]]$ put*
$$\mathsf{x} := t^{-1} + U(t) = t^{-1} + \sum_{n\geq 1} c_n t^n. \tag{1.50}$$
Then $t\mathsf{x} = 1 + tU \in 1 + tR[[t]]$ and $1/\mathsf{x} = t/(1+tU) \in tR[[t]]$, and the following identity holds in $R[[t]]$:
$$a\big(\mathsf{x} - t^{-1}\big) + b\log(t\mathsf{x}) + Q\big(1/\mathsf{x}\big) = 0. \tag{1.51}$$
Equivalently: writing $t = X^{-1}$ and $x = X + \sum_{n\geq 1} c_n X^{-n}$, the quantity $ax + b\log x + Q(1/x)$ equals $aX + b\log X$ to all orders.
5. Analytic version. *Let F be differentiably of exponential–power type with data $(C,a,1,b,1,Q)$ on $[x_0,\infty)$ (theorem 1.2), strictly increasing there, with $F(x) \to \infty$. For y in the range of F set $L = \log(y/C)$ and let $X = X(y)$ be the large core solution of $L = aX + b\log X$ furnished by theorem 1.20. Then $X(y) \to \infty$ and, for every fixed $N \geq 0$,*
$$F^{-1}(y) = X + \sum_{n=1}^{N}\frac{c_n}{X^n} + \mathcal{O}\big(X^{-N-1}\big) \qquad (y \to \infty). \tag{1.52}$$

Proof. (1) $\log(1+tu) = \sum_{k\geq 1}(-1)^{k+1}(tu)^k/k \in tR[[t,u]]$, and $t/(1+tu) = t\sum_{k\geq 0}(-tu)^k \in tR[[t,u]]$, so $(t/(1+tu))^j \in t^jR[[t,u]]$ and the sum over j is summable; hence $Q(t/(1+tu)) \in tR[[t,u]]$.

(2) is theorem 1.16.

(3) Put $w := tU = \sum_{r\ge 1} \gamma_r t^r$ with γ as stated; $\gamma_1 = 0$ because $U \in tR[[t]]$. Then $[t^n]\ \log(1+w)$ is given by (1.18), while

$$[t^n]\, Q\left(\frac{t}{1+w}\right) = \sum_{j\ge 1} q_j\, [t^{n-j}]\, (1+w)^{-j} = \sum_{j=1}^{n} q_j \sum_{k=0}^{n-j} (-1)^k \binom{j+k-1}{k} \widehat{B}_{n-j,k}(\gamma)$$

by (1.17). Since $c_n = [t^n]\ \Psi(t, U)$, this is (1.49). For triangularity, $\widehat{B}_{n,k}(\gamma)$ depends only on $\gamma_1, \dots, \gamma_{n-k+1}$, that is on $c_1, \dots, c_{n-k}$; in the first sum $k \ge 1$ and in the second $j \ge 1$, so c_n never occurs. (In fact $\widehat{B}_{n,k}(\gamma) = 0$ for $2k > n$, because $\gamma_1 = 0$.)

(4) With $\mathsf{x} = t^{-1}(1+w)$, $w = tU$, one has $a(\mathsf{x} - t^{-1}) = aU$, $\log(t\mathsf{x}) = \log(1+w)$ and $1/\mathsf{x} = t/(1+w)$, so the left-hand side of (1.51) equals $aU + b\log(1+tU) + Q(t/(1+tU)) = a\big(U - \Psi(t,U)\big) = 0$.

(5) Fix $N \ge 0$, set $t := 1/X$, $U_N(t) := \sum_{n=1}^{N} c_n t^n$ and $x_N := X + U_N(t) = X\big(1 + tU_N(t)\big)$. By theorem 1.20(3), $X \to \infty$ and $X \sim L/a$; since $tU_N(t) = \mathcal{O}(t^2)$ we get $x_N = X\big(1 + \mathcal{O}(X^{-2})\big)$, so $x_N \to \infty$ and $x_N \ge x_0$ for all large y.

Residual. Apply (1.3) with $M = N$:

$$\log F(x_N) = \log C + a x_N + b \log x_N + \sum_{j=1}^{N} q_j x_N^{-j} + E_N, \qquad |E_N| \le K_N x_N^{-N-1}.$$

Since $\log y = \log C + aX + b\log X$ and $\log x_N = \log X + \log(1 + tU_N(t))$,

$$r_N := \log F(x_N) - \log y = g(t) + E_N, \qquad g(t) := aU_N(t) + b\log\big(1 + tU_N(t)\big) + \sum_{j=1}^{N} q_j \left(\frac{t}{1 + tU_N(t)}\right)^j.$$

Here g is real-analytic on a neighbourhood of $t = 0$, being a finite composition of analytic maps with $1 + tU_N(t) \to 1$; its Taylor series at $t = 0$ is therefore the corresponding formal series. By (4) the formal series $aU + b\log(1 + tU) + Q(t/(1 + tU))$ vanishes identically. Replacing U by U_N changes it by an element of $t^{N+1}R[[t]]$, because $U - U_N \in t^{N+1}R[[t]]$ and the maps $u \mapsto \log(1 + tu)$, $u \mapsto (t/(1 + tu))^j$ are t-adically continuous; and discarding the terms of Q with $j > N$ changes it by an element of $t^{N+1}R[[t]]$ as well. Hence the Taylor series of g vanishes through order t^N, so $g(t) = \mathcal{O}(t^{N+1})$ and

$$|r_N| \le \big|g(1/X)\big| + K_N x_N^{-N-1} = \mathcal{O}\big(X^{-N-1}\big).$$

Slope. By (1.4) with $\alpha = 1$, $(\log F)'(x) = a + \mathcal{O}(x^{-1})$, so there is $x_1 \ge x_0$ with $(\log F)' \ge a/2$ on $[x_1, \infty)$. For all large y both x_N and $F^{-1}(y)$ lie in $[x_1, \infty)$. Theorem 1.42, applied to $\log F$ on that half-line, gives

$$\big|x_N - F^{-1}(y)\big| \le \frac{|r_N|}{a/2} = \mathcal{O}\big(X^{-N-1}\big),$$

which is (1.52). □

Remark 1.26 (No logarithms in the coefficients). Part (3) is the structural reason for centring at the Lambert core: the whole logarithmic tower has been resummed into X, so the c_n are constants of the coefficient ring. The alternative expansion of section 1.6 has coefficients that are polynomials in $\log L$ whose degree grows with the order. This is an algebraic statement and by itself says nothing about numerical accuracy; the numerical comparison is a separate, chapterwise matter (see theorem 1.36).

Proposition 1.27 (General exponent grid)**.** *Let $\delta > 0$ and keep the remaining hypotheses of theorem 1.25. Let s be a second formal variable and put*

$$\Psi_\delta(t, s, u) := -\frac{1}{a}\left[b\log(1 + tu) + Q\left(\frac{s}{(1 + tu)^\delta}\right)\right], \tag{1.53}$$

where $(1+tu)^{-\delta}$ *is the formal binomial series* (1.16). *Then* Ψ_δ *lies in the ideal* (t,s) *of* $R[[t,s,u]]$, *the equation* $U=\Psi_\delta(t,s,U)$ *has a unique solution* $U=\sum_{i+j\geq 1} c_{i,j}t^i s^j$ *in the ideal* (t,s) *of* $R[[t,s]]$, *and the substitution* $t=X^{-1}$, $s=X^{-\delta}$ *reverts* (1.2)*: formally*

$$ax+b\log x+Q\big(x^{-\delta}\big)=aX+b\log X,\qquad x=X+\sum_{i+j\geq 1} c_{i,j}X^{-i-j\delta}. \tag{1.54}$$

For $\delta=1$ *the two markers coincide and* $\Psi_\delta(t,t,u)=\Psi(t,u)$ *of* (1.47).

Proof. The membership $\Psi_\delta\in(t,s)$ and the existence and uniqueness of U follow exactly as in parts (1) and (2) of theorem 1.25, with the t-adic filtration replaced by the (t,s)-adic one: Ψ_δ lies in (t,s), so $\Psi_\delta(t,s,U)$ modulo $(t,s)^{d+1}$ depends only on U modulo $(t,s)^d$. The identity (1.54) is the computation of part (4) with $1/\mathsf{x}=t/(1+tU)$ replaced by $\mathsf{x}^{-\delta}=s(1+tU)^{-\delta}$. Finally the substitution is legitimate because, for each real exponent θ, only finitely many pairs (i,j) with $i,j\geq 0$ satisfy $i+j\delta=\theta$ when $\delta>0$. □

Remark 1.28 (When the grid degenerates). If δ is irrational the exponents $i+j\delta$ are pairwise distinct and the monomial scale is a free monoid on two generators. If $\delta=p/q$ in lowest terms, different pairs (i,j) can give the same exponent and the correct statement is that the reversion lives in $R[[X^{-1/q}]]$. Every chapter of this volume has $\delta/\alpha=1$ after theorem 1.4, so the degeneracy is never met here.

1.5.2 Reversion about an arbitrary admissible core

Theorem 1.29 (Reversion about an arbitrary core)**.** *Let* R *be a commutative* $\mathbb{Q}$*-algebra and* $\Lambda\in R$ *invertible. Consider*

$$\Lambda E+h(E)+f(t,E)=0,\qquad h\in u^2R[[u]],\quad f\in tR[[t,u]]. \tag{1.55}$$

Then (1.55) *has a unique solution* $E=\sum_{n\geq 1} e_nt^n\in tR[[t]]$*; it satisfies the triangular recurrence*

$$e_n=-\frac{1}{\Lambda}[t^n]\Big[h\big(E_{<n}\big)+f\big(t,E_{<n}\big)\Big],\qquad E_{<n}:=\sum_{j=1}^{n-1}e_jt^j, \tag{1.56}$$

and the closed formula

$$e_n=\sum_{m=1}^{2n-1}\frac{1}{m}[t^nu^{m-1}]\Big(-\Lambda^{-1}\big[h(u)+f(t,u)\big]\Big)^m. \tag{1.57}$$

If moreover $h(u)=\sum_{m\geq 2}h_mu^m$ *and* $f(t,u)=\sum_{k\geq 1}\phi_k(t)\,(1+u)^{-\mu_k}$ *with* $\phi_k\in tR[[t]]$ *and* $\mu_k\in\mathbb{N}$, *then*

$$e_n=-\frac{1}{\Lambda}\left[\sum_{m=2}^{n}h_m\,\widehat{B}_{n,m}(e)+\sum_{k\geq 1}\sum_{i=1}^{n}\big([t^i]\,\phi_k\big)\sum_{m=0}^{n-i}(-1)^m\binom{\mu_k+m-1}{m}\widehat{B}_{n-i,m}(e)\right], \tag{1.58}$$

in which only $e_1,\dots,e_{n-1}$ *occur.*

Proof. Existence, uniqueness and (1.57) are theorem 1.17 applied to $\Psi:=-\Lambda^{-1}[h+f]$. For (1.56), note as in the proof of that lemma that e_n occurs neither in $[t^n]\,h(E)$ nor in $[t^n]\,f(t,E)$, so both may be computed from $E_{<n}$. For (1.58) apply theorem 1.9 to h and (1.17) to each $(1+E)^{-\mu_k}$, then convolve with ϕ_k; the displayed ranges are those outside which the ordinary Bell polynomials vanish. □

Remark 1.30 (How the chapters instantiate it). For the linear–logarithmic core the shorter route of theorem 1.25 is available, because measuring the displacement in units of X makes the entire perturbation divisible by t. Part 3 needs the general form: there Λ is the normalized core slope $\log X$ of theorem 1.23, $h(E)=(1+E)\log(1+E)-E$ and $f(t,E)=\sum_{k\geq 1}\alpha_kt^{2k}(1+E)^{-(2k-1)}$, and (1.58) is exactly the Bell recurrence of double_factorial_transseries-2. The coefficient ring there is $\mathbb{Q}[\Lambda^{-1}]$: the e_n are *not* constants but polynomials in Λ^{-1}, of degree at most $2n-1$ and with no constant term.

1.5.3 The perturbation-operator form

Proposition 1.31 (Inverse perturbation operator). *Let Φ_0 be a phase with Φ_0' nonvanishing near the point considered, let X solve $\Phi_0(X) = L$, and consider*

$$L = \Phi_0(x) + \varepsilon\,\varrho(x). \tag{1.59}$$

Put $\mathscr{D}_0 := \Phi_0'(X)^{-1}\,\mathrm{d}/\mathrm{d}X$. Then, as a formal power series in ε,

$$x = X + \sum_{m\geq 1} \frac{(-\varepsilon)^m}{m!}\,\mathscr{D}_0^{m-1}\left[\frac{\varrho(X)^m}{\Phi_0'(X)}\right]. \tag{1.60}$$

Proof. Change coordinate to $\lambda := \Phi_0(x)$, so that $x = \Phi_0^{-1}(\lambda)$ and (1.59) becomes $L = \lambda + \varepsilon\varrho_0(\lambda)$ with $\varrho_0 := \varrho \circ \Phi_0^{-1}$. This is a near-identity reversion in λ with small parameter ε; Lagrange–Bürmann in the form (1.24), with the observable $G = \Phi_0^{-1}$, gives

$$\Phi_0^{-1}(\lambda) = \Phi_0^{-1}(L) + \sum_{m\geq 1} \frac{(-\varepsilon)^m}{m!}\left(\frac{\mathrm{d}}{\mathrm{d}L}\right)^{m-1}\Big[(\Phi_0^{-1})'(L)\,\varrho_0(L)^m\Big].$$

Finally $(\Phi_0^{-1})'(L) = 1/\Phi_0'(X)$, $\varrho_0(L) = \varrho(X)$ and $\mathrm{d}/\mathrm{d}L = \mathscr{D}_0$ by the chain rule. □

Remark 1.32 (What the operator form is for). This is the form in which an *exponentially small* forward correction is transported through the inverse map: with ϱ a Stokes discontinuity or a terminant remainder rather than a power series, $-\varepsilon\varrho(X)/\Phi_0'(X)$ is the leading displacement and the higher terms generate the nonlinear sectors. As a statement about a formal series in ε it is exact; as an asymptotic statement about a particular ϱ it needs the hypotheses of theorem 1.44. Expanding ϱ in inverse powers of X reproduces (1.48).

1.6 Flattening the core

Theorem 1.25 hides the explicit scale of the answer inside X. Unfolding X produces the polynomial–logarithmic transseries in which the four chapters state their headline formulae. Throughout this section H is an indeterminate; in the analytic application it is specialized to $\log(L/a)$.

Theorem 1.33 (Direct polynomial–logarithmic reversion). *Let R be a commutative $\mathbb{Q}$-algebra, $a \in R$ invertible, $b \in R$, $Q = \sum_{j\geq 1} q_j t^j \in tR[[t]]$, and let H be an indeterminate. Set*

$$\Phi(t, u; H) := -b\log\big(1 + t(u - bH)\big) - Q\left(\frac{at}{1 + t(u - bH)}\right) \in t\,R[H][[t, u]]. \tag{1.61}$$

1. *There is a unique $V = \sum_{n\geq 1} P_n(H)\,t^n \in t\,R[H][[t]]$ with $V = \Phi(t, V; H)$, and*

$$P_n(H) = \sum_{m=1}^{n} \frac{1}{m}[t^n u^{m-1}]\,\Phi(t, u; H)^m. \tag{1.62}$$

2. Triangular Bell recurrence. *Put $\gamma_1 := -bH$ and $\gamma_r := P_{r-1}(H)$ for $r \geq 2$. Then*

$$P_n(H) = -b\sum_{k=1}^{n} \frac{(-1)^{k+1}}{k}\widehat{B}_{n,k}(\gamma) - \sum_{j=1}^{n} q_j a^j \sum_{k=0}^{n-j}(-1)^k\binom{j+k-1}{k}\widehat{B}_{n-j,k}(\gamma), \tag{1.63}$$

in which only $P_1, \ldots, P_{n-1}$ occur.

3. Shape. $P_n \in R[H]$ *with*

$$\deg_H P_n \le n, \qquad [H^n]\, P_n = \frac{b^{n+1}}{n}, \qquad P_n(0) = a^{n+1} c_n, \tag{1.64}$$

where c_n are the Lambert-centred coefficients of theorem 1.25. In particular $\deg_H P_n = n$ exactly when b is not a zero divisor, and the coefficients of P_n lie in the $\mathbb{Q}$-subalgebra generated by a, a^{-1}, b, $q_1, \dots, q_n$.

4. Formal exactness and uniqueness. *Put $t = L^{-1}$ and $H = \log(L/a)$, and*

$$x := \frac{1}{a}\left[L - bH + \sum_{n\ge 1} \frac{P_n(H)}{L^n}\right]. \tag{1.65}$$

Then $ax + b\log x + Q(1/x) = L$ as an identity of formal series in L^{-1} with polynomial coefficients in H. Conversely, (1.65) *is the* only *expansion of the form $x = a^{-1}\big[L - bH + \sum_{n\ge1} \Pi_n(H) L^{-n}\big]$, with $\Pi_n \in R[H]$, that solves the equation coefficientwise.*

5. Analytic version. *Under the hypotheses of theorem 1.25(5), with $L = \log(y/C)$ and $H = \log(L/a)$, for every fixed $N \ge 0$,*

$$F^{-1}(y) = \frac{1}{a}\left[L - bH + \sum_{n=1}^{N} \frac{P_n(H)}{L^n}\right] + \mathcal{O}\left(\frac{(1+H)^{N+1}}{L^{N+1}}\right) \qquad (y \to \infty). \tag{1.66}$$

Proof. (1) Membership of Φ in $tR[H][[t,u]]$ is checked as in theorem 1.25(1), the extra additive term $-bH$ inside $1 + t(u - bH)$ being harmless because it is multiplied by t. Theorem 1.16 over the ring $R[H]$ then gives existence, uniqueness and (1.62).

(2) Put $w := t(V - bH) = \sum_{r\ge1} \gamma_r t^r$ with γ as stated; apply (1.18) to $\log(1+w)$ and (1.17) to $(1+w)^{-j}$, and convolve the latter with the factor $(at)^j$. Only $\gamma_1, \dots, \gamma_{n-k+1}$, that is H and $P_1, \dots, P_{n-k}$, occur.

(4) Set $t = L^{-1}$. With x as in (1.65),

$$x = \frac{1}{at}\big(1 + t(V - bH)\big) = \frac{1+w}{at}, \qquad ax = \frac{1}{t} + (V - bH), \qquad \log x = H + \log(1+w), \qquad \frac{1}{x} = \frac{at}{1+w},$$

where $\log x$ uses $\log\big(1/(at)\big) = \log(L/a) = H$. Hence

$$ax + b\log x + Q(1/x) - L = (V - bH) + b\big(H + \log(1+w)\big) + Q\left(\frac{at}{1+w}\right) = V - \Phi(t, V; H) = 0.$$

For uniqueness, substituting a general $\Pi = \sum \Pi_n(H) t^n$ in place of V turns the equation into $\Pi = \Phi(t, \Pi; H)$, and theorem 1.16 has a unique solution.

(3) We show by strong induction that $\deg_H P_n \le n$ and, in addition, that $\deg_H \gamma_r \le r - 1$ for $r \ge 2$. The base is $\gamma_1 = -bH$, of degree 1. Assume $\deg_H P_j \le j$ for all $j < n$, so $\deg_H \gamma_r = \deg_H P_{r-1} \le r - 1$ for $2 \le r \le n$. A monomial of $\widehat{B}_{n,k}(\gamma)$ is a product $\gamma_{r_1} \cdots \gamma_{r_k}$ with $r_1 + \cdots + r_k = n$ and $r_\nu \ge 1$, hence of H-degree at most $\sum_\nu r_\nu = n$; so the first sum of (1.63) has H-degree at most n. In the second sum the ordinary Bell polynomial has first index $n - j$ with $j \ge 1$, so its H-degree is at most $n - 1$. Hence $\deg_H P_n \le n$, completing the induction.

A monomial of H-degree exactly n requires $\deg_H \gamma_{r_\nu} = r_\nu$ for every ν, which forces $r_\nu = 1$ for all ν because $\deg_H \gamma_r \le r - 1$ for $r \ge 2$. Thus $k = n$, the monomial is $\gamma_1^n = (-bH)^n$, and $[H^n]\, \widehat{B}_{n,k}(\gamma) = 0$ for $k < n$ while $\widehat{B}_{n,n}(\gamma) = (-bH)^n$. Only the term $k = n$ of the first sum of (1.63) therefore survives, and

$$[H^n]\, P_n = -b \cdot \frac{(-1)^{n+1}}{n} \cdot (-b)^n = \frac{(-1)^{2n+2}\, b^{n+1}}{n} = \frac{b^{n+1}}{n}.$$

For the constant term put $H = 0$ in (1.61):

$$\Phi(t,u;0) = -b\log(1+tu) - Q\left(\frac{at}{1+tu}\right).$$

If $U(t) = \sum c_n t^n$ solves $U = \Psi(t,U)$ with Ψ of (1.47), put $\widetilde{V}(t) := a\,U(at)$. Then $t\widetilde{V}(t) = (at)U(at)$, so

$$\Phi\big(t,\widetilde{V}(t);0\big) = -b\log\big(1+(at)U(at)\big) - Q\left(\frac{at}{1+(at)U(at)}\right) = a\,\Psi\big(at,U(at)\big) = a\,U(at) = \widetilde{V}(t),$$

so $\widetilde{V}$ solves the $H=0$ fixed-point equation; by uniqueness $P_n(0) = [t^n]\,\widetilde{V} = a\cdot a^n c_n = a^{n+1}c_n$.

(5) Fix N. Put x_N for the truncation displayed in (1.66). Exactly as in the proof of theorem 1.25(5), the residual $r_N := \log F(x_N) - \log y$ splits as $g(t) + E_N$ with $|E_N| \le K_N x_N^{-N-1}$ and g analytic in $t = 1/L$ *with coefficients polynomial in* H; by (4) the Taylor coefficients of g vanish through order N, and the coefficient of t^{N+1} is a polynomial in H of degree at most $N+1$. Hence $|r_N| = \mathcal{O}\big((1+H)^{N+1}L^{-N-1}\big)$. The slope bound $(\log F)' \ge a/2$ and theorem 1.42 finish the proof, after noting $L \asymp aX$ and $H = \log(L/a)$ so that $x_N \to \infty$. □

Theorem 1.34 (The pure Lambert block in closed form)**.** *Let* $\left[{n \atop m}\right]$ *be the* unsigned *Stirling numbers of the first kind, so that the signed ones are* $(-1)^{n-m}\left[{n \atop m}\right]$ *and* $\log(1+z)^m = m!\sum_{n\ge m}(-1)^{n-m}\left[{n \atop m}\right]z^n/n!$*. When* $Q = 0$ *the polynomials of theorem 1.33 are*

$$\boxed{P_n^{(0)}(H) = b^{\,n+1}\sum_{m=1}^{n}(-1)^{m+1}\frac{\left[{n \atop m}\right]}{(n-m+1)!}\,H^{\,n-m+1}} \tag{1.67}$$

for $n \ge 1$*. Equivalently, clearing the binomial,*

$$P_n^{(0)}(H) = \frac{b^{\,n+1}}{n!}\sum_{m=1}^{n}(-1)^{m+1}(m-1)!\left[{n \atop m}\right]\binom{n}{m-1}H^{\,n-m+1}. \tag{1.68}$$

In particular

$$[H^n]\,P_n^{(0)} = \frac{b^{\,n+1}}{n}, \qquad P_n^{(0)}(0) = 0, \tag{1.69}$$

and the first four are

$$P_1^{(0)} = b^2H,\quad P_2^{(0)} = \frac{b^3}{2}\big(H^2-2H\big),\quad P_3^{(0)} = \frac{b^4}{6}\big(2H^3-9H^2+6H\big),\quad P_4^{(0)} = \frac{b^5}{12}\big(3H^4-22H^3+36H^2-12H\big). \tag{1.70}$$

Consequently the Lambert core itself unfolds as

$$X = \frac{1}{a}\left[L - bH + \sum_{n\ge1}\frac{P_n^{(0)}(H)}{L^n}\right], \qquad H = \log\frac{L}{a}, \tag{1.71}$$

formally, and with remainder $\mathcal{O}\big((1+H)^{N+1}L^{-N-1}\big)$ *after truncation at* $n = N$*.*

Proof. With $Q = 0$, (1.62) reads

$$P_n^{(0)}(H) = \sum_{m=1}^{n}\frac{(-b)^m}{m}[t^n u^{m-1}]\,\log\big(1+t(u-bH)\big)^m.$$

Insert $\log(1+z)^m = m!\sum_{r\ge m}(-1)^{r-m}\left[{r \atop m}\right]z^r/r!$ with $z = t(u-bH)$; the coefficient of t^n is $m!\,(-1)^{n-m}\left[{n \atop m}\right](u-bH)^n/n!$, and $[u^{m-1}]\,(u-bH)^n = \binom{n}{m-1}(-bH)^{n-m+1}$. Hence

$$P_n^{(0)}(H) = \sum_{m=1}^{n}\frac{(-b)^m}{m}\cdot\frac{m!\,(-1)^{n-m}\left[{n \atop m}\right]}{n!}\binom{n}{m-1}(-b)^{n-m+1}H^{n-m+1},$$

and $(-b)^m(-b)^{n-m+1}(-1)^{n-m} = (-1)^{n+1}(-1)^{n-m}b^{n+1} = (-1)^{m+1}b^{n+1}$, which gives (1.68). Since $(m-1)!\binom{n}{m-1}/n! = 1/(n-m+1)!$, the two displayed forms agree and (1.67) follows. Every term carries H^{n-m+1} with $m \le n$, so no constant term occurs and $P_n^{(0)}(0) = 0$; the term $m = 1$ has $\left[{n \atop 1}\right] = (n-1)!$ and gives $[H^n]\,P_n^{(0)} = b^{n+1}(n-1)!/n! = b^{n+1}/n$. Finally (1.71) is theorem 1.33(4) and (5) with $Q = 0$, in which case the unique solution of $ax + b\log x = L$ is the core X of theorem 1.20. □

Remark 1.35 (Three source formulae, one identity). The closed form (1.67) appears in the sources in three different dresses, all with the signed Stirling numbers $s(n,m) = (-1)^{n-m}\left[{n \atop m}\right]$. Swing_Factorial_Transseries_Article states (1.68); Partition_Number_Transseries_and_Asymptotic_Inverse states $P_j(L) = 2^{j+1}\sum_{q\le j} s(j,q)L^{j+1-q}/(j+1-q)!$; Partition_Number_Transseries_and_Inverse states $P_m(\ell) = \frac{1}{m!}\sum_j 2^j(j-1)!\,s(m,j)\binom{m}{j-1}(2\ell)^{m-j+1}$. The last two are the specialization $a = 1$, $b = -2$ of (1.67), and all three agree with it as exact rational polynomials for $1 \le n \le 8$, which was checked in exact rational arithmetic (theorem 1.48). The volume uses (1.67): it is the shortest of the four, it avoids the letter s, which Part 4 needs for Dedekind sums, and it makes the leading coefficient (1.69) a one-line consequence.

Remark 1.36 (Both normalizations, and what is and is not claimed). The Lambert-centred expansion (1.52) and the flattened expansion (1.66) are two truncations of the same object: by (1.71), expanding X and then each power X^{-n} in the flattened scale returns (1.66), and conversely collecting in the flattened series all terms generated by $b\log x$ alone rebuilds X. What differs is the coefficient ring: the c_n are constants, the P_n are polynomials of degree n in $H = \log(L/a)$. Several of the sources add that Lambert centring is numerically more accurate at equal truncation order and support this with tables. Part 0 makes no such claim: the comparison depends on the size of H relative to L and on the particular data, it is an empirical statement about specific truncations, and the supporting tables belong to the chapters, where they are reported with their provenance. What Part 0 does assert is the algebraic statement of theorem 1.26, and the remainder bounds (1.52) and (1.66), whose right-hand sides differ by the factor $(1 + H)^{N+1}$ — and that factor is the only rigorous difference between the two at fixed order.

1.7 The three inverses of a discrete sequence

Definition 1.37 (The three inverse objects)**.** Let $(A_n)_{n\ge n_0}$ be real numbers, strictly increasing for $n \ge n_1 \ge n_0$, with $A_n \to +\infty$.

1. The *integer staircase*, or generalized inverse, is

$$N_*(y) := \min\{n \ge n_1 : A_n \ge y\}, \qquad y \le \sup_n A_n. \tag{1.72}$$

   It is an integer-valued, nondecreasing step function, constant on each interval $(A_{n-1}, A_n]$.

2. Given a continuous, strictly increasing $\mathcal{A} : [n_1, \infty) \to \mathbb{R}$ with $\mathcal{A}(n) = A_n$ for every integer $n \ge n_1$ — an *admissible interpolation* — the *interpolated inverse* is $\nu_{\mathcal{A}} := \mathcal{A}^{-1}$, defined on $[A_{n_1}, \infty)$.

3. The *asymptotic inverse* is the formal transseries produced by theorem 1.25 or theorem 1.33 from a model F for which the chapter in question proves an asymptotic identity along the sequence. It is attached to no particular interpolation.

Remark 1.38 (The three are genuinely different). Object (1) is canonical but not smooth; object (2) is smooth but not canonical — distinct admissible interpolations agree at the integers and differ in between, hence have different inverses; object (3) is canonical as a formal series but determines a function only after a summation convention is fixed. The three coincide only in the following precise senses: $N_* = \lceil \nu_{\mathcal{A}} \rceil$ for *every* admissible $\mathcal{A}$ (theorem 1.39(1)), and $\nu_{\mathcal{A}}(A_n) = n$ for every admissible $\mathcal{A}$ and every integer $n \ge n_1$. The asymptotic series (3) is independent of the choice in $\mathcal{A}$ at every algebraic order, but the exponentially small layers are not: the partition articles make this explicit for the Rademacher amplitudes, which are periodic in the index and therefore genuinely depend on how the sequence is continued off the lattice.

Theorem 1.39 (Staircase recovery and the separation condition)**.** *Notation as in theorem 1.37, with $\mathcal{A}$ admissible and $\nu := \nu_{\mathcal{A}}$.*

1. Exact rounding. *For every y in the range of $\mathcal{A}$, that is for $A_{n_1} \le y < \sup_n A_n$,*

$$N_*(y) = \lceil \nu(y) \rceil, \tag{1.73}$$

with $\lceil m \rceil = m$ for integers m. The identity holds for every admissible interpolation.

2. Separation condition. *Let $x_* \in \mathbb{R}$ and $\eta \ge 0$ satisfy $|x_* - \nu(y)| \le \eta$. If*

$$\lceil x_* - \eta \rceil = \lceil x_* + \eta \rceil, \tag{1.74}$$

then $N_*(y)$ equals that common value. Condition (1.74) *holds if and only if the interval $[x_* - \eta, x_* + \eta]$ is contained in a single half-open interval $(m-1, m]$; it can fail for arbitrarily small $\eta > 0$, namely when $\nu(y)$ is within η of an integer from above.*

3. Inverse along the range. *If $y = A_n$ for an integer $n \ge n_1$ and $|x_* - \nu(y)| \le \eta < \frac{1}{2}$, then $n = \lfloor x_* + \frac{1}{2} \rfloor$. This case is unconditional: no separation hypothesis is needed, because $\nu(A_n) = n$ exactly.*

4. Residue classes. *Suppose $r \ge 1$ and that for each $\rho \in \{0, 1, \dots, r-1\}$ the subsequence $(A_n)_{n \equiv \rho\ (r)}$ is strictly increasing for $n \ge n_1$ and admits an admissible interpolation $\mathcal{A}_\rho$ on the class, with inverse ν_ρ. Then, for y large enough,*

$$N_*(y) = \min_{0 \le \rho < r} \left(\rho + r \left\lceil \frac{\nu_\rho(y) - \rho}{r} \right\rceil \right). \tag{1.75}$$

5. No uniform smooth representation. *For every Y in the range and every continuous $g : [Y, \infty) \to \mathbb{R}$,*

$$\sup_{y \ge Y} \bigl| N_*(y) - g(y) \bigr| \ \ge\ \tfrac{1}{2}. \tag{1.76}$$

Equivalently, the $\mathcal{O}(1)$ periodic layer $\lceil x \rceil - x - \frac{1}{2} = \pi^{-1} \sum_{m \ge 1} m^{-1} \sin(2\pi m x)$ (valid for $x \notin \mathbb{Z}$) is not a refinement that can be dropped: it is the leading discrete transmonomial of the generalized inverse.

Proof. (1) For an integer $n \ge n_1$, $A_n \ge y$ is equivalent to $\mathcal{A}(n) \ge y$, hence to $n \ge \nu(y)$ by strict monotonicity of $\mathcal{A}$. The least such integer is $\lceil \nu(y) \rceil$.

(2) The function $\lceil \cdot \rceil$ is constant on each $(m-1, m]$ and nondecreasing; since $\nu(y) \in [x_* - \eta, x_* + \eta]$, (1.74) forces $\lceil \nu(y) \rceil$ to equal the common value, and (1) applies. The characterization of (1.74) and the failure mode are immediate from the same description of $\lceil \cdot \rceil$.

(3) $\nu(A_n) = n$ because $\mathcal{A}(n) = A_n$; hence $|x_* - n| \le \eta < 1/2$ and rounding to the nearest integer returns n.

(4) Fix ρ. Among the integers $n \equiv \rho \pmod r$ with $n \ge n_1$, the condition $A_n \ge y$ is, by (1) applied to the class, equivalent to $n \ge \nu_\rho(y)$. Writing $n = \rho + rk$ with $k \in \mathbb{Z}$, the least admissible k is $\lceil (\nu_\rho(y) - \rho)/r \rceil$, so the least admissible n in the class is $\rho + r\lceil (\nu_\rho(y) - \rho)/r \rceil$. Taking the minimum over ρ gives the least admissible n without a congruence restriction, which is $N_*(y)$.

(5) Fix n with $A_n > Y$. On $(A_{n-1}, A_n]$ we have $N_* = n$ and on $(A_n, A_{n+1}]$ we have $N_* = n + 1$. For $\epsilon > 0$ small,

$$1 = \bigl| N_*(A_n + \epsilon) - N_*(A_n) \bigr| \le \bigl| N_*(A_n + \epsilon) - g(A_n + \epsilon) \bigr| + \bigl| g(A_n + \epsilon) - g(A_n) \bigr| + \bigl| g(A_n) - N_*(A_n) \bigr|.$$

Letting $\epsilon \downarrow 0$ and using continuity of g gives $1 \le 2 \sup_{y \ge Y} |N_* - g|$. For the Fourier identity, the function $x \mapsto \lceil x \rceil - x - \frac{1}{2}$ is 1-periodic, equal to $-(x - \lfloor x \rfloor) + \frac{1}{2}$ shifted, i.e. minus the standard sawtooth, whose Fourier series is the displayed one and converges pointwise off $\mathbb{Z}$. □

Proposition 1.40 (Certified discrete inversion)**.** *In the setting of theorem 1.39(4), suppose that for each ρ an approximation x_ρ with $|x_\rho - \nu_\rho(y)| \le \eta_\rho$ is available. Then $N_*(y)$ is determined by at most $\sum_\rho(\lfloor 2\eta_\rho/r \rfloor + 2)$ exact evaluations of the sequence: for each ρ list the integers $n \equiv \rho \pmod r$ in $[x_\rho - \eta_\rho,\, x_\rho + \eta_\rho + r]$, evaluate A_n, and take the least n over all classes with $A_n \ge y$.*

Proof. By theorem 1.39(4) the class-ρ candidate is $\rho + r\lceil(\nu_\rho(y) - \rho)/r\rceil$, and $\nu_\rho(y)$ lies in $[x_\rho - \eta_\rho, x_\rho + \eta_\rho]$; the least integer of the class that is $\ge \nu_\rho(y)$ therefore lies in $[x_\rho - \eta_\rho, x_\rho + \eta_\rho + r]$. Since the class subsequence is increasing, the test $A_n \ge y$ identifies it, and the outer minimum over ρ is (1.75). □

Remark 1.41 (The parity instances). Part 5 and Part 3 are the case $r = 2$. The formula $\rho + r\lceil(\nu_\rho - \rho)/r\rceil$ specializes to $2\lceil\nu_0/2\rceil$ and $2\lceil(\nu_1 - 1)/2\rceil + 1$, which is exactly the pair of parity candidates of double_factorial_transseries-2, and $N_* = \min$ of the two is its "exact generalized inverse" theorem. Swing_Factorial_Transseries_Article states the corresponding *floor* formulae for the threshold query "the greatest index in the class ρ whose sequence value is at most y"; those are $\rho + r\lfloor(\nu_\rho - \rho)/r\rfloor$, obtained from (1.75) by replacing $\lceil\cdot\rceil$ with $\lfloor\cdot\rfloor$ and $\min$ with $\max$, with the same proof. The partition chapter is the case $r = 1$, where (1.75) reduces to (1.73).

Source claim. Several sources write, or invite the reader to write, $N_*(y) = \nu(y) + o(1)$. **Defect.** This is false uniformly in y, by theorem 1.39(5). Butcher_Tree_Transseries-2 already flags it ("An expansion such as $N_*(y) = \nu(y) + o(1)$ is false uniformly in y"), and Partition_Number_Transseries_and_Asymptotic_Inverse likewise separates the three objects; but the qualification is absent from the summary statements of the other articles. **Correction.** The volume uses only the three statements theorem 1.39(1), (3), (4): an exact ceiling identity valid for every admissible interpolation, an unconditional nearest-integer recovery *along the range* $y = A_n$, and a residue-class refinement. Nothing else about N_* is asserted.

1.8 Error control, transport, and optimal truncation

1.8.1 Backward error certifies forward error

Theorem 1.42 (Residual-to-root conversion)**.** *Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ differentiable with*

$$0 < m \le f'(\xi) \le M \qquad (\xi \in I). \tag{1.77}$$

Let $x_\dagger \in I$ satisfy $f(x_\dagger) = \lambda$, let $x_* \in I$, and set the residual $r_* := f(x_*) - \lambda$. *Then*

$$\frac{|r_*|}{M} \;\le\; |x_* - x_\dagger| \;\le\; \frac{|r_*|}{m}. \tag{1.78}$$

Moreover, if f is continuous on I and $[x_* - \varrho,\, x_* + \varrho] \subseteq I$ with $\varrho := |r_*|/m$, then f takes the value λ somewhere in that interval; the residual is therefore a certificate*, not merely a bound conditional on the existence of a nearby root.*

Proof. By the mean value theorem, $r_* = f(x_*) - f(x_\dagger) = f'(\xi)(x_* - x_\dagger)$ for some ξ between the two points, and (1.77) gives (1.78). For the certificate, suppose $r_* > 0$ (the case $r_* < 0$ is symmetric and $r_* = 0$ is trivial). Then $f(x_*) = \lambda + r_* > \lambda$, while

$$f(x_* - \varrho) \le f(x_*) - m\varrho = \lambda + r_* - |r_*| = \lambda,$$

using $f' \ge m$. The intermediate value theorem supplies a point of $[x_* - \varrho, x_*]$ where $f = \lambda$. □

Remark 1.43 (Use in the logarithmic variable). In every application $f = \log F$ and $\lambda = \log y$: F itself overflows any fixed-precision arithmetic long before the asymptotic regime begins, whereas $\log F$ is computable from a digamma or Laplace representation throughout. The lower bound in (1.78) makes the residual test sharp — a small residual is necessary as well as sufficient.

1.8.2 Transport of a controlled forward remainder

Theorem 1.44 (Remainder transport). *Let $I = [x_1, \infty)$, let $f_0 : I \to \mathbb{R}$ be differentiable with $f_0' \geq m > 0$, and let $\varepsilon : I \to \mathbb{R}$ be differentiable with $\sup_I |\varepsilon'| \leq \theta < m$. Put $f := f_0 + \varepsilon$. For λ in the common range let $x_0 = x_0(\lambda) \in I$ solve $f_0(x_0) = \lambda$ and $x = x(\lambda) \in I$ solve $f(x) = \lambda$. Then*

1. Rigorous bound.
$$\big|x(\lambda) - x_0(\lambda)\big| \leq \frac{\big|\varepsilon\big(x_0(\lambda)\big)\big|}{m - \theta}. \tag{1.79}$$

2. First-order law with explicit error. *If moreover $f_0 \in C^2(I)$ with $\sup_I |f_0''| \leq M_2$, then*
$$x(\lambda) - x_0(\lambda) = -\frac{\varepsilon(x_0)}{f_0'(x_0)} + E, \qquad |E| \leq \frac{|\varepsilon(x_0)|}{m\,(m-\theta)} \left(\frac{M_2\,|\varepsilon(x_0)|}{m - \theta} + \theta \right). \tag{1.80}$$

In particular, if $\varepsilon(x) = \mathcal{O}(x^{-M-1})$ and $\varepsilon'(x) = o(1)$ as $x \to \infty$, then $x(\lambda) - x_0(\lambda) = -\varepsilon(x_0)/f_0'(x_0)\,(1 + o(1)) = \mathcal{O}(x_0^{-M-1})$.

Proof. Since $f' = f_0' + \varepsilon' \geq m - \theta > 0$, f is strictly increasing and $x(\lambda)$ is unique. The residual of x_0 in f is $f(x_0) - \lambda = f_0(x_0) + \varepsilon(x_0) - \lambda = \varepsilon(x_0)$, so theorem 1.42 applied to f with m replaced by $m - \theta$ gives (1.79).

For (2), write $d := x - x_0$. By the mean value theorem there is ξ between x_0 and x with $0 = f(x) - \lambda = \varepsilon(x_0) + f'(\xi)d$, so $d = -\varepsilon(x_0)/f'(\xi)$ and
$$d + \frac{\varepsilon(x_0)}{f_0'(x_0)} = \varepsilon(x_0) \left(\frac{1}{f_0'(x_0)} - \frac{1}{f'(\xi)} \right) = \varepsilon(x_0)\, \frac{f'(\xi) - f_0'(x_0)}{f_0'(x_0)\, f'(\xi)}.$$
Now $|f'(\xi) - f_0'(x_0)| \leq |f_0'(\xi) - f_0'(x_0)| + |\varepsilon'(\xi)| \leq M_2|d| + \theta \leq M_2|\varepsilon(x_0)|/(m-\theta) + \theta$ by (1.79), while $f_0'(x_0) \geq m$ and $f'(\xi) \geq m - \theta$. Substituting gives (1.80). The final assertion follows since then $\theta \to 0$ and $|\varepsilon(x_0)| \to 0$. □

Corollary 1.45 (Forward truncation to inverse accuracy). *Let F be differentiably of exponential–power type with data $(C, a, 1, b, 1, Q)$, strictly increasing on $[x_1, \infty)$ with $(\log F)' \geq m > 0$ there. Let x_* solve exactly the truncated equation*
$$\log C + a x_* + b \log x_* + \sum_{j=1}^{M} q_j x_*^{-j} = \log y.$$
Then $|x_* - F^{-1}(y)| \leq K_M x_*^{-M-1}/m$ with K_M the constant of (1.3).

Proof. The residual of x_* in $\log F$ is exactly the forward remainder, of modulus at most $K_M x_*^{-M-1}$; apply theorem 1.42. □

1.8.3 Least-term truncation and the beyond-all-orders floor

Theorem 1.46 (Optimal truncation and the exponential floor). *Let $A > 0$, $\beta \in \mathbb{R}$, $K > 0$, $x_0 \geq 1$, and let $\varphi : [x_0, \infty) \to \mathbb{R}$ satisfy the* Gevrey-1 remainder bound
$$\left| \varphi(x) - \sum_{j=1}^{M} q_j x^{-j} \right| \;\leq\; K\,\Gamma(M + \beta)\, A^{-M} x^{-M} \qquad (M \geq 1,\ x \geq x_0). \tag{1.81}$$

1. Least-term envelope. *There are $x_1 \geq x_0$ and $K' = K'(K, A, \beta)$ such that, choosing $M = M(x) := \lfloor Ax \rfloor$,*
$$\left| \varphi(x) - \sum_{j=1}^{M(x)} q_j x^{-j} \right| \;\leq\; K'\, x^{\beta - 1/2}\, \mathrm{e}^{-Ax} \qquad (x \geq x_1). \tag{1.82}$$

2. Transported inverse error. *Let F be as in theorem 1.45, with* $\log F(x) = \log C + ax + b\log x + \varphi(x)$ *and* $(\log F)' \geq m > 0$ *on* $[x_1, \infty)$. *If* x_* *solves the optimally truncated equation exactly, then*

$$\left|x_* - F^{-1}(y)\right| \;\leq\; \frac{K'}{m}\, x_*^{\beta-1/2}\, \mathrm{e}^{-Ax_*}. \tag{1.83}$$

3. The floor is real. *Suppose in addition that there are* $M_0 \geq 1$ *and* $0 < k \leq K''$ *with*

$$k\,\Gamma(M+\beta)A^{-M} \;\leq\; |q_M| \;\leq\; K''\,\Gamma(M+\beta)A^{-M} \qquad (M \geq M_0). \tag{1.84}$$

Then

$$\log \min_{M\geq M_0} \left|q_M\right| x^{-M} \;=\; -Ax + \mathcal{O}(\log x) \qquad (x\to\infty), \tag{1.85}$$

so that no *truncation of the algebraic series achieves an error of smaller exponential order than* e^{-Ax}.

Proof. (1) Stirling's formula gives constants C_β and $M_\beta \geq 1$ with $\Gamma(M+\beta) \leq C_\beta M^{M+\beta-1/2}\mathrm{e}^{-M}$ for all $M \geq M_\beta$. With $M = \lfloor Ax \rfloor$ and x so large that $M \geq M_\beta$,

$$\Gamma(M+\beta)A^{-M}x^{-M} \leq C_\beta M^{\beta-1/2}\mathrm{e}^{-M}\left(\frac{M}{Ax}\right)^M \leq C_\beta M^{\beta-1/2}\mathrm{e}^{-M},$$

because $M \leq Ax$. Since $M > Ax - 1$, we have $\mathrm{e}^{-M} < \mathrm{e}\cdot\mathrm{e}^{-Ax}$, and $M^{\beta-1/2} \leq C'_\beta x^{\beta-1/2}$ for $x \geq x_1$ (with C'_β depending only on A and β, since $Ax - 1 \leq M \leq Ax$). Combining with (1.81) gives (1.82).

(2) The residual of x_* in $\log F$ is exactly the left-hand side of (1.82) evaluated at x_*; apply theorem 1.42.

(3) Enlarging M_0 we may assume $M_0 \geq M_\beta$. Write $u := Ax$ and $g(M) := (M/u)^M\mathrm{e}^{-M}$, so that by Stirling there are $0 < c_\beta \leq C_\beta$ with

$$c_\beta M^{\beta-1/2}g(M) \;\leq\; \Gamma(M+\beta)(Ax)^{-M} \;\leq\; C_\beta M^{\beta-1/2}g(M) \qquad (M \geq M_\beta). \tag{1.86}$$

Upper bound. Take $M = \lfloor Ax \rfloor$ in the right-hand inequality of (1.84) and repeat the estimate of part (1); this gives $\min_{M\geq M_0}|q_M|x^{-M} \leq K'''x^{\beta-1/2}\mathrm{e}^{-Ax}$ once $\lfloor Ax \rfloor \geq M_0$.

Lower bound. Note first that $\log g(M) = M\log(M/u) - M$ has derivative $\log(M/u)$, so g decreases on $(0,u)$, increases on (u,∞), and

$$g(M) \;\geq\; g(u) = \mathrm{e}^{-u} = \mathrm{e}^{-Ax} \qquad (M > 0). \tag{1.87}$$

Set $h(M) := M^{\beta-1/2}g(M)$; then $(\log h)'(M) = (\beta - \frac{1}{2})/M + \log(M/u)$, which is positive as soon as $M \geq 2u$ and $2u \geq 2|\beta - \frac{1}{2}|/\log 2$, so h is increasing on $[2u,\infty)$ for all large x. Consequently

$$\min_{M\geq M_0} h(M) = \min_{M_0\leq M\leq 2u} h(M) \;\geq\; \Big(\min_{M_0\leq M\leq 2u} M^{\beta-1/2}\Big)\mathrm{e}^{-Ax} \;\geq\; c\,x^{-|\beta-1/2|}\,\mathrm{e}^{-Ax}$$

for $x \geq x_1$, using (1.87) and $\min_{M_0\leq M\leq 2u} M^{\beta-1/2} \geq \min\big(M_0^{\beta-1/2}, (2Ax)^{\beta-1/2}\big)$. Combining with the left-hand inequalities of (1.84) and (1.86), $|q_M|x^{-M} \geq k\,c_\beta\,c\,x^{-|\beta-1/2|}\mathrm{e}^{-Ax}$ for every $M \geq M_0$. Taking logarithms of the two bounds gives (1.85). □

Remark 1.47 (What may and may not be appended). Theorem 1.46 bounds the size of the optimal remainder; it does not produce it. In particular, adding a term $c\,\mathrm{e}^{-Ax}$ with an arbitrarily chosen constant c to the divergent series is *not* a valid transseries completion. The double factorial and swing sources make the point explicitly: on the positive real axis the Borel singularities of these phases lie at $\pm\mathrm{i}\pi m$, so the exponentially small sectors are terminant-weighted combinations of $\mathrm{e}^{\mp\mathrm{i}\pi m x}$, whose optimally truncated magnitude is $\mathrm{e}^{-\pi x}$ but which are not real exponentials. The unambiguous completion on the positive axis is the exact Borel–Laplace representation, when the chapter provides one. Where a chapter does supply a rigorously normalized exponential sector, its transport through the inverse map is theorem 1.31 at first order and theorem 1.44 with an explicit error.

Table 1.1: Model data of the four chapters, in the normal form of theorem 1.2 and after the α-reduction of theorem 1.4. “Core” names the admissible core of theorem 1.8 and the Lambert branch selected by theorem 1.21. The row for Part 3 is *not* an instance of theorem 1.2; see the repair box in section 1.2.

Chapter	sequence	variable x	α	b	core, branch
2	rooted trees, A000081	n	1	$-3/2$	$ax + b\log x$, W_{-1}
3	double factorial, A006882	$n+1$	—	—	$\frac{x}{2}(\log x - 1)$, W_0
4	partitions, A000041	$n - \frac{1}{24}$	$1/2$	-1	$ax^{1/2} + b\log x$, W_{-1}
5	swing factorial, A056040	n	1	$\sigma/2$	$ax + b\log x$, W_0 or W_{-1}

1.9 The parameter dictionary

The constants a are: $a = \log \alpha_{\mathrm{Otter}}$ for Part 2; $a = \pi\sqrt{2/3}$ for Part 4, in which $\alpha = 1/2$ and theorem 1.4 substitutes $\xi = (n - 1/24)^{1/2}$, after which $b/\alpha = -2$ and the core equation is $\xi \mapsto a\xi - 2\log\xi$ — the equation the three partition articles write as $\rho - 2\log\rho = L$ after rescaling a to 1; and $a = \log 2$ for Part 5, with $\sigma = (-1)^{n+1}$ the parity transmonomial. Each chapter verifies (1.4) directly and states its own q_j.

Remark 1.48 (Reproducibility of the Part 0 identities). Every coefficient identity of this chapter was checked in exact rational arithmetic, with no floating-point step, over six independent choices of the data $(a, b, q_1, \ldots, q_N)$ with $a, b, q_j \in \mathbb{Q}$, $N \in \{10, 12\}$, using a truncated-series engine over $\mathbb{Q}[H]$ built on Python’s `fractions.Fraction`. The identities checked were: the Lagrange closed formula (1.48) against the defining fixed point; the ordinary-Bell recurrence (1.49); the vanishing of the residual $ax + b\log x + Q(1/x) - L$ under the flattened ansatz (1.65) through order L^{-N}; the shape statements (1.64); and the closed form (1.67) together with its agreement with the three source formulae listed in theorem 1.35. All checks passed. No numerical constant is asserted anywhere in this chapter.

Chapter 2

Butcher–Pólya rooted trees (A000081)

This chapter merges the three independently written treatments *Butcher_Tree_Transseries*, *Butcher_Tree_Transseries-2* and *Butcher_Tree_Transseries-3* of the special-function-inversion draft set. Below they are called S1, S2 and S3. S2 is a book-length treatment and is the most complete on the inverse and on the inherited singularity at $-\sqrt{\rho}$; S3 is the most careful about what is and is not proved beyond the dominant sector; S1 has the shortest route to the universal Cayley coefficients and the only treatment of the compositional inverse of the generating function. Where they disagree the disagreement is stated in the text. Every numerical constant retained here has been recomputed from the exact divisor recurrence at 60–110 working decimal digits; see section 2.11.3.

The subject of this chapter is the sequence

$$0,\ 1,\ 1,\ 2,\ 4,\ 9,\ 20,\ 48,\ 115,\ 286,\ 719,\ 1842,\ 4766,\ 12486,\ 32973,\ 87811,\ \dots \tag{2.1}$$

(OEIS A000081), where for $n \geq 1$ the term a_n counts isomorphism classes of rooted, unlabelled, non-plane trees on n vertices. These are the *Pólya trees*; in numerical analysis the same shapes index Butcher series, elementary differentials and Runge–Kutta order conditions, which is the origin of the name *Butcher tree numbers.* We set $a_0 = 0$ throughout.

The sequence grows like $C\alpha^n n^{-3/2}$ with $\alpha = \rho^{-1}$ and ρ the Otter singularity. Thus, in the language of theorem 1.2, its asymptotic interpolation is an exponential–power model with

$$\boxed{\alpha_{\mathrm{model}} = 1, \qquad a_{\mathrm{model}} = \lambda := \log(1/\rho), \qquad b_{\mathrm{model}} = -\tfrac{3}{2}, \qquad \delta = 1.} \tag{2.2}$$

This chapter is therefore the *linear-phase* instance of the Part 0 apparatus: no α-reduction (theorem 1.4) is needed, the Lambert core (theorem 1.20) applies verbatim, and the entire reversion is a substitution into theorem 1.25. Everything specific to rooted trees is the computation of the model's data C, λ and Q; that computation is the substance of sections 2.1 to 2.5, and the inversion in section 2.7 is short precisely because Part 0 has already been proved.

2.1 The functional equation and the exact recurrence

2.1.1 Pólya's multiset equation

Let $\mathcal{T}$ be the class of unlabelled non-plane rooted trees and let $\mathcal{Z}$ be one atom (one vertex). Deleting the root of a tree leaves an unordered multiset of rooted trees, and grafting a multiset of rooted trees onto a fresh root inverts this. Hence

$$\mathcal{T} = \mathcal{Z} \times \mathrm{MSET}(\mathcal{T}). \tag{2.3}$$

Proposition 2.1 (Pólya's equation and the Euler product)**.** *Let* $T(z) = \sum_{n\geq 1} a_n z^n \in \mathbb{Z}[[z]]$*. Then, as identities of formal power series,*

$$T(z) = z \prod_{r\geq 1} (1 - z^r)^{-a_r}, \tag{2.4}$$

$$T(z) = z \exp\left(\sum_{k\geq 1} \frac{T(z^k)}{k} \right). \tag{2.5}$$

Proof. A multiset of rooted trees that contains m_j copies of the jth shape has size $\sum_j m_j |j|$ and contributes $z^{\sum_j m_j |j|}$. For one fixed shape of size r, unbounded multiplicity contributes $\sum_{m\geq 0} z^{rm} = (1 - z^r)^{-1}$; there are a_r shapes of size r, and distinct shapes are chosen independently, so the forest hanging from the root has generating function $\prod_{r\geq 1}(1 - z^r)^{-a_r}$. Every factor is a unit of $\mathbb{Z}[[z]]$ and the product is z-adically convergent because the rth factor is $1 + \mathcal{O}(z^r)$. Multiplying by the root gives eq. (2.4). Taking the formal logarithm and using $-\log(1 - w) = \sum_{k\geq 1} w^k/k$,

$$\log \frac{T(z)}{z} = \sum_{r\geq 1} a_r \sum_{k\geq 1} \frac{z^{rk}}{k} = \sum_{k\geq 1} \frac{1}{k} \sum_{r\geq 1} a_r (z^k)^r = \sum_{k\geq 1} \frac{T(z^k)}{k},$$

the rearrangement being legitimate because for each fixed power of z only finitely many pairs (r, k) contribute. Exponentiating gives eq. (2.5). □

Remark 2.2. Equation (2.5) is purely formal; no analytic hypothesis is used, and none is available at this stage. All three siblings state it correctly, but only S2 says explicitly that the identity holds in $\mathbb{Z}[[z]]$ before any radius of convergence has been produced. We keep that discipline: section 2.2 is the first place where analysis enters.

2.1.2 The divisor recurrence

Definition 2.3 (Divisor transform)**.** For $m \geq 1$ put

$$b_m := \sum_{d\mid m} d\, a_d. \tag{2.6}$$

Theorem 2.4 (Otter–Pólya recurrence)**.** *With* $a_1 = 1$ *one has, for every* $n \geq 2$,

$$(n - 1)\, a_n = \sum_{j=1}^{n-1} b_j\, a_{n-j}. \tag{2.7}$$

Proof. Write $G(z) = T(z)/z = \sum_{n\geq 0} a_{n+1} z^n$, a unit of $\mathbb{Q}[[z]]$ with $G(0) = 1$. From eq. (2.4) and the computation in the proof of theorem 2.1,

$$\log G(z) = \sum_{r\geq 1} a_r \sum_{k\geq 1} \frac{z^{rk}}{k} = \sum_{m\geq 1} \frac{z^m}{m} \sum_{d\mid m} d\, a_d = \sum_{m\geq 1} \frac{b_m}{m} z^m,$$

where the inner rearrangement groups the pairs (r, k) with $rk = m$ and uses $1/k = d/m$ for $d = r$. Formal logarithmic differentiation gives $G'(z) = G(z) \sum_{m\geq 1} b_m z^{m-1}$. Extracting $[z^{n-1}]$,

$$n\, a_{n+1} = \sum_{j=1}^{n} b_j\, a_{n-j+1},$$

and replacing n by $n - 1$ yields eq. (2.7). □

Remark 2.5 (Why the recurrence is triangular, and its cost). Although b_n contains the summand $n\,a_n$, the right-hand side of eq. (2.7) uses only b_j with $j < n$, so a_n is determined by $a_1, \dots, a_{n-1}$. Implemented as a sieve — after computing a_n, add $n\,a_n$ to $b_n, b_{2n}, b_{3n}, \dots$ — the divisor transform costs $\mathcal{O}(N \log N)$ additions and the convolutions cost $\mathcal{O}(N^2)$ integer operations for $a_1, \dots, a_N$. The integrality assertion $(n-1) \mid \sum_j b_j a_{n-j}$ is a free exact self-test on every step, and was used as such in the computations of section 2.11.3.

2.1.3 The self-interaction and its coefficients

Separating the $k = 1$ term of eq. (2.5) is the structural move on which everything else rests.

Definition 2.6 (Disturbance and Cayley argument)**.** Put

$$R(z) := \sum_{k\geq 2} \frac{T(z^k)}{k}, \qquad \zeta(z) := z\,\mathrm{e}^{R(z)}. \tag{2.8}$$

With this notation eq. (2.5) reads

$$T(z) = \zeta(z)\,\mathrm{e}^{T(z)}. \tag{2.9}$$

Lemma 2.7 (Coefficients of the disturbance)**.** *$R(z) = \sum_{r\geq 2} \varrho_r z^r$ with*

$$\varrho_r = \frac{1}{r} \sum_{\substack{d|r \\ d<r}} d\,a_d = \frac{b_r - r\,a_r}{r} \in \mathbb{Q}_{\geq 0}. \tag{2.10}$$

In particular ϱ_r depends only on a_d with $d \leq r/2$, and $\varrho_2 = \frac{1}{2}$, $\varrho_3 = \frac{1}{3}$, $\varrho_4 = \frac{3}{4}$, $\varrho_5 = \frac{1}{5}$, $\varrho_6 = \frac{3}{2}$, $\varrho_7 = \frac{1}{7}$, $\varrho_8 = \frac{19}{8}$.

Proof. $R(z) = \sum_{k\geq 2} \sum_{d\geq 1} \frac{a_d}{k} z^{kd}$; collecting the terms with $kd = r$ gives $\varrho_r = \sum_{d|r,\, d<r} a_d/(r/d) = \frac{1}{r} \sum_{d|r,\, d<r} d\,a_d$, which is $(b_r - ra_r)/r$ by eq. (2.6). Every divisor $d < r$ of r satisfies $d \leq r/2$. The listed values follow from $a_1 = a_2 = 1$, $a_3 = 2$, $a_4 = 4$: e.g. $\varrho_4 = (1 \cdot 1 + 2 \cdot 1)/4 = \frac{3}{4}$ and $\varrho_6 = (1 + 2 \cdot 1 + 3 \cdot 2)/6 = \frac{3}{2}$. □

Theorem 2.7 is due to S2; S1 and S3 carry R only through the double sum $\sum_{k\geq 2} \sum_n a_n \rho^{kn}(\cdots)$. The single-sum form is used throughout this chapter because it turns every quantity below into one geometrically convergent series in the exact integers a_r (theorem 2.20).

2.2 The Cayley–Lambert normal form and the Otter singularity

2.2.1 Factorisation through the Cayley tree function

Definition 2.8 (Cayley tree function)**.** Let $\mathcal{C}(w)$ be the unique formal solution of $\mathcal{C} = w\mathrm{e}^{\mathcal{C}}$ with $\mathcal{C}(0) = 0$. It is analytic on $|w| < \mathrm{e}^{-1}$ and

$$\mathcal{C}(w) = -W_0(-w), \tag{2.11}$$

where W_0 is the principal Lambert branch.

Comparing eq. (2.9) with theorem 2.8,

$$\boxed{T(z) = \mathcal{C}\big(\zeta(z)\big) = -W_0\big(-\zeta(z)\big).} \tag{2.12}$$

This holds as formal series, and as an identity of analytic functions on the common domain of the branches selected at the origin. The point of the factorisation is a clean separation of labour: *all* of the singular mechanism is the universal square-root branch point of $\mathcal{C}$ at $w = \mathrm{e}^{-1}$, and *all* of the tree-specific information is packed into the map ζ, which — as we now show — is analytic on a strictly larger disc than T itself.

Lemma 2.9 (Elementary bounds on the radius). *The radius of convergence ρ of T satisfies $\frac{1}{4} \le \rho \le \mathrm{e}^{-1}$.*

Proof. Since $a_n \ge 1$ for $n \ge 1$ we have $\rho \le 1$, so for $0 < x < \rho$ every x^k with $k \ge 2$ lies in $(0, \rho)$ as well. Upper bound: for such x the series $T(x)$ and $R(x)$ converge, all terms are nonnegative, and $\zeta(x) = x\mathrm{e}^{R(x)} \ge x$; so eq. (2.9) gives $T(x) \ge x\mathrm{e}^{T(x)}$, i.e. $x \le T(x)\mathrm{e}^{-T(x)} \le \sup_{u \ge 0} u\mathrm{e}^{-u} = \mathrm{e}^{-1}$. Letting $x \uparrow \rho$ gives $\rho \le \mathrm{e}^{-1}$. Lower bound: forgetting the identification of isomorphic children embeds the shapes counted by a_n into the rooted *plane* trees on n vertices, of which there are $\frac{1}{n}\binom{2n-2}{n-1} \le 4^n$; hence $\limsup a_n^{1/n} \le 4$ and $\rho \ge \frac{1}{4}$. □

Proposition 2.10 (Analyticity of the disturbance). *R, and hence ζ, is analytic on $|z| < \sqrt{\rho}$, and the series $\sum_{k \ge 2} T(z^k)/k$ converges normally on compact subsets together with all its derivatives. In particular R and ζ are analytic at $z = \rho$, since $\rho < \sqrt{\rho}$ by theorem 2.9.*

Proof. Fix $r < \sqrt{\rho}$, so $r^2 < \rho$. Since T has radius ρ and $T(w) = w + \mathcal{O}(w^2)$, there are $M > 0$ and $r_0 \in (0, 1)$ with $|T(w)| \le M|w|$ for $|w| \le r_0$. For $|z| \le r$, only finitely many k have $r^k > r_0$, and for each such k we have $r^k \le r^2 < \rho$, so $T(z^k)$ is defined and bounded there. For the remaining k, $|T(z^k)/k| \le Mr^k/k$, which is summable. Differentiating j times multiplies the kth tail term by a factor $\mathcal{O}(k^j r^{-j})$, still summable. Weierstrass' theorem gives analyticity and termwise differentiation. □

Proposition 2.11 (Critical characterisation of ρ). *With $\rho \in [\frac{1}{4}, \mathrm{e}^{-1}]$ the radius of convergence of T,*

$$T(\rho) = 1, \qquad \zeta(\rho) = \mathrm{e}^{-1}, \qquad \log\rho + 1 + R(\rho) = 0. \tag{2.13}$$

Moreover ζ is strictly increasing on $(0, \sqrt{\rho})$ and

$$\theta_1 := \mathrm{e}\rho\,\zeta'(\rho) = 1 + \rho R'(\rho) > 0. \tag{2.14}$$

Proof. All coefficients of T, hence of R and of $\zeta = z\mathrm{e}^R$, are nonnegative, and $\zeta(z) = z + \frac{1}{2}z^3 + \frac{1}{3}z^4 + \cdots$ has positive linear coefficient; therefore $\zeta'(x) > 0$ on $(0, \sqrt{\rho})$, and ζ is strictly increasing there.

Step 1: $T(x) < 1$ on $[0, \rho)$, and $T = \mathcal{C} \circ \zeta$ there. On $[0, \rho)$ the function T is continuous, increasing, with $T(0) = 0$, and eq. (2.9) gives $\zeta(x) = T(x)\mathrm{e}^{-T(x)} \le \mathrm{e}^{-1}$. Let $\rho_1 := \sup\{x \in [0, \rho) : T(x) < 1\}$. On $[0, \rho_1)$ we have $\zeta(x) < \mathrm{e}^{-1}$, so $\mathcal{C}$ is analytic at $\zeta(x)$ and $\mathcal{C} \circ \zeta$ is an analytic solution of eq. (2.9) agreeing with T at 0; by uniqueness of that germ and analytic continuation, $T = \mathcal{C} \circ \zeta$ on $[0, \rho_1)$. If $\rho_1 < \rho$ then $T(\rho_1) = 1$ and $\zeta(\rho_1) = \mathrm{e}^{-1}$; but ζ is strictly increasing, so $\zeta(x) > \mathrm{e}^{-1}$ for $x \in (\rho_1, \rho)$, contradicting $\zeta(x) \le \mathrm{e}^{-1}$. Hence $\rho_1 = \rho$.

Step 2: $\zeta(\rho) = \mathrm{e}^{-1}$. By theorem 2.10 the function ζ is analytic at ρ, and by Step 1 and monotonicity $\zeta(\rho) = \lim_{x \uparrow \rho} \zeta(x) \le \mathrm{e}^{-1}$. If the inequality were strict, $\mathcal{C}$ would be analytic at $\zeta(\rho)$ and $\mathcal{C} \circ \zeta$ would be analytic in a full neighbourhood of ρ and would agree with T on $[0, \rho)$; then T would continue analytically through ρ, contradicting Pringsheim's theorem, which makes the positive point $z = \rho$ a singularity of a series with nonnegative coefficients. Hence $\zeta(\rho) = \mathrm{e}^{-1}$ and, by theorem 2.8, $T(\rho) = \mathcal{C}(\mathrm{e}^{-1}) = 1$.

Step 3. Taking logarithms in $\zeta(\rho) = \rho\mathrm{e}^{R(\rho)} = \mathrm{e}^{-1}$ gives the scalar equation; equivalently $R(\rho) = \lambda - 1$ with $\lambda = -\log\rho$, and $R(\rho) > 0$ independently re-proves $\rho < \mathrm{e}^{-1}$. Finally $\zeta' = (1/z + R')\zeta$, so $\mathrm{e}\rho\zeta'(\rho) = \mathrm{e}\zeta(\rho)\bigl(1 + \rho R'(\rho)\bigr) = 1 + \rho R'(\rho)$, positive because R has nonnegative coefficients. □

Proposition 2.12 (Uniqueness of the dominant singularity). *$z = \rho$ is the only singularity of T on the circle $|z| = \rho$.*

Proof. Take $z = \rho\mathrm{e}^{\mathrm{i}\vartheta}$ with $\vartheta \not\equiv 0$. The series $\zeta(z) = z + \frac{1}{2}z^3 + \frac{1}{3}z^4 + \cdots$ has nonnegative coefficients, so $|\zeta(z)| \le \zeta(\rho) = \mathrm{e}^{-1}$, with equality only if the three displayed nonzero terms z, $\frac{1}{2}z^3$, $\frac{1}{3}z^4$ share one argument, i.e. $\mathrm{e}^{2\mathrm{i}\vartheta} = \mathrm{e}^{3\mathrm{i}\vartheta} = 1$, hence $\mathrm{e}^{\mathrm{i}\vartheta} = 1$ and $\vartheta \equiv 0$. So $|\zeta(z)| < \mathrm{e}^{-1}$ strictly, $\mathcal{C}$ is analytic at $\zeta(z)$, and ζ is analytic at z because $\rho < \sqrt{\rho}$ (theorem 2.10). Hence $T = \mathcal{C} \circ \zeta$ is analytic at z. □

Remark 2.13. This is S3's argument, and it is the only one of the three that is self-contained: S1 and S2 appeal to "positivity and aperiodicity" without exhibiting the aperiodicity witness. The three coefficients z, z^3, z^4 of ζ furnish that witness explicitly, because $\gcd(3-1, 4-1) = 1$.

2.2.2 The constants

Convention 2.14 (Growth constants). Throughout the chapter

$$\alpha := \rho^{-1}, \qquad \lambda := \log \alpha = -\log \rho, \qquad p := \tfrac{3}{2}. \tag{2.15}$$

Notation collision between the siblings. S2 and S3 write $\alpha = \rho^{-1}$ and $\lambda = \log \alpha$; S1 writes exactly the opposite, $\lambda = \rho^{-1}$ and $\alpha = -\log \rho$. The symbols therefore cross-denote between the drafts, and S1's Ω_-/α, $\mathrm{e}^{\alpha n}$ and d_m formulas must be read with $\alpha = \lambda_{\text{here}}$. This volume adopts the two-to-one majority reading eq. (2.15); every formula imported from S1 has been translated.

Solving $\log \rho + 1 + R(\rho) = 0$ from eq. (2.10) with 700 exact tree numbers at 110 working digits (see section 2.11.3) gives

$$\rho = 0.3383218568992076951961126257170170531837746075329677955723037 76\ldots, \tag{2.16}$$
$$\alpha = 2.955765285651994974714817524123194588375492304663596595350472478\ldots, \tag{2.17}$$
$$\lambda = 1.083757597244624660482711980896174279049662124202672234914427065\ldots, \tag{2.18}$$
$$R(\rho) = 0.083757597244624660482711980896174279049662124202672234914427065\ldots, \tag{2.19}$$

and, as an independent consistency check on the whole pipeline,

$$\lambda = 1 + R(\rho), \tag{2.20}$$

which is nothing but eq. (2.13) rearranged; the two computed values agree to all 70 digits carried. Two further constants are recorded here because they are quoted repeatedly:

$$\theta_1 = 1.216004561824048922157212619228187444\ldots, \qquad \zeta'(\rho) = 1.3222411426969302644512848 89\ldots. \tag{2.21}$$

2.3 The universal Cayley branch germ

The next result is universal: it depends on the Cayley function alone, not on rooted trees, and applies verbatim to *any* class satisfying $T = \zeta \mathrm{e}^{T}$ with analytic ζ.

Definition 2.15 (Cayley kernel). Put

$$\Phi(v) := \frac{2\big(1 - (1 - v)\mathrm{e}^{v}\big)}{v^2} = \sum_{m \ge 0} \frac{2(m+1)}{(m+2)!}\, v^m = \sum_{m \ge 0} \frac{2}{(m+2)\, m!}\, v^m = 1 + \frac{2}{3}v + \frac{1}{4}v^2 + \frac{1}{15}v^3 + \cdots. \tag{2.22}$$

The two displayed coefficient forms agree because $2(m+1)/(m+2)! = 2/((m+2)\,m!)$; S1 and S3 use the first, S2 the second.

Theorem 2.16 (Universal Cayley coefficients). *Set $w = (1 - v)/\mathrm{e}$ and $\mathcal{C}(w) = 1 - v$ on the branch reached from $0 < w < \mathrm{e}^{-1}$. Then $\upsilon = \frac{1}{2}v^2\Phi(v)$, the inverse germ is $v = \sum_{m \ge 1} \omega_m (2\upsilon)^{m/2}$, and for every $m \ge 1$*

$$\boxed{\omega_m = \frac{1}{m}\,[\upsilon^{m-1}]\,\Phi(v)^{-m/2} = \frac{1}{m}\sum_{k=0}^{m-1}\binom{-m/2}{k}\widehat{B}_{m-1,k}\big(\tfrac{2}{3}, \tfrac{1}{4}, \tfrac{1}{15}, \ldots\big) \in \mathbb{Q}.} \tag{2.23}$$

Consequently, on that branch,

$$\mathcal{C}\left(\frac{1 - \upsilon}{\mathrm{e}}\right) = 1 + \sum_{r \ge 1} c_r\, \upsilon^{r/2}, \qquad c_r = -2^{r/2}\,\omega_r. \tag{2.24}$$

Proof. From $\mathcal{C} = w\mathrm{e}^{\mathcal{C}}$ we get $\mathrm{e}w = \mathcal{C}\mathrm{e}^{1-\mathcal{C}}$, so with $\mathcal{C} = 1 - v$,

$$1 - v = \mathrm{e}w = (1-v)\mathrm{e}^{v}, \qquad \text{hence} \qquad v = 1 - (1-v)\mathrm{e}^{v} = \sum_{m\ge 2} \frac{m-1}{m!} v^m = \frac{v^2}{2}\Phi(v),$$

the middle identity by expanding e^v and collecting. Put $\varsigma = \sqrt{2v}$ with the determination $\varsigma/v \to 1$; then $\varsigma = v\,\Phi(v)^{1/2}$ is an ordinary power series in v with nonzero linear coefficient, so theorem 1.14 applies with $\phi(v) = \Phi(v)^{1/2}$ and yields

$$[\varsigma^m]\, v(\varsigma) = \frac{1}{m}[v^{m-1}]\, \phi(v)^{-m} = \frac{1}{m}[v^{m-1}]\, \Phi(v)^{-m/2} = \omega_m.$$

The Bell form is theorem 1.12 applied to the unit series Φ, whose coefficients are $2(m+1)/(m+2)!$ for $m \ge 1$. Rationality follows because $\binom{-m/2}{k} \in \mathbb{Q}$ and $\widehat{B}_{m-1,k}$ is a polynomial with rational coefficients in rational arguments. Finally $\mathcal{C} = 1 - v = 1 - \sum_m \omega_m \varsigma^m = 1 - \sum_m \omega_m 2^{m/2} v^{m/2}$, which is eq. (2.24). □

Remark 2.17 (Two normalisations, one object). S2 works with ω_m (expanding in $\varsigma = \sqrt{2v}$) while S1 and S3 work with c_r (expanding in $v^{1/2}$). The dictionary is $c_r = -2^{r/2}\omega_r$, verified here for $1 \le r \le 18$ by exact rational arithmetic. The ω_m are rational, the c_r alternate between rationals and rational multiples of $\sqrt{2}$; this is the only reason to prefer ω for tables.

Table 2.1: Exact universal Cayley coefficients ω_m of eq. (2.23), recomputed by rational arithmetic from eq. (2.22).

m	ω_m	m	ω_m
1	1	10	−5776369/1515591000
2	−1/3	11	169709463197/69528040243200
3	11/72	12	−1118511313/709296588000
4	−43/540	13	667874164916771/650782456676352000
5	769/17280	14	−500525573/744761417400
6	−221/8505	15	103663334225097487/234281684403486720000
7	680863/43545600	16	−466901817532379/1595278956070800000
8	−1963/204120	17	212352941850863050433/1092422025561400934400000
9	226287557/37623398400	18	−106040742894306601/818378104464320400000

The corresponding c_r are

$$\begin{aligned} c_1 &= -\sqrt{2}, & c_2 &= \tfrac{2}{3}, & c_3 &= -\tfrac{11\sqrt{2}}{36}, & c_4 &= \tfrac{43}{135}, \\ c_5 &= -\tfrac{769\sqrt{2}}{4320}, & c_6 &= \tfrac{1768}{8505}, & c_7 &= -\tfrac{680863\sqrt{2}}{5443200}, & c_8 &= \tfrac{3926}{25515}. \end{aligned} \tag{2.25}$$

Remark 2.18 (On the observed sign pattern). All three siblings display ω_m with strictly alternating signs and none proves it. The data in table 2.1 are consistent with $\operatorname{sgn}\omega_m = (-1)^{m+1}$ for $1 \le m \le 18$, and this is what one expects from the fact that the inverse function $v(\varsigma)$ of $\varsigma = v\Phi(v)^{1/2}$ has its nearest singularity on the negative ς-axis; but no proof is offered here and the statement is recorded as an observation, not as a result.

2.4 The Puiseux expansion at the Otter singularity

2.4.1 The local disturbance series

Definition 2.19 (Local coordinate and disturbance)**.** Write

$$s := 1 - \frac{z}{\rho}, \qquad \theta(s) := 1 - \mathrm{e}\,\zeta\big(\rho(1-s)\big) = \sum_{m\ge 1} \theta_m s^m. \tag{2.26}$$

The constant term vanishes by eq. (2.13). By theorem 2.10, θ is analytic on the image of $|z| < \sqrt{\rho}$ under $z \mapsto s$, namely the disc $|s-1| < \rho^{-1/2}$; in particular its Taylor series at $s = 0$ converges for $|s| < \rho^{-1/2} - 1 = 0.71923\ldots$.

Taylor's formula gives the derivative form

$$\theta_m = \frac{(-1)^{m+1}\mathrm{e}\rho^m}{m!}\,\zeta^{(m)}(\rho), \qquad m \geq 1, \tag{2.27}$$

consistent with eq. (2.14): $\theta_1 = \mathrm{e}\rho\zeta'(\rho)$. For computation the following coefficient form is preferable, since it needs nothing but the integers a_r and the single scalar ρ.

Lemma 2.20 (Local coefficients of R from the tree numbers)**.** *Let $\widehat{R}_m := [s^m]\, R\big(\rho(1-s)\big)$. Then*

$$\boxed{\widehat{R}_m = (-1)^m \sum_{r\geq 2} \varrho_r \binom{r}{m} \rho^r = (-1)^m\, \frac{\rho^m R^{(m)}(\rho)}{m!},} \tag{2.28}$$

with ϱ_r as in eq. (2.10)*. The series converges geometrically: for fixed m the general term is $\mathcal{O}\big(r^m \rho^{r/2}\big)$.*

Proof. By theorem 2.7, $R(\rho(1-s)) = \sum_{r\geq 2} \varrho_r \rho^r (1-s)^r$, and $[s^m]\,(1-s)^r = (-1)^m\binom{r}{m}$. The middle expression is a Taylor coefficient of R at ρ in the variable ρs, giving the right-hand form. For the convergence rate, every divisor $d < r$ of r satisfies $d \leq r/2$, so $\varrho_r \leq a_{\lfloor r/2 \rfloor}\, d(r)$ with $d(r)$ the number of divisors of r; since $\limsup_n a_n^{1/n} = \rho^{-1}$, for every $\epsilon > 0$ there is K_ϵ with $\varrho_r \rho^r \leq K_\epsilon(\sqrt{\rho} + \epsilon)^r$. With $\binom{r}{m} \leq r^m$ the claim follows. $\square$

Remark 2.21 (A dedup). S1 (its ℓ_m) and S3 (its $[s^m]\,R(\rho(1-s))$) both give the *double* sum $(-1)^m \sum_{k\geq 2} \frac{1}{k} \sum_{n\geq 1} a_n \binom{kn}{m} \rho^{kn}$. Equation (2.28) is the same number after the inner divisor collection of theorem 2.7, and is one sum rather than two. The two forms were checked against each other numerically to 70 digits for $0 \leq m \leq 30$.

Lemma 2.22 (The disturbance coefficients)**.** *With $\widehat{R}_m$ as in theorem 2.20, put $\widehat{R}_+(s) := \sum_{m\geq 1} \widehat{R}_m s^m$ and $E(s) := \exp \widehat{R}_+(s) = \sum_{m\geq 0} E_m s^m$. Then $E_0 = 1$,*

$$E_m = \sum_{k=0}^{m} \frac{1}{k!}\, \widehat{B}_{m,k}\big(\widehat{R}_1, \widehat{R}_2, \ldots\big) = \frac{1}{m} \sum_{k=1}^{m} k\, \widehat{R}_k\, E_{m-k}, \tag{2.29}$$

and

$$\boxed{\theta_m = E_{m-1} - E_m, \qquad m \geq 1.} \tag{2.30}$$

Proof. By eq. (2.13), $\mathrm{e}\zeta(\rho) = 1$, so

$$\mathrm{e}\,\zeta\big(\rho(1-s)\big) = \frac{\rho(1-s)\mathrm{e}^{R(\rho(1-s))}}{\rho\,\mathrm{e}^{R(\rho)}} = (1-s)\,\mathrm{e}^{\widehat{R}_+(s)} = (1-s)E(s),$$

because $\widehat{R}_0 = R(\rho)$. Hence $\theta(s) = 1 - (1-s)E(s)$ and $\theta_m = -(E_m - E_{m-1})$ for $m \geq 1$, while the constant term is $1 - E_0 = 0$. The two expressions for E_m are the ordinary Bell expansion of an exponential (theorem 1.9) and the standard triangular exponential recurrence obtained from $E' = \widehat{R}'_+ E$. $\square$

Remark 2.23 (The jet route). S2 computes θ_m instead from the *scaled logarithmic jets*

$$\ell_j := \rho^j\, \frac{\mathrm{d}^j}{\mathrm{d}z^j} \log\zeta(z)\bigg|_{z=\rho} = (-1)^{j-1}(j-1)! + \sum_{r\geq j} \varrho_r\, r^{\underline{j}}\, \rho^r, \tag{2.31}$$

via $\theta_m = \frac{(-1)^{m+1}}{m!} B_m(\ell_1, \ldots, \ell_m)$, where B_m is the complete exponential Bell polynomial and $r^{\underline{j}} = r(r-1)\cdots(r-j+1)$. This is equivalent to theorem 2.22 through $\widehat{R}_m = (-1)^m \rho^m R^{(m)}(\rho)/m!$ and

$\ell_j = (-1)^j j!\,\widehat{R}_j$ for $j \geq 1$ up to the $\log z$ term; both routes were implemented and agree. Theorem 2.22 is preferred because it avoids the factorial rescaling and its intermediate quantities stay $\mathcal{O}(1)$. S2's normalisation $s_j = 2\theta_j$ (it expands $2(1-\mathrm{e}\zeta)$ rather than $1-\mathrm{e}\zeta$) accounts for every factor of 2 that differs between the siblings' formulas for t_r.

The first disturbance coefficients, recomputed, are

$$\begin{aligned}
\theta_1 &= \ 1.2160045618240489221572126 19\ldots, & \theta_2 &= -0.4584157298563917191808477 17\ldots,\\
\theta_3 &= \ 0.4481047743895350436657341 22\ldots, & \theta_4 &= -0.3995616799562096774062319 80\ldots,\\
\theta_5 &= \ 0.3852513419466130007195463 96\ldots, & \theta_6 &= -0.3898065619592033419491646 40\ldots.
\end{aligned} \tag{2.32}$$

2.4.2 All local coefficients

Theorem 2.24 (Every Puiseux coefficient at ρ)**.** *Factor $\theta(s) = \theta_1 s\bigl(1+U(s)\bigr)$ with $U(s) = \sum_{j\geq 1} u_j s^j$, $u_j = \theta_{j+1}/\theta_1$. Then in a slit neighbourhood of $z=\rho$ there is a convergent Puiseux expansion*

$$T(z) = 1 + \sum_{n\geq 1} t_n\, s^{n/2}, \qquad s = 1 - \frac{z}{\rho}, \tag{2.33}$$

and for every $n \geq 1$

$$\boxed{t_n = \sum_{\substack{1\leq r\leq n\\ r\equiv n\ (2)}} c_r\,\theta_1^{r/2}\,[s^{(n-r)/2}]\,\bigl(1+U(s)\bigr)^{r/2} = \sum_{\substack{1\leq r\leq n\\ r\equiv n\ (2)}} c_r\,\theta_1^{r/2} \sum_{k=0}^{(n-r)/2} \binom{r/2}{k} \widehat{B}_{\frac{n-r}{2},k}(u_1,u_2,\ldots).} \tag{2.34}$$

Thus t_n is a finite expression in $\omega_1,\ldots,\omega_n$, $\theta_1^{\pm 1/2}$ and $\theta_2,\ldots,\theta_{\lceil (n+1)/2\rceil}$.

Proof. Near $z=\rho$ we have $\mathrm{e}\zeta = 1-\theta(s)$ with $\theta(0)=0$, so by eq. (2.12) and eq. (2.24), $T = 1 + \sum_{r\geq 1} c_r\theta(s)^{r/2}$, the branch being the one with $T\to 1^-$ as $s\to 0^+$. Substituting $\theta(s)^{r/2} = \theta_1^{r/2}s^{r/2}(1+U(s))^{r/2}$, a term of index r can contribute to $s^{n/2}$ only when $n-r$ is a nonnegative even integer. Extracting $[s^{(n-r)/2}]$ and applying theorem 1.12 to the unit series $1+U$ gives eq. (2.34). Convergence: θ is analytic and has a simple zero at $s=0$ with $\theta_1>0$ (eq. (2.14)), so U is analytic with $U(0)=0$; choosing a slit neighbourhood on which $|U|<1$ and a consistent square root, the convergent series eq. (2.24) may be composed with the analytic map $s\mapsto\theta(s)$. □

Corollary 2.25. *$t_1 = -\sqrt{2\theta_1}$, $t_2 = \frac{2}{3}\theta_1$, and $t_3 = -\frac{11\sqrt{2}}{36}\theta_1^{3/2} - \frac{\sqrt{2}}{2}\theta_2\theta_1^{-1/2}$.*

Proof. Take $n=1,2,3$ in eq. (2.34) with $c_1 = -\sqrt{2}$, $c_2 = \frac{2}{3}$, $c_3 = -\frac{11\sqrt{2}}{36}$ and $\widehat{B}_{0,0}=1$, $\widehat{B}_{1,1}(u) = u_1$, $\binom{1/2}{1} = \frac{1}{2}$. □

Remark 2.26 (Sibling cross-check). In S2's normalisation $s_1 = 2\theta_1$, $s_2 = 2\theta_2$, the same coefficient reads $t_3 = -(11s_1^2 + 36s_2)/(72\sqrt{s_1})$; substituting $s_j = 2\theta_j$ reproduces theorem 2.25 exactly. Likewise $t_1 = -\sqrt{s_1}$ and $t_2 = s_1/3$. Both were checked numerically.

Remark 2.27 (Analytic versus singular local terms). The even part $\sum_j t_{2j}s^j$ of eq. (2.33) is analytic at $z=\rho$. A fixed analytic monomial $(1-z/\rho)^j$ is a polynomial in z and contributes nothing to $[z^n]$ once $n>j$; hence only the odd coefficients t_{2j+1} enter the algebraic $\rho^{-n}n^{-\bullet}$ asymptotics of section 2.5. This is not a discarding of information: the analytic germ continues through ρ and reappears in the exponentially smaller sectors of section 2.9. The sign pattern of table 2.2 is not monotone and should not be over-read; these are local analytic data, not the final coefficient constants.

Table 2.2: Puiseux coefficients in $T(z) = 1 + \sum_{n\ge1} t_n(1 - z/\rho)^{n/2}$, recomputed from eq. (2.34) at 90 working digits.

n	t_n	n	t_n
1	$-1.55949002037464088554 2$	10	$0.20598483127790741873 6$
2	$0.81066970788269928143 8$	11	$-0.05272470849819056231 5$
3	$-0.28548702161284560530 4$	12	$0.01702656875495366944 6$
4	$0.16537236571208389347 7$	13	$-0.15237062436632539628 6$
5	$-0.34245997040215420104 2$	14	$0.17370288329985046289 4$
6	$0.31740722594652856356 8$	15	$-0.01447370373952704485 7$
7	$-0.10777880029163100918 2$	16	$-0.02189951761121556223 3$
8	$0.06138495705583510468 9$	17	$-0.14454719357090970547 7$
9	$-0.19521238359755646361 3$	18	$0.17607710888501777906 2$

In Otter's coordinate $\rho - z = \rho s$ the leading behaviour reads $T(z) = 1 - \mathsf{b}\,(\rho - z)^{1/2} + \mathcal{O}(\rho - z)$ with

$$\mathsf{b} = -\frac{t_1}{\sqrt{\rho}} = 2.68112814726711223857732878\ldots. \tag{2.35}$$

Theorem 2.28 (Local expansion with an explicit remainder). *For every $N \ge 1$ there are a dented ("Δ-domain") neighbourhood Δ of ρ and a constant K_N such that*

$$\left|T(z) - 1 - \sum_{n=1}^{N} t_n\left(1 - \frac{z}{\rho}\right)^{n/2}\right| \le K_N\left|1 - \frac{z}{\rho}\right|^{(N+1)/2}, \qquad z \in \Delta. \tag{2.36}$$

Proof. By theorem 2.10 the map $s \mapsto \theta(s)$ is analytic on a disc $|s| < \varepsilon_0$; by eq. (2.14) it has a simple zero at $s = 0$ with $\theta_1 > 0$. Shrinking ε_0, $|U(s)| < \frac{1}{2}$ there. The series eq. (2.24) converges for $|v|$ small; a Cauchy estimate on the analytic function $v \mapsto \mathcal{C}((1 - v)/\mathrm{e})$ restricted to a slit disc gives $|c_r| \le K\varepsilon_1^{-r/2}$. Composing convergent expansions and using Taylor's theorem with remainder on $(1 + U)^{r/2}$ gives eq. (2.36) on the slit disc $\{|s| < \varepsilon,\ |\arg s| < \pi - \eta\}$, which is a Δ-domain at ρ in the z-variable. □

2.5 Transfer to the all-orders coefficient asymptotics

2.5.1 Exact transfer of a singular monomial

For $\beta \notin \mathbb{N}_0$ and $n \ge 0$,

$$[z^n]\left(1 - \frac{z}{\rho}\right)^{\beta} = \rho^{-n}(-1)^n\binom{\beta}{n} = \rho^{-n}\frac{\Gamma(n - \beta)}{\Gamma(-\beta)\Gamma(n + 1)}. \tag{2.37}$$

For $\beta \in \mathbb{N}_0$ the left-hand side vanishes for $n > \beta$ — the analytic-germ statement of theorem 2.27. It remains to expand the gamma ratio to all orders.

Lemma 2.29 (Gamma-ratio coefficients). *Let $B_m(x)$ be the Bernoulli polynomials, and for $\beta \in \mathbb{C}$ set*

$$\varkappa_m(\beta) := \frac{(-1)^{m+1}}{m(m+1)}\big(B_{m+1}(-\beta) - B_{m+1}(1)\big), \qquad m \ge 1, \tag{2.38}$$

and define $g_r(\beta)$ by $\exp\big(\sum_{m\ge1}\varkappa_m(\beta)v^m\big) = \sum_{r\ge0} g_r(\beta)v^r$, equivalently

$$g_0(\beta) = 1, \qquad g_r(\beta) = \frac{1}{r}\sum_{m=1}^{r} m\,\varkappa_m(\beta)\,g_{r-m}(\beta) = \sum_{k=0}^{r}\frac{1}{k!}\widehat{B}_{r,k}\big(\varkappa_1(\beta), \varkappa_2(\beta), \ldots\big). \tag{2.39}$$

Then, for fixed β and $n \to \infty$ in any sector avoiding the negative real axis,

$$\frac{\Gamma(n-\beta)}{\Gamma(n+1)} \sim n^{-\beta-1} \sum_{r\geq 0} \frac{g_r(\beta)}{n^r}. \tag{2.40}$$

Each g_r is a polynomial of degree $2r$ in β with rational coefficients; the first five are

$$g_0 = 1, \qquad g_1(\beta) = \frac{\beta(\beta+1)}{2}, \quad g_2(\beta) = \frac{\beta(\beta+1)(\beta+2)(3\beta+1)}{24},$$
$$g_3(\beta) = \frac{\beta^2(\beta+1)^2(\beta+2)(\beta+3)}{48}, \tag{2.41}$$

$$g_4(\beta) = \frac{\beta(\beta+1)(\beta+2)(\beta+3)(\beta+4)\big(15\beta^3+30\beta^2+5\beta-2\big)}{5760}. \tag{2.42}$$

In particular $\big(g_r(\tfrac{1}{2})\big)_{r\geq 0} = 1, \frac{3}{8}, \frac{25}{128}, \frac{105}{1024}, \frac{1659}{32768}, \frac{6237}{262144}, \ldots$

Proof. The Euler–Maclaurin/Bernoulli form of Stirling's expansion gives, for fixed a, b,

$$\log \frac{\Gamma(n+a)}{\Gamma(n+b)\, n^{a-b}} \sim \sum_{m\geq 1} \frac{(-1)^{m+1}}{m(m+1)} \frac{B_{m+1}(a)-B_{m+1}(b)}{n^m}$$

as $n \to \infty$ in the stated sectors. Taking $a = -\beta, b = 1$ yields $\log\big(\Gamma(n-\beta)/\Gamma(n+1)\big) \sim -(\beta+1)\log n + \sum_m \varkappa_m(\beta) n^{-m}$, and exponentiating an asymptotic series term by term is legitimate, producing exactly the coefficients of $\exp\big(\sum \varkappa_m v^m\big)$. The recurrence follows from differentiating that generating identity; the Bell form is theorem 1.9. Degree: $\varkappa_m$ has degree $m+1$ in β and vanishes at $\beta = 0$, so a monomial $\prod \varkappa_{m_i}$ with $\sum m_i = r$ has degree $\leq r + \#\{i\} \leq 2r$; the displayed $g_1, \ldots, g_4$ were recomputed here from eq. (2.38) by exact rational arithmetic. ☐

Remark 2.30 (Part 0 candidate). Theorem 2.29 is not specific to rooted trees and is used verbatim wherever a chapter transfers an algebraic singularity to coefficient asymptotics. It is listed in the report as a candidate for promotion into Part 0. All three siblings state it; S2 and S3 give the same recurrence, S1 the same object under the name $\gamma_m(\beta)$ with $\eta_r(\beta)$ for $\varkappa_r(\beta)$.

2.5.2 The master coefficient formula

Theorem 2.31 (All-orders dominant expansion of A000081). *For every fixed $M \geq 0$,*

$$\boxed{a_n = \frac{\rho^{-n}}{\sqrt{\pi}\, n^{3/2}} \left(\sum_{m=0}^{M} \frac{\tau_m}{n^m} + \mathcal{O}(n^{-M-1}) \right), \qquad n \to \infty,} \tag{2.43}$$

where

$$\boxed{\tau_m = \sum_{j=0}^{m} \Delta_j\, t_{2j+1}\, g_{m-j}\big(j+\tfrac{1}{2}\big), \qquad \Delta_j := \frac{(-1)^{j+1}(2j+1)!!}{2^{j+1}} = \frac{\sqrt{\pi}}{\Gamma(-j-\frac{1}{2})}.} \tag{2.44}$$

Every ingredient is a finite coefficient extraction: t_{2j+1} by eq. (2.34), *g_r by eq.* (2.39), *θ_m by eq.* (2.30), *ω_m by eq.* (2.23), *ϱ_r by eq.* (2.10).

Proof. Fix M and apply theorem 2.28 with $N = 2M+2$: on a Δ-domain at ρ,

$$T(z) = A(z) + \sum_{j=0}^{M} t_{2j+1} \left(1 - \frac{z}{\rho}\right)^{j+1/2} + \mathcal{O}\left(\left|1 - \frac{z}{\rho}\right|^{M+3/2} \right),$$

with A the polynomial collecting the even (analytic) terms. By theorem 2.12 the only singularity on $|z| = \rho$ is ρ itself, so the Cauchy contour may be deformed into the standard Flajolet–Odlyzko Δ-contour. The polynomial A contributes nothing to $[z^n]$ for $n > \deg A$. Each retained monomial contributes exactly $\rho^{-n} t_{2j+1} \Gamma(n - j - \frac{1}{2}) / \big(\Gamma(-j - \frac{1}{2})\Gamma(n+1)\big)$ by eq. (2.37), whose asymptotic expansion is eq. (2.40); the $\mathcal{O}$-term transfers to $\mathcal{O}(\rho^{-n} n^{-M-5/2})$ by the standard Hankel-contour estimate. Collecting the coefficient of $\rho^{-n} n^{-m-3/2}$ from the pairs $j + r = m$ and multiplying through by $\sqrt{\pi}$ gives eq. (2.44), the identity $\Gamma(-j - \frac{1}{2}) = (-4)^{j+1}(j+1)!\sqrt{\pi}/(2j+2)! = (-1)^{j+1} 2^{j+1} \sqrt{\pi}/(2j+1)!!$ supplying Δ_j. □

Corollary 2.32 (Otter's constant)**.** *With* $C := \tau_0/\sqrt{\pi}$*,*

$$\tau_0 = -\frac{t_1}{2} = \sqrt{\frac{\theta_1}{2}}, \qquad C = \sqrt{\frac{\theta_1}{2\pi}} = \sqrt{\frac{\mathrm{e}\rho\,\zeta'(\rho)}{2\pi}} = \sqrt{\frac{1 + \rho R'(\rho)}{2\pi}}, \qquad a_n \sim C\,\alpha^n n^{-3/2}. \tag{2.45}$$

Numerically

$$\tau_0 = 0.7797450101873204427711036979\ldots, \qquad C = 0.4399240125710253040409033914345447648\ldots. \tag{2.46}$$

Proof. Set $m = 0$ in eq. (2.44): $\Delta_0 = -\frac{1}{2}$, $g_0 = 1$, so $\tau_0 = -t_1/2 = \sqrt{2\theta_1}/2 = \sqrt{\theta_1/2}$ by theorem 2.25. Divide by $\sqrt{\pi}$ and use eq. (2.14). □

The identity $C^2 = \theta_1/(2\pi)$ is a genuinely independent check: the left side comes through the whole Puiseux/transfer chain, the right side directly from one derivative of ζ. The two agree to all 55 digits carried.

Expanding eq. (2.44) for small m gives the transfer identities

$$\begin{aligned}
\tau_0 &= -\tfrac{1}{2} t_1, \\
\tau_1 &= -\tfrac{3}{16}\big(t_1 - 4t_3\big) = -\tfrac{3}{16} t_1 + \tfrac{3}{4} t_3, \\
\tau_2 &= -\tfrac{5}{256}\big(5t_1 - 72t_3 + 96t_5\big) = -\tfrac{25}{256} t_1 + \tfrac{45}{32} t_3 - \tfrac{15}{8} t_5, \\
\tau_3 &= -\tfrac{105}{2048}\big(t_1 - 44t_3 + 160t_5 - 128t_7\big) = -\tfrac{105}{2048} t_1 + \tfrac{1155}{512} t_3 - \tfrac{525}{64} t_5 + \tfrac{105}{16} t_7, \\
\tau_4 &= -\tfrac{21}{65536}\big(79t_1 - 10800t_3 + 81600t_5 - 161280t_7 + 92160t_9\big).
\end{aligned} \tag{2.47}$$

S2 states the right-hand forms, S3 the factored left-hand forms; they are identical after clearing denominators, and both were re-derived here symbolically from eq. (2.44). For any m beyond 4 the compact convolution is far preferable to the expanded form.

Table 2.3: Coefficients of the dominant expansion $a_n \sim \rho^{-n}(\pi n^3)^{-1/2} \sum_{m \geq 0} \tau_m n^{-m}$ and their normalisations $f_m = \tau_m/\tau_0$, recomputed here.

m	τ_m	$f_m = \tau_m/\tau_0$
0	0.7797450101873204427711	1
1	0.07828911261061096206128	0.1004034800964014347637
2	0.3929402676631860184743	0.5039343151023035331022
3	1.537879315978838091102	1.972284908382278469206
4	8.200844090435596159220	10.51734090413157399274
5	57.29291473494343787740	73.47647498402047106417
6	503.0445050262735818248	645.1397552456612928153
7	5359.600933884326033403	6873.530274463408619941
8	67342.06920114653048341	86364.21948371140797700
9	975425.4970695924732213	1250954.458605978293018
10	15996932.49293173312961	20515594.56480361813667
11	292822531.3353392231146	—
12	5914523441.293932519117	—

Higher values continue smoothly; for reference $\tau_{15} = 8.078305401468884289 \times 10^{13}$, $\tau_{18} = 2.073828085209893749 \times 10^{18}$.

Writing

$$F(v) := 1 + \sum_{m \geq 1} f_m v^m, \qquad f_m = \frac{\tau_m}{\tau_0}, \tag{2.48}$$

the inversion-ready normal form is

$$\boxed{a_n \sim C\,\mathrm{e}^{\lambda n}\, n^{-p}\, F(1/n), \qquad p = \tfrac{3}{2}.} \tag{2.49}$$

Remark 2.33 (Divergence and least-term truncation). Equation (2.43) asserts a Poincaré expansion at every fixed order; it does *not* assert convergence, and table 2.3 shows the coefficients growing rapidly. The optimal truncation and the size of the resulting floor are governed by theorem 1.46; the relevant nearest competing exponential scale is identified in section 2.9 as $\rho^{-n/2}$, so the least-term floor at index n is $\mathcal{O}(\rho^{n/2})$ relative to the leading term — consistent with the error tables of section 2.11.2, where the improvement stops at $n = 20$ around $M = 6$. S3 states this correctly as a numerical strategy rather than a theorem; S1 and S2 call the late-term growth "the perturbative shadow of farther singularities", which is the right heuristic but is not proved anywhere in the three drafts and is not proved here.

2.6 The multiplicative reciprocal

If "inverse" is read as the reciprocal sequence, the answer is immediate from eq. (2.49) and requires no inversion theory at all.

Proposition 2.34 (All-orders reciprocal)**.** *Write* $1/F(v) = 1 + \sum_{m\geq 1} \bar{f}_m v^m$*. Then*

$$\bar{f}_m = \sum_{k=1}^{m} (-1)^k\, \widehat{B}_{m,k}(f_1, f_2, \ldots) = -\sum_{k=1}^{m} f_k\, \bar{f}_{m-k} \qquad (\bar{f}_0 = 1), \tag{2.50}$$

and for every fixed $M \geq 0$

$$\frac{1}{a_n} = \frac{\sqrt{\pi}\,\rho^n n^{3/2}}{\tau_0} \left(\sum_{m=0}^{M} \frac{\bar{f}_m}{n^m} + \mathcal{O}(n^{-M-1}) \right). \tag{2.51}$$

Proof. The first equality in eq. (2.50) is theorem 1.12 with exponent -1; the second is the Cauchy product $F \cdot F^{-1} = 1$. For eq. (2.51), note that $a_n > 0$ and, by eq. (2.43), $a_n = C\alpha^n n^{-p}(1 + \varepsilon_n)$ with $\varepsilon_n \to 0$; hence $1/a_n$ is defined and $1/(1+\varepsilon_n)$ has the asymptotic expansion obtained by formally inverting the (asymptotic) series for $1 + \varepsilon_n$ — a legitimate operation because the leading term is $1 \neq 0$ and each truncation error is $\mathcal{O}(n^{-M-1})$. □

The first reciprocal coefficients, recomputed, are

$$\begin{aligned} \bar{f}_1 &= -0.1004034800964014347637, & \bar{f}_2 &= -0.4938534562868350540384, \\ \bar{f}_3 &= -1.872103543737176348103, & \bar{f}_4 &= -9.882481221441551170916, \\ \bar{f}_5 &= -69.51082491336461883211, & \bar{f}_6 &= -616.9168645978433816315. \end{aligned} \tag{2.52}$$

Remark 2.35 (Reciprocal of the full transseries). Writing $a_n = \mathcal{M}_n(1 + \mathcal{E}_n)$ with $\mathcal{M}_n$ the complete dominant sector and $\mathcal{E}_n$ the sum of relative exponentially small sectors of section 2.9, one has $1/a_n = \mathcal{M}_n^{-1} \sum_{q\geq 0} (-\mathcal{E}_n)^q$; the products generate exactly the additive semigroup of the singular actions, with coefficients given by finite Cauchy products. This is a formal statement about the transseries, and inherits whatever status theorem 2.55 has on the chosen continuation domain.

Warning 2.36. The reciprocal is *not* a functional inverse. Its exponential scale is ρ^n, while the functional inverse of section 2.7 grows only logarithmically in its argument. All three siblings make this point; it is repeated here because the ambiguity of the word "inverse" is exactly what theorem 1.37 exists to remove.

2.7 The asymptotic inverse: the Part 0 dictionary

2.7.1 The parameter dictionary

Everything needed to invert eq. (2.49) has been proved in Part 0. What this chapter must supply is only the dictionary and the verification of Part 0's hypotheses.

Proposition 2.37 (A000081 as an instance of the exponential–power model)**.** *Let*

$$\mathcal{A}(x) := C\,\mathrm{e}^{\lambda x} x^{-p} F(x^{-1}), \qquad p = \tfrac{3}{2}, \tag{2.53}$$

with C from theorem 2.32 and F from eq. (2.48)*. Then $\mathcal{A}$ is an instance of theorem 1.2 under*

Part 0	C	a	α	b	δ	$Q(t)$
here	$C = \sqrt{\theta_1/(2\pi)}$	$\lambda = \log(1/\rho)$	1	$-\frac{3}{2}$	1	$H(t) = \log F(t)$

(2.54)

with $H \in t\mathbb{R}[[t]]$. In particular $a = \lambda > 0$ and $\alpha = 1$, so $\mathcal{A}$ is already in the linear-phase normal form and theorem 1.4 is the identity substitution.

Proof. Equation (2.49) is theorem 1.2 with the stated substitutions; $\lambda > 0$ because $\rho < 1$ (theorem 2.11), $C > 0$ because $\theta_1 > 0$, and $H(0) = \log F(0) = \log 1 = 0$ so $H \in t\mathbb{R}[[t]]$. ☐

Definition 2.38 (Additive phase coefficients)**.** Write $H(v) = \log F(v) = \sum_{m\geq 1} \varphi_m v^m$. By theorem 1.12,

$$\varphi_m = \sum_{k=1}^{m} \frac{(-1)^{k-1}}{k} \widehat{B}_{m,k}(f_1, f_2, \dots) = f_m - \frac{1}{m} \sum_{k=1}^{m-1} k\,\varphi_k\, f_{m-k}. \tag{2.55}$$

Taking logarithms in eq. (2.43) gives the form used for inversion,

$$\log a_n = \lambda n - \tfrac{3}{2}\log n + \log C + \sum_{m=1}^{M} \frac{\varphi_m}{n^m} + \mathcal{O}(n^{-M-1}), \tag{2.56}$$

with, recomputed here,

$$\begin{aligned} \varphi_1 &= 0.10040348009640143476 37, & \varphi_2 &= 0.49889388569456929357 03,\\ \varphi_3 &= 1.92202553384182182159 1, & \varphi_4 &= 10.19739642337376887053,\\ \varphi_5 &= 71.47146706207073604412, & \varphi_6 &= 630.8599376073372478899. \end{aligned} \tag{2.57}$$

2.7.2 The exact Lambert carrier

Corollary 2.39 (Carrier; specialisation of theorem 1.20)**.** *Let $y > y_* := C\,(\mathrm{e}\lambda/p)^p = 1.2108232025837572603\ldots$. Then the equation $y = C\mathrm{e}^{\lambda x} x^{-p}$ has exactly one solution $x_0 = x_0(y) > p/\lambda$, namely*

$$\boxed{x_0(y) = -\frac{p}{\lambda}\, W_{-1}\!\left(-\frac{\lambda}{p}\left(\frac{C}{y}\right)^{1/p}\right) = -\frac{3}{2\lambda}\, W_{-1}\!\left(-\frac{2\lambda}{3}\left(\frac{C}{y}\right)^{2/3}\right).} \tag{2.58}$$

Proof. Raise $y = C\mathrm{e}^{\lambda x} x^{-p}$ to the power $-1/p$ and multiply by $-\lambda/p$:

$$-\frac{\lambda}{p}\left(\frac{C}{y}\right)^{1/p} = -\frac{\lambda x}{p}\,\mathrm{e}^{-\lambda x/p} = \xi \mathrm{e}^{\xi}, \qquad \xi := -\frac{\lambda x}{p}.$$

Write $\epsilon := -\frac{\lambda}{p}(C/y)^{1/p}$. The map $x \mapsto C\mathrm{e}^{\lambda x} x^{-p}$ is strictly increasing on $(p/\lambda, \infty)$ (its logarithmic derivative is $\lambda - p/x > 0$ there) with infimum y_* attained as $x \downarrow p/\lambda$, so $y > y_*$ gives a unique root $x_0 > p/\lambda$, i.e. $\xi < -1$; and $y > y_*$ is exactly $\epsilon \in (-\mathrm{e}^{-1}, 0)$. On that range $\xi \mapsto \xi\mathrm{e}^{\xi}$ has the two real inverse branches $W_0 \in (-1, 0)$ and $W_{-1} < -1$, and the required root is the second. This is theorem 1.20 with $L = \log(y/C)$, $a = \lambda$, $b = -p$ and branch index $k = -1$; the branch rule there selects $k = -1$ under exactly the condition $x > -b/a$. ☐

Remark 2.40. The principal branch W_0 returns the other, small root of the same equation, which is the spurious solution: all three siblings say so, and the threshold y_* makes the dichotomy quantitative. For A000081 the threshold lies below $a_3 = 2$, so eq. (2.58) may be used at every sequence value from $n = 3$ on.

2.7.3 All algebraic corrections

Theorem 2.41 (Corrections; specialisation of theorem 1.25). *Let*

$$\Psi(t,u) := \frac{p}{\lambda}\log(1+tu) - \frac{1}{\lambda}H\left(\frac{t}{1+tu}\right) \in t\mathbb{R}[[t,u]]. \tag{2.59}$$

Then there is a unique $D(t) = \sum_{m\geq 1} d_m t^m \in t\mathbb{R}[[t]]$ with $D(t) = \Psi\big(t, D(t)\big)$, its coefficients are given by theorem 1.16,

$$\boxed{d_m = \sum_{k=1}^{m} \frac{1}{k}\,[t^m u^{k-1}]\,\Psi(t,u)^k,} \tag{2.60}$$

equivalently by the triangular extraction

$$d_m = [t^m]\left\{\frac{p}{\lambda}\log\big(1+tD_{<m}(t)\big) - \frac{1}{\lambda}H\left(\frac{t}{1+tD_{<m}(t)}\right)\right\}, \qquad D_{<m} = \sum_{j<m} d_j t^j, \tag{2.61}$$

or by the completely explicit ordinary-Bell form

$$d_m = \frac{p}{\lambda}\sum_{r=1}^{m}\frac{(-1)^{r-1}}{r}\widehat{B}_{m-r,r}(d_1,d_2,\ldots) - \frac{1}{\lambda}\sum_{k=1}^{m}\varphi_k\sum_{r=0}^{m-k}(-1)^r\binom{k+r-1}{r}\widehat{B}_{m-k-r,r}(d_1,d_2,\ldots), \tag{2.62}$$

in which the index constraints force only $d_1,\ldots,d_{m-1}$ to occur on the right. With x_0 from eq. (2.58) *the complete algebraic inverse is*

$$\boxed{\nu(y) \sim x_0(y) + \sum_{m\geq 1}\frac{d_m}{x_0(y)^m}.} \tag{2.63}$$

Proof. Theorem 1.25 states exactly this for a model $Ce^{ax}x^b e^{Q(x^{-\delta})}$ with $\Psi(t,u) = -\frac{1}{a}\big[b\log(1+tu) + Q\big(t/(1+tu)\big)\big]$; substituting the dictionary eq. (2.54) ($a=\lambda$, $b=-p$, $\delta=1$, $Q=H$) gives eq. (2.59). Concretely: putting $x = x_0 + D$ into $\log(y/C) = \lambda x - p\log x + H(x^{-1})$ and subtracting the carrier equation $\log(y/C) = \lambda x_0 - p\log x_0$ gives, with $t = x_0^{-1}$,

$$\lambda D - p\log(1+tD) + H\left(\frac{t}{1+tD}\right) = 0,$$

which is $D = \Psi(t,D)$. Since $\Psi \in t\mathbb{R}[[t,u]]$, the formal implicit-function theorem gives a unique solution in $t\mathbb{R}[[t]]$; theorem 1.16 converts it into eq. (2.60), and eq. (2.61) is the same statement read as a recursion. Equation (2.62) follows by expanding $\log(1+tD) = \sum_r \frac{(-1)^{r-1}}{r}t^r D^r$ and $H\big(t/(1+tD)\big) = \sum_k \varphi_k t^k \sum_r (-1)^r\binom{k+r-1}{r}t^r D^r$ and applying theorem 1.9 to the powers D^r. ☐

A formal-to-analytic slide in S1. S1's "General coefficient formula for the inverse" is stated as $x(y) \sim X(y) + \sum_m d_m X(y)^{-m}$, an asymptotic assertion, but its proof ends with "substitution … proves formal inversion order by order" and supplies no remainder. As stated, theorem 2.41 is a theorem about formal series only; the asymptotic reading requires theorem 2.42 below, whose argument is S3's — the only one of the three drafts that separates the two statements. The volume keeps them separate.

Theorem 2.42 (Remainder; specialisation of theorem 1.44). *Fix $M \geq 1$ and set $\nu_M(y) := x_0(y) + \sum_{m=1}^{M-1} d_m x_0(y)^{-m}$. Let $f(x) = \log C + \lambda x - p\log x + H(x^{-1})$ be the (asymptotic) logarithm of the forward interpolant. Then*

$$f\big(\nu_M(y)\big) - \log y = \mathcal{O}\big(x_0(y)^{-M}\big), \qquad \nu(y) - \nu_M(y) = \mathcal{O}\big(x_0(y)^{-M}\big) = \mathcal{O}\big((\log y)^{-M}\big). \tag{2.64}$$

Proof. By theorem 2.41 the coefficients of $t^1, \dots, t^{M-1}$ in $D - \Psi(t, D)$ vanish, so truncating leaves a residual $\mathcal{O}(t^M)$; this is the first claim. Since $f'(x) = \lambda - p/x + \mathcal{O}(x^{-2})$ is bounded away from 0 for large x, theorem 1.42 converts the residual into a forward error of the same order divided by f'; this is theorem 1.44 with $a = \lambda$. Finally $x_0(y) = \lambda^{-1}\log y\,(1 + o(1))$ by eq. (2.58). □

2.7.4 Coefficients

Expanding eq. (2.61) through t^5 — and verifying the result symbolically — gives the closed forms

$$d_1 = -\frac{\varphi_1}{\lambda}, \tag{2.65}$$

$$d_2 = -\frac{p\varphi_1 + \lambda\varphi_2}{\lambda^2}, \tag{2.66}$$

$$d_3 = -\frac{\lambda\varphi_1^2 + p^2\varphi_1 + p\lambda\varphi_2 + \lambda^2\varphi_3}{\lambda^3}, \tag{2.67}$$

$$d_4 = -\frac{1}{2\lambda^4}\Big(5p\lambda\varphi_1^2 + 6\lambda^2\varphi_1\varphi_2 + 2p^3\varphi_1 + 2p^2\lambda\varphi_2 + 2p\lambda^2\varphi_3 + 2\lambda^3\varphi_4\Big), \tag{2.68}$$

$$\begin{aligned} d_5 = -\frac{1}{2\lambda^5}\Big(&4\lambda^2\varphi_1^3 + 9p^2\lambda\varphi_1^2 + 14p\lambda^2\varphi_1\varphi_2 + 8\lambda^3\varphi_1\varphi_3 + 4\lambda^3\varphi_2^2 \\ &+ 2p^4\varphi_1 + 2p^3\lambda\varphi_2 + 2p^2\lambda^2\varphi_3 + 2p\lambda^3\varphi_4 + 2\lambda^4\varphi_5\Big). \end{aligned} \tag{2.69}$$

Beyond $m = 5$ the expanded expressions grow fast, and eqs. (2.60) and (2.61) are both shorter and less error-prone.

Table 2.4: Additive phase coefficients φ_m and Lambert-carrier corrections d_m, recomputed here. The last column is S2's normalisation $\mathrm{D}_m = \lambda^{m+1} d_m$ (see theorem 2.43).

m	φ_m	d_m	$\mathrm{D}_m = \lambda^{m+1} d_m$
1	0.1004034800964014348	−0.09264385352561312984	−0.1088130343442745145
2	0.4988938856945692936	−0.5885630399313472628	−0.7491856512881932495
3	1.922025533841821822	−2.596680167407531021	−3.582177326832277789
4	10.19739642337376887	−13.14905339996706569	−19.65872121138776004
5	71.47146706207073604	−85.49870151462556652	−138.5327478518560861
6	630.8599376073372479	−711.2629148329465609	−1248.979326371692854
7	6753.629059387619229	−7306.788270284967257	−13905.40884474814835
8	85171.04383262353790	−89561.76088263724860	−184719.1911265080905
9	1236987.135642947528	−1274829.503083708903	–
10	20325777.10708850337	−20641099.00032138544	–

Numerical error in S2's audit appendix. S2's appendix table of η_j and D_j reports $D_3 = -3.582177326832123\ldots$, disagreeing with its own main text ($-3.58217732683227778910300984$4) from the fourteenth significant digit. Recomputation confirms the main-text value, $\mathrm{D}_3 = -3.58217732683227778910301\ldots$; the appendix entry is wrong. Every other entry of that table was checked and is correct to the digits shown.

Remark 2.43 (Three normalisations of the same inverse). The siblings work in three different variables, which is why their coefficient lists look different:

- S1 and S3 expand in the index variable x itself and obtain the d_m of table 2.4; their carriers agree with eq. (2.58).
- S2 rescales to $X = \lambda x$, so that its carrier is $X_0 = -pW_{-1}\bigl(-\frac{1}{p}\mathrm{e}^{-Z/p}\bigr)$ with $Z = \log\bigl(y/(C\lambda^p)\bigr)$, its corrections are $\mathrm{D}_m = \lambda^{m+1}d_m$, and its phase coefficients are $\eta_m^{\mathrm{S2}} = \lambda^m\varphi_m$. Both dictionaries were verified numerically for $1 \le m \le 8$, and $X_0 = \lambda x_0$ holds identically because the two carrier equations differ by that substitution.
- S3 also uses x but writes the polynomial–logarithmic form in terms of $\mathsf{q} = p/\lambda$ and $\eta_m^{\mathrm{S3}} = \varphi_m/\lambda$.

The constant appearing in S2's Z is $C\lambda^p = 0.49633614165959974747273 69464\ldots$.

2.7.5 Polynomial–logarithmic form

Eliminating the Lambert function gives the expansion in the canonical logarithmic scale. That is theorem 1.33; only the dictionary is recorded here.

Corollary 2.44 (Flattened inverse). *Put*

$$\mathcal{X} := \frac{1}{\lambda}\log\frac{y}{C}, \qquad L := \log\mathcal{X}, \qquad \mathsf{q} := \frac{p}{\lambda}, \qquad \eta_k := \frac{\varphi_k}{\lambda}. \tag{2.70}$$

Then the inverse problem is $x - \mathsf{q}\log x + \sum_{k\ge1}\eta_k x^{-k} = \mathcal{X}$, and theorem 1.33 gives

$$\boxed{\nu(y) \sim \mathcal{X} + \mathsf{q}L + \sum_{m\ge1}\frac{P_m(L)}{\mathcal{X}^m},} \tag{2.71}$$

where each P_m is a polynomial of degree m over $\mathbb{Q}[\mathsf{q},\eta_1,\ldots,\eta_m]$ with

$$[L^m]\,P_m(L) = \frac{(-1)^{m+1}}{m}\mathsf{q}^{m+1}, \qquad P_m(0) = d_m. \tag{2.72}$$

The identity $P_m(0) = d_m$ is not an accident: setting $L = 0$ in the flattening recursion reduces it to the fixed-point equation solved by the Lambert carrier, and formal uniqueness in theorem 1.25 identifies the two. The first two polynomials are

$$P_1(L) = \mathsf{q}^2L - \eta_1 = 1.9156590172195452616\,L - 0.0926438535256131298, \tag{2.73}$$

$$\begin{aligned} P_2(L) &= -\tfrac{1}{2}\mathsf{q}^3L^2 + \bigl(\mathsf{q}^3 + \mathsf{q}\eta_1\bigr)L - \mathsf{q}\eta_1 - \eta_2 \\ &= -1.3257062894576032164\,L^2 + 2.7896427837264004647\,L - 0.5885630399313472628. \end{aligned} \tag{2.74}$$

Remark 2.45 (The two flattenings are not the same polynomials). S2 flattens in $Z = \log\bigl(y/(C\lambda^p)\bigr)$ and $\log Z$; S3 flattens in $\mathcal{X} = \lambda^{-1}\log(y/C)$ and $\log\mathcal{X}$. The two recursions are formally the *same* recursion with different parameters — S3's is S2's under $(p,\eta_k^{\mathrm{S2}}) \mapsto (\mathsf{q},\eta_k^{\mathrm{S3}})$ — but they expand the same function in *different* variables, so their coefficient polynomials genuinely differ. Neither draft is in error; the caution is only that the two lists must not be compared term by term. We adopt S3's normalisation because it keeps the index variable x throughout and so matches eq. (2.63).

Retaining only the first two scales,

$$\nu(y) = \frac{\log y}{\lambda} + \frac{p}{\lambda}\log\log y - \frac{1}{\lambda}\log\bigl(C\lambda^p\bigr) + o(1), \tag{2.75}$$

that is

$$\nu(y) = 0.9227155616186015248\log y + 1.3840733424279022872\log\log y + 0.6463639827002088073 + o(1). \tag{2.76}$$

Equation (2.63) is preferable numerically because W_{-1} resums the entire logarithmic tower exactly before any truncation is made.

2.8 The discrete inverse

The objects distinguished in theorem 1.37 are, for this sequence: the smooth inverse ν of section 2.7; the integer staircase

$$N(y) := \min\{n \geq 1 : a_n \geq y\}; \tag{2.77}$$

and the round-to-nearest generalized inverse. Applying theorem 1.39 requires its separation hypothesis, which for A000081 is supplied by the next two lemmas.

Lemma 2.46 (Strict monotonicity). *$a_1 = a_2 = 1$ and $a_{n+1} > a_n$ for every $n \geq 2$.*

Proof. Monotonicity: the map sending a rooted tree $\mathfrak{t}$ on n vertices to the rooted tree whose root has the single child subtree $\mathfrak{t}$ is an injection from the n-vertex shapes into the $(n+1)$-vertex shapes, so $a_{n+1} \geq a_n$. Strictness for $n \geq 2$: when $n+1 \geq 3$, the star $K_{1,n}$ rooted at its centre has $n \geq 2$ children and is therefore not in the image of that injection, whence $a_{n+1} \geq a_n + 1$. The values $a_1 = a_2 = 1$ are immediate. ☐

Lemma 2.47 (Separation). *$a_{n+1}/a_n \to \alpha = 2.9557\ldots > 1$; more precisely*

$$\log \frac{a_{n+1}}{a_n} = \lambda - \frac{3}{2n} + \frac{\frac{3}{4} - \varphi_1}{n^2} + \frac{-\frac{1}{2} + \varphi_1 - 2\varphi_2}{n^3} + \mathcal{O}(n^{-4}). \tag{2.78}$$

Proof. Subtract eq. (2.56) at n from the same relation at $n+1$, and expand $\log(n+1) = \log n + n^{-1} - \frac{1}{2}n^{-2} + \frac{1}{3}n^{-3} - \cdots$ and $(n+1)^{-m} = n^{-m}(1 - mn^{-1} + \cdots)$. Collecting powers of n^{-1} gives eq. (2.78); the limit statement is its leading term. ☐

Corollary 2.48 (Staircase recovery). *Let ν_M be as in theorem 2.42. For every $M \geq 2$ and all sufficiently large n, $\left|\nu_M(a_n) - n\right| < \frac{1}{2}$, so n is recovered from a_n by rounding. For a general real target $y > 1$, $N(y) = \nu(y) + \mathcal{O}(1)$, and certifying $N(y)$ exactly requires the two-sided test $a_{m-1} < y \leq a_m$, which eq.* (2.63) *reduces to a bounded neighbourhood of $m \approx \nu(y)$.*

Proof. The first claim is theorem 2.42 with $M \geq 2$, since $\mathcal{O}\big((\log a_n)^{-2}\big) = \mathcal{O}(n^{-2})$. For the second, apply theorem 1.39, whose separation hypothesis holds by theorems 2.46 and 2.47: the logarithmic derivative of the interpolation tends to $\lambda > 0$, so multiplying y by a bounded factor moves $\nu(y)$ by a bounded amount, while consecutive sequence values differ by the asymptotically fixed factor α. The final sentence is the definition of N. ☐

Warning 2.49 (The $\mathcal{O}(1)$ rounding layer is not a small correction). An assertion of the form $N(y) = \nu(y) + o(1)$ *uniformly in real y* is false, and no descending power–logarithmic series can be true uniformly: arbitrarily small changes of y across a value a_n change $N(y)$ by one. Once a monotone interpolation $\mathcal{A}_{\mathrm{ex}}$ with $\mathcal{A}_{\mathrm{ex}}(n) = a_n$ has been fixed, one has exactly $N(y) = \lceil \nu_{\mathrm{ex}}(y) \rceil$ with ν_{ex} its ordinary inverse, and for non-integral χ

$$\lceil \chi \rceil = \chi + \frac{1}{2} + \frac{1}{\pi} \sum_{m \geq 1} \frac{\sin(2\pi m \chi)}{m}, \tag{2.79}$$

an $\mathcal{O}(1)$ periodic layer that never tends to zero. The correct statements are: about ν; or at the special values $y = a_n$, where $\nu(a_n) = n + \mathcal{O}(n^{-M})$ for every fixed M; or about a rounded exact interpolation. S2 states this sharply and S3 restates it; S1's phrase "can never be resolved more accurately than one unit" is the same point stated loosely, and is corrected here. The layer eq. (2.79) must not be confused with the exponentially small oscillation of section 2.9: the latter is genuinely $o(1)$ and has an entirely different origin.

Remark 2.50 (A definitional disagreement). S1 and S2 define the staircase as $\min\{n \geq 2 : a_n \geq y\}$, S3 as $\min\{n \geq 1 : a_n \geq y\}$. The two differ only at $y \leq 1$, where the tie $a_1 = a_2 = 1$ makes the choice a pure convention. This volume uses eq. (2.77) and confines all statements to $y > 1$, where the definitions agree.

2.9 Exponentially small sectors

2.9.1 Why one exponential scale is not the whole story

Equation (2.43) is complete in inverse powers of n at the scale ρ^{-n}. It cannot see a term σ^{-n} with $|\sigma| > \rho$, because $(\rho/|\sigma|)^n$ decays faster than every power of n^{-1}. Such terms come from singularities of the analytic continuation of T, and the functional equation exposes exactly two mechanisms for producing them:

1. **inherited preimages**: if T is singular at τ and $\sigma^k = \tau$ for some $k \geq 2$, the summand $T(z^k)/k$ of R can make R, hence T, singular at σ;
2. **new Cayley critical points**: on a continued sheet where R is analytic, any σ with $\zeta(\sigma) = \mathrm{e}^{-1}$ and $\zeta'(\sigma) \neq 0$ is a square-root branch point of T by theorem 2.16.

Iterating the first mechanism from the dominant point produces the *root-lift skeleton*

$$\mathscr{R} := \left\{\rho^{1/m}\mathrm{e}^{2\pi\mathrm{i}j/m} \ : \ m \geq 1,\ 0 \leq j < m\right\}, \tag{2.80}$$

whose radii increase to 1 and whose arguments become dense. Relative to the dominant sector the level-m points carry the exponential weight

$$\sigma_{m,j}^{-n} = \rho^{-n/m}\mathrm{e}^{-2\pi\mathrm{i}jn/m}, \qquad \text{i.e.} \qquad \exp\left[-\lambda\left(1-\frac{1}{m}\right)n\right] \text{ relative to } \rho^{-n}. \tag{2.81}$$

Warning 2.51 (What the skeleton is and is not). Equation (2.80) is a list of *candidate* inherited germs generated by the recursion. It is not a theorem that every listed point is singular on a given sheet, that every singularity is listed, or that a listed point is reachable along a prescribed path. Continuation can also meet solutions of $\zeta(z) = \mathrm{e}^{-1}$ that are not in $\mathscr{R}$, and a critical point can coalesce with an inherited germ (which, by eq. (2.89) below, halves an exponent and can create quarter powers).

Remark 2.52 (A disagreement about the unit circle). S1 and S2 assert that $|z| = 1$ is a natural boundary for T and that the continued singularities accumulate at every point of it, citing Drmota–Gittenberger. S3 makes no such assertion and states only the candidate skeleton with the caveat of theorem 2.51. This volume records the natural-boundary claim as *reported* and does not use it: every statement in this section is conditioned on the explicit finite-radius hypothesis $\mathsf{H}(\mathcal{R})$ of theorem 2.54, and nothing below depends on what happens as $\mathcal{R} \uparrow 1$.

2.9.2 Transfer from a general singular germ

Theorem 2.53 (Universal sector coefficients)**.** *Let $\sigma \neq 0$ be an accessible singularity on a specified sheet, put $\mathsf{u} = 1 - z/\sigma$, and suppose that in a Δ-domain at σ*

$$T(z) = A_\sigma(\mathsf{u}) + \sum_{\beta\in E_\sigma} c_{\sigma,\beta}\,\mathsf{u}^\beta + \mathcal{O}\big(|\mathsf{u}|^{\beta_{\max}}\big), \tag{2.82}$$

with A_σ analytic, $E_\sigma \subset \mathbb{Q}_{>0} \setminus \mathbb{N}_0$ finite and $\beta_{\max} > \max E_\sigma$. Then the contribution of eq. (2.82) *to* $[z^n]\,T$ *is*

$$\boxed{\mathfrak{S}_\sigma(n) \sim \sigma^{-n}\sum_{\beta\in E_\sigma}\frac{c_{\sigma,\beta}}{\Gamma(-\beta)}\,n^{-\beta-1}\sum_{r\geq 0}\frac{g_r(\beta)}{n^r},} \tag{2.83}$$

so the coefficient of $\sigma^{-n}n^{-\beta-1-r}$ is exactly $c_{\sigma,\beta}\,g_r(\beta)/\Gamma(-\beta)$.

Proof. Apply eq. (2.37) with ρ replaced by σ to each displayed monomial and expand the gamma ratio by theorem 2.29. The analytic part A_σ has no Hankel jump and contributes no large-n tail; the $\mathcal{O}$-term transfers by the usual Hankel estimate. □

Equation (2.83) is the exact analogue of eq. (2.44) and allows arbitrary ramification exponents, not only half-integers. Relative to the dominant sector one introduces the *action*

$$\Omega_\sigma := \operatorname{Log}\frac{\sigma}{\rho}, \qquad \text{so that} \qquad \sigma^{-n} = \rho^{-n}\mathrm{e}^{-n\Omega_\sigma}. \tag{2.84}$$

Definition 2.54 (Continuation hypothesis $\mathsf{H}(\mathcal{R})$)**.** Fix $\rho < \mathcal{R} < 1$. We say $\mathsf{H}(\mathcal{R})$ holds for a chosen continuation of T if that continuation is analytic on $|z| < \mathcal{R}$ after removing finitely many pairwise disjoint cuts issuing from a finite set $\Sigma_\mathcal{R}$; every $\sigma \in \Sigma_\mathcal{R}$ carries a germ of the form eq. (2.82) in a Δ-domain; and the deformed coefficient contour may be taken on a circle of radius at least $\mathcal{R}$, up to harmless indentations, on which T has at most polynomial growth.

Theorem 2.55 (Finite-level coefficient transseries)**.** *Assume* $\mathsf{H}(\mathcal{R})$ *and let* $\mathfrak{S}_\sigma$ *be the multiplier-weighted sector of eq.* (2.83) *attached to the local Hankel cycle actually traversed. Then, for any finite truncation of the exponent and inverse-power sums,*

$$a_n = \sum_{\sigma\in\Sigma_\mathcal{R}} \mathfrak{S}_\sigma(n) + \mathcal{O}\big(\mathcal{R}^{-n}n^K\big) \tag{2.85}$$

for a finite K determined by the boundary growth.

Proof. Deform the Cauchy contour from a small circle about 0 into the union of small Hankel contours around the cuts issuing from $\Sigma_\mathcal{R}$ plus an outer contour of radius $\mathcal{R}$. On each Hankel contour apply theorem 2.53; the outer contour is bounded by $\mathcal{R}^{-n}$ times the allowed polynomial factor. Only finitely many singularities and finitely many local terms are retained, so the deformation and the term-by-term summation are legitimate. Passing to a different sheet changes the homology coefficients of the local cycles; those coefficients are the Stokes/intersection multipliers absorbed into $\mathfrak{S}_\sigma$. □

S1's global transseries is not a theorem. S1 displays $a_n \sim \mathcal{A}_\rho(n) + \sum_{\sigma\in\mathfrak{S}\setminus\{\rho\}} S_\sigma\mathcal{A}_\sigma(n)$ as a numbered displayed result over "the singular set encountered by the chosen contour continuation", with no hypothesis attached, while conceding in its introduction that globally the formula "is best understood as a formal resurgent/Darboux transseries". The volume replaces it by theorem 2.55, which is S3's formulation and the only one of the three carrying an explicit hypothesis; the unrestricted sum over an infinite $\mathfrak{S}$ is not asserted anywhere here. The correct reading of a "global" transseries in the presence of accumulating singularities is the compatible family of the finite-radius expansions eq. (2.85) as $\mathcal{R}$ grows, which is exactly what S2's "projective meaning" remark says.

2.9.3 The local recursion produces every germ

Lemma 2.56 (Pull-back coordinates)**.** *Let $z = \sigma(1-\mathsf{u})$ and $\tau = \sigma^k$. Then*

$$1 - \frac{z^k}{\tau} = 1-(1-\mathsf{u})^k = k\,\mathsf{u}\left(1 + \sum_{r\ge1} p_{k,r}\mathsf{u}^r\right), \qquad p_{k,r} = \frac{(-1)^r}{k}\binom{k}{r+1}, \tag{2.86}$$

and consequently, for a parent monomial,

$$\left(1-\frac{z^k}{\tau}\right)^\beta = (k\mathsf{u})^\beta \sum_{N\ge0}\mathsf{u}^N\sum_{j=0}^N\binom{\beta}{j}\widehat{B}_{N,j}(p_{k,1},p_{k,2},\ldots). \tag{2.87}$$

Proof. $1-(1-\mathsf{u})^k = -\sum_{r\ge1}\binom{k}{r}(-\mathsf{u})^r = k\mathsf{u} - \sum_{r\ge2}\binom{k}{r}(-1)^r\mathsf{u}^r$; dividing by $k\mathsf{u}$ and re-indexing $r\mapsto r+1$ gives $p_{k,r}$. Equation (2.87) is theorem 1.12 applied to the unit series $1+\sum_r p_{k,r}\mathsf{u}^r$. □

Theorem 2.57 (Recursive completeness of the local calculus). *Fix a sheet and a point $\sigma \neq 0$, and choose a local uniformiser $z = \sigma(1 - v^{\mathsf{L}})$ in which every parent germ needed in $R(z)$ is an ordinary power series in v. Then the local germ of T at σ is computable to any finite order by the following finite operations, and only these:*

1. *pull back each singular summand $T(z^k)/k$ by eq.* (2.87)*;*
2. *add the regular Taylor parts to obtain R, and exponentiate by eq.* (2.29) *to obtain $\zeta = z\mathrm{e}^R$;*
3. *if $\zeta_0 := \zeta(\sigma) \neq \mathrm{e}^{-1}$ on the reached branch, so that the selected Cayley branch $\mathcal{C}_\flat$ is analytic at ζ_0, compose regularly:*

$$T = \sum_{r\geq 0} \frac{\mathcal{C}_\flat^{(r)}(\zeta_0)}{r!}\big(\zeta - \zeta_0\big)^r, \qquad \mathcal{C}_\flat'(w) = \frac{\mathcal{C}_\flat(w)}{w\big(1 - \mathcal{C}_\flat(w)\big)}, \tag{2.88}$$

   the higher derivatives following by Faà di Bruno;

4. *if $\zeta_0 = \mathrm{e}^{-1}$ and the reached branch has $T(\sigma) = 1$, compose critically with the universal germ of theorem 2.16:*

$$T = 1 + \sum_{r\geq 1} c_r\, \theta_\sigma^{r/2}, \qquad \theta_\sigma := 1 - \mathrm{e}\zeta. \tag{2.89}$$

If θ_σ vanishes to order ς in v, then $\theta_\sigma^{1/2}$ begins at $v^{\varsigma/2}$ and the ramification index doubles when ς is odd; this is the only new ramification mechanism.

Proof. Induction on the depth of σ in the pull-back graph. At depth 0, theorem 2.24 gives the dominant germ. At the induction step, $z \mapsto z^k$ converts each already-known germ into a Puiseux series in a common uniformiser by eq. (2.87); the ring $\mathbb{C}[[v]]$ is closed under the sum defining R, under exponentiation and under multiplication by $z = \sigma(1 - v^{\mathsf{L}})$; and the outer composition is either eq. (2.88) or eq. (2.89), both of which determine every coefficient at finite order by theorems 1.9 and 1.10. The derivative formula in eq. (2.88) follows by differentiating $\mathcal{C} = w\mathrm{e}^{\mathcal{C}}$. ☐

Remark 2.58 (A dedup). S1 introduces a "generalized Bell algebra" $\mathfrak{B}_{\gamma,m}$ on a semigroup-graded power-series ring for this step. Theorem 2.57 avoids it: after a common uniformiser v has been chosen, everything is an *ordinary* power series in v and the standard $\widehat{B}$, B of theorems 1.9 and 1.10 suffice. That is S3's organisation and is adopted here; nothing is lost, because the choice of L is exactly what the semigroup grading was encoding.

2.9.4 The first inherited sector, at $-\sqrt{\rho}$

The point

$$\sigma_- := -\sqrt{\rho} = -0.58165441363339425381 01\ldots \tag{2.90}$$

is reached by continuing the germ at 0 along the negative real axis, which never meets the dominant positive singularity. At $z = \sigma_-$ the summand $T(z^2)/2$ of R reaches $T(\rho)/2$ and inherits the dominant square-root branch, while every $T(z^k)$ with $k \geq 3$ is evaluated at $|\sigma_-|^k \leq \rho^{3/2} < \rho$, strictly inside the original disc. Take the uniformiser

$$z = \sigma_-(1 - v^2), \qquad v = \Big(1 - \tfrac{z}{\sigma_-}\Big)^{1/2}, \qquad \text{so} \qquad 1 - \frac{z^2}{\rho} = v^2(2 - v^2). \tag{2.91}$$

Proposition 2.59 (Complete local data at σ_-). *Write $R\big(\sigma_-(1-v^2)\big) = \sum_{m\geq 0} r_m^- v^m$ and $\zeta\big(\sigma_-(1-v^2)\big) = \sum_{m\geq 0} \zeta_m^- v^m$. Then, with $t_0 := 1$,*

$$r_m^- = \frac{1}{2} \sum_{\substack{0\leq r\leq m\\ r\equiv m\,(2)}} t_r (-1)^{\frac{m-r}{2}} 2^{\frac{r}{2}-\frac{m-r}{2}} \binom{r/2}{\frac{m-r}{2}} + \mathbf{1}_{2|m} (-1)^{m/2} \sum_{k\geq 3} \frac{1}{k} \sum_{n\geq 1} a_n \sigma_-^{kn} \binom{kn}{m/2}, \tag{2.92}$$

both parts converging geometrically. Defining $\mathcal{E}_m^-$ by $\sum_{m\ge0}\mathcal{E}_m^- v^m = \exp\bigl(\sum_{m\ge0} r_m^- v^m\bigr)$, so that $\mathcal{E}_m^- = \mathrm{e}^{r_0^-}\sum_{k\ge0}\frac{1}{k!}\widehat{B}_{m,k}(r_1^-, r_2^-, \ldots)$, one has

$$\zeta_m^- = \sigma_-\bigl(\mathcal{E}_m^- - \mathcal{E}_{m-2}^-\bigr), \qquad \mathcal{E}_{<0}^- := 0. \tag{2.93}$$

Setting $y_- := \mathcal{C}_0(\zeta_0^-) = -W_0(-\zeta_0^-)$, the outer composition is regular because $y_- \neq 1$, and writing $T\bigl(\sigma_-(1-v^2)\bigr) = y_- + \sum_{m\ge1} t_m^- v^m$ one has the closed form

$$\boxed{r_1^- = \frac{t_1}{\sqrt{2}}, \qquad t_1^- = \frac{y_-}{1-y_-}\cdot\frac{t_1}{\sqrt{2}}.} \tag{2.94}$$

Proof. The $k=2$ contribution to R is $\frac12 T\bigl(\rho(1-v^2(2-v^2))\bigr)$; inserting eq. (2.33) with $s = v^2(2-v^2)$, so $s^{r/2} = v^r 2^{r/2}(1-v^2/2)^{r/2}$, and expanding $(1-v^2/2)^{r/2}$ binomially gives the first sum. The $k\ge3$ contributions are $\sum_{k\ge3}\frac1k\sum_n a_n\sigma_-^{kn}(1-v^2)^{kn}$, whose v^{2j} coefficient is the second sum; convergence is geometric already at $k=3$ because $|\sigma_-|^3 < \rho$. Equation (2.93) is $\zeta = \sigma_-(1-v^2)\mathrm{e}^R$. For eq. (2.94): only $r=1$ contributes to v^1 in the first sum, giving $r_1^- = \frac12 t_1\sqrt{2} = t_1/\sqrt{2}$; hence $\zeta_1^- = \zeta_0^- r_1^-$ and, by eq. (2.88) with $\mathcal{C}_0'(w) = \mathcal{C}_0(w)/\bigl(w(1-\mathcal{C}_0(w))\bigr)$, $t_1^- = \mathcal{C}_0'(\zeta_0^-)\zeta_1^- = \frac{y_-}{\zeta_0^-(1-y_-)}\zeta_0^- r_1^-$, which is the claim. □

Recomputed at 80 working digits (see the repair below),

$$\begin{aligned} r_0^- &= 0.4686366279248455480872, & \zeta_0^- &= -0.9293757352012050888501,\\ y_- &= -0.5410329414204534926205, & t_1^- &= 0.3871501110302429034772, \end{aligned} \tag{2.95}$$

and the first local coefficients are

$$\begin{aligned} t_2^- &= -0.0325649185156476341473, & t_3^- &= 0.0172004216031220914830,\\ t_4^- &= 0.1040770395183844385254, & t_5^- &= 0.1619473607727023611895,\\ t_6^- &= -0.0629259981879930868969, & t_7^- &= -0.2440310392314213513551. \end{aligned} \tag{2.96}$$

S2's minus-sector constants are printed past their accuracy. S2 prints r_0^-, z_0^-, y_- and b_1^- to 40 significant digits, e.g. $r_0^- = 0.4686366279248455480872026033545827113234$. Two independent recomputations here — one through the germ recursion at 90 digits, one by direct summation of $\frac12 + \sum_{k\ge3} T(\sigma_-^k)/k$ at 80 digits with 1400 exact tree numbers and no truncation of the inner sums — agree with each other and give $r_0^- = 0.4686366279248455480872026219318193 14\ldots$, which departs from S2's value at the *twenty-sixth* significant digit. The same 25-digit agreement, and no more, is found for ζ_0^-, y_-, t_1^- and every derived quantity, consistent with a single truncation of the $k\ge3$ tail in S2's computation. All minus-sector constants are therefore quoted here to at most 22 digits.

Only the odd t_{2j+1}^- are singular in z at σ_-, so theorem 2.53 with $\beta = j+\frac12$ gives

$$\boxed{a_n^{(-)} \sim \mathfrak{s}_-\,\sigma_-^{-n} n^{-3/2}\sum_{k\ge0}\frac{B_k^-}{n^k}, \qquad B_k^- = \sum_{j=0}^{k}\frac{t_{2j+1}^-}{\Gamma(-j-\frac12)}g_{k-j}\bigl(j+\tfrac12\bigr),} \tag{2.97}$$

where $\mathfrak{s}_-$ is the multiplier of the local cycle; for the direct negative-axis continuation it is 1, and it is displayed only to keep the contour dependence visible in larger deformations. Since $\sigma_-^{-n} = (-1)^n\rho^{-n/2}$, the sector is alternating, with

$$\begin{aligned} B_0^- &= -0.1092130299563101757054 9, & B_1^- &= -0.0336766622077828533671 9,\\ B_2^- &= -0.1790009011730463163890 3, & B_3^- &= -1.6423420049532799830 0,\\ B_4^- &= -18.926645863292083967 3, & B_5^- &= -277.723089846678995649. \end{aligned} \tag{2.98}$$

Relative to the leading dominant term its amplitude is

$$\frac{B_0^-}{C} = -0.2482543049151649529916\ldots, \qquad \frac{a_n^{(-)}}{a_n^{(0)}} \sim -0.2482543049\ldots \left(-\sqrt{\rho}\right)^n, \tag{2.99}$$

which is smaller than every power of $1/n$: this is the first explicitly computed genuinely beyond-all-orders correction to A000081. Its action is

$$\Omega_- = \operatorname{Log}\frac{\sigma_-}{\rho} = \frac{\lambda}{2} + \mathrm{i}\pi \pmod{2\pi\mathrm{i}}, \qquad \frac{\Omega_-}{\lambda} = \frac{1}{2} + \mathrm{i}\frac{\pi}{\lambda}, \qquad \frac{\pi}{\lambda} = 2.8987964297339787377\ldots. \tag{2.100}$$

Remark 2.60. The point $+\sqrt{\rho}$ lies behind the dominant branch point on real continuation and belongs to sheet-dependent continuations; complex critical points and overlapping root lifts may occur farther out. Their local data are produced by theorem 2.57 and their multipliers depend on the chosen contour. Equation (2.97) is one explicit sector, not an assertion that it exhausts modulus $\sqrt{\rho}$.

2.9.5 Transport of the sectors through the inverse

Write the completed forward interpolation as

$$\mathfrak{a}(x) = \mathcal{A}(x)\big(1 + \mathcal{E}(x)\big), \qquad \mathcal{E}(x) = \sum_{\Omega\neq 0} \mathrm{e}^{-\Omega x} x^{\mu_\Omega} \sum_{r\geq 0} e_{\Omega,r} x^{-r}, \tag{2.101}$$

with $\mathcal{A}$ from eq. (2.53) and the support of $\mathcal{E}$ contained in the additive semigroup generated by the actions eq. (2.84). For non-integer x a factor σ^{-x} requires a choice of $\operatorname{Log}\sigma$; different choices agree at integers and differ between them, so the exponentially small part of a *functional* inverse is not canonical until a continuation and pairing convention has been fixed. Fix one, pairing conjugate branches so that $\mathfrak{a}$ is real on the positive axis.

Proposition 2.61 (Sector transport)**.** *Let ν_0 be the complete algebraic inverse of $\mathcal{A}$ (eq.* (2.63)*) and put $\nu_{\text{full}} = \nu_0 + \Delta$. Then Δ is the unique solution, in the transseries field ordered by the semigroup of actions, of*

$$\Delta = \Psi_{\nu_0}(\Delta), \qquad \Psi_{\nu_0}(w) := -\frac{\log\big(1 + \mathcal{E}(\nu_0 + w)\big)}{\mathcal{Q}_{\nu_0}(w)}, \qquad \mathcal{Q}_x(w) := \frac{f(x+w) - f(x)}{w}, \tag{2.102}$$

with $f = \log\mathcal{A}$ and $\mathcal{Q}_x(0) = f'(x)$; its coefficients are given by theorem 1.16, so that in particular

$$\Delta = -\frac{\mathcal{E}(\nu_0)}{f'(\nu_0)} + \mathcal{O}(\mathcal{E}^2) \sim -\frac{\mathcal{E}(\nu_0)}{\lambda}, \tag{2.103}$$

and a single relative forward sector $\mathcal{E}_\Omega \sim e_{\Omega,0}\mathrm{e}^{-\Omega x}x^{\mu_\Omega}$ maps to

$$\Delta_\Omega(y) \sim -\frac{e_{\Omega,0}}{\lambda}\, C^{\Omega/\lambda}\, y^{-\Omega/\lambda}\, \nu_0^{\mu_\Omega - p\Omega/\lambda}. \tag{2.104}$$

Proof. $f(\nu_0{+}\Delta){-}f(\nu_0){+}\log(1{+}\mathcal{E}(\nu_0{+}\Delta)) = 0$ is the exact equation, and $f(\nu_0{+}w){-}f(\nu_0) = w\,\mathcal{Q}_{\nu_0}(w)$ by definition; dividing gives eq. (2.102). Since $f'(\nu_0)$ has leading term $\lambda \neq 0$ it is invertible in the coefficient field, and every occurrence of Δ on the right is either nonlinear or multiplied by a positive transmonomial, so the support is well ordered and each requested transmonomial receives finitely many contributions; theorem 1.16 then gives the coefficients. Equation (2.103) is the first term. For eq. (2.104), use $y \sim C\mathrm{e}^{\lambda\nu_0}\nu_0^{-p}$ to write $\mathrm{e}^{-\Omega\nu_0} = (y/C)^{-\Omega/\lambda}\nu_0^{-p\Omega/\lambda}$. □

For the first inherited sector, $\Omega_- = \frac{\lambda}{2} + \mathrm{i}\pi$ and $e_{-,0} = B_0^-/C$. Pairing the two lateral determinations $\pm\mathrm{i}\pi$ replaces $\mathrm{e}^{-\mathrm{i}\pi x}$ by $\cos(\pi x)$, and $\mathrm{e}^{-\lambda\nu_0/2} = C^{1/2}y^{-1/2}\nu_0^{-3/4}$, so

$$\boxed{\Delta_-(y) \sim 0.15193347365953098826\ldots\; y^{-1/2}\,\nu_0(y)^{-3/4}\cos\big(\pi\nu_0(y)\big),} \tag{2.105}$$

the amplitude being $\sqrt{C}\cdot\big(-B_0^-/(\lambda C)\big)$ with $-B_0^-/(\lambda C) = 0.22906811038403197857\ldots$. The phase is

$$\pi\nu_0(y) = \frac{\pi}{\lambda}\log y + \frac{3\pi}{2\lambda}\log\log y + \mathcal{O}(1), \qquad \frac{\pi}{\lambda} = 2.89879642973397873771\ldots, \qquad \frac{3\pi}{2\lambda} = 4.34819464460096810657\ldots \tag{2.106}$$

so the first nonperturbative correction to the inverse has envelope $y^{-1/2}(\log y)^{-3/4}$ and a logarithmic oscillation with a slower $\log\log y$ drift.

Warning 2.62 (Two different oscillations). Equation (2.105) tends to zero and comes from a complex singular action. The sawtooth of eq. (2.79) is $\mathcal{O}(1)$, never tends to zero, and comes from integer rounding. They have different origins, different scales, and must not be conflated.

2.10 Other inverse notions: the compositional inverse of T

Since $T(z) = z + \mathcal{O}(z^2)$ with integer coefficients and unit linear coefficient, T has a compositional inverse $Z := T^{\langle -1\rangle} \in z\mathbb{Z}[[z]]$, unrelated to the growth inversion of section 2.7.

Proposition 2.63 (Coefficients of Z). *For $n \geq 1$,*

$$[w^n]\,Z(w) = \frac{1}{n}[z^{n-1}]\left(\frac{z}{T(z)}\right)^n = \frac{1}{n}[z^{n-1}]\,\exp\left(-n\sum_{k\geq 1}\frac{T(z^k)}{k}\right), \tag{2.107}$$

a finite ordinary-Bell expression in $a_1, \ldots, a_{n-1}$. The first sixteen values, recomputed by exact rational arithmetic, are

$$1,\ -1,\ 0,\ 1,\ -1,\ 1,\ -4,\ 11,\ -18,\ 18,\ 0,\ -60,\ 189,\ -360,\ 453,\ -373. \tag{2.108}$$

Proof. The first equality is theorem 1.14 with $\phi(z) = T(z)/z$, a unit series. The second is eq. (2.5) rearranged, and theorem 1.9 converts the exponential into a finite Bell sum. Integrality is automatic for the compositional inverse of a series in $z + z^2\mathbb{Z}[[z]]$. □

Remark 2.64. S1 displays $Z(w) = w - w^2 + w^4 - w^5 + w^6 - 4w^7 + 11w^8 - \cdots$, which agrees with eq. (2.108), and identifies the coefficients with OEIS A050395. The expansion is confirmed here; the OEIS identification is reported from S1 and has not been re-checked in this volume.

Near the dominant value $w = 1$ the branch of Z through $Z(1) = \rho$ is *analytic* in $1 - w$, because it undoes the square root: from $T = \zeta e^T$ one has $\zeta = Te^{-T}$, hence

$$Z(w) = \zeta^{\langle -1\rangle}\big(we^{-w}\big) \tag{2.109}$$

on that branch, ζ being injective on $(0, \sqrt{\rho})$ by theorem 2.11. Since $we^{-w} - e^{-1}$ has a double zero at $w = 1$, the local expansion of Z at $w = 1$ has no linear term. Its coefficients follow from one more Lagrange extraction: writing $\zeta(\rho + x) - e^{-1} = x\,\psi(x)$ with $\psi(0) = \zeta'(\rho) \neq 0$,

$$[\mathsf{v}^m]\,\big(\zeta^{\langle -1\rangle}(e^{-1} + \mathsf{v}) - \rho\big) = \frac{1}{m}[x^{m-1}]\,\psi(x)^{-m}, \tag{2.110}$$

followed by composition with $(1-\varsigma)e^{-(1-\varsigma)} - e^{-1} = -e^{-1}\sum_{m\geq 2}\frac{m-1}{m!}\varsigma^m = -e^{-1}\frac{\varsigma^2}{2}\Phi(\varsigma)$, the same kernel eq. (2.22) that governed the forward branch.

2.11 Algorithms, validation and audit

2.11.1 The pipeline

Algorithm 2.65 (All dominant and inverse coefficients through order M).

1. Generate $a_1, \dots, a_N$ and $b_1, \dots, b_N$ exactly by eq. (2.7) with a divisor sieve, checking the divisibility assertion at each step; form ϱ_r by eq. (2.10).
2. Solve $\log x + 1 + \sum_{r=2}^{N} \varrho_r x^r = 0$ for ρ by safeguarded Newton or interval bisection. Choose N so that $\rho^{N/2}$ is below the target precision; the omitted tail is geometrically small because R is analytic on $|z| < \sqrt{\rho}$ (theorem 2.10).
3. Compute $\widehat{R}_0, \dots, \widehat{R}_{2M+1}$ by eq. (2.28); exponentiate by eq. (2.29) and form $\theta_1, \dots, \theta_{2M+2}$ by eq. (2.30), then $u_j = \theta_{j+1}/\theta_1$.
4. Compute $\omega_1, \dots, \omega_{2M+1}$ exactly by eq. (2.23) (rationals), set $c_r = -2^{r/2}\omega_r$, and form $t_1, t_3, \dots, t_{2M+1}$ by eq. (2.34).
5. Compute $g_r(j + \frac{1}{2})$ exactly for $j + r \le M$ by eq. (2.39), then $\tau_0, \dots, \tau_M$ by eq. (2.44), then $C = \tau_0/\sqrt{\pi}$ and $f_m = \tau_m/\tau_0$.
6. Reciprocal: $\bar{f}_m$ by eq. (2.50).
7. Inverse: φ_m by eq. (2.55), then d_m by eq. (2.61), or $P_m(L)$ by the flattening recursion of theorem 1.33.
8. Sectors: apply theorem 2.57 outward in modulus, transfer each germ by eq. (2.83), and invert by theorem 2.61.

At every stage the coefficient of order m depends only on data of order at most m; no nonlinear global solve occurs after ρ is known. The useful power-series primitives are the three dense recurrences

$$[v^n]\,F^\gamma = \frac{1}{n}\sum_{k=1}^{n}\big((\gamma+1)k - n\big)F_k[v^{n-k}]\,F^\gamma, \quad E_n = \frac{1}{n}\sum_{k=1}^{n} kA_kE_{n-k}, \quad H_n = F_n - \frac{1}{n}\sum_{k=1}^{n-1} kH_kF_{n-k}, \tag{2.111}$$

for F^γ with $F_0 = 1$, for $E = \mathrm{e}^A$ with $A_0 = 0$, and for $H = \log F$ with $F_0 = 1$. They are equivalent to the Bell forms used in the statements and are what an implementation should actually run.

Warning 2.66 (Precision discipline). The coefficients grow rapidly and eq. (2.44) combines alternating contributions from several Puiseux coefficients, so obtaining d reliable digits of τ_M can require substantially more than d working digits. The tables in this chapter were produced with 90–110 decimal working digits, with universal objects (ω_m, $g_r(\beta)$, a_n) held in exact rational or integer arithmetic. The repair recorded at section 2.9.4 is exactly what happens when this discipline lapses.

2.11.2 Numerical validation

Let $A_M(n)$ denote eq. (2.43) truncated after $\tau_M n^{-M}$. Table 2.5 reports the signed relative error $\big(A_M(n) - a_n\big)/a_n$ against exact integers from eq. (2.7); table 2.6 reports the signed index error $\nu_M(a_n) - n$ for the Lambert-normalised inverse $\nu_M = x_0 + \sum_{m \le M} d_m x_0^{-m}$. Both tables were regenerated here and agree entry by entry with the corresponding tables of S1 and S2 at the displayed precision.

Table 2.5: Signed relative error $(A_M(n) - a_n)/a_n$ of the forward expansion.

n	$M=0$	$M=2$	$M=4$	$M=6$	$M=8$	$M=10$
10	$-1.52 \cdot 10^{-2}$	$-3.49 \cdot 10^{-4}$	$2.63 \cdot 10^{-3}$	$3.99 \cdot 10^{-3}$	$5.52 \cdot 10^{-3}$	$8.77 \cdot 10^{-3}$
20	$-6.58 \cdot 10^{-3}$	$-3.42 \cdot 10^{-4}$	$-3.18 \cdot 10^{-5}$	$1.04 \cdot 10^{-6}$	$9.73 \cdot 10^{-6}$	$1.41 \cdot 10^{-5}$
50	$-2.22 \cdot 10^{-3}$	$-1.77 \cdot 10^{-5}$	$-2.88 \cdot 10^{-7}$	$-1.20 \cdot 10^{-8}$	$-9.84 \cdot 10^{-10}$	$-1.36 \cdot 10^{-10}$
100	$-1.06 \cdot 10^{-3}$	$-2.08 \cdot 10^{-6}$	$-8.06 \cdot 10^{-9}$	$-7.88 \cdot 10^{-11}$	$-1.50 \cdot 10^{-12}$	$-4.73 \cdot 10^{-14}$
200	$-5.15 \cdot 10^{-4}$	$-2.53 \cdot 10^{-7}$	$-2.40 \cdot 10^{-10}$	$-5.73 \cdot 10^{-13}$	$-2.66 \cdot 10^{-15}$	$-2.04 \cdot 10^{-17}$
500	$-2.03 \cdot 10^{-4}$	$-1.59 \cdot 10^{-8}$	$-2.39 \cdot 10^{-12}$	$-9.02 \cdot 10^{-16}$	$-6.62 \cdot 10^{-19}$	$-8.01 \cdot 10^{-22}$

Two features are worth naming. First, at $n = 10$ and $n = 20$ the error stops improving and then worsens: the expansion is asymptotic, and the least-term truncation of theorem 1.46 has been passed.

Table 2.6: Signed index error $\nu_M(a_n) - n$ of the Lambert-normalised inverse; $M = 0$ is the bare carrier x_0.

n	$M=0$	$M=1$	$M=2$	$M=4$	$M=6$	$M=8$
10	$1.64 \cdot 10^{-2}$	$7.15 \cdot 10^{-3}$	$1.29 \cdot 10^{-3}$	$-2.61 \cdot 10^{-3}$	$-4.16 \cdot 10^{-3}$	$-5.76 \cdot 10^{-3}$
20	$6.55 \cdot 10^{-3}$	$1.91 \cdot 10^{-3}$	$4.44 \cdot 10^{-4}$	$3.76 \cdot 10^{-5}$	$-2.02 \cdot 10^{-7}$	$-9.39 \cdot 10^{-6}$
50	$2.11 \cdot 10^{-3}$	$2.59 \cdot 10^{-4}$	$2.32 \cdot 10^{-5}$	$3.32 \cdot 10^{-7}$	$1.26 \cdot 10^{-8}$	$9.98 \cdot 10^{-10}$
100	$9.88 \cdot 10^{-4}$	$6.16 \cdot 10^{-5}$	$2.74 \cdot 10^{-6}$	$9.34 \cdot 10^{-9}$	$8.35 \cdot 10^{-11}$	$1.53 \cdot 10^{-12}$
200	$4.78 \cdot 10^{-4}$	$1.50 \cdot 10^{-5}$	$3.33 \cdot 10^{-7}$	$2.79 \cdot 10^{-10}$	$6.09 \cdot 10^{-13}$	$2.71 \cdot 10^{-15}$
500	$1.88 \cdot 10^{-4}$	$2.38 \cdot 10^{-6}$	$2.10 \cdot 10^{-8}$	$2.78 \cdot 10^{-12}$	$9.59 \cdot 10^{-16}$	$6.75 \cdot 10^{-19}$

Second, the inverse errors track the forward relative errors divided by $f' \approx \lambda$, exactly as theorem 2.42 predicts; this is why tables 2.5 and 2.6 are so nearly proportional.

Remark 2.67 (On comparisons with the literature). S1 states that its first five coefficients “agree with Genitrini’s independently computed values” and S3 that its Puiseux table “agrees with the independent table in [Genitrini 2016]”. Those comparisons are reported from the drafts and have *not* been re-performed here; what has been done is a full independent recomputation of every constant from the exact recurrence, described in section 2.11.3.

2.11.3 Audit

Every numerical value printed in this chapter was produced by an independent implementation of theorem 2.65 written for this volume: exact integer generation of $a_1, \dots, a_N$ with N between 700 and 1400; exact rational ϱ_r, ω_m and $g_r(\beta)$ (the last from an exact Bernoulli-number table); and 60–110 decimal working digits for everything else. The following independent identities were checked and hold to all digits carried.

1. $(n-1) \mid \sum_{j<n} b_j a_{n-j}$ at every step, for $n \le 1400$; $a_{10} = 719$, $a_{20} = 12\,826\,228$, $a_{50} = 425\,976\,989\,835\,141\,038\,353$.
2. $\lambda = 1 + R(\rho)$, i.e. eq. (2.20).
3. $\theta_0 = 0$, the vanishing of the constant term in eq. (2.26) predicted by $\mathrm{e}\zeta(\rho) = 1$.
4. $C^2 = \theta_1/(2\pi)$, i.e. eq. (2.45) — the two sides travel through disjoint parts of the pipeline.
5. $c_r = -2^{r/2}\omega_r$ for $1 \le r \le 18$ in exact arithmetic, and $t_1 = -\sqrt{2\theta_1}$, $t_2 = \frac{2}{3}\theta_1$.
6. Equation (2.47) re-derived symbolically from eq. (2.44), and the $g_r(\beta)$ of eqs. (2.41) and (2.42) re-derived from eq. (2.38).
7. $\mathrm{D}_m = \lambda^{m+1} d_m$ and $\eta_m^{\mathrm{S2}} = \lambda^m \varphi_m$ for $1 \le m \le 8$, reconciling S1/S3 with S2.
8. $P_1(0) = d_1$, $P_2(0) = d_2$ (a case of eq. (2.72)).
9. $r_1^- = t_1/\sqrt{2}$ and the closed form eq. (2.94), each against the germ recursion; and r_0^- by a second, wholly independent summation (see the repair in section 2.9.4).
10. The single-sum eq. (2.28) against the siblings’ double sum, for $0 \le m \le 30$.

2.12 Summary

The Pólya equation becomes transparent after the factorisation $T = \mathcal{C} \circ \zeta$ with $\mathcal{C}(w) = -W_0(-w)$: the universal layer is one Lagrange coefficient $[v^{r-1}]\,\Phi(v)^{-r/2}$, the problem-specific layer is ordinary Bell

Table 2.7: Principal symbols of this chapter.

Symbol	Meaning
a_n, $T(z)$	A000081 and its ordinary generating function.
b_m, ϱ_r	Divisor transform $\sum_{d\mid m} da_d$; coefficients of R, $\varrho_r = (b_r - ra_r)/r$.
R, ζ	Disturbance $\sum_{k\geq 2} T(z^k)/k$ and Cayley argument $z\mathrm{e}^R$; $T = -W_0(-\zeta)$.
ρ, α, λ, p	Otter singularity; $\alpha = \rho^{-1}$, $\lambda = \log \alpha$, $p = \frac{3}{2}$.
Φ, ω_m, c_r	Universal Cayley kernel and its two coefficient normalisations, $c_r = -2^{r/2}\omega_r$.
s, θ, θ_m, u_j	Local coordinate $1 - z/\rho$; disturbance $1 - \mathrm{e}\zeta$; its coefficients; $u_j = \theta_{j+1}/\theta_1$.
$\widehat{R}_m$, ℓ_j	$[s^m]\, R(\rho(1-s))$; scaled logarithmic jets $\rho^j (\log\zeta)^{(j)}(\rho)$.
t_n	Puiseux coefficients, $T = 1 + \sum t_n s^{n/2}$.
$\varkappa_m(\beta)$, $g_r(\beta)$	Bernoulli increments and gamma-ratio transfer coefficients.
τ_m, C, f_m, $\bar{f}_m$	Forward coefficients; $C = \tau_0/\sqrt{\pi}$; $f_m = \tau_m/\tau_0$; reciprocal coefficients.
F, H, φ_m	$F = 1 + \sum f_m v^m$; $H = \log F = \sum \varphi_m v^m$; this H is the Part 0 datum Q.
x_0, d_m, ν	Lambert carrier; its corrections; the smooth asymptotic inverse.
$\mathcal{X}$, L, q, η_k, P_m	Flattening variables and coefficient polynomials.
$N(y)$	Integer staircase $\min\{n \geq 1 : a_n \geq y\}$.
σ_-, t_m^-, B_k^-, Ω_σ	The inherited singularity $-\sqrt{\rho}$, its local and sector coefficients, and the actions.

composition of the analytic disturbance ζ, and coefficient transfer is Bernoulli polynomials in the gamma ratio. The chain is

$$a_n \;\to\; \varrho_r \;\to\; \rho \;\to\; \widehat{R}_m \;\to\; \theta_m \;\to\; t_n \;\to\; \tau_m \;\to\; f_m \;\to\; \varphi_m \;\to\; d_m \text{ or } P_m(L), \tag{2.112}$$

every arrow a finite Bell- or Bernoulli-polynomial formula. Inversion adds nothing new: A000081 is the $\alpha = 1$ case of theorem 1.2 with $a = \lambda$, $b = -\frac{3}{2}$, $Q = H$, so theorem 1.20 supplies the carrier exactly, theorem 1.25 every algebraic correction, theorem 1.33 the logarithmic form, theorem 1.44 the error, and theorem 1.39 the passage to the integer inverse. Beyond all orders, the recursion supplies a singularity grammar whose first explicit instance is the alternating sector of relative size $0.2483\ldots\,(-\sqrt{\rho})^n$ at $\sigma_- = -\sqrt{\rho}$, transported by theorem 2.61 into an inverse correction of envelope $y^{-1/2}(\log y)^{-3/4}$ with a $\log y$-periodic phase. What remains open is not coefficient algebra but analysis: the classification of accessible sheets, of critical preimages, and of their Stokes connections — and, prior to that, a proof of $\mathsf{H}(\mathcal{R})$ for $\mathcal{R}$ close to 1.

Chapter 3

The double factorial

This chapter merges three independent treatments of $n!!$: *Double_Factorial_Transseries* (henceforth S1), *Double_Factorial_Transseries-3* (S2), and *double_factorial_transseries-2* (S3). All three reach the same forward and inverse coefficients; they differ in organisation, in what is proved, and — decisively — in notation, where the letter C carries three incompatible meanings. The chapter adopts S3's *exact* centred normal form as its spine, S2's Borel kernel and sharp remainder, S2's graded closed coefficient formula, and S1/S2's parity-mean centring of the periodic interpolation. Every displayed rational number below was regenerated with exact rational arithmetic (`sympy`) from the defining recurrences, and every numerical table was recomputed at 220-digit precision (`mpmath`); see section 3.14.

3.1 Orientation: which inverse?

For $n \in \mathbb{N}$ the double factorial is

$$0!! = 1!! = 1, \qquad n!! = n\,(n-2)!! \quad (n \geq 2), \tag{3.1}$$

so that $n!!$ is the product of the positive integers not exceeding n of the same parity as n. The sequence $1, 1, 2, 3, 8, 15, 48, 105, 384, 945, \ldots$ is OEIS A006882. Its elementary closed forms already contain the whole difficulty:

$$(2m)!! = 2^m\, m!, \qquad (2m+1)!! = \frac{2^{m+1}\Gamma\left(m+\frac{3}{2}\right)}{\sqrt{\pi}}. \tag{3.2}$$

The even subsequence is a gamma function of the halved argument; the odd subsequence is the *same* gamma function of the halved argument times a different constant. The two constants do not agree, and no amount of reindexing makes them agree.

Warning 3.1. $n!!$ is not the restriction to $\mathbb{N}$ of one canonical slowly varying function. Its even and odd subsequences are two gamma-type branches whose quotient is a nonzero constant $\sqrt{2/\pi} \neq 1$. Consequently *there is no canonical real inverse of* $n!!$ *until an interpolation is fixed*, and different admissible interpolations give inverses that differ at an order *larger* than the entire Bernoulli correction grid (theorem 3.8). Every statement in this chapter that begins "the inverse" names its interpolation first.

Following theorem 1.37 we therefore distinguish, for a target $y \to +\infty$:

(i) *the two parity-resolved functional inverses*, obtained by inverting the canonical gamma continuation of one parity subsequence (section 3.6);

(ii) *the functional inverse of a single all-integer interpolation*, for instance the periodic one recorded by OEIS (section 3.8);

(iii) *the integer staircase inverses*, the least n with $n!! \geq y$ and the greatest n with $n!! \leq y$ (section 3.9).

Objects (i) and (iii) are intrinsic to the sequence; object (ii) is not. This is the chapter's spine, and theorem 3.8 makes it quantitative.

Two asymptotic scales must also be kept apart. Taking logarithms converts the growth of $n!!$ into a phase $\frac{1}{2}u(\log u - 1)$ whose inversion is *exactly* a Lambert expression (theorem 1.23); expanding that Lambert function produces an *outer* layer in powers of $1/\log\log y$ with polynomial dependence on $\log\log\log y$ (theorem 1.33). The Bernoulli corrections generate a far finer *inner* grid in odd inverse powers of the Lambert core. The fixed shift -1 lies strictly between the two levels. Writing all of this as one informal series in $1/\log y$ destroys the ordering; see theorem 3.42.

3.2 Exact continuations and the parity split

3.2.1 The two parity branches

Definition 3.2 (Parity branches)**.** For the parity label $\delta \in \{0, 1\}$ put

$$\kappa_\delta = \left(\frac{2}{\pi}\right)^{\delta/2}, \qquad \mathcal{D}_\delta(x) = \kappa_\delta\, 2^{x/2}\, \Gamma\Big(\frac{x}{2} + 1\Big) \qquad (x > -2), \tag{3.3}$$

and write

$$A_\delta = \log\big(\kappa_\delta \sqrt{\pi}\big) = \begin{cases} \frac{1}{2}\log \pi, & \delta = 0, \\ \frac{1}{2}\log 2, & \delta = 1, \end{cases} \qquad \mathrm{e}^{A_0} = \sqrt{\pi}, \quad \mathrm{e}^{A_1} = \sqrt{2}. \tag{3.4}$$

Here $\delta = 0$ labels the even subsequence and $\delta = 1$ the odd one.

Proposition 3.3 (Exact parity interpolation and monotonicity)**.** *For every integer $n \geq 0$ with $n \equiv \delta \pmod 2$ one has $\mathcal{D}_\delta(n) = n!!$. Moreover each $\mathcal{D}_\delta$ is positive and strictly increasing on $[0, \infty)$, so it has a single-valued inverse $\mathcal{D}_\delta^{-1}$ on its range; and*

$$\mathcal{D}_1(x) = \sqrt{\frac{2}{\pi}}\, \mathcal{D}_0(x), \qquad \mathcal{D}_1^{-1}(y) = \mathcal{D}_0^{-1}\bigg(\sqrt{\frac{\pi}{2}}\, y\bigg). \tag{3.5}$$

Proof. If $n = 2m$ then $\mathcal{D}_0(2m) = 2^m\Gamma(m+1) = 2^m m! = (2m)!!$ by eq. (3.2). If $n = 2m + 1$ then

$$\mathcal{D}_1(2m+1) = \sqrt{\frac{2}{\pi}}\, 2^{m+1/2}\Gamma\big(m + \tfrac{3}{2}\big) = \frac{2^{m+1}}{\sqrt{\pi}}\Gamma\big(m + \tfrac{3}{2}\big) = (2m+1)!!.$$

Positivity is clear because $\Gamma > 0$ on $(0, \infty)$. For monotonicity,

$$\frac{\mathcal{D}_\delta'(x)}{\mathcal{D}_\delta(x)} = \frac{1}{2}\left(\log 2 + \psi\Big(\frac{x}{2} + 1\Big)\right), \qquad \psi = \Gamma'/\Gamma, \tag{3.6}$$

and ψ is strictly increasing on $(0, \infty)$ because $\psi'(z) = \sum_{m \geq 0}(z+m)^{-2} > 0$. At $x = 0$ the bracket equals $\log 2 - \gamma = 0.11593\ldots > 0$, hence it is positive for all $x \geq 0$. Equation (3.5) is immediate from eq. (3.3), since $\kappa_1/\kappa_0 = \sqrt{2/\pi}$. □

Remark 3.4. The constant κ_δ disappears from eq. (3.6). Both branches therefore have the same local conditioning, the same forward correction series, and — as theorem 3.29 shows — literally the same inverse coefficient polynomials. Parity survives only as the additive constant A_δ subtracted from $\log y$.

3.2.2 The periodic OEIS interpolation, and why it is not canonical

Definition 3.5 (Periodic interpolation)**.** Put

$$\mathscr{D}(x) = 2^{x/2}\, \Gamma\Big(\frac{x}{2} + 1\Big) \left(\frac{2}{\pi}\right)^{(1 - \cos \pi x)/4}. \tag{3.7}$$

Because $\frac{1}{2}\sin^2(\pi x/2) = \frac{1}{4}(1 - \cos\pi x)$, this is the interpolation recorded in the OEIS entry for A006882, written there with the half-argument sine.

Proposition 3.6 (Interpolation and eventual monotonicity)**.** *$\mathscr{D}(n) = n!!$ for every integer $n \geq 0$, and $\mathscr{D}$ is strictly increasing on $[1.04, \infty)$. Its inverse on that ray, denoted $\mathscr{D}^{-1}$, is what "the inverse of $\mathscr{D}$" means below; it is not a globally single-valued inverse near the origin.*

Proof. The periodic factor equals 1 at even integers and $\sqrt{2/\pi}$ at odd integers, so $\mathscr{D}$ agrees with $\mathcal{D}_0$ at even and with $\mathcal{D}_1$ at odd integers; theorem 3.3 finishes the first claim. Next,

$$\frac{\mathscr{D}'(x)}{\mathscr{D}(x)} = \frac{1}{2}\left(\log 2 + \psi\left(\frac{x}{2}+1\right)\right) + \frac{\pi}{4}\log\left(\frac{2}{\pi}\right)\sin\pi x. \tag{3.8}$$

For $z > \frac{1}{2}$ one has $\psi(z) > \log\bigl(z - \frac{1}{2}\bigr)$, hence $\psi(x/2+1) > \log((x+1)/2)$ and the first summand exceeds $\frac{1}{2}\log(x+1)$. The second summand has modulus at most πa, where

$$a := \frac{1}{4}\log\frac{\pi}{2} = 0.112895676322\ldots > 0. \tag{3.9}$$

Thus $\mathscr{D}'/\mathscr{D} > \frac{1}{2}\log(x+1) - \pi a$, which is positive as soon as $x + 1 > \mathrm{e}^{2\pi a} = 2.03265\ldots$, i.e. for $x \geq 1.04$. ☐

Remark 3.7. The bound in the proof is not sharp: numerically the last zero of eq. (3.8) is at $x = 0.6970947\ldots$, so $\mathscr{D}$ is in fact strictly increasing on $[0.70, \infty)$. Only the proved threshold is used below.

The next theorem is the precise form of theorem 3.1. None of the three sources states it; all three assert the phenomenon informally.

Theorem 3.8 (The inverse transseries is not determined by the sequence)**.** *For $\lambda \in \mathbb{R}$ set $\mathscr{D}_\lambda(x) = \mathscr{D}(x)\,\mathrm{e}^{\lambda\sin\pi x}$. Then:*

(a) *$\mathscr{D}_\lambda(n) = n!!$ for every integer $n \geq 0$, and $\mathscr{D}_\lambda$ is real-analytic on $\mathbb{R}$;*

(b) *$\mathscr{D}_\lambda$ is strictly increasing on $[\,\mathrm{e}^{2\pi(a+|\lambda|)} - 1, \infty)$, so it has a large real inverse;*

(c) *with T and Λ the Lambert data of eq.* (3.32) *built from $R_* = \log y - \frac{1}{4}\log(2\pi)$, the leading displacement of that inverse beyond the core is*

$$\mathscr{D}_\lambda^{-1}(y) - (T-1) = \frac{2\sqrt{a^2+\lambda^2}}{\Lambda}\cos\left(\pi T - \arctan\frac{\lambda}{a}\right) + \mathcal{O}\bigl(\Lambda^{-2}\bigr)\,. \tag{3.10}$$

In particular the amplitude $2\sqrt{a^2+\lambda^2}/\Lambda$ of the leading oscillatory sector is an unbounded function of the interpolation and is larger, by the factor T, than the entire Bernoulli grid $\mathcal{O}(T^{-1}\Lambda^{-1})$ *of theorem 3.29.*

Proof. (a) $\sin\pi n = 0$ for integer n, and $\mathrm{e}^{\lambda\sin\pi x}$ is entire.

(b) Differentiating logarithmically adds $\lambda\pi\cos\pi x$ to eq. (3.8); repeating the estimate in the proof of theorem 3.6 gives $\mathscr{D}_\lambda'/\mathscr{D}_\lambda > \frac{1}{2}\log(x+1) - \pi(a+|\lambda|)$.

(c) By theorem 3.44 below, with $u = x+1$,

$$\log\mathscr{D}_\lambda(u-1) = \frac{u}{2}(\log u - 1) + \frac{1}{4}\log(2\pi) - a\cos\pi u - \lambda\sin\pi u + \Omega(u),$$

because $\cos\pi x = -\cos\pi u$ and $\sin\pi x = -\sin\pi u$. Subtracting the core identity $\frac{1}{2}T(\log T - 1) = R_*$, putting $u = T + \Delta$ and multiplying by 2 exactly as in theorem 3.45 gives

$$\Lambda\Delta + T\,\varphi(\Delta/T) + 2\Omega(T+\Delta) - 2a\cos(\theta+\pi\Delta) - 2\lambda\sin(\theta+\pi\Delta) = 0, \qquad \theta = \pi T.$$

All terms except the last two are $\mathcal{O}(T^{-1})$ once $\Delta = \mathcal{O}(\Lambda^{-1})$, so at leading order $\Lambda\Delta_0 = 2a\cos(\theta + \pi\Delta_0) + 2\lambda\sin(\theta+\pi\Delta_0) = 2\sqrt{a^2+\lambda^2}\cos(\theta+\pi\Delta_0 - \arctan(\lambda/a))$. Since $\Delta_0 = \mathcal{O}(\Lambda^{-1})$, replacing $\theta + \pi\Delta_0$ by θ costs $\mathcal{O}(\Lambda^{-2})$, which is eq. (3.10). ☐

Remark 3.9. The family $e^{\lambda \sin \pi x}$ is only the simplest witness. Any real-analytic g vanishing on $\mathbb{N}$ and with g' bounded produces the same conclusion, with $2a\cos\theta$ replaced by the value of $-2g$ transported to the core. What *is* intrinsic is the pair of parity-branch transseries and the integer staircase; the periodic sector is a property of the chosen interpolation and nothing else.

3.2.3 The centred coordinate and the exact normal form

Definition 3.10 (Centred coordinate). Throughout, $u := x + 1$ and

$$\Phi(u) := \frac{u}{2}(\log u - 1), \qquad \Omega(u) := \log\Gamma\left(\frac{u+1}{2}\right) - \frac{u}{2}\log\frac{u}{2} + \frac{u}{2} - \frac{1}{2}\log(2\pi). \tag{3.11}$$

Theorem 3.11 (Exact centred normal form). *For every $x > -1$ and every $\delta \in \{0, 1\}$, with $u = x + 1$,*

$$\boxed{\mathcal{D}_\delta(x) = e^{A_\delta}\left(\frac{u}{e}\right)^{u/2} e^{\Omega(u)}, \qquad \textit{equivalently} \qquad \log\mathcal{D}_\delta(x) = \Phi(u) + A_\delta + \Omega(u).} \tag{3.12}$$

This is an identity, not an asymptotic statement, and Ω does not depend on δ.

Proof. Insert $x = u - 1$ into eq. (3.3):

$$\log\mathcal{D}_\delta(x) = \log\kappa_\delta + \frac{u-1}{2}\log 2 + \log\Gamma\left(\frac{u+1}{2}\right).$$

Subtract $A_\delta + \Phi(u)$ and use $A_\delta = \log\kappa_\delta + \frac{1}{2}\log\pi$:

$$\begin{aligned}\log\mathcal{D}_\delta(x) - A_\delta - \Phi(u) &= \frac{u-1}{2}\log 2 + \log\Gamma\left(\frac{u+1}{2}\right) - \frac{1}{2}\log\pi - \frac{u}{2}\log u + \frac{u}{2}\\ &= \log\Gamma\left(\frac{u+1}{2}\right) - \frac{u}{2}(\log u - \log 2) + \frac{u}{2} - \frac{1}{2}\log 2 - \frac{1}{2}\log\pi,\end{aligned}$$

which is exactly $\Omega(u)$ as defined in eq. (3.11). □

Remark 3.12 (Why $u = x + 1$). The shift is not cosmetic. It turns the gamma argument into $\frac{1}{2}(u+1) = \frac{u}{2} + \frac{1}{2}$, and Bernoulli polynomials evaluated at $\frac{1}{2}$ annihilate every odd index; consequently Ω has only *odd* inverse powers of u (theorem 3.16) and the explicit $\frac{1}{2}\log x$ of the uncentred Stirling form (theorem 3.23) is absorbed into the dominant phase. This sparse support is precisely why the inverse lives on the grid $T^{-1}, T^{-3}, T^{-5}, \ldots$.

At an integer n the two constants A_0, A_1 combine into one parity transmonomial: writing $\chi_n = (-1)^n$,

$$A(n) = \frac{1}{4}\log(2\pi) + \frac{\chi_n}{4}\log\frac{\pi}{2}, \qquad e^{A(n)} = (2\pi)^{1/4}\left(\frac{\pi}{2}\right)^{\chi_n/4}, \tag{3.13}$$

so that for every integer $n \geq 0$

$$n!! = \left(\frac{n+1}{e}\right)^{(n+1)/2} \exp\bigl(A(n) + \Omega(n+1)\bigr). \tag{3.14}$$

Equation (3.14) is exact. The factor $(-1)^n$ is a genuine transmonomial of the sequence's leading behaviour, not an error term.

Remark 3.13 (Notation across the three sources). The letter C means three different things in the siblings, and two Greek letters collide:

this chapter	S1	S2	S3
$\kappa_\delta = (2/\pi)^{\delta/2}$	κ_δ	κ_r	C_ε
$\mathrm{e}^{A_\delta} \in \{\sqrt{\pi}, \sqrt{2}\}$	C_δ	A_r	—
$A_\delta \in \{\frac{1}{2}\log\pi, \frac{1}{2}\log 2\}$	$\log C_\delta$	C_r	A_ε
c_k (centred log. coefficient)	c_k	β_k	c_k
$a_k = 2c_k$	$2c_k$	α_k	a_k
$a = \frac{1}{4}\log(\pi/2)$	$\rho/2$	a	—
$P_r(L_2)$ (Lambert polynomial)	P_r	u_{r+1}	α_{r+1}
$\tau_m(L_2)$ (reciprocal poly.)	h_m	τ_m	β_m

Part 0 adds one further collision: theorem 1.8 writes the factorial core as $\Phi_0(X) = X(\kappa \log X + d)$, and that κ is *not* the branch prefactor κ_δ of S1 and S2. This chapter therefore never abbreviates the core parameters, always writing $\Phi_0(u) = u\big(\frac{1}{2}\log u - \frac{1}{2}\big)$ in full; see theorem 1.24. In particular S2's α_k, β_k are Bernoulli data while S3's α_j, β_n are outer Lambert polynomials. Readers comparing the sources should translate before comparing formulae; the numbers themselves agree everywhere (section 3.14).

3.3 Parameter dictionary for the Part 0 apparatus

The chapter uses the Part 0 toolkit rather than re-deriving it. Its dominant core is the *factorial* one of theorem 1.8, not the exponential–power model theorem 1.2; see theorem 3.14. The specialisation is the following.

Part 0 datum	even branch $\delta = 0$	odd branch $\delta = 1$
interpolation	$2^{x/2}\Gamma(\frac{x}{2}+1)$	$\sqrt{2/\pi}\; 2^{x/2}\Gamma(\frac{x}{2}+1)$
affine shift x_* of theorem 1.8	$x_* = -1$, taken first	$x_* = -1$, taken first
centred coordinate $u = x - x_*$	$u = x + 1$	$u = x + 1$
admissible core Φ_0 of theorem 1.8	$\Phi_0(u) = u\big(\frac{1}{2}\log u - \frac{1}{2}\big)$	same
additive constant	$A_0 = \frac{1}{2}\log\pi$	$A_1 = \frac{1}{2}\log 2$
perturbation $Q(s) = \sum_{k\geq 1} a_k s^{2k-1}$	$a_k = 2c_k$, eq. (3.19)	same
target after constant removal	$R_0 = \log y - \frac{1}{2}\log\pi$	$R_1 = \log y - \frac{1}{2}\log 2$
core (theorem 1.23)	$T_\delta = 2R_\delta / W_0(2R_\delta/\mathrm{e})$	same formula
core slope parameter	$\Lambda_\delta = \log T_\delta$	same formula
inner grid	$q = T_\delta^{-1}$, $z = q^2$	same

Concretely: theorems 1.9 and 1.10 supply the Bell polynomials used throughout; theorem 1.12 supplies the coefficients of $(1+w)^{-j}$ and $\log(1+w)$; theorem 1.23 inverts the dominant phase exactly by W_0; theorem 1.29, whose formal engine is theorem 1.16, generates the correction blocks and is specialised once in theorem 3.28; theorem 1.33 unfolds W_0 into the outer layer of section 3.7; theorems 1.37 and 1.39 govern section 3.9; theorems 1.42 and 1.44 convert forward residuals into inverse errors; and theorem 1.46 fixes the least-term scale of section 3.10.

Warning 3.14 (The double factorial is not of exponential–power type). It is tempting — and wrong — to file $\mathcal{D}_\delta$ under theorem 1.2, $F(x) = C\exp(ax^\alpha)x^b \exp Q(x^{-\delta})$ with $\alpha \in (0,1]$. By theorem 3.11, $\log \mathcal{D}_\delta(x) = \frac{u}{2}(\log u - 1) + A_\delta + \Omega(u)$ with $u = x+1$, so the dominant phase is $\frac{1}{2}u\log u$. For every $\alpha \leq 1$,

$$\frac{u \log u}{u^\alpha} \longrightarrow \infty \qquad (u \to +\infty),$$

so no admissible choice of $(C, a, \alpha, b, \delta, Q)$ reproduces it: the double factorial lies outside the exponential–power family, in the *factorial* class of theorem 1.8, with the parameters that theorem 1.24 records for this chapter. This is the structural difference from Parts 2, 4 and 5, whose sequences have $\log a_n = an^\alpha + b\log n + \cdots$ with $\alpha \leq 1$. What survives from the exponential–power theory is the *shape* of the reversion, not the model: theorem 1.29 covers both cores, and the resulting fixed-point map is eq. (3.37). Note also that the shift $x_* = -1$ must be applied *before* any further reduction: it is what removes the

$\frac{1}{2}\log x$ of eq. (3.30) and produces the sparse odd-power perturbation (theorem 3.12). None of S1, S2, S3 claims membership of the exponential–power family; all three invert $\frac{1}{2}u(\log u - 1)$ directly.

3.4 Borel structure of the centred correction

The exact Borel kernel is due to S2 and S3, independently and in the same form; S1 has only the formal Bernoulli series. The sharp first-omitted-term bound of theorem 3.17 is S2's; S3 proves the weaker bound with $\eta(2N+2)$ replaced by 1, and that weaker form is not used here.

Theorem 3.15 (Borel–Laplace representation)**.** *For* $\Re u > 0$,

$$\Omega(u) = \int_0^\infty \mathrm{e}^{-us}\,\widehat{\Omega}(s)\,\mathrm{d}s, \qquad \widehat{\Omega}(s) = \frac{1}{2s^2}\left(\frac{s}{\sinh s} - 1\right), \tag{3.15}$$

the singularity of $\widehat{\Omega}$ *at* $s = 0$ *being removable. Moreover*

$$\widehat{\Omega}(s) = \sum_{m\ge 1} \frac{(-1)^m}{s^2 + \pi^2 m^2}, \tag{3.16}$$

so the Borel singularities of Ω *are exactly the simple poles* $s = \pm \mathrm{i}\pi m$, $m \ge 1$, *with residues*

$$\mathrm{Res}_{s=\mathrm{i}\pi m}\,\widehat{\Omega}(s) = \frac{(-1)^{m+1}\mathrm{i}}{2\pi m}, \qquad \mathrm{Res}_{s=-\mathrm{i}\pi m}\,\widehat{\Omega}(s) = \frac{(-1)^m\mathrm{i}}{2\pi m}. \tag{3.17}$$

Proof. Put $t = u/2$ and $\widetilde{\Omega}(t) = \log\Gamma(t+\frac{1}{2}) - t\log t + t - \frac{1}{2}\log(2\pi)$, so that $\Omega(u) = \widetilde{\Omega}(u/2)$ by eq. (3.11). Binet's first integral and Frullani's formula give, for $t > 0$,

$$\psi\big(t+\tfrac{1}{2}\big) - \log t = \int_0^\infty \mathrm{e}^{-tr}\left(\frac{1}{r} - \frac{1}{2\sinh(r/2)}\right)\mathrm{d}r,$$

the integrand being $\mathcal{O}(r)$ at the origin and exponentially damped at infinity. On the other hand

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_0^\infty \mathrm{e}^{-tr}\,\frac{r/(2\sinh(r/2)) - 1}{r^2}\,\mathrm{d}r = \int_0^\infty \mathrm{e}^{-tr}\left(\frac{1}{r} - \frac{1}{2\sinh(r/2)}\right)\mathrm{d}r,$$

differentiation under the integral being justified by the same bounds. Since $\widetilde{\Omega}'(t) = \psi(t+\frac{1}{2}) - \log t$ and both $\widetilde{\Omega}(t)$ and the displayed integral tend to 0 as $t \to +\infty$, the two functions coincide. Substituting $r = 2s$ and $t = u/2$ turns the integral into eq. (3.15); analytic continuation in u extends it to $\Re u > 0$. The expansion $s/\sinh s = 1 - s^2/6 + \mathcal{O}(s^4)$ removes the origin.

For eq. (3.16), the elementary Mittag–Leffler expansion $\dfrac{s}{\sinh s} = 1 + 2s^2\sum_{m\ge1}\dfrac{(-1)^m}{s^2+\pi^2m^2}$ gives the claim after division by $2s^2$. The residues follow from $1/(s^2+\pi^2m^2) = \big[(s-\mathrm{i}\pi m)(s+\mathrm{i}\pi m)\big]^{-1}$. □

Theorem 3.16 (The centred logarithmic coefficients)**.** *As* $u \to \infty$ *in any closed subsector of* $|\arg u| < \pi/2$,

$$\Omega(u) \sim \sum_{k\ge1} \frac{c_k}{u^{2k-1}}, \tag{3.18}$$

where the following three expressions coincide:

$$\boxed{c_k = \frac{2^{2k-1}B_{2k}\big(\frac{1}{2}\big)}{2k(2k-1)} = \frac{\big(1-2^{2k-1}\big)B_{2k}}{2k(2k-1)} = (-1)^k\,\frac{(2k-2)!\,\eta(2k)}{\pi^{2k}}} \tag{3.19}$$

with $B_m(z)$ *the Bernoulli polynomials,* $B_m = B_m(0)$, *and* $\eta(s) = (1-2^{1-s})\zeta(s)$ *the Dirichlet eta function. There are no even inverse powers.*

Proof. Expand each summand of eq. (3.16) by

$$\frac{1}{s^2+A^2} = \sum_{j=0}^{N-1} \frac{(-1)^j s^{2j}}{A^{2j+2}} + \frac{(-1)^N s^{2N}}{A^{2N}(s^2+A^2)}, \qquad A>0, \tag{3.20}$$

with $A = \pi m$, sum over m, and integrate term by term against e^{-us}, using $\int_0^\infty \mathrm{e}^{-us} s^{2j} \mathrm{d}s = (2j)!\, u^{-2j-1}$. The coefficient of $u^{-(2j+1)}$ is

$$(2j)! \sum_{m\geq 1} \frac{(-1)^m(-1)^j}{(\pi m)^{2j+2}} = (-1)^{j+1} \frac{(2j)!\, \eta(2j+2)}{\pi^{2j+2}},$$

which is the third expression in eq. (3.19) with $k = j+1$. Euler's evaluation $\zeta(2k) = (-1)^{k+1}(2\pi)^{2k} B_{2k}/\big(2(2k)!\big)$ together with $\eta(2k) = (1-2^{1-2k})\zeta(2k)$ and $B_{2k}(\frac{1}{2}) = (2^{1-2k}-1)B_{2k}$ converts it into the first two. Only odd powers $u^{-(2k-1)}$ occur because $\widehat{\Omega}$ is even. □

Theorem 3.17 (Sharp alternating remainder). *For real $u > 0$ and $N \geq 0$ put $R_N(u) = \Omega(u) - \sum_{k=1}^N c_k u^{-(2k-1)}$ (empty sum for $N=0$). Then $R_N(u)$ has the sign of the first omitted term and is smaller than it in modulus:*

$$0 < (-1)^{N+1} R_N(u) < \frac{|c_{N+1}|}{u^{2N+1}} = \frac{(2N)!\, \eta(2N+2)}{\pi^{2N+2} u^{2N+1}}. \tag{3.21}$$

Proof. Insert eq. (3.20) with $A = \pi m$ into eq. (3.16) and Laplace-transform. The residual kernel is $(-1)^{N+1} s^{2N} S_N(s)$ with

$$S_N(s) = \sum_{m\geq 1} \frac{(-1)^{m-1}}{(\pi m)^{2N}\big((\pi m)^2 + s^2\big)}.$$

Writing $\big((\pi m)^2+s^2\big)^{-1} = \int_0^\infty \mathrm{e}^{-s^2 v} \mathrm{e}^{-\pi^2 m^2 v} \mathrm{d}v$ gives

$$S_N(s) = \int_0^\infty \mathrm{e}^{-s^2 v} \sum_{m\geq 1} \frac{(-1)^{m-1} \mathrm{e}^{-\pi^2 m^2 v}}{(\pi m)^{2N}} \mathrm{d}v.$$

For each $v \geq 0$ the inner series is alternating with strictly decreasing positive terms, hence strictly positive. Therefore $S_N(s) > 0$ and S_N is strictly decreasing in s^2, so $0 < S_N(s) \leq S_N(0) = \eta(2N+2)/\pi^{2N+2}$. Multiplying by s^{2N}, integrating against e^{-us} and using $\int_0^\infty \mathrm{e}^{-us} s^{2N} \mathrm{d}s = (2N)! u^{-(2N+1)}$ gives eq. (3.21). □

Corollary 3.18 (Certified forward logarithm). *For $x \to +\infty$, $u = x+1$ and either parity,*

$$\log \mathcal{D}_\delta(x) = \Phi(u) + A_\delta + \sum_{k=1}^{N} \frac{c_k}{u^{2k-1}} + \theta_N \frac{c_{N+1}}{u^{2N+1}} \qquad \textit{for some } \theta_N = \theta_N(u) \in (0,1). \tag{3.22}$$

Thus truncating after any number of terms overshoots or undershoots by strictly less than the first omitted term, with the sign of that term. For the integer sequence replace A_δ by $A(n)$ of eq. (3.13).

Proof. Combine theorems 3.11 and 3.17. □

Table 3.1: The centred logarithmic coefficients c_k of eq. (3.19) and the doubled coefficients $a_k = 2c_k$ used in the inverse problem. Regenerated in exact rational arithmetic from B_{2k}.

k	c_k	$a_k = 2c_k$
1	$-1/12$	$-1/6$

k	c_k	$a_k = 2c_k$
2	$7/360$	$7/180$
3	$-31/1260$	$-31/630$
4	$127/1680$	$127/840$
5	$-511/1188$	$-511/594$
6	$1414477/360360$	$1414477/180180$
7	$-8191/156$	$-8191/78$
8	$118518239/122400$	$118518239/61200$
9	$-5749691557/244188$	$-5749691557/122094$

Remark 3.19 (Late coefficients). From the third form in eq. (3.19) and $1 < \eta(2k) < 1 + 2^{-2k} \cdot \frac{4}{3}$,

$$c_k = (-1)^k \frac{(2k-2)!}{\pi^{2k}} \big(1 + \mathcal{O}(4^{-k})\big), \tag{3.23}$$

so eq. (3.18) is Gevrey-1 in the variable u^{-2}. The ratio of consecutive term moduli is $\sim 2k(2k-1)/(\pi u)^2$; the least term occurs near $2k = \pi u$ and has size $\asymp u^{-1/2} \mathrm{e}^{-\pi u}$, matching the distance π from the origin to the nearest Borel poles $\pm \mathrm{i}\pi$. This is the instance of theorem 1.46 relevant here; see section 3.10.

3.5 The forward transseries

3.5.1 Multiplicative, reciprocal, and arbitrary-power coefficients

Exponentiation of eq. (3.18) is an instance of the exponential formula, so theorem 1.10 does all the work at once for every power.

Definition 3.20. Embed the c_k sparsely on the odd indices,

$$\gamma_r = \begin{cases} c_{(r+1)/2}, & r \text{ odd}, \\ 0, & r \text{ even}, \end{cases} \qquad r \geq 1, \tag{3.24}$$

and for a scalar α define $s_m^{(\alpha)}$ by

$$\exp\left(\alpha \sum_{r \geq 1} \gamma_r \, u^{-r} \right) = \sum_{m \geq 0} \frac{s_m^{(\alpha)}}{u^m}, \qquad s_m := s_m^{(1)}. \tag{3.25}$$

Proposition 3.21 (All multiplicative coefficients at once). *For every α and every $m \geq 0$,*

$$s_m^{(\alpha)} = \frac{1}{m!} B_m\big(\alpha\, 1!\, \gamma_1, \alpha\, 2!\, \gamma_2, \ldots, \alpha\, m!\, \gamma_m\big) \tag{3.26}$$

$$= \sum_{\substack{k_1, k_2, \ldots \geq 0 \\ \sum_{r \geq 1} r k_r = m}} \prod_{r \geq 1} \frac{(\alpha \gamma_r)^{k_r}}{k_r!} = \sum_{\substack{k_1, k_2, \ldots \geq 0 \\ \sum_{j \geq 1} (2j-1) k_j = m}} \prod_{j \geq 1} \frac{(\alpha c_j)^{k_j}}{k_j!}, \tag{3.27}$$

and equivalently

$$s_0^{(\alpha)} = 1, \qquad m\, s_m^{(\alpha)} = \alpha \sum_{1 \leq j \leq (m+1)/2} (2j-1)\, c_j\, s_{m-2j+1}^{(\alpha)} \qquad (m \geq 1). \tag{3.28}$$

Consequently, for $x \to +\infty$, $u = x + 1$ and every fixed $M \geq 0$,

$$\mathcal{D}_\delta(x)^\alpha = \mathrm{e}^{\alpha A_\delta} \left(\frac{u}{\mathrm{e}} \right)^{\alpha u/2} \left(\sum_{m=0}^{M} \frac{s_m^{(\alpha)}}{u^m} + \mathcal{O}\Big(u^{-(M+1)}\Big) \right). \tag{3.29}$$

The cases $\alpha = 1$ and $\alpha = -1$ give $\mathcal{D}_\delta$ itself and its reciprocal.

Proof. Put $t = u^{-1}$ and $S_\alpha(t) = \exp(\alpha \sum_r \gamma_r t^r)$. Equation (3.26) is the complete-Bell generating identity of theorem 1.10; multinomial expansion of the exponential gives eq. (3.27), and the second form there uses $\gamma_{2j-1} = c_j$, $\gamma_{2j} = 0$. Logarithmic differentiation $tS'_\alpha = \alpha\big(\sum_r r\gamma_r t^r\big)S_\alpha$ and comparison of coefficients give eq. (3.28). For eq. (3.29), take N with $2N - 1 \geq M$; by theorems 3.11 and 3.17, $\alpha\Omega(u) = \alpha \sum_{k\leq N} c_k u^{1-2k} + \mathcal{O}(u^{-(2N+1)})$, and exp of a finite sum of $\mathcal{O}(u^{-1})$ terms plus a $\mathcal{O}(u^{-(2N+1)})$ error equals $\sum_{m\leq M} s_m^{(\alpha)} u^{-m} + \mathcal{O}(u^{-(M+1)})$, the exponential being Lipschitz on bounded sets. □

Corollary 3.22 (Reciprocal sign symmetry). $s_m^{(-\alpha)} = (-1)^m s_m^{(\alpha)}$ *for all* $m \geq 0$*; in particular the reciprocal coefficients are* $s_m^{(-1)} = (-1)^m s_m$.

Proof. Only odd r carry a nonzero γ_r, so $\sum_r \gamma_r(-t)^r = -\sum_r \gamma_r t^r$ and hence $S_\alpha(-t) = S_{-\alpha}(t)$. Reading off $[t^m]$ gives the claim. □

Table 3.2: Centred multiplicative coefficients $s_m = s_m^{(1)}$ of eq. (3.29) and the uncentred coefficients g_m of theorem 3.23. Regenerated from eq. (3.28) in exact rational arithmetic.

m	s_m (centred)	g_m (uncentred)
0	1	1
1	$-1/12$	$1/6$
2	$1/288$	$1/72$
3	$1003/51840$	$-139/6480$
4	$-4027/2488320$	$-571/155520$
5	$-5128423/209018880$	$163879/6531840$
6	$168359651/75246796800$	$5246819/1175731200$
7	$68168266699/902961561600$	$-534703531/7054387200$
8	$-587283555451/86684309913600$	$-4483131259/338610585600$

3.5.2 The uncentred normal form and the centre-shift identity

Proposition 3.23 (Uncentred expansion). *As* $x \to +\infty$,

$$\log \mathcal{D}_\delta(x) \sim \frac{x}{2}(\log x - 1) + \frac{1}{2}\log x + A_\delta + \sum_{k\geq 1} \frac{b_k}{x^{2k-1}}, \qquad b_k = \frac{2^{2k-1}B_{2k}}{2k(2k-1)} = (-1)^{k-1}\frac{(2k-2)!\,\zeta(2k)}{\pi^{2k}}, \tag{3.30}$$

with $b_1 = \frac{1}{6}, b_2 = -\frac{1}{45}, b_3 = \frac{8}{315}, b_4 = -\frac{8}{105}, b_5 = \frac{128}{297}$*. Exponentiating,* $\mathcal{D}_\delta(x) \sim \mathrm{e}^{A_\delta}\sqrt{x}\,(x/\mathrm{e})^{x/2} \sum_m g_m x^{-m}$*, where* g_m *is obtained from eq.* (3.28) *on replacing* c_j *by* b_j*; this is the familiar form, with amplitudes* $\sqrt{\pi}$ *(even) and* $\sqrt{2}$ *(odd). The exact Borel kernel of the uncentred correction is* $\widehat{\mathcal{U}}(s) = \frac{\coth s}{2s} - \frac{1}{2s^2} = \sum_{m\geq 1}\big(s^2 + \pi^2 m^2\big)^{-1}$*, with the* same *pole lattice* $\pm\mathrm{i}\pi m$ *but all residues of one sign pattern.*

Proof. Apply the standard Stirling expansion to $\log\Gamma(z+1)$ with $z = x/2$ and add $(x/2)\log 2 + \log\kappa_\delta$; the elementary terms give the first three summands of eq. (3.30), and the corrections scale as $z^{-(2k-1)} = 2^{2k-1}x^{-(2k-1)}$. Euler's evaluation of $\zeta(2k)$ gives the second form of b_k. For the kernel, the uncentred correction is $\mathcal{U}(x) = \Omega_\Gamma(x/2)$ with Ω_Γ Binet's function; substituting $t = 2s$ in $\Omega_\Gamma(z) = \int_0^\infty \mathrm{e}^{-zt}\big(\frac{\coth(t/2)}{2t} - \frac{1}{t^2}\big)\mathrm{d}t$ gives the stated kernel, and $\frac{\coth s}{s} = \frac{1}{s^2} + 2\sum_{m\geq 1}(s^2 + \pi^2 m^2)^{-1}$ gives the partial fractions. □

Warning 3.24. The two statements $n!! \sim \mathrm{e}^{A(n)}\sqrt{n}\,(n/\mathrm{e})^{n/2}$ and $n!! \sim \mathrm{e}^{A(n)}\big((n+1)/\mathrm{e}\big)^{(n+1)/2}$ are both correct and are not in conflict: they use different normal forms. The first leaves a factor $\sqrt{n}$ and an *uncentred* correction series; the second absorbs that factor into the phase. Only the centred form leads to the sparse inverse grid, and it is the one used from section 3.6 onwards.

Proposition 3.25 (Centre-shift identity). *For every $r \geq 1$,*

$$\frac{(-1)^{r+1}}{2r(r+1)} + (-1)^{r+1} \sum_{k=1}^{\lfloor (r+1)/2 \rfloor} \binom{r-1}{2k-2} c_k = \begin{cases} b_{(r+1)/2}, & r \text{ odd}, \\ 0, & r \text{ even}. \end{cases} \tag{3.31}$$

Proof. Substitute $u = x + 1 = x(1+w)$, $w = x^{-1}$, into $\Phi(u) + \sum_k c_k u^{1-2k}$ and compare with eq. (3.30). First, with $\log u = \log x + \log(1+w)$,

$$\Phi(u) = \frac{x}{2}(\log x - 1) + \frac{1}{2}(\log x - 1) + \frac{x+1}{2}\log(1+w).$$

Expanding $\log(1+w) = \sum_{j\geq 1}(-1)^{j-1}w^j/j$ and splitting the prefactor $\frac{x+1}{2} = \frac{x}{2} + \frac{1}{2}$ gives

$$\frac{x+1}{2}\log(1+w) = \frac{1}{2} + \frac{1}{2}\sum_{r\geq 1}(-1)^r x^{-r}\left(\frac{1}{r+1} - \frac{1}{r}\right) = \frac{1}{2} + \sum_{r\geq 1}\frac{(-1)^{r+1}}{2r(r+1)}x^{-r},$$

so that $\Phi(u) = \frac{x}{2}(\log x - 1) + \frac{1}{2}\log x + \sum_{r\geq 1}(-1)^{r+1}\big(2r(r+1)\big)^{-1}x^{-r}$; the two halves $\pm\frac{1}{2}$ cancel. Second,

$$\sum_{k\geq 1}\frac{c_k}{u^{2k-1}} = \sum_{k\geq 1} c_k\, x^{-(2k-1)}(1+w)^{-(2k-1)} = \sum_{r\geq 1} x^{-r} \sum_{k\leq (r+1)/2} c_k \binom{-(2k-1)}{r-2k+1},$$

and $\binom{-(2k-1)}{r-2k+1} = (-1)^{r-2k+1}\binom{r-1}{r-2k+1} = (-1)^{r+1}\binom{r-1}{2k-2}$. Adding the two coefficient lists and matching against eq. (3.30) — where the coefficient of x^{-r} is $b_{(r+1)/2}$ for odd r and 0 for even r — gives eq. (3.31). ☐

Remark 3.26. Equation (3.31) was verified symbolically for $1 \leq r \leq 9$: the even cases return 0 and the odd cases return $\frac{1}{6}, -\frac{1}{45}, \frac{8}{315}, -\frac{8}{105}, \frac{128}{297}$. It is the cheapest available certificate that the centring is consistent, because it fails immediately under any sign or index slip in eq. (3.19). S3 states the identity without proof; the proof above is supplied here.

3.6 The branchwise functional inverse

All three siblings reach the same objects in this section. The presentation follows S3 for the exact normal form and S2 for the graded closed formula and the sign law; S1's "scaling from the half-shifted inverse Gamma problem" is subsumed by theorem 3.61. The proofs of theorems 3.35 and 3.37 are new: the corresponding source arguments are sketches (see the chapter report).

3.6.1 The Lambert core

Lemma 3.27 (Exact inversion of the dominant phase). *Let $R > 0$. The equation $\Phi(T) = \frac{1}{2}T(\log T - 1) = R$ has a unique solution $T > \mathrm{e}$, namely*

$$w = W_0\left(\frac{2R}{\mathrm{e}}\right), \qquad T = \frac{2R}{w} = \mathrm{e}^{w+1}, \qquad \Lambda := w + 1 = \log T, \tag{3.32}$$

and then

$$\Phi'(T) = \frac{\Lambda}{2}, \qquad \frac{\mathrm{d}T}{\mathrm{d}R} = \frac{2}{\Lambda}, \qquad T(\Lambda - 1) = 2R. \tag{3.33}$$

Proof. This is theorem 1.23 for the parameters of theorem 1.24; for completeness, setting $w = \log T - 1$ turns $\Phi(T) = R$ into $w\mathrm{e}^{w+1} = 2R$, i.e. $w\mathrm{e}^w = 2R/\mathrm{e} > 0$, whose unique real solution is $w = W_0(2R/\mathrm{e}) > 0$, the principal branch being the only real branch on $(0, \infty)$. Φ is strictly increasing on $(1, \infty)$ with range $(-\frac{1}{2}, \infty)$, which gives uniqueness of T. The identities in eq. (3.33) follow from $\Phi'(T) = \frac{1}{2}\log T$, implicit differentiation of $\Phi(T) = R$, and $T = 2R/w = 2R/(\Lambda - 1)$. ☐

For the parity branch δ one takes

$$R_\delta = \log y - A_\delta, \qquad w_\delta,\ T_\delta,\ \Lambda_\delta \text{ as in eq. (3.32).} \tag{3.34}$$

The subscript δ is suppressed inside coefficient calculations, where it never appears.

3.6.2 The normalised displacement equation

Lemma 3.28 (The factorial-core reversion kernel)**.** *Write $u = T(1+E)$, $q = T^{-1}$, $z = q^2$, and*

$$\varphi(v) = (1+v)\log(1+v) - v = \sum_{m\ge 2} \frac{(-1)^m}{m(m-1)} v^m, \qquad a_k = 2c_k. \tag{3.35}$$

Then the equation $\Phi(u) + \Omega(u) = R$, with Ω replaced by its asymptotic series, is equivalent to the formal normal form

$$\boxed{\Lambda E + \varphi(E) + \sum_{k\ge 1} a_k\, z^k (1+E)^{-(2k-1)} = 0.} \tag{3.36}$$

Equivalently, in absolute displacement $U = TE$ with $t = q$,

$$U = \Psi(t, U), \qquad \Psi(t,v) = -\frac{1}{\Lambda}\left[\frac{\varphi(tv)}{t} + \sum_{k\ge 1} a_k\, t^{2k-1}(1+tv)^{-(2k-1)}\right] \in t\,\mathbb{Q}(\Lambda)[[t,v]], \tag{3.37}$$

which is the specialisation of theorem 1.29 to the factorial core, so that theorem 1.16 applies verbatim. There is a unique formal solution, and it has the form

$$E = \sum_{n\ge 1} e_n(\Lambda)\, z^n, \qquad \textit{hence} \qquad u - T = \sum_{n\ge 1} \frac{e_n(\Lambda)}{T^{2n-1}}: \tag{3.38}$$

only odd *inverse powers of the Lambert core occur.*

Proof. Because $\Phi(T) = R$,

$$\Phi\big(T(1+E)\big) - \Phi(T) = \frac{T(1+E)}{2}\big(\log T + \log(1+E) - 1\big) - \frac{T}{2}(\log T - 1) = \frac{T}{2}\big[\Lambda E + \varphi(E)\big],$$

using $\Lambda = \log T$ and $(1+E)\log(1+E) = \varphi(E) + E$. Also $\Omega(u) = \sum_k c_k T^{-(2k-1)}(1+E)^{-(2k-1)}$ formally. Multiplying the whole equation by $2/T = 2q$ and using $a_k = 2c_k$ and $q \cdot q^{2k-1} = z^k$ gives eq. (3.36). Dividing by T and solving for $U = TE$ gives eq. (3.37); the bracket lies in $t\,\mathbb{Q}[[t,v]]$ because $\varphi(tv) = t^2v^2/2 + \mathcal{O}(t^3)$, so $\varphi(tv)/t = \mathcal{O}(t)$, and every Bernoulli term carries t^{2k-1} with $k \ge 1$. Existence and uniqueness of the fixed point are theorem 1.16. Finally, if E contains only even powers of q then so does every term of eq. (3.36); an induction on the q-valuation therefore forces $E \in z\,\mathbb{Q}(\Lambda)[[z]]$. ☐

3.6.3 The all-orders inverse and its remainder

Theorem 3.29 (Complete branchwise inverse transseries)**.** *Fix $\delta \in \{0,1\}$ and let $R_\delta, T_\delta, \Lambda_\delta$ be as in eq. (3.34). Write $d_{2n-1} := e_n$ for the coefficients of eq. (3.38). Then for every fixed $N \ge 0$, as $y \to +\infty$,*

$$\boxed{\mathcal{D}_\delta^{-1}(y) = T_\delta - 1 + \sum_{n=1}^{N} \frac{d_{2n-1}(\Lambda_\delta)}{T_\delta^{2n-1}} + \mathcal{O}\left(\frac{1}{T_\delta^{2N+1}\Lambda_\delta}\right),} \tag{3.39}$$

with an implied constant that may be chosen uniformly in $\delta \in \{0,1\}$. All parity dependence sits in $R_\delta = \log y - A_\delta$; the coefficient functions d_{2n-1} are universal.

Proof. Let $u_N = T + \sum_{n=1}^{N} e_n(\Lambda)T^{1-2n}$ and $E_N = (u_N - T)/T$. By construction, substituting E_N into eq. (3.36) annihilates the coefficients of $z^1, \ldots, z^N$, so the formal residual is $\mathcal{O}(z^{N+1})$ with coefficients rational in Λ and bounded as $\Lambda \to \infty$. Undoing the normalisation (multiplying by $T/2$) therefore bounds the exact forward residual

$$\rho_N := \Phi(u_N) + A_\delta + \Omega(u_N) - \log y$$

by $\mathcal{O}\big(T \cdot z^{N+1}\big) = \mathcal{O}\big(T^{-(2N+1)}\big)$, where replacing Ω by its N-term truncation costs at most $|c_{N+1}|u_N^{-(2N+1)}$ by theorem 3.17, of the same order. This is the hypothesis of theorem 1.42. For the slope, on the interval between u_N and the exact root,

$$\frac{\mathrm{d}}{\mathrm{d}u}\big[\Phi(u) + \Omega(u)\big] = \frac{1}{2}\log u + \Omega'(u) = \frac{\Lambda}{2}\big(1 + o(1)\big),$$

because $u = T\big(1+\mathcal{O}(T^{-2})\big)$ and, differentiating eq. (3.15) under the integral sign, $\Omega'(u) = -\int_0^\infty s\mathrm{e}^{-us}\widehat{\Omega}(s)\mathrm{d}s = \mathcal{O}(u^{-2})$. The mean-value theorem now divides ρ_N by a quantity asymptotic to $\Lambda/2$, which is the instance of theorem 1.44 for the factorial core and gives eq. (3.39). The parity constant A_δ disappears on differentiation, so the estimate is uniform in δ. ☐

3.6.4 Two generators for the coefficients

Equation (3.36) is exactly eq. (1.55), the reversion problem of theorem 1.29, with the instance data

$$h(v) = \varphi(v), \quad h_m = \frac{(-1)^m}{m(m-1)}, \qquad f(z,v) = \sum_{k\geq 1} \phi_k(z)(1+v)^{-\mu_k}, \quad \phi_k(z) = a_k z^k, \quad \mu_k = 2k-1. \tag{3.40}$$

Its triangular recurrence eq. (1.56) and its Bell form eq. (1.58) therefore apply verbatim; written out, the former is

$$e_n(\Lambda) = -\frac{1}{\Lambda}\,[z^n]\left\{\varphi(E_{n-1}) + \sum_{k=1}^{n} a_k\, z^k\big(1 + E_{n-1}\big)^{-(2k-1)}\right\}, \qquad E_{n-1} = \sum_{j<n} e_j z^j. \tag{3.41}$$

Only the coefficient *structure* is subject-specific.

Theorem 3.30 (Structure of the inverse blocks)**.** *For every $n \geq 1$,*

$$e_n(\Lambda) \in \Lambda^{-1}\mathbb{Q}[\Lambda^{-1}], \qquad \deg_{\Lambda^{-1}} e_n \leq 2n-1, \qquad e_n(\Lambda) = -\frac{a_n}{\Lambda} + \mathcal{O}(\Lambda^{-2}). \tag{3.42}$$

Proof. Specialising eq. (1.58) to eq. (3.40) gives

$$e_n = -\frac{1}{\Lambda}\left[\sum_{m=2}^{n} \frac{(-1)^m}{m(m-1)}\,\widehat{B}_{n,m}(e) + \sum_{k=1}^{n} a_k \sum_{m=0}^{n-k} (-1)^m \binom{2k+m-2}{m}\widehat{B}_{n-k,m}(e)\right], \tag{3.43}$$

in which every occurring e_j has $j < n$. A monomial of $\widehat{B}_{r,m}$ is a product $e_{j_1}\cdots e_{j_m}$ with $j_1+\cdots+j_m = r$; by induction its Λ^{-1}-degree lies between m and $\sum_\nu(2j_\nu - 1) = 2r - m$, and the outer factor Λ^{-1} raises this by one. In the first sum $r = n$ and $m \geq 2$, giving degree at most $2n-1$; in the second $r = n-k$, giving at most $2(n-k) - m + 1 \leq 2n-1$. Rationality is preserved throughout. Finally the only contribution of Λ^{-1}-degree exactly 1 is the term $k = n$, $m = 0$, namely $-a_n/\Lambda$, because every Bell monomial contains at least one $e_j = \mathcal{O}(\Lambda^{-1})$. ☐

Theorem 3.31 (Graded closed formula)**.** *Let $A(z) = \sum_{k\geq 1} a_k z^k$ and put*

$$A_{n,p} := [z^n]\,A(z)^p = \widehat{B}_{n,p}(a), \qquad K_{n;j,p} := [v^{j-1}]\,\varphi(v)^{\,j-p}(1+v)^{-(2n-p)}. \tag{3.44}$$

Expand $d_{2n-1}(\Lambda) = \sum_{j\ge 1} c_{n,j}\Lambda^{-j}$. *Then*

$$\boxed{c_{n,j} = \frac{(-1)^j}{j}\sum_{p=\lceil (j+1)/2\rceil}^{\min(j,n)} \binom{j}{p} A_{n,p}\, K_{n;j,p}} \qquad (1 \le j \le 2n-1), \tag{3.45}$$

and $c_{n,j} = 0$ *for* $j > 2n-1$. *The right-hand side is a finite expression in Bernoulli numbers, binomial coefficients and ordinary partial Bell polynomials; it computes any single scalar coefficient without generating the lower blocks.*

Proof. Put $H(z,v) = \varphi(v) + \sum_{k\ge 1} a_k z^k (1+v)^{-(2k-1)}$, so that eq. (3.36) reads $E = -\Lambda^{-1}H(z,E)$. The closed formula eq. (1.57) of theorem 1.29 gives

$$e_n = \sum_{j\ge 1}\frac{1}{j}\,[z^n v^{j-1}]\left(-\frac{H(z,v)}{\Lambda}\right)^j = \sum_{j\ge 1}\frac{(-1)^j}{j\Lambda^j}\,[z^n v^{j-1}]\,H(z,v)^j,$$

which already exhibits the grading by powers of Λ^{-1}. It remains to evaluate $[z^n v^{j-1}]\,H^j$. Expand H^j binomially, taking p of the j factors from the Bernoulli sector:

$$H^j = \sum_{p=0}^{j}\binom{j}{p}\varphi(v)^{j-p}\left(\sum_{k\ge 1} a_k z^k (1+v)^{-(2k-1)}\right)^p.$$

Extracting z^n from the p-th power, the surviving terms have $k_1 + \cdots + k_p = n$; their coefficients sum to $A_{n,p}$ and they share the factor $(1+v)^{-\sum_i (2k_i-1)} = (1+v)^{-(2n-p)}$. Hence $[z^n]\,H^j = \sum_p \binom{j}{p} A_{n,p}\varphi(v)^{j-p}(1+v)^{-(2n-p)}$, and extracting v^{j-1} gives eq. (3.45). (Equation (1.57) caps the outer sum at $j \le 2n-1$; the support argument below recovers that bound independently.)

For the summation range: $A_{n,p} = 0$ unless $1 \le p \le n$, while $\varphi(v)^{j-p}$ has v-valuation $2(j-p)$, so a nonzero v^{j-1}-coefficient forces $2(j-p) \le j-1$, i.e. $p \ge \lceil (j+1)/2\rceil$. Combining, $j \le 2p-1 \le 2n-1$; this re-proves the degree bound of eq. (3.42) and shows the expansion terminates. □

Remark 3.32. Equation (3.45) was checked against eq. (3.41) for all $1 \le n \le 5$ and all $1 \le j \le 2n-1$ in exact rational arithmetic: all 25 scalars agree. For explicit evaluation, $K_{n;j,p}$ unfolds by theorems 1.9 and 1.12 into

$$K_{n;j,p} = \sum_{h=2(j-p)}^{j-1}\widehat{B}_{h,\,j-p}(\nu)\,(-1)^{j-1-h}\binom{2n-p+j-2-h}{j-1-h}, \qquad \nu_1 = 0, \quad \nu_m = \frac{(-1)^m}{m(m-1)}\ (m\ge 2),$$

with the conventions $\widehat{B}_{0,0} = 1$ and $\widehat{B}_{h,0} = 0$ for $h > 0$.

Proposition 3.33 (A second, ungraded closed formula)**.** *Let* $S = A^{\langle -1\rangle}$ *be the formal compositional inverse of* $A(z) = \sum_k a_k z^k$, *which exists because* $a_1 = -\frac{1}{6} \neq 0$, *with*

$$[w^m]\,S(w) = \frac{1}{m}\,[t^{m-1}]\left(\frac{t}{A(t)}\right)^m. \tag{3.46}$$

Put

$$G_\Lambda(E) = (1+E)^2\,S\!\left(-\frac{\Lambda E + \varphi(E)}{1+E}\right) \in E\,\mathbb{Q}(\Lambda)[[E]], \qquad G_\Lambda'(0) = -\frac{\Lambda}{a_1} = 6\Lambda \neq 0. \tag{3.47}$$

Then

$$e_n(\Lambda) = \frac{1}{n}\,[E^{n-1}]\left(\frac{E}{G_\Lambda(E)}\right)^n. \tag{3.48}$$

Proof. Since $\sum_k a_k z^k(1+E)^{-(2k-1)} = (1+E)\,A\big(z/(1+E)^2\big)$, eq. (3.36) reads $A\big(z/(1+E)^2\big) = -\big(\Lambda E + \varphi(E)\big)/(1+E)$. Applying S and multiplying by $(1+E)^2$ gives $z = G_\Lambda(E)$; differentiating at $E=0$, where $\varphi(0)=\varphi'(0)=0$, gives $G'_\Lambda(0) = S'(0)\cdot(-\Lambda) = -\Lambda/a_1$. Equations (3.46) and (3.48) are then two applications of theorem 1.14. Although A is a divergent Gevrey series, every finite jet uses only finitely many a_k, so this is a formal identity requiring no convergence hypothesis. □

Remark 3.34. Equation (3.48) was verified against eq. (3.41) for $n \le 4$. It is less useful than eq. (3.45) in practice: it does not separate the powers of Λ^{-1}, and it requires reverting A first. Its interest is conceptual — it exhibits the inverse blocks as a *double* Lagrange–Bürmann extraction, the inner one reverting the pure Bernoulli series.

3.6.5 Boundary coefficients and the global sign law

Corollary 3.35 (The two edges of the coefficient triangle)**.** *For every $n \ge 1$,*

$$c_{n,1} = -a_n = -2c_n, \qquad c_{n,2n-1} = \frac{1}{3^n}\binom{1/2}{n} = -\frac{1}{2^{n-1}(2n-1)}\binom{2n-1}{n}a_1^n. \tag{3.49}$$

In particular the two closed forms recorded separately by S1 and S2 for the deepest coefficient are the same number.

Proof. For $j=1$ the range in eq. (3.45) forces $p=1$; then $A_{n,1}=a_n$, $K_{n;1,1} = [v^0]\,(1+v)^{-(2n-1)} = 1$, and the prefactor is -1.

For $j = 2n-1$, the range forces $p=n$; then $A_{n,n} = a_1^n$ and

$$K_{n;2n-1,n} = [v^{2n-2}]\,\varphi(v)^{\,n-1}(1+v)^{-n} = [v^{2n-2}]\,\left(\frac{v^2}{2}\right)^{n-1} = 2^{1-n},$$

because $\varphi(v)^{n-1}$ has valuation $2n-2$. With $\frac{(-1)^{2n-1}}{2n-1}\binom{2n-1}{n}$ as prefactor this is the second expression in eq. (3.49).

Equality of the two expressions is elementary. Using $\binom{1/2}{n} = (-1)^{n-1}\binom{2n}{n}\big(4^n(2n-1)\big)^{-1}$,

$$\frac{1}{3^n}\binom{1/2}{n} = (-1)^{n-1}\frac{\binom{2n}{n}}{12^n(2n-1)},$$

while $a_1 = -\frac{1}{6}$ and $\binom{2n-1}{n} = \frac12\binom{2n}{n}$ give

$$-\frac{\binom{2n-1}{n}a_1^n}{2^{n-1}(2n-1)} = (-1)^{n+1}\frac{\binom{2n}{n}}{2\cdot 2^{n-1}6^n(2n-1)} = (-1)^{n+1}\frac{\binom{2n}{n}}{12^n(2n-1)}.$$

□

Remark 3.36 (The deepest edge as a square root). Let $t_n = c_{n,2n-1}$. Only the quadratic part of φ and the single coefficient a_1 can reach $\Lambda^{-(2n-1)}$: in eq. (3.43) the degree bound $2r-m+1$ attains $2n-1$ only for $r=n$, $m=2$ in the first sum, since in the second sum $2(n-k)-m+1 = 2n-1$ forces $k=1$, $m=0$, where $\widehat{B}_{n-1,0}=0$ for $n\ge 2$. Hence $t_1 = -a_1 = \frac16$ and $t_n = -\frac12\sum_{j=1}^{n-1} t_j t_{n-j}$ for $n \ge 2$, i.e. $T(z) = \sum_n t_n z^n$ satisfies $T + \frac12 T^2 = \frac16 z$, whence $T(z) = -1+\sqrt{1+z/3}$ and $t_n = 3^{-n}\binom{1/2}{n}$. Equivalently, the maximal-pole part of eq. (3.36) is $\Lambda E + \frac12 E^2 - \frac16 z = 0$ with root $E = -\Lambda + \sqrt{\Lambda^2 + z/3}$. S1 asserts this without proof; the degree bookkeeping above supplies it.

Theorem 3.37 (Coefficientwise sign law)**.** *For every $n \ge 1$,*

$$(-1)^{n+1} d_{2n-1}(\Lambda) \in \Lambda^{-1}\mathbb{Q}_{>0}[\Lambda^{-1}], \qquad \deg_{\Lambda^{-1}} d_{2n-1} = 2n-1 \textit{ exactly.} \tag{3.50}$$

That is, $c_{n,j} \ne 0$ with sign $(-1)^{n+1}$ for every $1 \le j \le 2n-1$, and $c_{n,j} = 0$ beyond.

Proof. Write $a_k = (-1)^k a_k^+$ with $a_k^+ = 2(2k-2)!\,\eta(2k)\pi^{-2k} > 0$ by eq. (3.19). Substituting $z = -\zeta$ and $E = -P$ into eq. (3.36) and using $\varphi(-P) = \sum_{m\geq 2} P^m/\big(m(m-1)\big)$ turns it into

$$\Lambda P = \sum_{m\geq 2} \frac{P^m}{m(m-1)} + \sum_{k\geq 1} a_k^+\,\zeta^k(1-P)^{-(2k-1)}. \tag{3.51}$$

Every coefficient on the right is *strictly positive*: $1/(m(m-1)) > 0$, $a_k^+ > 0$, and $(1-P)^{-(2k-1)} = \sum_{m\geq 0}\binom{2k+m-2}{m}P^m$ has positive coefficients. Writing $P = \sum_n p_n(\Lambda)\zeta^n$, the triangular recurrence obtained from eq. (3.51) expresses p_n as Λ^{-1} times a positive-coefficient polynomial in $p_1, \dots, p_{n-1}$; by induction each $p_n \in \Lambda^{-1}\mathbb{Q}_{\geq 0}[\Lambda^{-1}]$, with no cancellation possible anywhere. Since $E(z) = -P(-z)$, we get $d_{2n-1} = e_n = (-1)^{n+1}p_n$, which gives the sign statement once we show that p_n has a *strictly* positive coefficient at every Λ^{-j} with $1 \leq j \leq 2n-1$.

We prove that by induction, exhibiting one contributing configuration for each j; because all contributions are nonnegative, one suffices. For $n = 1$, $p_1 = a_1^+/\Lambda$, and $2n-1 = 1$. Let $n \geq 2$ and assume the claim for all smaller indices.

- $j = 1$: take the forcing term $k = n$, $m = 0$ in eq. (3.51), contributing $a_n^+\Lambda^{-1}$.
- $2 \leq j \leq 2n-2$: take the forcing term $k = 1$ together with the single factor $m = 1$ of $(1-P)^{-1}$, so that the contribution is $\Lambda^{-1}a_1^+\binom{1}{1}$ times a coefficient of p_{n-1}. By induction p_{n-1} has strictly positive coefficients at $\Lambda^{-j'}$ for every $1 \leq j' \leq 2n-3$, and $j-1$ runs over exactly that range.
- $j = 2n-1$: take the nonlinear term $m = 2$, i.e. a product $p_{n_1}p_{n_2}$ with $n_1 + n_2 = n$, $n_i \geq 1$, at their top degrees $2n_i - 1$; the resulting Λ^{-1}-degree is $1 + (2n_1 - 1) + (2n_2 - 1) = 2n - 1$.

Hence every $c_{n,j}$, $1 \leq j \leq 2n-1$, is nonzero of sign $(-1)^{n+1}$; theorem 3.31 already showed $c_{n,j} = 0$ for $j > 2n-1$. ☐

Remark 3.38. The sign law was confirmed symbolically for $n \leq 7$: after multiplication by $(-1)^{n+1}$ all $1, 3, 5, 7, 9, 11, 13$ coefficients of the respective blocks are positive rationals, and the degree is exactly $2n-1$ in each case. S2 states the theorem but its proof gestures at "a positive rooted-tree contribution" without exhibiting one; the attainability induction above replaces that step. The law is also the cheapest runtime invariant for a symbolic implementation (theorem 3.64).

3.6.6 The explicit blocks

Repeated use of eq. (3.41) gives, in exact rational arithmetic,

$$d_1(\Lambda) = \frac{1}{6\Lambda}, \tag{3.52}$$

$$d_3(\Lambda) = -\frac{14\Lambda^2 + 10\Lambda + 5}{360\Lambda^3}, \tag{3.53}$$

$$d_5(\Lambda) = \frac{2232\Lambda^4 + 1176\Lambda^3 + 714\Lambda^2 + 350\Lambda + 105}{45360\Lambda^5}, \tag{3.54}$$

$$\begin{aligned} d_7(\Lambda) = -\frac{1}{5443200\Lambda^7}\big(&822960\Lambda^6 + 292536\Lambda^5 + 136956\Lambda^4 \\ &+ 68040\Lambda^3 + 32970\Lambda^2 + 12250\Lambda + 2625\big), \end{aligned} \tag{3.55}$$

$$\begin{aligned} d_9(\Lambda) = \frac{1}{359251200\Lambda^9}\big(&309052800\Lambda^8 + 77920128\Lambda^7 + 26024592\Lambda^6 \\ &+ 10019592\Lambda^5 + 4516028\Lambda^4 + 2023560\Lambda^3 \\ &+ 812350\Lambda^2 + 242550\Lambda + 40425\big), \end{aligned} \tag{3.56}$$

$$d_{11}(\Lambda) = -\frac{1}{5884534656000\Lambda^{11}}\big(46195687238400\Lambda^{10} + 8854370366400\Lambda^9$$

$$
\begin{aligned}
&+ 2171643318000\Lambda^8 + 605515022112\Lambda^7 + 212869408752\Lambda^6 \\
&+ 84136852800\Lambda^5 + 35589153600\Lambda^4 + 14234019800\Lambda^3 \\
&+ 4868113250\Lambda^2 + 1213962750\Lambda + 165540375\big).
\end{aligned} \tag{3.57}
$$

The rapidly growing numerators are an argument for regenerating blocks from eq. (3.41) rather than storing a table; d_{13}, which S3 also displays, is omitted here for that reason. The alternating block signs and the coefficientwise positivity after multiplication by $(-1)^{n+1}$ illustrate theorem 3.37; the leading and trailing coefficients illustrate theorem 3.35. In separated form,

$$
d_1 = \frac{1}{6\Lambda}, \qquad d_3 = -\frac{7}{180\Lambda} - \frac{1}{36\Lambda^2} - \frac{1}{72\Lambda^3},
$$

$$
d_5 = \frac{31}{630\Lambda} + \frac{7}{270\Lambda^2} + \frac{17}{1080\Lambda^3} + \frac{5}{648\Lambda^4} + \frac{1}{432\Lambda^5}, \tag{3.58}
$$

$$
d_7 = -\frac{127}{840\Lambda} - \frac{4063}{75600\Lambda^2} - \frac{11413}{453600\Lambda^3} - \frac{1}{80\Lambda^4} - \frac{157}{25920\Lambda^5} - \frac{35}{15552\Lambda^6} - \frac{5}{10368\Lambda^7}. \tag{3.59}
$$

The common-denominator forms are less prone to transcription error; the separated form makes $c_{n,1} = -a_n$ visible. Through three blocks,

$$
\begin{aligned}
\mathcal{D}_\delta^{-1}(y) = T - 1 + \frac{1}{6T\Lambda} - \frac{14\Lambda^2 + 10\Lambda + 5}{360\,T^3\Lambda^3} \\
+ \frac{2232\Lambda^4 + 1176\Lambda^3 + 714\Lambda^2 + 350\Lambda + 105}{45360\,T^5\Lambda^5} + \mathcal{O}\left(\frac{1}{T^7\Lambda}\right).
\end{aligned} \tag{3.60}
$$

Remark 3.39 (Lambert-only form, and a correction to the usual first guess). Because $T = 2R_\delta/w$ and $\Lambda = w + 1$, the expansion can be written without T. The first three terms are

$$
\begin{aligned}
\mathcal{D}_\delta^{-1}(y) \sim \frac{2R_\delta}{w} - 1 + \frac{w}{12R_\delta(w+1)} - \frac{w^3\big(14(w+1)^2 + 10(w+1) + 5\big)}{2880\,R_\delta^3(w+1)^3} \\
+ \frac{w^5\big(2232(w+1)^4 + 1176(w+1)^3 + 714(w+1)^2 + 350(w+1) + 105\big)}{1451520\,R_\delta^5(w+1)^5} + \cdots,
\end{aligned} \tag{3.61}
$$

with $w = W_0(2R_\delta/\mathrm{e})$. A frequently used rough inversion is $x \approx 2R_\delta/W_0(2R_\delta/\mathrm{e}) - 1$; its leading correction is *not* of order T^{-1} alone but

$$
\frac{1}{6\Lambda T} = \frac{w}{12R_\delta(w+1)} = \frac{1}{12R_\delta}\left(1 - \frac{1}{w+1}\right) \sim \frac{1}{12\log y}.
$$

The factor Λ^{-1} is forced by the slope $\Phi'(T) = \Lambda/2$ of eq. (3.33).

3.7 The outer iterated-logarithmic layer

Keeping W_0 unexpanded is both more accurate and algebraically cleaner. To place the result inside a conventional polynomial–logarithmic transseries one applies theorem 1.33. Set

$$
Z = \frac{2R_\delta}{\mathrm{e}}, \qquad L_1 = \log Z, \qquad L_2 = \log L_1. \tag{3.62}
$$

Proposition 3.40 (Outer data). *With $\left[{n \atop k}\right]$ the unsigned Stirling numbers of the first kind, theorem 1.33 gives, for $Z \to +\infty$,*

$$
W_0(Z) \sim L_1 - L_2 + \sum_{r \ge 1} \frac{P_r(L_2)}{L_1^r}, \qquad P_r(t) = \sum_{m=1}^{r} (-1)^{r-m} \left[{r \atop r-m+1}\right] \frac{t^m}{m!}. \tag{3.63}
$$

Write $W_0(Z)/L_1 = 1 + \sum_{j\ge1} \upsilon_j(L_2)L_1^{-j}$ *with* $\upsilon_1 = -L_2$ *and* $\upsilon_j = P_{j-1}$ *for* $j \ge 2$, *and define the reciprocal polynomials*

$$\tau_0 = 1, \qquad \tau_m = -\sum_{j=1}^{m} \upsilon_j\, \tau_{m-j} \quad (m \ge 1). \tag{3.64}$$

Then

$$T \sim \frac{2R_\delta}{L_1} \sum_{m\ge0} \frac{\tau_m(L_2)}{L_1^m}, \qquad \Lambda = 1 + W_0(Z). \tag{3.65}$$

The first data, recomputed from eqs. (3.63) and (3.64), are

$$P_1 = t, \qquad P_2 = \tfrac{1}{2}t^2 - t, \qquad P_3 = \tfrac{1}{3}t^3 - \tfrac{3}{2}t^2 + t, \tag{3.66}$$

$$\begin{aligned} &\tau_0 = 1, && \tau_1 = L_2, && \tau_2 = L_2^2 - L_2,\\ &\tau_3 = L_2^3 - \tfrac{5}{2}L_2^2 + L_2, && \tau_4 = L_2^4 - \tfrac{13}{3}L_2^3 + \tfrac{9}{2}L_2^2 - L_2, &&\\ &\tau_5 = L_2^5 - \tfrac{77}{12}L_2^4 + \tfrac{71}{6}L_2^3 - 7L_2^2 + L_2. && && \end{aligned} \tag{3.67}$$

Theorem 3.41 (Every coefficient of the fully expanded transseries). *Put* $t = L_1^{-1}$,

$$\mathcal{W}(t; L_2) = 1 + \sum_{j\ge1} \upsilon_j(L_2)t^j, \qquad \mathcal{L}(t; L_2) = \mathcal{W}(t; L_2) + t, \tag{3.68}$$

so that $W_0/L_1 = \mathcal{W}$, $\Lambda/L_1 = \mathcal{L}$ *and* $T = (2R_\delta/L_1)\mathcal{W}^{-1}$. *With* $e_n(\Lambda) = \sum_{p=1}^{2n-1} c_{n,p}\Lambda^{-p}$,

$$\mathcal{D}_\delta^{-1}(y) \sim \frac{2R_\delta}{L_1} \sum_{r\ge0} \frac{\tau_r(L_2)}{L_1^r} - 1 + \sum_{n\ge1} \left(\frac{L_1}{2R_\delta}\right)^{2n-1} \sum_{r\ge1} \frac{D_{n,r}(L_2)}{L_1^r}, \tag{3.69}$$

where every $D_{n,r} \in \mathbb{Q}[L_2]$ *is given by the finite extraction*

$$\boxed{D_{n,r}(L_2) = [t^r] \left\{ \mathcal{W}(t; L_2)^{2n-1} \sum_{p=1}^{2n-1} c_{n,p}\, t^p\, \mathcal{L}(t; L_2)^{-p} \right\}.} \tag{3.70}$$

Together with eqs. (3.45), (3.63) *and* (3.64) *this is a general formula for every coefficient at every level. The leading term of the* n*-th inverse-power sector is* $-a_n L_1^{2n-2}(2R_\delta)^{-(2n-1)}$.

Proof. The core term is eq. (3.65), and eq. (3.64) is reciprocal series division. For the n-th block,

$$e_n(\Lambda)\, T^{-(2n-1)} = \left(\frac{L_1}{2R_\delta}\right)^{2n-1} \mathcal{W}^{2n-1} \sum_{p=1}^{2n-1} c_{n,p} L_1^{-p} \mathcal{L}^{-p},$$

and setting $t = L_1^{-1}$ and extracting $[t^r]$ gives eq. (3.70); at fixed r only finitely many υ_j and $c_{n,p}$ occur, so the extraction is finite. The first nonzero power is $r = 1$, from $p = 1$ and the constant terms $\mathcal{W}(0) = \mathcal{L}(0) = 1$, with $c_{n,1} = -a_n$ by theorem 3.35. □

Proposition 3.42 (Ordering of the scales). *For every fixed* M, $T/L_1^M \to \infty$ *as* $y \to \infty$. *Consequently*

$$\frac{T}{L_1^M} \gg 1 \gg \frac{1}{T\Lambda} \gg \frac{1}{T^3\Lambda} \gg \frac{1}{T^5\Lambda} \gg \cdots, \tag{3.71}$$

so the constant shift -1 *is beyond every fixed order of the outer* L_1^{-1}*-expansion, yet dominates the entire inner grid. In particular the first Bernoulli displacement satisfies* $1/(6T\Lambda) \asymp 1/(12R_\delta) \asymp 1/(12\log y)$.

Proof. $T = e^\Lambda$ with $\Lambda = 1 + W_0(Z) \sim L_1$, while $L_1^M = e^{ML_2}$ with $L_2 = \log L_1 = o(L_1)$; hence $\log(T/L_1^M) = \Lambda - ML_2 \to \infty$. The remaining comparisons are immediate from $T \to \infty$ and eq. (3.33), which give $T\Lambda = 2R_\delta\Lambda/(\Lambda - 1) \sim 2R_\delta$. □

Remark 3.43. This ordering is destroyed if all terms are written as one informal series in $1/\log y$: the shift -1, the outer polynomials $\tau_m(L_2)/L_1^m$ and the inner blocks $d_{2n-1}(\Lambda)T^{1-2n}$ then become indistinguishable. S1, S2 and S3 all make this point; the quantitative version is eq. (3.71).

3.8 The periodic interpolation and its oscillatory inverse

S1 and S2 centre the periodic problem at the parity-mean constant $\frac{1}{4}\log(2\pi)$; S3 centres it at the even-branch constant $\frac{1}{2}\log\pi$ and consequently reports a leading displacement $2a(1+\cos\theta)/\Lambda$ rather than $2a\cos\theta/\Lambda$. The two are the same function expanded about different cores (they differ by $2a/\Lambda + \mathcal{O}(\Lambda^{-2})$, which is exactly $h' \cdot a$ for the core map h). This chapter adopts the mean centring, which makes the carrier a mean-zero oscillation with the cleanest harmonic content.

Lemma 3.44 (Centred exact form of $\mathscr{D}$). *With $u = x+1$, $a = \frac{1}{4}\log(\pi/2)$ as in eq.* (3.9) *and*

$$A_* := \frac{1}{4}\log(2\pi) = 0.4594692666\ldots, \qquad \mathrm{e}^{A_*} = (2\pi)^{1/4} = 1.5832334871\ldots, \tag{3.72}$$

one has, exactly and for all $x > -1$,

$$\log\mathscr{D}(x) = \Phi(u) + A_* - a\cos(\pi u) + \Omega(u). \tag{3.73}$$

At an integer $x = n$, $\cos\pi u = (-1)^{n+1}$ and $A_* - a\cos\pi u = A(n)$ of eq. (3.13)*: the periodic factor freezes to the correct parity constant.*

Proof. By eq. (3.7) and theorem 3.11 with $\delta = 0$, $\log\mathscr{D}(x) = \Phi(u) + A_0 + \Omega(u) + P(x)$ where $P(x) = \frac{1}{4}(1-\cos\pi x)\log(2/\pi) = -a(1-\cos\pi x)$. Since $\cos\pi x = \cos\pi(u-1) = -\cos\pi u$, $P = -a(1+\cos\pi u) = -a - a\cos\pi u$, and $A_0 - a = \frac{1}{2}\log\pi - \frac{1}{4}\log(\pi/2) = \frac{1}{4}\log(2\pi) = A_*$. ☐

Lemma 3.45 (Exact displacement equation). *Let $R_* = \log y - A_*$ and let T, Λ be the core data of eq.* (3.32) *built from R_*; put $q = T^{-1}$, $\theta = \pi T$. Then $x = \mathscr{D}^{-1}(y)$ satisfies $x = T - 1 + \Delta$ where Δ is the unique small root of*

$$\Lambda\Delta + \frac{\varphi(q\Delta)}{q} + 2\,\Omega(T+\Delta) - 2a\cos(\theta + \pi\Delta) = 0. \tag{3.74}$$

Replacing Ω by its asymptotic series turns eq. (3.74) *into the formal equation $\mathcal{F}(q,\Delta) = 0$ with*

$$\mathcal{F}(q,v) = \Lambda v + \sum_{m\ge 2}\frac{(-1)^m}{m(m-1)}v^m q^{m-1} - 2a\cos(\theta+\pi v) + 2\sum_{k\ge 1} c_k q^{2k-1}(1+qv)^{-(2k-1)}. \tag{3.75}$$

Since the periodic term is $\mathcal{O}(1)$ while every other nonlinear term carries a positive power of q, one has $\Delta = \mathcal{O}(\Lambda^{-1})$, not *$\mathcal{O}(q\Lambda^{-1})$.*

Proof. Evaluate eq. (3.73) at $u = T+\Delta$, set it equal to $\log y = R_* + A_*$, subtract $\Phi(T) = R_*$, and multiply by 2. By the computation in theorem 3.28 with $E = q\Delta$, $2\big[\Phi(T+\Delta) - \Phi(T)\big] = T\big[\Lambda q\Delta + \varphi(q\Delta)\big] = \Lambda\Delta + \varphi(q\Delta)/q$, since $T = q^{-1}$. This gives eq. (3.74). The last sum in eq. (3.75) is $2\Omega(T(1+qv))$ expanded. Finally $\varphi(q\Delta)/q = q\Delta^2/2 + \mathcal{O}(q^2)$, so at $q = 0$ the equation reduces to $\Lambda\Delta_0 = 2a\cos(\theta + \pi\Delta_0)$, forcing $\Delta_0 \asymp \Lambda^{-1}$. ☐

3.8.1 The periodic carrier

Theorem 3.46 (Existence, uniqueness, and the closed carrier expansion). *Fix $\theta \in \mathbb{R}$ and put $\mu = 2a/\Lambda$.*

(a) *If $\Lambda > 2\pi a = 0.70934\ldots$, the equation*

$$\Delta_0 = \mu\cos(\theta + \pi\Delta_0) \tag{3.76}$$

has exactly one real solution, and $|\Delta_0| \le \mu$.

(b) *If moreover $\mu \le \frac{1}{5}$, equivalently $\Lambda \ge 10a = 1.12896\ldots$, that solution is the sum of the convergent Lagrange–Bürmann series*

$$\boxed{\Delta_0 = \sum_{m\ge1} \frac{\mu^m}{m!} \frac{\mathrm{d}^{m-1}}{\mathrm{d}v^{m-1}} \cos^m(\theta+\pi v)\bigg|_{v=0} = \sum_{m\ge1} \frac{a^m}{m!\,\Lambda^m} \sum_{j=0}^{m} \binom{m}{j} \big(\mathrm{i}\pi(m-2j)\big)^{m-1} \mathrm{e}^{\mathrm{i}(m-2j)\theta}.} \tag{3.77}$$

The second expression is real. At inverse-logarithmic order m only the Fourier modes $m, m-2, \ldots, -m$ occur.

(c) *The first three terms are*

$$\Delta_0 = \mu\cos\theta - \frac{\pi}{2}\mu^2 \sin 2\theta + \frac{\pi^2}{2}\mu^3 \cos\theta\,(2-3\cos^2\theta) + \mathcal{O}(\mu^4). \tag{3.78}$$

Proof. (a) The map $g(v) = \mu\cos(\theta+\pi v)$ satisfies $|g'| \le \pi\mu = 2\pi a/\Lambda < 1$ and $|g| \le \mu$, so it is a contraction of $\mathbb{R}$ into $[-\mu,\mu]$; Banach's theorem gives a unique fixed point there, and any real fixed point lies in $[-\mu,\mu]$.

(b) Equation (3.76) is $v = \mu\phi(v)$ with $\phi(v) = \cos(\theta+\pi v)$ entire. By the classical convergence criterion for Lagrange's series, if there is $r > 0$ with $|\mu| \max_{|v|=r} |\phi(v)| < r$, then the series $\sum_m \frac{\mu^m}{m!} \left(\frac{\mathrm{d}}{\mathrm{d}v}\right)^{m-1} \phi(v)^m|_{v=0}$ converges and its sum is the unique root of $v = \mu\phi(v)$ in $|v| < r$. Here $|\cos(\theta+\pi v)| \le \cosh(\pi|\Im v|) \le \cosh(\pi r)$ on $|v| = r$; with $r = 0.38$ and $\mu \le \frac{1}{5}$, $\mu\cosh(0.38\pi) \le 0.3603 < 0.38$. (The optimal choice $\max_{r>0} r/\cosh(\pi r) = 0.21095\ldots$ at $r = 0.382$ shows $\mu < 0.2109$ already suffices.) For the Fourier form, insert $\cos^m v = 2^{-m}\sum_j \binom{m}{j} \mathrm{e}^{\mathrm{i}(m-2j)v}$ into the derivative:

$$\frac{\mathrm{d}^{m-1}}{\mathrm{d}v^{m-1}} \cos^m(\theta+\pi v)\bigg|_{v=0} = 2^{-m} \sum_{j=0}^{m} \binom{m}{j} \big(\mathrm{i}\pi(m-2j)\big)^{m-1} \mathrm{e}^{\mathrm{i}(m-2j)\theta},$$

and $\mu^m 2^{-m} = a^m \Lambda^{-m}$. Pairing j with $m-j$ replaces $m-2j$ by its negative, conjugating each summand, so the sum is real. Only $|m-2j| \le m$ occurs, with the same parity as m.

(c) For $m = 1$ the derivative is $\cos\theta$. For $m = 2$, $\frac{\mathrm{d}}{\mathrm{d}v}\cos^2 = -2\pi\cos\sin = -\pi\sin 2(\cdot)$, giving $-\frac{\pi}{2}\mu^2 \sin 2\theta$. For $m = 3$, $\frac{\mathrm{d}^2}{\mathrm{d}v^2} \cos^3(\theta+\pi v)|_0 = 3\pi^2 \cos\theta(2-3\cos^2\theta)$, and division by 3! gives the third term. □

S1 states the carrier expansion as "convergent for sufficiently large Λ" with no threshold and no criterion, and S2 proves only the contraction estimate, from which convergence of the Lagrange series does not follow. Theorem 3.46 separates the two claims and supplies an explicit sufficient condition, $\mu \le \frac{1}{5}$. Numerically at $\Lambda = 3$, $\theta = 2$, the partial sums $m \le 1, \ldots, 5$ of eq. (3.77) approximate the exact fixed point -0.0256993556 with errors $5.6\cdot10^{-3}, 1.1\cdot10^{-3}, 1.8\cdot10^{-4}, 1.9\cdot10^{-5}, 1.9\cdot10^{-6}$, consistent with a convergence ratio $\approx \mu/0.211$.

3.8.2 Algebraic corrections around the carrier

Theorem 3.47 (Triangular periodic-grid recurrence). *Let*

$$M := \partial_v \mathcal{F}(0,\Delta_0) = \Lambda + 2\pi a \sin(\theta + \pi\Delta_0), \tag{3.79}$$

which satisfies $M \ge \Lambda - 2\pi a > 0$ for $\Lambda > 2\pi a$. Seek $\Delta = \Delta_0 + \sum_{n\ge1} \Delta_n q^n$ with each Δ_n a formal series in Λ^{-1} with trigonometric-polynomial coefficients, and put $\Delta_{<n} = \Delta_0 + \sum_{j<n} \Delta_j q^j$. Then the solution of $\mathcal{F} = 0$ is unique and generated by

$$\boxed{\Delta_n = -\frac{1}{M}\,[q^n]\ \mathcal{F}\big(q, \Delta_{<n}(q)\big), \qquad n \ge 1.} \tag{3.80}$$

The first two are

$$\Delta_1 = \frac{\frac{1}{6} - \frac{1}{2}\Delta_0^2}{M}, \qquad \Delta_2 = -\frac{1}{M}\left[a\pi^2\cos(\theta + \pi\Delta_0)\,\Delta_1^2 + \Delta_0\Delta_1 + \frac{\Delta_0 - \Delta_0^3}{6}\right]. \tag{3.81}$$

Proof. In $[q^n]\,\mathcal{F}(q,\Delta)$ the unknown Δ_n appears only linearly, through the v-derivative of the q-free part $\Lambda v - 2a\cos(\theta + \pi v)$ evaluated at $v = \Delta_0$, which is M; every other contribution involves only Δ_j with $j < n$. This is theorem 1.16 applied in the variable q over the coefficient ring generated by Λ^{-1} and $\mathrm{e}^{\pm\mathrm{i}\theta}$, and M is invertible there because $M = \Lambda(1 + \mathcal{O}(\Lambda^{-1}))$.

For $n = 1$: the $m = 2$ term of eq. (3.75) gives $\frac{1}{2}\Delta_0^2$, the cosine gives $2a\pi\sin(\theta + \pi\Delta_0)\Delta_1$, and the $k = 1$ Bernoulli term gives $2c_1 = -\frac{1}{6}$; collecting gives $M\Delta_1 + \frac{1}{2}\Delta_0^2 - \frac{1}{6} = 0$. For $n = 2$: the $m = 2$ term gives $\Delta_0\Delta_1$, the $m = 3$ term gives $-\frac{1}{6}\Delta_0^3$, the second-order Taylor term of the cosine gives $a\pi^2\cos(\theta + \pi\Delta_0)\Delta_1^2$, and expanding $2c_1 q(1 + q\Delta)^{-1}$ gives $-2c_1\Delta_0 = \frac{1}{6}\Delta_0$; collecting gives eq. (3.81). ☐

Corollary 3.48 (Asymptotic meaning)**.** *Fix $N, K \geq 0$. Truncate Δ after q^N and expand each Δ_n in Λ^{-1} through Λ^{-K}, obtaining $\widetilde{\Delta}$. Then, as $y \to +\infty$,*

$$\mathscr{D}^{-1}(y) = T - 1 + \widetilde{\Delta} + \mathcal{O}\!\left(\frac{1}{T^{N+1}\Lambda}\right) + \mathcal{O}\!\left(\frac{1}{\Lambda^{K+2}}\right), \tag{3.82}$$

uniformly in the phase $\theta = \pi T$.

Proof. By construction eq. (3.80) cancels the residual of $\mathcal{F}$ through q^N, and theorem 3.46 cancels it through Λ^{-K} at each fixed q-order; the replacement of Ω by its truncation is controlled by theorem 3.17. The exact v-derivative of the left side of eq. (3.74) equals $M + \mathcal{O}(q)$ and is bounded below by $\Lambda - 2\pi a - \mathcal{O}(q)$, a positive multiple of Λ for large T; all trigonometric coefficients and their derivatives are bounded, so theorem 1.42 converts the residual into a root error after division by a quantity $\asymp \Lambda$, uniformly in θ. ☐

Proposition 3.49 (The parity branches resum the periodic sector)**.** *At an integer $u = n + 1$ one has $A_* - a\cos\pi u = A_\delta$ with $\delta \equiv n \pmod 2$. The branchwise normalisation therefore absorbs the whole $\mathcal{O}(1)$ periodic constant into the target R_δ before the core is formed, so the $\mathcal{O}(\Lambda^{-1})$ carrier disappears and the first correction is the much smaller $1/(6T\Lambda)$. In this precise sense the two parity transseries are subsequence resummations of the periodic one: they build the parity phase into the core instead of expanding it perturbatively.*

Proof. Immediate from theorem 3.44 and theorems 3.29 and 3.46. ☐

Proposition 3.50 (Phase-coordinate reversion, formal)**.** *Let $h(R) = 2R/W_0(2R/\mathrm{e})$ be the inverse of Φ (theorem 3.27), and set $\mathcal{A}(R) = -a\cos\big(\pi h(R)\big) + \Omega\big(h(R)\big)$. Then the equation $\Phi(u) + \mathcal{A}(\Phi(u)) = R_*$ — which is eq. (3.73) rewritten in the phase coordinate $\sigma = \Phi(u)$ — has the formal reversion*

$$\sigma = R_* + \sum_{m\geq 1}\frac{(-1)^m}{m!}\left(\frac{\mathrm{d}}{\mathrm{d}R_*}\right)^{m-1}\mathcal{A}(R_*)^m, \qquad \mathscr{D}^{-1}(y) = h(\sigma) - 1. \tag{3.83}$$

Proof. Write $\sigma = R_* + \varsigma$; then $\varsigma = -\mathcal{A}(R_* + \varsigma)$, and Lagrange's formula for $\varsigma = \tau\phi(\varsigma)$ with $\phi(\varsigma) = -\mathcal{A}(R_* + \varsigma)$, $\tau = 1$, together with the translation invariance of $\mathrm{d}/\mathrm{d}\varsigma$, gives eq. (3.83). ☐

S3 states eq. (3.83) as a boxed asymptotic expansion, writing "setting $\tau = 1$ as an asymptotic expansion gives". That is a formal-to-analytic slide: $\mathcal{A}$ is not small in any norm that makes the reversion series converge, and m-fold differentiation with respect to R_* produces factors $h'(R_*) = 2/\Lambda$ whose accumulated effect is exactly what needs to be estimated. The statement is demoted here to a *formal* proposition. An asymptotic claim would require, at each order m, a bound on $\big(\mathrm{d}/\mathrm{d}R_*\big)^{m-1}\mathcal{A}^m$ uniform in the oscillatory phase — available for the truncated grid of theorem 3.48, which is therefore the statement to use. Comparing the two routes at leading order: the $m = 1$ term of eq. (3.83) contributes $h'(R_*) \cdot (-\mathcal{A}) = 2a\cos(\pi T)/\Lambda + \mathcal{O}(T^{-1}\Lambda^{-1})$, in agreement with eq. (3.78).

Warning 3.51. The carrier of theorem 3.46 records the choice eq. (3.7), not a property of $n!!$. By theorem 3.8 another interpolation, agreeing with $n!!$ at every nonnegative integer, changes the amplitude of this sector from $2a$ to $2\sqrt{a^2+\lambda^2}$ and shifts its phase, while leaving every integer value and both parity transseries untouched.

3.9 The discrete inverses

Objects (i) and (iii) of section 3.1 are intrinsic to the sequence. They are the residue-class staircase of theorems 1.37 and 1.39 at modulus $r = 2$, and nothing new has to be proved: theorem 1.41 already records this chapter as that instance. Only the dictionary and the arithmetic caveat are subject-specific.

Proposition 3.52 (Specialisation of the staircase)**.** *Take $r = 2$, residue classes $\rho \in \{0,1\}$, $A_n = n!!$, and $\nu_\rho(y) = \mathcal{D}_\rho^{-1}(y)$, which is legitimate because $\mathcal{D}_\rho$ interpolates the class-ρ subsequence and is strictly increasing (theorem 3.3). Then eq.* (1.75) *reads*

$$N(y) := \min\{n \in \mathbb{N} : n!! \geq y\} = \min\left\{2\left\lceil \frac{\nu_0(y)}{2}\right\rceil,\ 2\left\lceil \frac{\nu_1(y)-1}{2}\right\rceil + 1\right\} \qquad (y > 1), \tag{3.84}$$

and its floor counterpart, obtained from theorem 1.39 by replacing $\lceil\cdot\rceil$ with $\lfloor\cdot\rfloor$ and min *with* max*, is*

$$J(y) := \max\{n \in \mathbb{N} : n!! \leq y\} = \max_{\rho\in\{0,1\}} \left\{\rho + 2\left\lfloor \frac{\nu_\rho(y)-\rho}{2}\right\rfloor\right\}. \tag{3.85}$$

Replacing ν_ρ by the truncation of eq. (3.39) *with a certified error bound η_ρ (for instance from eq.* (3.100)*) determines $N(y)$ and $J(y)$ exactly under the separation condition of theorem 1.39, and theorem 1.40 bounds the number of exact evaluations of $n!!$ needed by $\sum_\rho(\lfloor\eta_\rho\rfloor + 2)$. Those evaluations should be made as $\log \mathcal{D}_\rho(n) = \log\kappa_\rho + \frac{n}{2}\log 2 + \log\Gamma(\frac{n}{2}+1)$, or by eq.* (3.1) *in exact integers.*

Proof. Everything is theorem 1.39 and theorem 1.40 with $r = 2$; the hypotheses hold by theorem 3.3 and, for the asymptotic input, by theorem 3.29. ☐

Warning 3.53 (Never form y). Equation (3.84), evaluated with four inverse blocks, was tested at $y = n!!$ for every $6 \leq n \leq 199$ and returned the correct $N(y) = n$ in all 194 cases — but only when the comparison $A_n \geq y$ is made on exact integers or on $\log y$. Converting $n!!$ to a fixed-precision float first destroys the comparison long before the asymptotics fail: at 80 working digits this already produces spurious off-by-one answers from $n \approx 120$ onwards, because 120!! has more than 100 digits. The failure is arithmetic, not asymptotic, and it is easy to misdiagnose as a defect of the expansion.

3.10 Beyond all algebraic orders

Proposition 3.54 (Optimal truncation, forward and inverse)**.** *Let $u > 0$. The terms of eq.* (3.18) *have moduli $|c_k|u^{1-2k}$ with consecutive ratio $\sim 2k(2k-1)/(\pi u)^2$; the least term occurs at $2k \approx \pi u$ and*

$$\min_k \left|\frac{c_k}{u^{2k-1}}\right| = \mathcal{O}\left(u^{-1/2}\mathrm{e}^{-\pi u}\right). \tag{3.86}$$

Consequently, by theorem 1.46 and the slope $\Phi'(T) = \Lambda/2$, the optimally truncated branchwise inverse has error

$$\mathcal{O}\left(\frac{\mathrm{e}^{-\pi T}}{\sqrt{T}\,\Lambda}\right). \tag{3.87}$$

Proof. Equation (3.23) gives $|c_k| = (2k-2)!\pi^{-2k}(1+\mathcal{O}(4^{-k}))$, whence the ratio. At $2k = \pi u$, Stirling's formula for $(2k-2)!$ gives eq. (3.86). For the inverse, theorem 3.17 bounds the forward logarithmic residual at optimal truncation by eq. (3.86) with $u = T(1+\mathcal{O}(T^{-2}))$; dividing by the slope $\frac{1}{2}\Lambda + \mathcal{O}(T^{-2})$ via theorems 1.42 and 1.44 gives eq. (3.87). □

Proposition 3.55 (Transport of a forward exponential sector)**.** *Suppose an exponentially improved representation of the centred forward phase retains the algebraic terms $c_1, \dots, c_K$ and has an additional residual $\mathcal{R}$, and let u_{alg} solve the correspondingly truncated algebraic inverse problem. Then*

$$u - u_{\mathrm{alg}} = -\frac{2\,\mathcal{R}(u_{\mathrm{alg}})}{\log u_{\mathrm{alg}} - 2\sum_{k=1}^{K}(2k-1)c_k\,u_{\mathrm{alg}}^{-2k}} + \mathcal{O}\biggl(\frac{\sup|F''|\,\mathcal{R}^2}{\Lambda^3} + \frac{|\mathcal{R}\,\mathcal{R}'|}{\Lambda^2}\biggr), \tag{3.88}$$

where F is the doubled algebraic phase and the supremum is over the interval between u_{alg} and the exact root; the estimate assumes the displayed denominator is $\asymp \Lambda$ and $|\mathcal{R}'| = o(\Lambda)$. At leading order this is $-2\mathcal{R}(T)/\Lambda$.

Proof. Let F be the doubled algebraic phase minus $2R$, so $F(u_{\mathrm{alg}}) = 0$ and the exact phase equals $F + 2\mathcal{R}$. Taylor expansion of $0 = F(u_{\mathrm{alg}} + \Delta) + 2\mathcal{R}(u_{\mathrm{alg}} + \Delta)$ about u_{alg}, with $F'(u_{\mathrm{alg}})$ the stated denominator, gives the first Newton correction; replacing $F' + 2\mathcal{R}'$ by F' and iterating once produces the two displayed error terms. □

Warning 3.56 (What the exponential sector is, and is not). The Borel poles of eq. (3.16) lie at $s = \pm\mathrm{i}\pi m$, so the exponentials associated with them are $\mathrm{e}^{\mp\mathrm{i}\pi m u}$ together with direction-dependent terminant multipliers, not real exponentials. On the positive axis the optimally truncated remainder has modulus $\asymp u^{-1/2}\mathrm{e}^{-\pi u}$, but this is the size of a *terminant-weighted combination*: appending a bare term $\mathrm{const}\cdot\mathrm{e}^{-\pi u}$ to the divergent Bernoulli series is not a valid transseries completion, and conflating the two gives incorrect Stokes statements. The unambiguous positive-real completion is the exact integral eq. (3.15) itself, which by theorem 3.17 even provides two-sided bounds at every order; eq. (3.88) then carries any rigorously normalised forward completion through the inverse map. All three sources make this point; S3 states it most sharply.

3.11 Complex inverse sheets

Proposition 3.57 (Local sheets)**.** *Fix a branch of* $\log y$ *and integers ℓ (logarithmic winding) and k (Lambert branch). Put*

$$R_{\delta,\ell} = \log y + 2\pi\mathrm{i}\ell - A_\delta, \quad w_{k,\ell} = W_k\biggl(\frac{2R_{\delta,\ell}}{\mathrm{e}}\biggr), \quad T_{k,\ell} = \frac{2R_{\delta,\ell}}{w_{k,\ell}}, \quad \Lambda_{k,\ell} = w_{k,\ell} + 1. \tag{3.89}$$

In any sector in which $T_{k,\ell} \to \infty$, the Stirling expansion for $\log\Gamma$ *is uniform, and the corresponding solution branch of $\mathcal{D}_\delta(x) = y$ is simple, the* same *universal polynomials give*

$$x_{\delta;k,\ell}(y) \sim T_{k,\ell} - 1 + \sum_{n\ge 1}\frac{d_{2n-1}(\Lambda_{k,\ell})}{T_{k,\ell}^{2n-1}}. \tag{3.90}$$

Proof. Every step of theorems 3.28 and 3.30 is an identity of formal series over $\mathbb{C}(\Lambda)$ and uses only $\Phi(T) = R$, $\Lambda = \log T$; eq. (3.89) secures those. The analytic statement is the sectorial version of theorem 3.29, with the mean-value step replaced by the standard implicit-function estimate in the sector. □

Remark 3.58 (Branch caveats). The pair (k,ℓ) is an asymptotic label, not a global classification: distinct labels can be joined by analytic continuation around the poles of Γ or the logarithmic cut. Branch collisions occur exactly where $\mathcal{D}_\delta' = 0$, i.e. where $\log 2 + \psi(x/2+1) = 0$; by theorem 3.3 this never happens for real $x \ge 0$, but it does for complex and for negative real x, and near such points eq. (3.90) must be replaced by a Puiseux-type local analysis.

3.12 Multifactorials

Only S3 treats multifactorials. Its normal form is reproduced here with a proof, and theorem 3.61 is new: it subsumes the "half-shifted inverse Gamma" scaling that S1 and S2 record separately, by identifying the Gamma case as the member $p = 1$.

Proposition 3.59 (Residue-class branches). *Let $p \geq 1$ and $r \in \{1, \ldots, p\}$. For integers $x \equiv r \pmod p$, $x \geq r$, set $x!_{(p)} = x(x-p)(x-2p)\cdots r$. Then, with*

$$\mathcal{D}_{p,r}(x) = C_{p,r}\, p^{x/p}\, \Gamma\!\left(\frac{x}{p}+1\right), \qquad C_{p,r} = \frac{p^{1-r/p}}{\Gamma(r/p)}, \tag{3.91}$$

one has $\mathcal{D}_{p,r}(x) = x!_{(p)}$ for every such x. For $p = 2$, $C_{2,2} = 1 = \kappa_0$ and $C_{2,1} = \sqrt{2/\pi} = \kappa_1$; for $p = 1$, $C_{1,1} = 1$ and $\mathcal{D}_{1,1}(x) = \Gamma(x+1)$.

Proof. Write $x = r + mp$. Then $x!_{(p)} = \prod_{i=0}^{m}(r+ip) = p^{m+1}\prod_{i=0}^{m}(\frac{r}{p}+i) = p^{m+1}\Gamma(\frac{r}{p}+m+1)/\Gamma(\frac{r}{p})$, while $\mathcal{D}_{p,r}(x) = C_{p,r}p^{r/p+m}\Gamma(\frac{r}{p}+m+1)$. Equating gives $C_{p,r} = p^{1-r/p}/\Gamma(r/p)$. ☐

Theorem 3.60 (Centred multifactorial normal form). *Put $u = x + \frac{p}{2}$ and $A_{p,r} = \log C_{p,r} + \frac{1}{2}\log(2\pi/p)$. Then, as $x \to +\infty$,*

$$\log \mathcal{D}_{p,r}(x) = \frac{u}{p}(\log u - 1) + A_{p,r} + \Omega_p(u), \qquad \Omega_p(u) \sim \sum_{k\geq 1} \frac{c_k^{(p)}}{u^{2k-1}}, \quad c_k^{(p)} = \frac{p^{2k-1}B_{2k}(\frac{1}{2})}{2k(2k-1)}, \tag{3.92}$$

and Ω_p has the exact Borel–Laplace representation

$$\Omega_p(u) = \int_0^\infty \mathrm{e}^{-us}\, \frac{\dfrac{ps}{2\sinh(ps/2)} - 1}{ps^2}\, \mathrm{d}s, \tag{3.93}$$

whose singularities are the simple poles $s = \pm 2\pi \mathrm{i} m/p$. Hence the optimally truncated forward remainder has scale $\mathrm{e}^{-2\pi u/p}$. For $p = 2$ this is theorems 3.11 and 3.15; for $p = 1$ it is the half-shifted Stirling expansion of $\log\Gamma(x+1) = \log\Gamma(u+\frac{1}{2})$.

Proof. Put $t = u/p$, so $x = pt - \frac{p}{2}$ and $\log \mathcal{D}_{p,r} = \log C_{p,r} + (t - \frac{1}{2})\log p + \log\Gamma(t+\frac{1}{2})$. The half-shifted Stirling expansion $\log\Gamma(t+\frac{1}{2}) \sim t\log t - t + \frac{1}{2}\log(2\pi) + \sum_k B_{2k}(\frac{1}{2})\big(2k(2k-1)\big)^{-1}t^{1-2k}$ – the case $a = \frac{1}{2}$ of $\log\Gamma(t+a)$, in which every odd-index term vanishes because $B_{2j+1}(\frac{1}{2}) = 0$ – gives

$$\log \mathcal{D}_{p,r} \sim \log C_{p,r} - \tfrac{1}{2}\log p + \tfrac{1}{2}\log(2\pi) + \frac{u}{p}(\log u - 1) + \sum_k \frac{p^{2k-1}B_{2k}(\frac{1}{2})}{2k(2k-1)u^{2k-1}},$$

since $(t-\frac{1}{2})\log p + t\log t = (u/p)\log u - \frac{1}{2}\log p$ and $t^{1-2k} = p^{2k-1}u^{1-2k}$. The constant is $A_{p,r}$. For eq. (3.93), $\Omega_p(u) = \widetilde{\Omega}(u/p)$ with $\widetilde{\Omega}$ as in the proof of theorem 3.15, where $\widetilde{\Omega}(t) = \int_0^\infty \mathrm{e}^{-tr}\big(r/(2\sinh(r/2)) - 1\big)r^{-2}\mathrm{d}r$; substituting $r = ps$ gives eq. (3.93), and $\sinh(ps/2) = 0$ at $s = 2\pi \mathrm{i} m/p$. ☐

Theorem 3.61 (Universal multifactorial inverse and its scaling). *Let $R = \log y - A_{p,r}$, $v = W_0(pR/\mathrm{e})$, $X = pR/v$, $\Lambda = v + 1 = \log X$, so that $\frac{X}{p}(\log X - 1) = R$. With $u = X(1+E)$, $q = X^{-1}$, $z = q^2$, the normalised equation is*

$$\Lambda E + \varphi(E) + \sum_{k\geq 1} a_k^{(p)} z^k (1+E)^{-(2k-1)} = 0, \qquad a_k^{(p)} := p\, c_k^{(p)} = p^{2k}\, \frac{B_{2k}(\frac{1}{2})}{2k(2k-1)}. \tag{3.94}$$

Every statement of theorems 3.30, 3.31, 3.33, 3.35 and 3.37 holds verbatim with a_k replaced by $a_k^{(p)}$, and for every fixed N

$$\mathcal{D}_{p,r}^{-1}(y) = X - \frac{p}{2} + \sum_{n=1}^{N} \frac{d_{2n-1}^{(p)}(\Lambda)}{X^{2n-1}} + \mathcal{O}\left(\frac{1}{X^{2N+1}\Lambda}\right). \tag{3.95}$$

Moreover the coefficient polynomials of all multifactorials are one family:

$$\boxed{d_{2n-1}^{(p)}(\Lambda) = p^{2n}\, d_{2n-1}^{(1)}(\Lambda) \qquad \text{for every } p \geq 1,\ n \geq 1,} \tag{3.96}$$

where $d^{(1)}$ are the blocks of the half-shifted inverse-Gamma problem. In particular $d_{2n-1} = d_{2n-1}^{(2)} = 4^n\, d_{2n-1}^{(1)}$.

Proof. The derivation of eq. (3.94) repeats theorem 3.28 with $\frac{2}{T}$ replaced by $\frac{p}{X}$: the phase difference is $\frac{X}{p}[\Lambda E + \varphi(E)]$, and $\frac{p}{X}\Omega_p(u) = \sum_k p\, c_k^{(p)} q^{2k}(1+E)^{-(2k-1)}$. Since eq. (3.94) differs from eq. (3.36) only in the values of the forcing coefficients, all structural statements transfer. Now $a_k^{(p)} = p^{2k} a_k^{(1)}$ by eq. (3.92), so substituting $q \mapsto pq$ (equivalently $z \mapsto p^2 z$) carries the $p = 1$ equation into the general one with Λ untouched. Uniqueness of the formal solution gives $E^{(p)}(q) = E^{(1)}(pq)$, i.e. $e_n^{(p)} = p^{2n} e_n^{(1)}$, which is eq. (3.96). The remainder in eq. (3.95) follows from theorems 1.42 and 1.44 exactly as in theorem 3.29, the slope now being $\frac{1}{p}\log X = \Lambda/p$. $\square$

Remark 3.62. Equation (3.96) explains and replaces the "scaling from the half-shifted inverse Gamma problem" of S1 ($p_n = 4^n d_{2n-1}$) and the identity $d_{2n-1}^{!!} = 4^n d_{2n-1}^{\Gamma}$ of S2: both are the case $(p, 1) = (2, 1)$. It was verified symbolically for $n \leq 4$ and $p \in \{2, 3\}$. As a coefficient-polynomial identity it says nothing about the numerical Lambert data, which differ between the two problems; the identity is in Λ, not in y. It also supplies a strong regression test: an implementation storing only $d^{(1)}$ reproduces every multifactorial.

p	object	centre u	phase	$a_1^{(p)}$	Borel poles
1	$\Gamma(x+1)$	$x + \frac{1}{2}$	$u(\log u - 1)$	$-\frac{1}{24}$	$\pm 2\pi i m$
2	$n!!$	$x + 1$	$\frac{u}{2}(\log u - 1)$	$-\frac{1}{6}$	$\pm \pi i m$
3	$n!!!$	$x + \frac{3}{2}$	$\frac{u}{3}(\log u - 1)$	$-\frac{3}{8}$	$\pm \frac{2\pi}{3} i m$
p	$x!_{(p)}$	$x + \frac{p}{2}$	$\frac{u}{p}(\log u - 1)$	$-\frac{p^2}{24}$	$\pm \frac{2\pi}{p} i m$

3.13 Algorithms and certification

Algorithm 3.63 (Forward coefficient table). Given M and a power $\alpha \in \mathbb{Q}$: compute c_k from eq. (3.19) for $1 \leq k \leq \lceil M/2 \rceil$; set $\gamma_{2k-1} = c_k$, $\gamma_{2k} = 0$; run eq. (3.28). This costs $\mathcal{O}(M^2)$ rational operations and never forms a symbolic exponential.

Algorithm 3.64 (Inverse blocks). On the variable $z = q^2$, with Λ symbolic:

```
input: N
E := 0
for n := 1,...,N:
    H := (1+E)*log(1+E) - E
    for k := 1,...,n:
        H := H + a[k]*z^k*(1+E)^(-(2*k-1))
    H := truncate(H, z^(n+1))
    e[n] := - coefficient(H, z^n) / Lambda
    E := E + e[n]*z^n
    assert truncate(residual(E), z^(n+1)) = 0       # (2.residual)
    assert sign_invariant(e[n], n)                  # Theorem sign law
return e[1],...,e[N]
```

Every product, logarithm and negative power may be truncated above z^n at step n; no full symbolic logarithm is ever needed. Representing each e_n as a dense polynomial in Λ^{-1} of degree exactly $2n-1$ (theorem 3.37) gives an exact allocation size.

The two assertions are worth more than a comparison against a stored table. The first is the *formal residual certificate*: with $E_N = \sum_{n\le N} e_n z^n$,

$$\Lambda E_N + \varphi(E_N) + \sum_{k=1}^{N+1} a_k z^k (1+E_N)^{-(2k-1)} = \mathcal{O}\big(z^{N+1}\big)\,, \tag{3.97}$$

and the coefficient of z^{N+1}, divided by $-\Lambda$, *is* the next block. This single identity catches sign, indexing and centring mistakes at once. The second is theorem 3.37: after multiplication by $(-1)^{n+1}$ every one of the $2n-1$ coefficients must be a positive rational. Independent checks available in addition:

(1) expanding the exact gamma ratio of theorem 3.11 reproduces c_k;

(2) expanding the Borel kernel eq. (3.16) reproduces c_k from η-values, and theorem 3.23 reproduces b_k from ζ-values;

(3) the centre-shift identity eq. (3.31) links the two;

(4) eq. (3.45) and eq. (3.48) must agree with eq. (3.41) coefficient by coefficient;

(5) eq. (3.96) compares every block with the independently generated $p=1$ family.

Every displayed block $d_1,\dots,d_{13}$ of this chapter passes all five.

Algorithm 3.65 (Stable evaluation from a logarithmic target)**.** Never form y; accept $\ell = \log y$. For branch δ:

$$R = \ell - A_\delta, \quad w = W_0(2R/\mathrm{e}), \quad T = 2R/w, \quad \Lambda = w+1, \quad x \leftarrow T - 1 + \sum_{n=1}^{N} \frac{d_{2n-1}(\Lambda)}{T^{2n-1}}. \tag{3.98}$$

Refine by Newton on the logarithmic residual $F_\delta(x) = \log \kappa_\delta + \frac{x}{2}\log 2 + \log\Gamma(\frac{x}{2}+1) - \ell$,

$$x_{\text{new}} = x - \frac{F_\delta(x)}{\frac{1}{2}\big(\log 2 + \psi(\frac{x}{2}+1)\big)}, \tag{3.99}$$

whose denominator is positive by eq. (3.6); convergence is quadratic once the asymptotic value is in the basin, which a few blocks always achieve in the ranges tested. For $\mathscr{D}$, add $\frac{1}{4}(1-\cos\pi x)\log(2/\pi)$ to F_0 and its derivative to the denominator, and start from eq. (3.82).

Proposition 3.66 (A posteriori certificate)**.** *Let $\widetilde{x}$ be any approximation and $\rho = F_\delta(\widetilde{x})$. If I is an interval containing $\widetilde{x}$ and $\mathcal{D}_\delta^{-1}(y)$, then*

$$\big|\widetilde{x} - \mathcal{D}_\delta^{-1}(y)\big| \le \frac{2|\rho|}{\inf_{\xi\in I}\big(\log 2 + \psi(\xi/2+1)\big)}, \tag{3.100}$$

and since ψ is increasing the infimum is attained at the left endpoint. At large arguments the denominator is $\sim \Lambda$, matching eq. (3.39).

Proof. This is theorem 1.42 with the explicit slope eq. (3.6); the mean-value theorem applied to F_δ on I gives eq. (3.100). □

Table 3.3: Forward: centred logarithmic expansion, $\Phi(u) + A_\delta + \sum_{k\le N} c_k u^{1-2k} - \log \mathcal{D}_\delta(n)$, $u = n+1$. Signs alternate, confirming theorem 3.17.

δ	n	$N=0$	$N=1$	$N=2$	$N=3$	$N=4$
0	20	$3.96616\cdot 10^{-3}$	$-2.09362\cdot 10^{-6}$	$5.98269\cdot 10^{-9}$	$-4.14413\cdot 10^{-11}$	$5.30661\cdot 10^{-13}$
1	21	$3.78606\cdot 10^{-3}$	$-1.82137\cdot 10^{-6}$	$4.74399\cdot 10^{-9}$	$-2.99567\cdot 10^{-11}$	$3.49749\cdot 10^{-13}$
0	100	$8.25064\cdot 10^{-4}$	$-1.88702\cdot 10^{-8}$	$2.34020\cdot 10^{-12}$	$-7.04697\cdot 10^{-16}$	$3.92938\cdot 10^{-19}$
1	101	$8.16975\cdot 10^{-4}$	$-1.83207\cdot 10^{-8}$	$2.22773\cdot 10^{-12}$	$-6.57742\cdot 10^{-16}$	$3.59602\cdot 10^{-19}$

Table 3.4: Forward: relative error $|F_M/\mathcal{D}_\delta - 1|$ of the centred multiplicative expansion eq. (3.29) truncated after $s_M u^{-M}$, with $\delta \equiv x \pmod 2$.

x	δ	$M=0$	$M=1$	$M=3$	$M=5$	$M=9$
10	0	$7.590\cdot 10^{-3}$	$4.330\cdot 10^{-5}$	$2.599\cdot 10^{-7}$	$4.974\cdot 10^{-9}$	$1.388\cdot 10^{-11}$
11	1	$6.957\cdot 10^{-3}$	$3.538\cdot 10^{-5}$	$1.751\cdot 10^{-7}$	$2.782\cdot 10^{-9}$	$5.454\cdot 10^{-12}$
20	0	$3.974\cdot 10^{-3}$	$9.988\cdot 10^{-6}$	$1.432\cdot 10^{-8}$	$6.756\cdot 10^{-11}$	$1.314\cdot 10^{-14}$
21	1	$3.793\cdot 10^{-3}$	$9.014\cdot 10^{-6}$	$1.166\cdot 10^{-8}$	$4.972\cdot 10^{-11}$	$7.956\cdot 10^{-15}$
50	0	$1.635\cdot 10^{-3}$	$1.483\cdot 10^{-6}$	$3.106\cdot 10^{-10}$	$2.113\cdot 10^{-13}$	$9.594\cdot 10^{-19}$
51	1	$1.604\cdot 10^{-3}$	$1.424\cdot 10^{-6}$	$2.861\cdot 10^{-10}$	$1.866\cdot 10^{-13}$	$7.800\cdot 10^{-19}$

Table 3.5: Branchwise inverse: $x_N - x$ with $y = \mathcal{D}_\delta(x)$ exact and x_N the truncation of eq. (3.39) after N blocks. Signs alternate, as theorem 3.37 predicts for the dominant term.

δ	x	$N=0$	$N=1$	$N=2$	$N=3$	$N=5$
0	10	$-6.3074\cdot 10^{-3}$	$1.6444\cdot 10^{-5}$	$-1.6097\cdot 10^{-7}$	$3.6890\cdot 10^{-9}$	$1.1386\cdot 10^{-11}$
1	11	$-5.5808\cdot 10^{-3}$	$1.2104\cdot 10^{-5}$	$-9.9815\cdot 10^{-8}$	$1.9338\cdot 10^{-9}$	$4.2632\cdot 10^{-12}$
0	20	$-2.6055\cdot 10^{-3}$	$1.7520\cdot 10^{-6}$	$-4.7751\cdot 10^{-9}$	$3.1029\cdot 10^{-11}$	$7.6506\cdot 10^{-15}$
1	21	$-2.4497\cdot 10^{-3}$	$1.4956\cdot 10^{-6}$	$-3.7170\cdot 10^{-9}$	$2.2041\cdot 10^{-11}$	$4.5233\cdot 10^{-15}$
0	50	$-8.3109\cdot 10^{-4}$	$8.9794\cdot 10^{-8}$	$-4.1941\cdot 10^{-11}$	$4.7156\cdot 10^{-14}$	$3.4429\cdot 10^{-19}$
1	51	$-8.1110\cdot 10^{-4}$	$8.4219\cdot 10^{-8}$	$-3.7845\cdot 10^{-11}$	$4.0942\cdot 10^{-14}$	$2.7669\cdot 10^{-19}$
0	100	$-3.5755\cdot 10^{-4}$	$9.5805\cdot 10^{-9}$	$-1.1469\cdot 10^{-12}$	$3.3159\cdot 10^{-16}$	$1.5896\cdot 10^{-22}$
1	101	$-3.5329\cdot 10^{-4}$	$9.2785\cdot 10^{-9}$	$-1.0892\cdot 10^{-12}$	$3.0878\cdot 10^{-16}$	$1.4232\cdot 10^{-22}$

Table 3.6: Inverse of the periodic interpolation: $\big(T - 1 + \widetilde{\Delta}\big) - x$ with $y = \mathscr{D}(x)$ exact, successively adding the carrier Δ_0 (computed as the exact fixed point of eq. (3.76)), Δ_1/T and Δ_2/T^2 from eq. (3.81). The core error matches the predicted carrier amplitude $2a/\Lambda$: at $x = 20$, $2a/\Lambda = 0.074080$ against an observed 0.071518; at $x = 100$, 0.048919 against 0.048564. No source validates this sector numerically.

x	core $T-1$	$+\Delta_0$	$+\Delta_1/T$	$+\Delta_2/T^2$
20	$7.1518\cdot 10^{-2}$	$-2.5603\cdot 10^{-3}$	$-1.2693\cdot 10^{-5}$	$1.6637\cdot 10^{-6}$
21	$-7.5539\cdot 10^{-2}$	$-2.4124\cdot 10^{-3}$	$1.4359\cdot 10^{-5}$	$1.4221\cdot 10^{-6}$
20.5	$-2.5254\cdot 10^{-3}$	$-3.2850\cdot 10^{-3}$	$2.0627\cdot 10^{-6}$	$2.2808\cdot 10^{-6}$
100	$4.8564\cdot 10^{-2}$	$-3.5495\cdot 10^{-4}$	$-2.3071\cdot 10^{-7}$	$9.4103\cdot 10^{-9}$
100.25	$3.4218\cdot 10^{-2}$	$-3.9845\cdot 10^{-4}$	$-1.8912\cdot 10^{-7}$	$1.0779\cdot 10^{-8}$

Table 3.7: Multifactorial inverse eq. (3.95), using only the $p = 1$ blocks rescaled by eq. (3.96). Entries are $x_N - x$ with $y = \mathcal{D}_{p,r}(x)$ exact; the $p = 2$ row reproduces table 3.5.

p	r	x	$N=0$	$N=1$	$N=2$	$N=3$
2	2	20	$-2.60549\cdot 10^{-3}$	$1.75197\cdot 10^{-6}$	$-4.77514\cdot 10^{-9}$	$3.10287\cdot 10^{-11}$
3	1	22	$-5.05001\cdot 10^{-3}$	$6.03949\cdot 10^{-6}$	$-2.95265\cdot 10^{-8}$	$3.44131\cdot 10^{-10}$
3	2	20	$-5.67882\cdot 10^{-3}$	$8.16471\cdot 10^{-6}$	$-4.75923\cdot 10^{-8}$	$6.60178\cdot 10^{-10}$
4	3	23	$-8.27262\cdot 10^{-3}$	$1.54424\cdot 10^{-5}$	$-1.18248\cdot 10^{-7}$	$2.15627\cdot 10^{-9}$
5	2	27	$-1.04166\cdot 10^{-2}$	$2.15596\cdot 10^{-5}$	$-1.85311\cdot 10^{-7}$	$3.79755\cdot 10^{-9}$

3.14 Numerical validation

All tables in this section were recomputed at 220-digit precision with `mpmath`: the exact target $\ell = \log \mathcal{D}_\delta(n)$ is formed from $\log \Gamma$, the Lambert data from eq. (3.98), and the blocks from eq. (3.41) in exact rational arithmetic. No rounded value of y is ever formed. Entries are *signed* (approximation minus exact), which the source tables partly suppress.

Remark 3.67. The near-geometric improvement at these moderate arguments is not a claim of convergence: at fixed x the Bernoulli-driven expansion diverges, and by theorem 3.54 the attainable accuracy is bounded below by $\asymp \mathrm{e}^{-\pi T}/(\sqrt{T}\Lambda)$. In high-precision work, truncate at the least term or finish with eq. (3.99). Note also that the parity dependence is confined to the core: once the correct A_δ is used, the two subsequences show nearly identical error patterns, as theorem 3.4 predicts.

3.15 Logical status

Three kinds of statement have been used, and the chapter keeps them apart.

Exact identities. Equations (3.3), (3.7) and (3.91) and the interpolation statements theorems 3.3, 3.6 and 3.59; the centred normal forms eqs. (3.12), (3.14) and (3.73); the Lambert identities eq. (3.33); the Borel–Laplace representations eqs. (3.15) and (3.93) and the Mittag–Leffler decomposition eq. (3.16). These hold wherever the functions are defined.

Formal coefficient identities. The Bell formulae eqs. (3.26) and (3.43), the recurrences eqs. (3.28), (3.41) and (3.80), the closed formulae eqs. (3.45), (3.48) and (3.70), the boundary and sign statements theorems 3.35 and 3.37, the centre-shift identity eq. (3.31), the scaling eq. (3.96), and theorem 3.50. These are identities of formal series over $\mathbb{Q}(\Lambda)$, or over the ring generated by Λ^{-1} and $\mathrm{e}^{\pm \mathrm{i}\theta}$ in the periodic case. No convergence of the Bernoulli series is asserted anywhere.

Poincaré asymptotics with remainders. Theorems 3.16 to 3.18 for the forward direction — where the remainder is in fact two-sided and sharp; theorems 3.29 and 3.48 and eq. (3.95) for the inverse, each obtained from a forward residual by theorems 1.42 and 1.44 together with the explicit slope eq. (3.6); and theorems 3.54 and 3.55 for the beyond-all-orders layer. Theorem 3.46(b) is a genuine convergence statement, with an explicit threshold.

Remark 3.68 (Practical summary). For symbolic work, centre at $u = x+1$, keep W_0 exact, generate c_k from Bernoulli numbers, exponentiate with complete Bell polynomials, and invert with the triangular $z = T^{-2}$ recurrence, checking eq. (3.97) and the sign law at every step. Expand W_0 into iterated logarithms only as a final presentation layer. When the parity of n is known, use the branch inverse; when it is not, state the interpolation — by theorem 3.8 its choice changes the leading sector of the answer.

Chapter 4

The partition numbers (A000041)

This chapter merges three independent treatments of the partition numbers filed in the `special-function-inversion` collection: *Partition_Number_Transseries_and_Asymptotic_Inverse* (below: **S1**), *Partition_Number_Transseries_and_Inverse* (**S2**), and *Partition_Numbers_Transseries_and_Inverse* (**S3**). All three take Rademacher's convergent series as analytic input and build a coefficient calculus on top of it; they differ in notation, in how the non-integral arithmetic phase is defined, and in which tail estimate is proved. Everything below that is not attributed is common to all three, restated once. Numerical constants, coefficient tables and error tables have been recomputed here (exact rationals through `fractions`/`sympy`; 140-digit `mpmath` for the tables); the recomputation is described in section 4.12.

4.1 Orientation: what is specific to $p(n)$

The partition function $p(n)$ is defined by Euler's product

$$\sum_{n\geq 0} p(n)q^n = \prod_{m\geq 1} \frac{1}{1-q^m}, \qquad |q|<1, \tag{4.1}$$

and is OEIS A000041. Hardy and Ramanujan proved

$$p(n) \sim \frac{1}{4\sqrt{3}\,n} \exp\left(\pi\sqrt{\frac{2n}{3}}\right), \tag{4.2}$$

and Rademacher replaced the asymptotic formula by an exactly convergent series. That single fact makes this chapter structurally different from Parts 2, 3 and 5: the exponentially small sectors are *given*, not reconstructed from the late behaviour of a divergent perturbative series. Three features are genuinely specific to $p(n)$ and are the subject of this chapter.

1. **An arithmetic sector index.** The sectors are labelled by a positive integer k (a root of unity $\mathrm{e}^{2\pi \mathrm{i} h/k}$) and a sign σ, and their amplitudes are the Rademacher sums $A_k(n)$ built from Dedekind sums. These amplitudes are *periodic* in n with period k; they are not constants, and after inversion they become bounded oscillations whose phase is quadratic in $\log y$.

2. **Non-well-ordered actions.** Relative to the dominant sector, the exponential actions are $1-1/k$ (increasing to 1) and $1+1/k$ (decreasing to 1). The second family has no least element. The support of the completed object is therefore not a locally finite Hahn grid, and the two signs may not be separated before summing over k.

3. **Convergence where one expects divergence.** Every fixed sector's algebraic series in $n^{-1/2}$ has radius of convergence exactly $\sqrt{24}$ in the variable $n^{-1/2}$, and the reversion series of the dominant sector converges as well. Nothing in this chapter is a divergent asymptotic series; consequently the least-term machinery of theorem 1.46 degenerates here, as recorded in theorem 4.37.

Everything else — the Lambert core, the reversion around it, the flattening into logarithms, the passage from a smooth inverse to an integer staircase, and the transport of remainders — is Part 1 material and is *cited*, never re-proved.

4.1.1 The exponential–power reduction and the Part 0 dictionary

Fix throughout

$$N = n - \frac{1}{24}, \qquad \lambda = \pi\sqrt{\frac{2}{3}} = \frac{\pi\sqrt{6}}{3}, \qquad \mu = \lambda\sqrt{N} = \frac{\pi}{6}\sqrt{24n-1}, \qquad \kappa = \frac{\pi^2}{6\sqrt{3}}. \tag{4.3}$$

The dominant Rademacher sector, derived in theorem 4.14, is

$$P_{1,+}(n) = \frac{e^{\lambda\sqrt{N}}}{4\sqrt{3}\,N}\left(1 - \frac{1}{\lambda\sqrt{N}}\right). \tag{4.4}$$

Read as a function of N, eq. (4.4) is exactly an instance of the exponential–power model theorem 1.2, $F(x) = C\exp(ax^\alpha)x^b\exp Q(x^{-\delta})$, with

$$x = N, \quad C = \frac{1}{4\sqrt{3}}, \quad a = \lambda, \quad \alpha = \frac{1}{2}, \quad b = -1, \quad \delta = \frac{1}{2}, \quad Q(t) = \log\left(1 - \frac{t}{\lambda}\right) = -\sum_{r\geq 1}\frac{t^r}{r\lambda^r}. \tag{4.5}$$

So the partition problem is the case $\alpha = 1/2$: a *square-root* phase, not a linear one. It is reduced to the linear case in two steps: the monomial substitution $\xi = x^\alpha = \sqrt{N}$ of theorem 1.4, followed by the elementary linear rescaling $\mu = \lambda\xi$ that normalises the exponential rate to 1.

Lemma 4.1 (The α-reduction for $p(n)$). (i) *Theorem 1.4 applied to eq.* (4.5) *with $\xi = \sqrt{N}$ gives an $\alpha = 1$ model with data $\left(\frac{1}{4\sqrt{3}},\ \lambda,\ 1,\ -2,\ 1,\ Q\right)$, $Q(t) = \log(1 - t/\lambda)$; note $b/\alpha = -2$ and $\delta/\alpha = 1$, so this is the normalisation $\delta = 1$ assumed from Part 1 onwards.* (ii) *The further substitution $\mu = \lambda\xi$ rescales any $\alpha = 1$ model with data $(C, a, 1, b, 1, Q)$ to one with data $\left(Ca^{-b},\ 1,\ 1,\ b,\ 1,\ Q(a\,\cdot)\right)$. Here the two steps give*

$$P_{1,+} = \kappa\,\frac{e^\mu}{\mu^2}\left(1 - \frac{1}{\mu}\right) = C' e^{a'\mu}\mu^{b'}\exp Q'(\mu^{-\delta'}), \tag{4.6}$$

with the dictionary

$$C' = \kappa = \frac{\pi^2}{6\sqrt{3}}, \qquad a' = 1, \qquad \alpha' = 1, \qquad b' = -2, \qquad \delta' = 1, \qquad Q'(t) = \log(1 - t).$$

The inverse map back to the index is exact and quadratic:

$$n = \frac{1}{24} + \frac{3\mu^2}{2\pi^2}. \tag{4.7}$$

Proof. (i) is theorem 1.4 at $\alpha = \delta = 1/2$, $b = -1$, which leaves C, a and the coefficients of Q unchanged and replaces (b, δ) by $(b/\alpha, \delta/\alpha)$. For (ii), substituting $\xi = \mu/a$ in $Ce^{a\xi}\xi^b\exp Q(\xi^{-1})$ gives $Ca^{-b}e^\mu\mu^b\exp Q(a/\mu)$, and $t \mapsto Q(at)$ again lies in $t\,\mathbb{R}[[t]]$. Here $a = \lambda$, $b = -2$, so $Ca^{-b} = \lambda^2/(4\sqrt{3}) = \pi^2/(6\sqrt{3}) = \kappa$ (using $\lambda^2 = 2\pi^2/3$) and $Q'(t) = Q(\lambda t) = \log(1 - t)$. Composing, $\mu = \lambda\sqrt{N}$ and eq. (4.4) become eq. (4.6). Inverting $\mu = \lambda\sqrt{n - 1/24}$ gives eq. (4.7), again by $\lambda^2 = 2\pi^2/3$. □

Remark 4.2 (The model hypotheses hold exactly, and Q converges). Theorem 1.2 asks for the analytic estimates eq. (1.3) and eq. (1.4) in addition to the formal shape. For the partition envelope both hold *exactly and trivially*: $F(\mu) = \kappa e^{\mu}\mu^{-2}(1-1/\mu)$ is an elementary closed form, so $\log F(\mu) - \log\kappa - \mu + 2\log\mu = \log(1-1/\mu)$ is literally the sum of the series Q', and $(\log F)' = \Phi'$ is computed in closed form in theorem 4.49; no asymptotic estimate is being differentiated. In particular $Q'(t) = \log(1-t)$ is *convergent*, with radius 1. Theorem 1.3 states that Q is divergent of Gevrey type in all four chapters of this volume; that is not so here, and the discrepancy should be read as a further instance of theorem 4.37 — the partition chapter is the one place in the volume where the whole apparatus applies to a convergent object.

Remark 4.3 (Why this is the point of the whole construction). After theorem 4.1 nothing in the inversion of the dominant sector is partition-specific. The three sources each rediscover, in their own notation, the Lambert core of theorem 1.20, the centred reversion of theorem 1.25 and the logarithmic flattening of theorem 1.33. We record the dictionary once (table 4.1) and cite. What is *not* generic — and occupies the rest of the chapter — is the arithmetic: the sector index, the Dedekind-sum amplitudes, the accumulation of actions at 1, and the interaction between the periodic amplitudes and the $\mathcal{O}(1)$ shift produced by the reversion.

Table 4.1: Parameter dictionary from Part 1 to the partition chapter. All Part 0 statements are used at these values.

Part 0 object	Partition value	Reference
model variable x	$\mu = \pi\sqrt{2(n-1/24)/3}$	eq. (4.3)
C	$\kappa = \pi^2/(6\sqrt{3})$	theorem 4.1
$a,\ \alpha$	$1,\ 1$ (after reduction and rescaling; $\lambda,\ 1/2$ before)	theorem 4.1
b	-2	theorem 4.1
$Q(t),\ \delta$	$\log(1-t),\ 1$	theorem 4.1
Lambert core X	$-2W_{-1}\left(-\frac{1}{2}\sqrt{\kappa/y}\right)$	theorem 4.46
$\Psi(t,u)$ of theorem 1.25	$3\log(1+tu) - \log(1-t+tu)$	theorem 4.50
reversion coefficients c_m	$1, \frac{5}{2}, \frac{13}{3}, \frac{53}{12}, -\frac{14}{5}, \ldots$	table 4.2
index map $x \mapsto n$	$n = \frac{1}{24} + 3\mu^2/(2\pi^2)$	eq. (4.7)
staircase separation	unit spacing of $\mathbb{Z}$; see theorem 4.83	theorem 1.39

Remark 4.4 (Notation across the three sources). The sources disagree on symbols in a way that has to be stated once, because two of the clashes are dangerous.

1. **The letter x.** S1 writes $N = n - 1/24$ for the shifted index and x for the phase $\pi\sqrt{2N/3}$; S2 writes $x = n - 1/24$ for the shifted index and μ for the phase; S3 writes N and μ. This volume uses N for the shifted index and μ for the phase, i.e. the S3 convention, which is the intersection of the other two.

2. **The shift itself.** Every displayed *exact* identity in this chapter is in the shifted variable $N = n - 1/24$. Expansions in $n^{-1/2}$ (section 4.5) re-expand the shift and are the only place where unshifted n appears inside a coefficient. Confusing the two is the single most common error in this subject: e.g. replacing N by n in eq. (4.4) costs a relative $\mathcal{O}(n^{-1/2})$ and destroys every statement about exponentially small sectors.

3. **The normalising constant** is C in S1, κ in S2, D in S3; all equal $\pi^2/(6\sqrt{3})$. We write κ, reserving D for nothing and Φ' for the slope.

4. **The logarithmic variable.** S1 and S2 expand in $R = \log(y/\kappa)$; S3 expands in $H = \log(4y/\kappa) = R + 2\log 2$. The two expansions have the *same* coefficient polynomials (see theorem 4.48) but are different series and must not be mixed.

4.2 Dedekind sums and the arithmetic amplitudes

4.2.1 Definitions

Definition 4.5 (Sawtooth and Dedekind sum)**.** For $u \in \mathbb{R}$ put

$$((u)) = \begin{cases} u - \lfloor u \rfloor - \frac{1}{2}, & u \notin \mathbb{Z}, \\ 0, & u \in \mathbb{Z}, \end{cases} \tag{4.8}$$

and for $k \geq 1$, $h \in \mathbb{Z}$,

$$s(h,k) = \sum_{r=1}^{k-1} \left(\left(\frac{r}{k}\right)\right) \left(\left(\frac{hr}{k}\right)\right), \tag{4.9}$$

with the empty-sum convention $s(h,1) = 0$.

Throughout, $U_k = \{h : 0 \leq h < k,\ \gcd(h,k) = 1\}$, so that $U_1 = \{0\}$.

Definition 4.6 (Rademacher amplitude)**.** For $k \geq 1$ and $n \in \mathbb{Z}$,

$$A_k(n) = \sum_{h \in U_k} \exp\left(\pi \mathrm{i}\, s(h,k) - \frac{2\pi \mathrm{i} h n}{k}\right). \tag{4.10}$$

Lemma 4.7 (Elementary properties)**.** *For every $k \geq 1$ and $n \in \mathbb{Z}$:*

1. *$A_k(n) \in \mathbb{R}$, and $A_k(n) = \sum_{h \in U_k} \cos\bigl(\pi s(h,k) - 2\pi h n/k\bigr)$;*
2. *$A_k(n+k) = A_k(n)$;*
3. *$|A_k(n)| \leq \varphi(k) \leq k$;*
4. *$A_1(n) = 1$ and $A_2(n) = (-1)^n$.*

Proof. (i) For $k \geq 2$ the involution $h \mapsto k - h$ is a bijection of U_k (note $k - h \neq h$ unless $k = 2h$, impossible for $\gcd(h,k) = 1$, $k \geq 3$; for $k = 2$ the set is $\{1\}$ and $s(1,2) = 0$ makes the single term real). The reciprocity $s(-h,k) = -s(h,k)$, immediate from $((-u)) = -((u))$, shows that the terms at h and $k - h$ are complex conjugates. Hence the sum is real and equals the sum of the real parts, which is the stated cosine sum. (ii) Replacing n by $n + k$ changes each exponent by $-2\pi\mathrm{i}h$, an integer multiple of $2\pi\mathrm{i}$. (iii) Triangle inequality on $\varphi(k)$ unimodular terms. (iv) $U_1 = \{0\}$ and $s(0,1) = 0$ give $A_1 \equiv 1$; $U_2 = \{1\}$, $s(1,2) = 0$ give $A_2(n) = \mathrm{e}^{-\pi \mathrm{i} n} = (-1)^n$. ☐

Only the trivial bound (iii) is used anywhere in this chapter. Weil-type bounds for these Kloosterman-like sums would sharpen the constants in theorems 4.23 and 4.78 but change no structure; see theorem 4.26.

Proposition 4.8 (The first amplitudes)**.** *With* $s(1,k) = \dfrac{(k-1)(k-2)}{12k}$,

$$\begin{aligned} A_1(n) &= 1, & A_2(n) &= (-1)^n, \\ A_3(n) &= 2\cos\left(\frac{\pi}{18} - \frac{2\pi n}{3}\right), & A_4(n) &= 2\cos\left(\frac{\pi}{8} - \frac{\pi n}{2}\right), \\ A_5(n) &= 2\cos\left(\frac{\pi}{5} - \frac{2\pi n}{5}\right) + 2\cos\frac{4\pi n}{5}, & A_6(n) &= 2\cos\left(\frac{5\pi}{18} - \frac{\pi n}{3}\right). \end{aligned} \tag{4.11}$$

Proof. $s(1,k) = \sum_{r=1}^{k-1} ((r/k))^2 = \sum_{r=1}^{k-1} (r/k - 1/2)^2 = \frac{(k-1)(2k-1)}{6k} - \frac{k-1}{2} + \frac{k-1}{4} = \frac{(k-1)(k-2)}{12k}$. Hence $s(1,3) = \frac{1}{18}$, $s(1,4) = \frac{1}{8}$, $s(1,5) = \frac{1}{5}$, $s(1,6) = \frac{5}{18}$, and $s(k-1,k) = -s(1,k)$. For $k = 3, 4, 6$ the set U_k is $\{1, k-1\}$, and the two terms are conjugate, giving twice the real part of the $h = 1$ term after reducing $2\pi(k-1)n/k \equiv -2\pi n/k \pmod{2\pi}$. For $k = 5$, $U_5 = \{1,2,3,4\}$ and a direct evaluation of eq. (4.9) gives $s(2,5) = s(3,5) = 0$: with $((1/5)), ((2/5)), ((3/5)), ((4/5))) = (-\frac{3}{10}, -\frac{1}{10}, \frac{1}{10}, \frac{3}{10})$, the four products in $s(2,5)$ are $\frac{3}{100}, -\frac{3}{100}, -\frac{3}{100}, \frac{3}{100}$. Pairing $h = 1$ with $h = 4$ and $h = 2$ with $h = 3$ gives the stated two cosines. ☐

4.2.2 The real-analytic interpolation of the amplitudes

The inverse problem needs the amplitudes at non-integral arguments: the smooth inverse of a smooth interpolation evaluates A_k at a real point. The finite Fourier shape of eq. (4.10) supplies a canonical choice.

Definition 4.9 (Entire amplitude interpolation)**.** For $k \geq 1$ and $\nu \in \mathbb{C}$ set

$$\widetilde{A}_k(\nu) = \sum_{h \in U_k} \cos\left(\pi s(h,k) - \frac{2\pi h \nu}{k}\right). \tag{4.12}$$

Lemma 4.10. *$\widetilde{A}_k$ is entire, k-periodic, real on $\mathbb{R}$, satisfies $\widetilde{A}_k(n) = A_k(n)$ for $n \in \mathbb{Z}$, and $|\widetilde{A}_k^{(r)}(\nu)| \leq \varphi(k)(2\pi/k)^r \cosh(2\pi|\Im\nu|) \leq k\,(2\pi)^r \mathrm{e}^{2\pi|\Im\nu|}$ for every $r \geq 0$. Moreover $\widetilde{A}_1 \equiv 1$ and $\widetilde{A}_2(\nu) = \cos \pi\nu$, while*

$$\widetilde{A}_3(\nu) = 2\cos(\pi\nu)\,\cos\left(\frac{\pi}{18} + \frac{\pi\nu}{3}\right). \tag{4.13}$$

Proof. Each summand is entire and k-periodic in ν; realness on $\mathbb{R}$ is manifest; agreement at integers is theorem 4.7(i). The derivative bound is termwise, using $|h/k| \leq 1$ and $|\cos^{(r)}(z)| \leq \cosh(\Im z)$. For $k = 1$, $U_1 = \{0\}$ makes the single term $\cos 0 = 1$ identically. For $k = 3$, sum-to-product on $A = \frac{\pi}{18} - \frac{2\pi\nu}{3}$ and $B = -\frac{\pi}{18} - \frac{4\pi\nu}{3}$ gives $\frac{A+B}{2} = -\pi\nu$ and $\frac{A-B}{2} = \frac{\pi}{18} + \frac{\pi\nu}{3}$, whence eq. (4.13). □

S2, Definition "Real Rademacher amplitude". S2 defines the interpolation as a sum over $1 \leq h \leq k$ with $\gcd(h,k) = 1$. For $k \geq 2$ this is the same index set as U_k, but for $k = 1$ it produces the single term $\cos(2\pi\nu)$ instead of the constant 1. S2 nevertheless uses $\mathcal{A}_1 \equiv 1$ throughout (its dominant sector carries no amplitude, and its relative-correction formula separates the $k = 1$ terms with amplitude 1), so the definition contradicts its own use: with S2's index set the interpolating function $\mathcal{P}$ would not be an interpolation of p — its dominant term would oscillate off-lattice with period 1 at full relative strength, and the "eventual monotonicity" proposition would be false. **Fix.** Take h over $U_k = \{0 \leq h < k : \gcd(h,k) = 1\}$ with $U_1 = \{0\}$, i.e. theorem 4.9; equivalently, decree $\widetilde{A}_1 \equiv 1$ separately, as S3 does. All of S2's later formulas are then correct as printed.

Remark 4.11 (Non-uniqueness, honestly stated). theorem 4.9 is *a* choice, not *the* choice. Any strictly increasing smooth $\mathcal{P}$ with $\mathcal{P}(n) = p(n)$ has an inverse, and the algebraic (Lambert–logarithmic) part of that inverse is the same for all of them, because it is determined by the dominant sector, which is interpolation-independent once N is treated as a real variable. The exponentially small sectors are *not* interpolation-independent. theorem 4.9 is singled out only because it is entire, finite-Fourier, real on $\mathbb{R}$, and literally the analytic continuation of the amplitudes already present in Rademacher's theorem. section 4.10 gives the alternative that avoids interpolation altogether.

4.3 Rademacher's formula in exact transseries normal form

4.3.1 The analytic input

Theorem 4.12 (Hardy–Ramanujan–Rademacher; quoted)**.** *For every integer $n \geq 1$,*

$$p(n) = \frac{1}{\pi\sqrt{2}} \sum_{k=1}^{\infty} A_k(n)\sqrt{k}\,\frac{\mathrm{d}}{\mathrm{d}n}\left[\frac{\sinh\left(\frac{\lambda}{k}\sqrt{n - \frac{1}{24}}\right)}{\sqrt{n - \frac{1}{24}}}\right], \tag{4.14}$$

the series converging when summed in increasing k.

This is the only unproved input of the chapter. It is Rademacher's theorem (1937, 1938, 1943); a textbook proof is in Apostol, and a machine-checked proof exists in Isabelle/HOL (Eberl–Paulson, 2026);

see theorem 4.100. All three sources take it as given and so do we. Note the factor $\sqrt{k}$: some normalisations absorb it into the definition of the arithmetic sum, and mixing conventions is a standard source of a spurious factor k. Theorem 4.87 records the cross-checks that detect such a slip.

4.3.2 Exact two-exponential sectors

Lemma 4.13 (Elementary derivative). *For $a > 0$ and $v > 0$,*

$$\frac{\mathrm{d}}{\mathrm{d}v}\Big(v^{-1/2}\sinh(a\sqrt{v})\Big) = \frac{1}{4v^{3/2}}\Big[(a\sqrt{v}-1)\mathrm{e}^{a\sqrt{v}} + (a\sqrt{v}+1)\mathrm{e}^{-a\sqrt{v}}\Big]. \tag{4.15}$$

Proof. With $w = a\sqrt{v}$ one has $\mathrm{d}w/\mathrm{d}v = w/(2v)$ and $v^{-1/2}\sinh w = a\,w^{-1}\sinh w$, so

$$\frac{\mathrm{d}}{\mathrm{d}v}\big(v^{-1/2}\sinh(a\sqrt{v})\big) = a\,\frac{\mathrm{d}}{\mathrm{d}w}\left(\frac{\sinh w}{w}\right)\frac{w}{2v} = \frac{a}{2v}\left(\cosh w - \frac{\sinh w}{w}\right) = \frac{1}{2v^{3/2}}\big(w\cosh w - \sinh w\big),$$

using $a = w/\sqrt{v}$ in the last step. Finally $2(w\cosh w - \sinh w) = (w-1)\mathrm{e}^{w} + (w+1)\mathrm{e}^{-w}$. ☐

Theorem 4.14 (Exact sector decomposition). *Let N, λ, μ, κ be as in eq. (4.3) and set, for $k \geq 1$ and $\sigma \in \{+1, -1\}$,*

$$P_{k,\sigma}(n) = \frac{A_k(n)}{4\sqrt{3k}\,N}\,\mathrm{e}^{\sigma\mu/k}\left(1 - \frac{\sigma k}{\mu}\right) = \frac{\kappa}{\mu^2}\frac{A_k(n)}{\sqrt{k}}\,\mathrm{e}^{\sigma\mu/k}\left(1 - \frac{\sigma k}{\mu}\right). \tag{4.16}$$

Then, for every integer $n \geq 1$,

$$p(n) = \sum_{k\geq 1}\big(P_{k,+}(n) + P_{k,-}(n)\big), \tag{4.17}$$

the two signs being paired before the sum over k is formed. Equivalently, with the entire paired kernel

$$g(z) = \cosh z - \frac{\sinh z}{z} = \sum_{m\geq 1}\frac{2m}{(2m+1)!}z^{2m}, \qquad g(0) = 0, \tag{4.18}$$

one has the manifestly convergent form

$$p(n) = \frac{2\kappa}{\mu^2}\sum_{k\geq 1}\frac{A_k(n)}{\sqrt{k}}\,g\Big(\frac{\mu}{k}\Big). \tag{4.19}$$

Proof. Apply theorem 4.13 with $v = N$ and $a = \lambda/k$, so that $a\sqrt{v} = \mu/k$, inside eq. (4.14). The kth summand becomes

$$\frac{A_k(n)\sqrt{k}}{\pi\sqrt{2}}\cdot\frac{1}{4N^{3/2}}\Big[\Big(\frac{\mu}{k}-1\Big)\mathrm{e}^{\mu/k}+\Big(\frac{\mu}{k}+1\Big)\mathrm{e}^{-\mu/k}\Big] = \frac{A_k(n)\sqrt{k}}{\pi\sqrt{2}}\cdot\frac{\mu}{4kN^{3/2}}\Big[\mathrm{e}^{\mu/k}\Big(1-\frac{k}{\mu}\Big)+\mathrm{e}^{-\mu/k}\Big(1+\frac{k}{\mu}\Big)\Big].$$

Now $\mu/(\pi\sqrt{2}\sqrt{N}) = \lambda/(\pi\sqrt{2}) = 1/\sqrt{3}$, so the prefactor is $A_k(n)/(4\sqrt{3k}\,N)$, which is eq. (4.16) summed over σ. Since $\mu^2 = \lambda^2 N = (2\pi^2/3)N$, we get $1/(4\sqrt{3k}\,N) = \kappa/(\sqrt{k}\,\mu^2)$, the second form of eq. (4.16). Adding the two signs and using $\mathrm{e}^{z}(1-1/z) + \mathrm{e}^{-z}(1+1/z) = 2g(z)$ with $z = \mu/k$ gives eq. (4.19). All manipulations are termwise algebra, so the sum is unchanged. ☐

Remark 4.15 (The three sources' normal forms are the same). S1 writes $p(n) = \frac{\mathcal{C}}{x^2}\sum_k \frac{A_k}{\sqrt{k}}\{\mathrm{e}^{x/k}(1 - k/x) + \mathrm{e}^{-x/k}(1+k/x)\}$ (its x is our μ); S2 writes the sectors $P_{k,\sigma} = \frac{A_k}{4\sqrt{3k}\,x}\mathrm{e}^{\sigma\mu/k}(1-\sigma k/\mu)$ (its x is our N); S3 writes $p(n) = D\mu^{-3}\sum_k A_k\sqrt{k}[(\frac{\mu}{k}-1)\mathrm{e}^{\mu/k} + (\frac{\mu}{k}+1)\mathrm{e}^{-\mu/k}]$. Factoring μ/k out of S3's bracket and using $\sqrt{k}\cdot(1/k) = 1/\sqrt{k}$ turns S3 into S1, and $\kappa/\mu^2 = 1/(4\sqrt{3k}\,N)\cdot\sqrt{k}$ turns S1 into S2. All three are eq. (4.16). We adopt the second form of eq. (4.16) because it displays the Part 0 model $\kappa\mathrm{e}^{\mu}\mu^{-2}(1-1/\mu)$ at $k=1$, $\sigma=+$ verbatim.

4.3.3 The dominant envelope and the exact relative factorisation

Definition 4.16 (Dominant envelope, actions)**.** Put

$$F(\mu) = P_{1,+} = \kappa\,\frac{\mathrm{e}^{\mu}}{\mu^2}\left(1-\frac{1}{\mu}\right) = \kappa\,\mathrm{e}^{\mu}\,\frac{\mu-1}{\mu^3}, \tag{4.20}$$

and for $k \geq 1$, $\sigma = \pm 1$ (excluding $(1,+)$) the *action*

$$\alpha_{k,\sigma} = 1 - \frac{\sigma}{k} \qquad \text{so that} \qquad \alpha_{1,-} = 2, \quad \alpha_{k,+} = 1 - \frac{1}{k}, \quad \alpha_{k,-} = 1 + \frac{1}{k}. \tag{4.21}$$

Proposition 4.17 (Exact relative factorisation)**.** *For $\mu > 1$,*

$$p(n) = F(\mu)\big(1 + \mathcal{R}(\mu; n)\big), \tag{4.22}$$

where, with the amplitude ratios

$$r_{k,\sigma}(\mu; n) = \frac{A_k(n)}{\sqrt{k}}\,\frac{1-\sigma k/\mu}{1-1/\mu}, \tag{4.23}$$

one has the finite-level resolved form, valid for each $K \geq 1$,

$$\mathcal{R} = r_{1,-}\,\mathrm{e}^{-2\mu} + \sum_{k=2}^{K}\sum_{\sigma=\pm 1} r_{k,\sigma}(\mu; n)\,\mathrm{e}^{-\alpha_{k,\sigma}\mu} + \mathcal{R}_{>K}(\mu; n), \tag{4.24}$$

and the exact paired form

$$\mathcal{R}(\mu; n) = \frac{1+\mu^{-1}}{1-\mu^{-1}}\mathrm{e}^{-2\mu} + \frac{2\mathrm{e}^{-\mu}}{1-\mu^{-1}}\sum_{k\geq 2}\frac{A_k(n)}{\sqrt{k}}\,g\Big(\frac{\mu}{k}\Big), \qquad \mathcal{R}_{>K} = \frac{2\mathrm{e}^{-\mu}}{1-\mu^{-1}}\sum_{k>K}\frac{A_k(n)}{\sqrt{k}}g\Big(\frac{\mu}{k}\Big). \tag{4.25}$$

Proof. Divide eq. (4.16) by eq. (4.20): $P_{k,\sigma}/F = \frac{A_k}{\sqrt{k}}\frac{1-\sigma k/\mu}{1-1/\mu}\mathrm{e}^{(\sigma/k-1)\mu}$, which is eq. (4.23) times $\mathrm{e}^{-\alpha_{k,\sigma}\mu}$; the case $k = 1, \sigma = -1$ has $A_1 = 1$ and gives the first term of eq. (4.25). For $k \geq 2$ the paired sum $P_{k,+} + P_{k,-} = \frac{2\kappa}{\mu^2}\frac{A_k}{\sqrt{k}}g(\mu/k)$ from eq. (4.19), divided by $F = \kappa\mathrm{e}^{\mu}\mu^{-2}(1-1/\mu)$, gives the second term. $\square$

Remark 4.18 (What the Hardy–Ramanujan formula omits). Equation (4.2) is $F(\mu)$ after replacing N by n and discarding $1 - 1/\mu$. Each of those two simplifications costs a relative $\mathcal{O}(n^{-1/2})$; together they account for the entire algebraic half-power series of section 4.5. Equation (4.22) loses nothing: it records every Ford-circle arc, in both signs, with its exact algebraic amplitude.

4.4 Ordering of the sectors and tail estimates

4.4.1 The action-one accumulation

The relative actions of eq. (4.21) are

$$\tfrac{1}{2},\ \tfrac{2}{3},\ \tfrac{3}{4},\ \ldots \uparrow 1 \qquad \text{and} \qquad 2,\ \tfrac{3}{2},\ \tfrac{4}{3},\ \ldots \downarrow 1.$$

The first family is increasing with supremum 1; the second is decreasing with infimum 1 and *has no least element.* Two consequences must be stated before any transseries language is used.

Definition 4.19 (Rademacher transseries)**.** A *Rademacher transseries* for p is the datum of

1. for each $K \geq 1$, the finite family of resolved sectors $\{(k,\sigma) : k \leq K\}$ with the coefficients of section 4.5, treated as independent formal exponential variables (so that coefficient extraction takes place in a finitely generated multivariate power-series ring), together with

2. the exact paired remainder $\mathcal{R}_{>K}$ of eq. (4.25), retained as one analytic block.

The limit $K \to \infty$ is taken analytically, through the convergence in theorem 4.12, and never by claiming that the whole support is a locally finite Hahn grid.

Remark 4.20 (Why the pairing is mandatory). For $k \gg \mu$ each of $P_{k,+}$ and $P_{k,-}$ separately contains a term proportional to $\sqrt{k}\,/\mu$, and $\sum_k \sqrt{k}/\mu$ diverges. Only in the sum do those pieces cancel: $2g(\mu/k) = \frac{2}{3}(\mu/k)^2 + O((\mu/k)^4)$, which makes the kth paired term $\mathcal{O}(k^{-3/2})$. Thus eq. (4.17) prescribes an order of summation; it is not an absolutely rearrangeable double series. All three sources say this; S2 and S3 say it most sharply, and theorem 4.19 follows S2.

Remark 4.21 (Truncating "through $\mathrm{e}^{-\mu}$" is not a prescription). Because the positive actions accumulate at 1, an instruction such as "retain all terms down to relative order $\mathrm{e}^{-\mu}$" selects infinitely many sectors and is meaningless. A legitimate truncation names either a Rademacher level K, or an action cutoff $\eta < 1$ (which then names finitely many positive sectors, namely $k \le (1-\eta)^{-1}$), or a target accuracy together with theorem 4.23.

4.4.2 The tail bound

Lemma 4.22 (Kernel bound)**.** *For $z \ge 0$,*

$$0 \le g(z) \le \frac{z^2}{3}\cosh z \le \frac{z^2}{3}\mathrm{e}^z, \qquad \textit{hence} \qquad g(z) \le \frac{\cosh 1}{3} z^2 \quad \textit{for } 0 \le z \le 1. \tag{4.26}$$

The factor $\cosh z$ *may not be dropped:* $g(1) = \mathrm{e}^{-1} = 0.3679\ldots > \frac{1}{3}$.

Proof. By eq. (4.18), $g(z) = \sum_{m\ge1} \frac{2m}{(2m+1)!} z^{2m}$ has non-negative coefficients, whence $g \ge 0$. For the upper bound compare with $\frac{z^2}{3}\cosh z = \sum_{m\ge1} \frac{z^{2m}}{3(2m-2)!}$ termwise: the claim is $\frac{2m}{(2m+1)!} \le \frac{1}{3(2m-2)!}$, i.e. $6m(2m-2)! \le (2m+1)!$, i.e. $6m \le (2m+1)2m(2m-1)$, which holds for all $m \ge 1$ with equality at $m = 1$. Monotonicity of cosh gives the last inequality on $[0,1]$, and $g(1) = \cosh 1 - \sinh 1 = \mathrm{e}^{-1}$ shows the factor cannot be dropped. $\square$

Theorem 4.23 (Explicit finite-K tail bound)**.** *Let $\mu > 1$ and $K \ge 1$. Then*

$$\left|\mathcal{R}_{>K}(\mu; n)\right| \le \frac{4\mu^2}{3\,(1-\mu^{-1})\sqrt{K}} \exp\left[-\left(1 - \frac{1}{K+1}\right)\mu\right]. \tag{4.27}$$

The same bound holds with $A_k(n)$ replaced by $\widetilde{A}_k(\nu)$ for any real ν, since $|\widetilde{A}_k| \le \varphi(k) \le k$ as well.

Proof. From eq. (4.25), $|A_k| \le k$ and theorem 4.22,

$$|\mathcal{R}_{>K}| \le \frac{2\mathrm{e}^{-\mu}}{1-\mu^{-1}} \sum_{k>K} \sqrt{k}\, g\!\left(\frac{\mu}{k}\right) \le \frac{2\mathrm{e}^{-\mu}}{1-\mu^{-1}} \cdot \frac{\mu^2}{3} \sum_{k>K} k^{-3/2}\mathrm{e}^{\mu/k} \le \frac{2\mu^2\mathrm{e}^{-\mu+\mu/(K+1)}}{3(1-\mu^{-1})} \sum_{k>K} k^{-3/2},$$

and $\sum_{k>K} k^{-3/2} \le \int_K^\infty t^{-3/2}\mathrm{d}t = 2K^{-1/2}$. $\square$

Theorem 4.24 (Sharper fixed-K tail)**.** *For each fixed $K \ge 1$, as $\mu \to \infty$,*

$$\sum_{k>K} \left(P_{k,+} + P_{k,-}\right) = \mathcal{O}_K\!\left(\mu^{-1/2}\mathrm{e}^{\mu/(K+1)}\right), \qquad \mathcal{R}_{>K} = \mathcal{O}_K\!\left(\mu^{3/2}\mathrm{e}^{-(1-1/(K+1))\mu}\right). \tag{4.28}$$

Proof. Split at $k = \mu$. For $K < k \le \mu$ use $|A_k| \le k$ and $2g(\mu/k) \le 2\cosh(\mu/k) \le 2\mathrm{e}^{\mu/(K+1)}$ in eq. (4.19), so those terms contribute at most $\frac{2\kappa}{\mu^2} \cdot 2\mathrm{e}^{\mu/(K+1)} \sum_{k\le\mu} \sqrt{k} = \mathcal{O}(\mu^{-1/2}\mathrm{e}^{\mu/(K+1)})$ since $\sum_{k\le\mu}\sqrt{k} = \mathcal{O}(\mu^{3/2})$. For $k > \mu$ put $z = \mu/k \in (0,1)$; then $g(z) \le \frac{1}{3}z^2\cosh 1$ by theorem 4.22, and the kth term is $\mathcal{O}(\mu^{-2}\sqrt{k}\cdot\mu^2 k^{-2}) = \mathcal{O}(k^{-3/2})$, whose tail is $\mathcal{O}(\mu^{-1/2})$. Dividing by $F \asymp \kappa\mu^{-2}\mathrm{e}^\mu$ gives the relative statement. $\square$

Remark 4.25 (Which bound to use). Theorem 4.23 (from S2) is explicit and uniform in K; theorem 4.24 (from S1 and S3) is sharper in μ by a factor $\mu^{-1/2}$ but only for fixed K. Both are needed: the first to certify a computation, the second to state clean asymptotic orders. S1 and S3 state only the second, S2 only the first.

Remark 4.26 (Sharper amplitude bounds). All constants above come from $|A_k| \le k$. Lehmer's explicit remainder estimates and the effective error bounds of Banerjee–Paule–Radu–Schneider (theorem 4.100) are much sharper, and Weil-type bounds for the Kloosterman-like sums A_k give $|A_k(n)| \ll_\varepsilon k^{1/2+\varepsilon}$ for many k. Substituting any such bound in the proofs of theorems 4.23 and 4.24 improves the powers of μ but changes no statement of this chapter, all of which are of the form "a fixed Rademacher level is exponentially accurate". For a *specific* residue class some A_k vanish, so the first non-zero omitted sector can be much smaller than the generic estimate.

4.4.3 An exact condensation at the accumulation level

Right at the accumulation action 1, an alternative to sector-by-sector bookkeeping is to sum the kernel expansion first. This is S3's contribution and has no counterpart in S1 or S2.

Definition 4.27 (Rademacher–Kloosterman zeta)**.** For $\Re s > 2$ and $n \in \mathbb{Z}$ put $\mathcal{Z}_n(s) = \sum_{k\ge1} \dfrac{A_k(n)}{k^s}$; the series converges absolutely there by theorem 4.7(iii). The same definition with $\widetilde{A}_k(\nu)$ defines $\mathcal{Z}_\nu(s)$ for real ν.

Theorem 4.28 (Exact zeta condensation)**.** *For every integer* $n \ge 1$,

$$p(n) = \kappa \sum_{r\ge1} \frac{4r}{(2r+1)!}\, \mu^{2r-2}\, \mathcal{Z}_n\!\left(2r + \frac{1}{2}\right), \tag{4.29}$$

and the double series obtained by expanding $\mathcal{Z}_n$ *converges absolutely.*

Proof. Insert $2g(z) = \sum_{r\ge1} \frac{4r}{(2r+1)!} z^{2r+1} \cdot z^{-1}$ — more precisely $2g(z) = \sum_{r\ge1} \frac{4r}{(2r+1)!} z^{2r}$ — into eq. (4.19) with $z = \mu/k$:

$$p(n) = \frac{\kappa}{\mu^2} \sum_{k\ge1} \frac{A_k(n)}{\sqrt{k}} \sum_{r\ge1} \frac{4r}{(2r+1)!}\frac{\mu^{2r}}{k^{2r}} = \kappa \sum_{k,r} \frac{4r}{(2r+1)!} \mu^{2r-2} \frac{A_k(n)}{k^{2r+1/2}}.$$

Absolute convergence: $\sum_k |A_k| k^{-2r-1/2} \le \sum_k k^{-2r+1/2} < \infty$ for $r \ge 1$, and $\sum_{r\ge1} \frac{4r}{(2r+1)!}\mu^{2r-2}\zeta(2r - \frac{1}{2})$ converges for every fixed μ because $\zeta(2r-\frac{1}{2}) \to 1$ and $\sum_r \frac{4r\mu^{2r-2}}{(2r+1)!} < \infty$. Tonelli's theorem justifies the interchange, giving eq. (4.29). □

Remark 4.29. Equation (4.29) orders nothing by asymptotic size — it is an exact analytic rearrangement, not a transseries. Its use is precisely where the transseries language fails: it packages the whole action-one accumulation into Dirichlet values $\mathcal{Z}_n(2r+\frac{1}{2})$, which can be used as a single block in the inverse formulas of section 4.9.

4.5 All-order coefficients in the unshifted variable

The shifted phase μ makes each sector elementary; the price is that the conventional scale is $n^{-1/2}$. The entire infinite algebraic array of this subject comes from re-expanding the single shift $N = n - 1/24$. This section gives all of it, in one formula valid for every sector.

4.5.1 The sector generating function

Theorem 4.30 (Sectorwise convergent half-power expansion). *Fix $k \geq 1$ and $\sigma \in \{+1, -1\}$, and set $t = n^{-1/2}$. Then*

$$P_{k,\sigma}(n) = \frac{A_k(n)}{4\sqrt{3k}\,n} \exp\left(\frac{\sigma\lambda\sqrt{n}}{k}\right) \sum_{m\geq 0} c_m^{(k,\sigma)}\, n^{-m/2}, \tag{4.30}$$

where the $c_m^{(k,\sigma)}$ are the Taylor coefficients at $t = 0$ of

$$\Omega_{k,\sigma}(t) = \frac{1}{1 - t^2/24} \exp\left[\frac{\sigma\lambda}{kt}\left(\sqrt{1 - \frac{t^2}{24}} - 1\right)\right]\left(1 - \frac{\sigma kt}{\lambda\sqrt{1 - t^2/24}}\right). \tag{4.31}$$

The function $\Omega_{k,\sigma}$ is holomorphic on $|t| < \sqrt{24}$, so the series in eq. (4.30) *converges for every real $n > 1/24$; in particular $c_0^{(k,\sigma)} = 1$.*

Proof. With $t = n^{-1/2}$ we have $N = n(1 - t^2/24)$ and $\mu = \lambda\sqrt{N} = (\lambda/t)\sqrt{1 - t^2/24}$. Substituting in eq. (4.16),

$$P_{k,\sigma} = \frac{A_k(n)}{4\sqrt{3k}\,n} \cdot \frac{1}{1 - t^2/24} \cdot \mathrm{e}^{\sigma\mu/k}\left(1 - \frac{\sigma kt}{\lambda\sqrt{1 - t^2/24}}\right),$$

and $\mathrm{e}^{\sigma\mu/k} = \mathrm{e}^{\sigma\lambda/(kt)} \exp[\frac{\sigma\lambda}{kt}(\sqrt{1 - t^2/24} - 1)]$ with $\mathrm{e}^{\sigma\lambda/(kt)} = \mathrm{e}^{\sigma\lambda\sqrt{n}/k}$. This is eq. (4.30) with eq. (4.31). Holomorphy: the only candidate singularities of eq. (4.31) in $|t| < \sqrt{24}$ are $t = 0$ and the zeros of $\sqrt{1 - t^2/24}$, and the latter lie on $|t| = \sqrt{24}$. At $t = 0$ the exponent is $\frac{\sigma\lambda}{kt}(\sqrt{1 - t^2/24} - 1) = -\frac{\sigma\lambda t}{48k} + \mathcal{O}(t^3)$, so the singularity is removable, and the last factor is holomorphic there as well. Setting $t = 0$ gives $\Omega_{k,\sigma}(0) = 1$. ☐

S1, Theorem "Universal sector coefficient generator". S1 states only that the series "converges for all sufficiently small $|t|$ and supplies an asymptotic expansion to arbitrary order". This understates the truth and, read literally, invites the error the next remark forbids: it suggests the half-power series is a merely asymptotic object. S2 and S3 both identify the radius as exactly $\sqrt{24}$, i.e. convergence for every real $n > 1/24$; theorem 4.30 adopts their statement with the branch-point argument made explicit. **Consequence.** Nothing in this chapter is a divergent series in $n^{-1/2}$, and no least-term truncation is available; see theorem 4.37.

4.5.2 A finite closed formula for every sector coefficient

Write

$$\beta = \beta_{k,\sigma} = \frac{\sigma\pi}{6k}, \qquad \text{so that} \qquad \frac{\sigma\lambda}{k} = 2\sqrt{6}\,\beta, \qquad \frac{\sigma k}{\lambda} = \frac{1}{12\beta} \cdot \sqrt{6}\,, \tag{4.32}$$

the two identities following from $\lambda = \pi\sqrt{6}/3$ and $k = \sigma\pi/(6\beta)$. Thus each sector is the *same* function of the single parameter β; the sector label enters nowhere else.

Theorem 4.31 (Finite coefficient formula, every sector). *For every $k \geq 1$, $\sigma = \pm 1$ and $m \geq 0$,*

$$c_m^{(k,\sigma)} = \frac{1}{(-4\sqrt{6})^m} \sum_{j=0}^{\lfloor (m+1)/2 \rfloor} \binom{m+1}{j} \frac{m + 1 - j}{(m + 1 - 2j)!}\, \beta_{k,\sigma}^{m-2j}, \tag{4.33}$$

terms with a negative factorial index being absent. In particular $c_m^{(k,\sigma)} \in \mathbb{Q}[\beta, \beta^{-1}] \cdot \sqrt{6}^{\,m \bmod 2}$, and the dependence on the sector is only through $\beta_{k,\sigma} = \sigma\pi/(6k)$.

Proof. Put $u = -t/(4\sqrt{6})$, so that $t^2/24 = 4u^2$ and, by eq. (4.32), $\frac{\sigma\lambda}{kt} = \frac{2\sqrt{6}\beta}{-4\sqrt{6}\,u} = -\frac{\beta}{2u}$ and $\frac{\sigma kt}{\lambda} = -\frac{4\sqrt{6}\,u\sqrt{6}}{12\beta} = -\frac{2u}{\beta}$. Writing $s = \sqrt{1-4u^2}$, eq. (4.31) becomes

$$G(u) := \Omega_{k,\sigma}(-4\sqrt{6}\,u) = \frac{e^{\beta T}}{s^2}\left(1 + \frac{2u}{\beta s}\right), \qquad T = T(u) = \frac{1-s}{2u}, \tag{4.34}$$

where we used $-\frac{\beta}{2u}(s-1) = \beta\frac{1-s}{2u} = \beta T$ and $s^2 = 1-4u^2$. The Catalan variable T satisfies

$$T = u(1+T^2), \qquad u = \frac{T}{1+T^2}, \qquad s = \frac{1-T^2}{1+T^2}, \qquad du = \frac{1-T^2}{(1+T^2)^2}\,dT. \tag{4.35}$$

(The first identity follows from $2uT = 1-s$ and $s^2 = 1-4u^2$; the rest are algebra.) Substituting eq. (4.35) into eq. (4.34) gives the exact differential identity

$$G(u)\,du = e^{\beta T}\left(\frac{1}{1-T^2} + \frac{2T}{\beta(1-T^2)^2}\right)dT = \frac{1}{\beta}\,d\!\left(\frac{e^{\beta T}}{1-T^2}\right), \tag{4.36}$$

because $\frac{d}{dT}\frac{e^{\beta T}}{1-T^2} = e^{\beta T}\big(\frac{\beta}{1-T^2} + \frac{2T}{(1-T^2)^2}\big)$. Now let $D_m = [u^m]\,G(u) = (-4\sqrt{6})^m c_m^{(k,\sigma)}$. Then

$$D_m = \operatorname*{Res}_{u=0}\frac{G(u)du}{u^{m+1}} = \frac{1}{\beta}\operatorname*{Res}_{T=0}\frac{(1+T^2)^{m+1}}{T^{m+1}}\,d\!\left(\frac{e^{\beta T}}{1-T^2}\right),$$

using $u^{-m-1} = T^{-m-1}(1+T^2)^{m+1}$. The residue of an exact differential vanishes, so integrating by parts,

$$D_m = -\frac{1}{\beta}\operatorname*{Res}_{T=0}\frac{e^{\beta T}}{1-T^2}\,d\big(T^{-m-1}(1+T^2)^{m+1}\big) = \frac{m+1}{\beta}\operatorname*{Res}_{T=0}e^{\beta T}(1+T^2)^m\,\frac{dT}{T^{m+2}},$$

where we used $d(T^{-m-1}(1+T^2)^{m+1}) = (1+T^2)^m T^{-m-2}\big(-(m+1)(1+T^2) + 2(m+1)T^2\big)dT = -(m+1)(1+T^2)^m(1-T^2)T^{-m-2}dT$, and the factor $1-T^2$ cancels. Hence

$$D_m = \frac{m+1}{\beta}[T^{m+1}]\,e^{\beta T}(1+T^2)^m = \frac{m+1}{\beta}\sum_{j\ge 0}\binom{m}{j}\frac{\beta^{m+1-2j}}{(m+1-2j)!},$$

the sum running over $0 \le j \le \lfloor (m+1)/2 \rfloor$. Finally $(m+1)\binom{m}{j} = (m+1-j)\binom{m+1}{j}$ and division by $(-4\sqrt{6})^m$ gives eq. (4.33). $\square$

Remark 4.32 (Why a finite sum exists). A square-root shift would normally give an infinite convolution at each order. The Catalan coordinate T absorbs the square root, and the derivative already present in Rademacher's formula turns the integrand into the exact differential eq. (4.36). After the change-of-variable Jacobian cancels the denominator $1-T^2$, only the polynomial $(1+T^2)^m$ survives.

Remark 4.33 (Provenance and the scope of the generalisation). For $(k,\sigma) = (1,+1)$, eq. (4.33) at $\beta = \pi/6$ is O'Sullivan's coefficient formula, also used by Banerjee, Paule, Radu and Schneider in their error analysis. S1 proves only that case, by the same Catalan substitution. S2 and S3 both observe that the proof never uses $k=1$ or $\sigma=+1$ — the sector enters only through β — and state the general formula; S3 gives a second, generating-function proof through the identity $\sum_{j\ge 0}\binom{m+2j}{j}q^j = C(q)^m/\sqrt{1-4q}$ for the Catalan series C. The residue proof above is S2's, written out. We record the generalisation as the principal coefficient statement of the chapter, since it covers every exponentially small sector at no extra cost.

The first coefficients, as functions of β, are

$$c_0 = 1, \qquad c_1 = -\frac{\sqrt{6}\,(\beta^2+2)}{24\beta}, \qquad c_2 = \frac{\beta^2+12}{192},$$
$$c_3 = -\frac{\sqrt{6}\,(\beta^4+36\beta^2+72)}{13824\,\beta}, \quad c_4 = \frac{\beta^4+80\beta^2+720}{221184}, \quad c_5 = -\frac{\sqrt{6}\,(\beta^6+150\beta^4+3600\beta^2+7200)}{26542080\,\beta}, \tag{4.37}$$

$$c_6 = \frac{\beta^6 + 252\beta^4 + 12600\beta^2 + 100800}{637009920}, \qquad c_7 = -\frac{\sqrt{6}\big(\beta^8 + 392\beta^6 + 35280\beta^4 + 705600\beta^2 + 1411200\big)}{107017666560\,\beta}. \tag{4.38}$$

These were recomputed here by expanding eq. (4.31) symbolically to order t^9 and comparing with eq. (4.33) coefficient by coefficient in exact arithmetic; the two agree identically. The parity pattern visible in eq. (4.37) — c_m is $\sqrt{6}$ times an odd Laurent polynomial in β for odd m, and a polynomial in β^2 for even m — is immediate from eq. (4.33), since $m - 2j \equiv m \pmod 2$ and the prefactor contributes $\sqrt{6}^{\,m}$.

4.5.3 The Bell-polynomial route

Equation (4.33) is the fastest generator. For symbolic systems, or when the sector coefficients are wanted as a byproduct of a general composition engine, the following route uses only the primitives of theorems 1.9 and 1.10.

Proposition 4.34 (Logarithmic coefficients of a sector)**.** *Write* $\log \Omega_{k,\sigma}(t) = \sum_{j\ge 1} \ell_j t^j$ *(suppressing the sector labels) and put* $a = 1/24$*. Then, for* $j \ge 1$*,*

$$\ell_j = \mathbf{1}_{2|j}\,\frac{a^{j/2}}{j/2} + \mathbf{1}_{2\nmid j}\,\frac{\sigma\lambda}{k}\binom{1/2}{(j+1)/2}(-a)^{(j+1)/2} - \sum_{\substack{1\le q\le j\\ 2|(j-q)}} \frac{1}{q}\left(\frac{\sigma k}{\lambda}\right)^{q} \frac{(q/2)_{(j-q)/2}}{\big((j-q)/2\big)!}\, a^{(j-q)/2}, \tag{4.39}$$

where $(x)_r = x(x+1)\cdots(x+r-1)$*. Consequently the sector coefficients are the complete exponential Bell polynomials*

$$c_m = \frac{1}{m!}\, B_m\big(1!\,\ell_1, 2!\,\ell_2, \ldots, m!\,\ell_m\big), \tag{4.40}$$

equivalently the triangular recurrence

$$c_0 = 1, \qquad c_m = \frac{1}{m}\sum_{j=1}^{m} j\,\ell_j\, c_{m-j} \quad (m \ge 1). \tag{4.41}$$

Proof. Taking logarithms in eq. (4.31),

$$\log\Omega = -\log(1-at^2) + \frac{\sigma\lambda}{kt}\big(\sqrt{1-at^2} - 1\big) + \log\left(1 - \frac{\sigma k}{\lambda} t(1-at^2)^{-1/2}\right).$$

The first term contributes a^r/r at degree $2r$. For the second, $\sqrt{1-at^2} - 1 = \sum_{r\ge1}\binom{1/2}{r}(-a)^r t^{2r}$, so dividing by t contributes $\frac{\sigma\lambda}{k}\binom{1/2}{r}(-a)^r$ at degree $2r-1$. For the third, expand $\log(1-w) = -\sum_{q\ge1} w^q/q$ with $w = \frac{\sigma k}{\lambda} t(1-at^2)^{-1/2}$ and use $(1-at^2)^{-q/2} = \sum_{r\ge0} \frac{(q/2)_r}{r!} a^r t^{2r}$; the term of degree $j = q + 2r$ is the displayed sum. Equation (4.40) is the exponential formula of theorem 1.10, and eq. (4.41) follows by comparing coefficients in $\Omega' = (\log\Omega)'\,\Omega$. □

For orientation, at general (k,σ) and $a = 1/24$,

$$\begin{aligned} \ell_1 &= -\frac{\sigma\lambda a}{2k} - \frac{\sigma k}{\lambda}, & \ell_2 &= a - \frac{k^2}{2\lambda^2},\\ \ell_3 &= -\frac{\sigma\lambda a^2}{8k} - \frac{\sigma k a}{2\lambda} - \frac{\sigma k^3}{3\lambda^3}, & \ell_4 &= \frac{a^2}{2} - \frac{k^2 a}{2\lambda^2} - \frac{k^4}{4\lambda^4}. \end{aligned} \tag{4.42}$$

4.5.4 The dominant Poincaré expansion, and what it does not say

Definition 4.35. $\omega_m = c_m^{(1,+1)}$, i.e. eq. (4.33) at $\beta = \pi/6$.

Corollary 4.36. *For each fixed $M \geq 1$,*

$$p(n) = \frac{e^{\lambda\sqrt{n}}}{4\sqrt{3}\,n}\left(\sum_{m=0}^{M-1}\omega_m n^{-m/2} + \mathcal{O}_M(n^{-M/2})\right), \tag{4.43}$$

with

$$\omega_0 = 1, \quad \omega_1 = -\frac{\sqrt{6}\,(\pi^2+72)}{144\pi}, \quad \omega_2 = \frac{\pi^2+432}{6912}, \quad \omega_3 = -\frac{\sqrt{6}\,(\pi^4+1296\pi^2+93312)}{2985984\,\pi}, \tag{4.44}$$

$$\omega_4 = \frac{\pi^4+2880\pi^2+933120}{286654464}, \tag{4.45}$$

numerically $\omega_1 = -0.443287976873582391\ldots, \omega_2 = 0.063927894155250197\ldots, \omega_3 = -0.027730933907464250\ldots,$ $\omega_4 = 0.003354707463289919\ldots$.

Proof. By theorem 4.30 with $(k,\sigma) = (1,+1)$, the displayed sum plus a convergent tail is exactly $P_{1,+}(n)$ divided by $e^{\lambda\sqrt{n}}/(4\sqrt{3}\,n)$; the tail after M terms is $\mathcal{O}_M(n^{-M/2})$ because $\Omega_{1,+}$ is holomorphic at 0. The difference $p(n) - P_{1,+}(n)$ is $\mathcal{O}(\mu^{3/2}e^{-\mu/2})$ relative to $P_{1,+}$ by theorem 4.24 with $K = 1$, hence beyond every algebraic order. The closed forms follow from eq. (4.33) at $\beta = \pi/6$; they were recomputed here in exact arithmetic and agree with the tables of S1 and S2 (which print the same numbers in two different but equal groupings). ☐

Convergent sector versus asymptotic total. This is the trap of the subject and all three sources are, at some point, loose about it. The three statements below are *different*:

1. $\sum_{m\geq 0}\omega_m n^{-m/2}$ *converges* for every $n > 1/24$ (theorem 4.30);
2. its sum is $P_{1,+}(n) \cdot 4\sqrt{3}\,n\,e^{-\lambda\sqrt{n}}$, *not* $p(n) \cdot 4\sqrt{3}\,n\,e^{-\lambda\sqrt{n}}$;
3. eq. (4.43) is an *asymptotic* statement about $p(n)$, whose error at every fixed order is a sum of two utterly different things: a convergent algebraic tail of size $\mathcal{O}(n^{-M/2})$ and an exponentially small Rademacher remainder of size $\mathcal{O}(\mu^{3/2}e^{-\mu/2})$ relative to the dominant sector.

S1's "Dominant Poincaré expansion" subsection writes $p(n) \sim \cdots$ without separating (ii) from (iii) and, in its Theorem on the sector generating function, weakens (i) to "sufficiently small $|t|$". S3's remark after its dominant Poincaré display states the correct trichotomy in one sentence. S2 states it as a standalone remark and is the presentation adopted here. **The practical consequence:** summing the ω_m series to convergence does *not* compute $p(n)$ to arbitrary accuracy; it computes $P_{1,+}(n)$, and the residual error stalls at the $e^{-\mu/2}$ floor.

Remark 4.37 (Least-term truncation degenerates here). Theorem 1.46 determines the optimal truncation point of a *factorially divergent* algebraic series and the size of the resulting beyond-all-orders floor. Its hypothesis fails for $p(n)$: by theorem 4.30 the algebraic series of each sector has a finite positive radius of convergence, so there is no least term and no truncation-induced floor. The floor that does exist — the $e^{-\mu/2}$ Rademacher gap — is not produced by resummation ambiguity but is supplied explicitly by the next sector. The partition function is thus the degenerate, best possible case of theorem 1.46: the exponentially small sectors are known exactly and independently of the perturbative data, and the optimal algebraic truncation is simply "as many terms as one wishes". This is the structural reason the numerical accuracy in table 4.3 improves so violently with K but not at all with extra algebraic orders past the $e^{-\mu/2}$ level.

4.6 The multiplicative reciprocal $1/p(n)$

The phrase "inverse of the partition numbers" is ambiguous. The reciprocal is the easy reading and is disposed of first; it needs no reversion at all.

Proposition 4.38 (Exact reciprocal transseries)**.** *Let $\mu > 1$ be large enough that $|\mathcal{R}(\mu; n)| < 1$, with $\mathcal{R}$ as in eq.* (4.25)*. Then*

$$\frac{1}{p(n)} = \frac{\mu^2}{\kappa} \frac{e^{-\mu}}{1-\mu^{-1}} \sum_{r\geq 0} (-1)^r \mathcal{R}(\mu; n)^r, \tag{4.46}$$

the series converging. As a formal identity in theorem 4.19 it holds without the inequality. If $\mathcal{R}$ is written at finite level as a linear form $\sum_{j\in J_K} r_j(\mu; n)\, q_j$ in independent exponential variables $q_j = e^{-\alpha_j \mu}$, then for every multi-index $\boldsymbol{\alpha} = (\alpha_j)_{j\in J_K} \neq \mathbf{0}$,

$$[\boldsymbol{q}^{\boldsymbol{\alpha}}]\left(1+\mathcal{R}\right)^{-1} = (-1)^{|\boldsymbol{\alpha}|} \frac{|\boldsymbol{\alpha}|!}{\boldsymbol{\alpha}!} \prod_{j\in J_K} r_j(\mu; n)^{\alpha_j}. \tag{4.47}$$

Proof. Invert eq. (4.22) and expand $(1+\mathcal{R})^{-1}$ geometrically; $1/F = \mu^2 e^{-\mu}/(\kappa(1-\mu^{-1}))$. For eq. (4.47), only the term $(-1)^{|\boldsymbol{\alpha}|}\mathcal{R}^{|\boldsymbol{\alpha}|}$ of the geometric series can produce $\boldsymbol{q}^{\boldsymbol{\alpha}}$, and the multinomial theorem supplies the coefficient. ☐

Corollary 4.39 (Reciprocal Poincaré coefficients)**.** *Define η_m by $\left(\sum_{m\geq 0}\omega_m t^m\right)^{-1} = \sum_{m\geq 0}\eta_m t^m$, i.e.*

$$\eta_0 = 1, \qquad \eta_m = -\sum_{j=1}^{m}\omega_j \eta_{m-j} \quad (m\geq 1), \qquad \textit{equivalently} \qquad \eta_m = \frac{1}{m!}B_m\left(-1!\ell_1, \ldots, -m!\ell_m\right) \tag{4.48}$$

with $\ell_j = \ell_j^{(1,+1)}$ of eq. (4.39)*. Then for each fixed M,*

$$\frac{1}{p(n)} = 4\sqrt{3}\, n\, e^{-\lambda\sqrt{n}} \left(\sum_{m=0}^{M-1}\eta_m n^{-m/2} + \mathcal{O}_M(n^{-M/2})\right), \tag{4.49}$$

and $\sum_{m\geq 0}\eta_m t^m$ converges on $|t| < \sqrt{24}$ minus the zero set of $\Omega_{1,+}$, to $1/\Omega_{1,+}(t)$.

Proof. The recurrence is Cauchy inversion of a power series with unit constant term; the Bell form is the exponential formula applied to $\Omega_{1,+}^{-1} = \exp(-\sum \ell_j t^j)$. Convergence is inherited from theorem 4.30 and $\Omega_{1,+}(0) = 1$. The asymptotic statement follows from theorem 4.36 exactly as there; the same caveats of the repair box after theorem 4.36 apply verbatim, with "$P_{1,+}$" replaced by "$1/P_{1,+}$". ☐

Remark 4.40. The reciprocal therefore requires no new analytic input and no series reversion. It is recorded because two of the three sources devote a section to it, and because it is the natural place to notice that the multinomial formula eq. (4.47) is the $|\boldsymbol{\alpha}|$-linear shadow of the far harder reversion formulas of section 4.9.

4.7 What "the inverse" of a discrete sequence means here

Proposition 4.41 (Strict monotonicity)**.** *$p(n) < p(n+1)$ for every $n \geq 1$.*

Proof. Adjoining a part 1 injects the partitions of n into those of $n+1$, and the one-part partition $(n+1)$ is not in the image. ☐

By theorem 1.37 there are three distinct objects, and this chapter needs all three.

1. **The integer staircase** (the generalised inverse)

$$p^{\leftarrow}(y) = \min\{n \geq 1 : p(n) \geq y\}, \qquad y \geq 1. \tag{4.50}$$

Well defined by theorem 4.41. It is integer valued with unit jumps, so no smooth expansion can represent it to $o(1)$.

2. **The functional inverse of a chosen interpolation.** Any strictly increasing smooth $\mathcal{P}$ with $\mathcal{P}(n) = p(n)$ has an inverse $\mathcal{N}$, and $p^{\leftarrow}(y) = \lceil \mathcal{N}(y) \rceil$ (theorem 4.44). theorem 4.9 supplies the canonical choice used here.

3. **The inverse along the range.** If one knows in advance that $y = p(n)$ for an integer n, the target is that integer. A smooth formula that is accurate to $o(1/2)$ recovers it by rounding (theorem 4.83), and the accuracy available is exponentially small, not merely $o(1)$.

Definition 4.42 (The Rademacher interpolation)**.** For real $\nu > 1/24$ put $\mu(\nu) = \frac{\pi}{6}\sqrt{24\nu - 1}$ and

$$\mathcal{P}(\nu) = \frac{2\kappa}{\mu(\nu)^2} \sum_{k\geq 1} \frac{\widetilde{A}_k(\nu)}{\sqrt{k}}\, g\biggl(\frac{\mu(\nu)}{k}\biggr), \tag{4.51}$$

with g as in eq. (4.18) and $\widetilde{A}_k$ as in theorem 4.9.

Proposition 4.43 (Interpolation, regularity, eventual monotonicity)**.** *The series eq.* (4.51) *and all its derivatives converge locally uniformly on* $(1/24, \infty)$*;* $\mathcal{P}$ *is real analytic there and* $\mathcal{P}(n) = p(n)$ *for every integer* $n \geq 1$*. Moreover there is* ν_0 *with* $\mathcal{P}'(\nu) > 0$ *for* $\nu \geq \nu_0$*, so* $\mathcal{P}$ *has a real inverse* $\mathcal{N}$ *on* $[\mathcal{P}(\nu_0), \infty)$*.*

Proof. Fix a compact $I \subset (1/24, \infty)$ and a complex neighbourhood of it on which μ is holomorphic and bounded. By theorem 4.10 every ν-derivative of $\widetilde{A}_k$ is $\mathcal{O}_r(k)$ uniformly in k there, while $g(\mu/k) = \mathcal{O}(k^{-2})$ uniformly with all parameter derivatives (from eq. (4.18) and μ bounded). Hence the kth summand of eq. (4.51) and each of its derivatives is $\mathcal{O}(k \cdot k^{-1/2} \cdot k^{-2}) = \mathcal{O}(k^{-3/2})$, and the Weierstrass M-test gives locally uniform convergence and legitimises termwise differentiation. Equality at integers is theorem 4.10 combined with theorem 4.14.

Monotonicity: write $\mathcal{P}(\nu) = F(\mu(\nu))(1 + \mathcal{R}(\mu(\nu); \nu))$ as in theorem 4.17 with $\widetilde{A}_k$ in place of A_k. We have $\mathrm{d}\mu/\mathrm{d}\nu = \lambda^2/(2\mu) > 0$ and $\mathrm{d}\log F/\mathrm{d}\mu = 1 - 3/\mu + 1/(\mu - 1) \to 1$. Differentiating $\mathcal{R}$ term by term and applying theorem 4.23 with $K = 1$ together with theorem 4.10 shows $\mathcal{R} + \partial_\mu \mathcal{R} = \mathcal{O}(\mu^3 \mathrm{e}^{-\mu/2})$; note that differentiating $\widetilde{A}_k(\nu(\mu))$ costs a factor $\mathcal{O}(k\mu)$, absorbed by replacing $k^{-3/2}$ with $k^{-1/2}$ in the tail sum and hence by one more power of μ. Thus $\mathcal{P}'(\nu) = F'(\mu)\mu'(\nu)\bigl(1 + \mathcal{O}(\mu^3 \mathrm{e}^{-\mu/2})\bigr)$, which is positive for large ν. ☐

Theorem 4.44 (Staircase from the smooth inverse)**.** *For all sufficiently large real* y*,*

$$p^{\leftarrow}(y) = \bigl\lceil \mathcal{N}(y) \bigr\rceil, \tag{4.52}$$

and $\mathcal{N}(p(n)) = n$ *exactly for every integer* $n \geq \nu_0$*.*

Proof. This is the specialisation of theorem 1.39 to the unit-spaced lattice $\mathbb{Z}$, whose separation hypothesis is theorem 4.41. Concretely: let $m = p^{\leftarrow}(y)$, so $p(m-1) < y \leq p(m)$. Since $\mathcal{P}$ is increasing on $[\nu_0, \infty)$ and $\mathcal{P}(j) = p(j)$, we get $m - 1 < \mathcal{N}(y) \leq m$, whence $\lceil \mathcal{N}(y) \rceil = m$. ☐

Remark 4.45 (The unavoidable sawtooth). For general real y the difference $p^{\leftarrow}(y) - \mathcal{N}(y)$ is the ceiling defect, of size $\mathcal{O}(1)$, with Fourier expansion

$$\lceil v \rceil - v = \frac{1}{2} + \frac{1}{\pi} \sum_{r\geq 1} \frac{\sin(2\pi r v)}{r} \qquad (v \notin \mathbb{Z}), \tag{4.53}$$

the series converging to the midpoint $1/2$ at integers where the true defect is 0. Since the first arithmetic correction to $\mathcal{N}$ is of size $y^{-1/2}$ (theorem 4.73), *exponential precision in* $\mathcal{N}$ *buys nothing for* $p^{\leftarrow}$ *at generic* y: the order-one sawtooth is an intrinsic final sector of the discrete problem, not an error to be removed. It matters only along the range $y = p(n)$, where the defect vanishes identically. This warning is S3's and is the sharpest of the three.

4.8 The Lambert core and the algebraic inverse

Throughout this section and the next, y denotes a large ordinate and

$$R = \log \frac{y}{\kappa}, \qquad \ell = \log R. \tag{4.54}$$

4.8.1 The carrier

Proposition 4.46 (Lambert carrier; specialisation of theorem 1.20). *The large solution $X = X(y)$ of $\kappa e^X X^{-2} = y$, equivalently of*

$$X - 2\log X = R, \tag{4.55}$$

is

$$X(y) = -2\, W_{-1}\!\left(-\frac{1}{2}\sqrt{\frac{\kappa}{y}}\right) = -2\, W_{-1}\!\left(-\frac{\pi}{\sqrt{24\sqrt{3}\, y}}\right). \tag{4.56}$$

Proof. This is theorem 1.20 at the dictionary values $a = 1$, $b = -2$, $C = \kappa$ of table 4.1: $X = (b/a)W_k((a/b)e^{R/b}) = -2W_k(-\frac{1}{2}e^{-R/2})$ and $e^{-R/2} = \sqrt{\kappa/y}$. The branch rule of theorem 1.20 selects $k = -1$ because $X \to +\infty$ forces $-X/2 \le -1$. The second form uses $\frac{1}{2}\sqrt{\kappa} = \frac{1}{2}\pi/\sqrt{6\sqrt{3}} = \pi/\sqrt{24\sqrt{3}}$. □

4.8.2 Flattening into logarithms

Theorem 4.47 (Carrier coefficient polynomials). *As $y \to \infty$,*

$$X \sim R + 2\ell + \sum_{j\ge 1} \frac{\Pi_j(\ell)}{R^j}, \tag{4.57}$$

in the sense of theorem 1.33, where

$$\Pi_j(\ell) = \frac{2^{j+1}}{j+1}\,[v^j]\,\big(\ell + \log(1+v)\big)^{j+1} = 2^{j+1}\sum_{q=1}^{j} \frac{s(j,q)}{(j+1-q)!}\,\ell^{\,j+1-q}, \tag{4.58}$$

with $s(j,q)$ the signed Stirling numbers of the first kind, $v(v-1)\cdots(v-j+1) = \sum_q s(j,q)v^q$. For each fixed J,

$$X = R + 2\ell + \sum_{j=1}^{J} \frac{\Pi_j(\ell)}{R^j} + \mathcal{O}\!\left(\frac{\ell^{\,J+1}}{R^{J+1}}\right). \tag{4.59}$$

Moreover $\deg \Pi_j = j$ with leading coefficient $(-1)^{j-1}2^{j+1}/j$, and $\Pi_j(0) = 0$.

Proof. Write $X = R(1+v)$ in eq. (4.55): $R + Rv - 2\log R - 2\log(1+v) = R$, i.e.

$$v = z\big(\ell + \log(1+v)\big), \qquad z = \frac{2}{R}, \tag{4.60}$$

a Lagrange fixed point $v = z\phi(v)$ with $\phi(v) = \ell + \log(1+v)$ and $\phi(0) = \ell \ne 0$. Theorem 1.14 gives $v = \sum_{m\ge 1} \frac{z^m}{m}\,[v^{m-1}]\,\phi(v)^m$; multiplying by R and putting $m = j+1$ (so $z^m R = 2^{j+1}R^{-j}$) yields the first equality of eq. (4.58). For the second, expand $(\ell + \log(1+v))^{j+1} = \sum_q \binom{j+1}{q}\ell^{\,j+1-q}\log(1+v)^q$ and use $\log(1+v)^q = q!\sum_{m\ge q} s(m,q)v^m/m!$; extracting $[v^j]$ keeps only $1 \le q \le j+1$, and the term $q = j+1$ contributes $s(j,j+1) = 0$. The surviving coefficient of $\ell^{\,j+1-q}$ is $\frac{2^{j+1}}{j+1}\binom{j+1}{q}q!\,\frac{s(j,q)}{j!} = 2^{j+1}\frac{s(j,q)}{(j+1-q)!}$, as claimed. Since $q \ge 1$, every monomial carries a positive power of ℓ, so $\Pi_j(0) = 0$; the top power is $q = 1$, with $s(j,1) = (-1)^{j-1}(j-1)!$, giving leading coefficient $2^{j+1}(-1)^{j-1}(j-1)!/j! = (-1)^{j-1}2^{j+1}/j$. The remainder statement eq. (4.59) is the general estimate of theorem 1.33 applied on the scale $\ell/R \to 0$; concretely, eq. (4.60) has an analytic solution $v = v(z,\ell)$ near $z = 0$ for ℓ in any fixed compact set, and $\ell/R \to 0$ lets Taylor's theorem with remainder be applied at order J. □

The first polynomials, recomputed here from the Lagrange form and independently from the Stirling form (they agree identically), are

$$\begin{aligned}
\Pi_1 &= 4\ell, & \Pi_2 &= -4\ell(\ell-2),\\
\Pi_3 &= \tfrac{8}{3}\ell\,(2\ell^2-9\ell+6), & \Pi_4 &= -\tfrac{8}{3}\ell\,(3\ell^3-22\ell^2+36\ell-12),\\
\Pi_5 &= \tfrac{16}{15}\ell\,(12\ell^4-125\ell^3+350\ell^2-300\ell+60), & \Pi_6 &= -\tfrac{16}{15}\ell\,(20\ell^5-274\ell^4+1125\ell^3-1700\ell^2+900\ell-120),
\end{aligned} \tag{4.61}$$

$$\Pi_7 = \tfrac{32}{105}\ell\left(120\ell^6-2058\ell^5+11368\ell^4-25725\ell^3+24500\ell^2-8820\ell+840\right), \tag{4.62}$$

so that, explicitly,

$$X = R + 2\ell + \frac{4\ell}{R} + \frac{-4\ell^2+8\ell}{R^2} + \frac{\frac{16}{3}\ell^3-24\ell^2+16\ell}{R^3} + \frac{-8\ell^4+\frac{176}{3}\ell^3-96\ell^2+32\ell}{R^4} + \cdots. \tag{4.63}$$

Remark 4.48 (Three printings of one polynomial family). The sources print eq. (4.58) in three different shapes, and it is worth recording once that they are the same family, because a reader comparing them will otherwise suspect an error.

- S1: $P_j(L) = 2^{j+1}\sum_{q=1}^{j}\frac{s(j,q)}{(j+1-q)!}L^{j+1-q}$ — identical to eq. (4.58).
- S2: $P_m(\ell) = \frac{1}{m!}\sum_{q=1}^{m}2^q(q-1)!\,s(m,q)\binom{m}{q-1}(2\ell)^{m-q+1}$. The coefficient of ℓ^{m-q+1} is $\frac{2^q(q-1)!}{m!}\binom{m}{q-1}2^{m-q+1} = \frac{2^{m+1}s(m,q)}{(m-q+1)!}$ after cancelling $\binom{m}{q-1} = \frac{m!}{(q-1)!(m-q+1)!}$. Identical.
- S3: $P_r(\ell) = \frac{1}{r!}\sum_{q=1}^{r}(q-1)!\,s(r,q)\binom{r}{q-1}\ell^{r-q+1}$, appearing in the expansion as $2^{r+1}P_r(\ell)$. The same cancellation gives $2^{r+1}P_r = \Pi_r$. Identical.

But the variables differ, and this is a genuine disagreement. S1 and S2 expand in $R = \log(y/\kappa)$ with $\ell = \log R$; S3 expands in $H = \log(4y/\kappa) = R + 2\log 2$ with $\ell_{S3} = \log(H/2)$, which is the natural variable for the substitution $X = 2w$, $w - \log w = H/2$. The two series

$$X = R + 2\log R + \sum_j \frac{\Pi_j(\log R)}{R^j} \qquad \text{and} \qquad X = H + 2\log\tfrac{H}{2} + \sum_j \frac{\Pi_j(\log\frac{H}{2})}{H^j}$$

are *different* asymptotic expansions of the same function, agreeing order by order only after re-expanding $H = R + 2\log 2$. Both are correct; mixing them is not. This volume uses the S1/S2 variables R, ℓ throughout, because R makes the phase equation exactly eq. (4.55) with no additive constant. We verified all of S3's displayed coefficients (its u- and N_0-expansions in H, through H^{-4} and H^{-3} respectively) by symbolic re-expansion; they are correct as printed in S3's own variables.

4.8.3 Restoring the dominant amplitude: the centred reversion

The carrier X solves $y = \kappa e^X X^{-2}$; the *dominant core* $\mu_0 = \mu_0(y)$ solves the full dominant sector,

$$y = F(\mu_0) = \kappa\, e^{\mu_0}\mu_0^{-2}\left(1 - \frac{1}{\mu_0}\right), \qquad \text{i.e.} \qquad \Phi(\mu_0) = R, \tag{4.64}$$

where the *dominant phase* is

$$\Phi(\mu) = \mu - 2\log\mu + \log\left(1 - \frac{1}{\mu}\right) = \mu - 3\log\mu + \log(\mu-1), \qquad \mu > 1. \tag{4.65}$$

Lemma 4.49 (Derivatives of the dominant phase)**.** *For* $\mu > 1$*,*

$$\Phi'(\mu) = 1 - \frac{3}{\mu} + \frac{1}{\mu-1} = 1 - \frac{2}{\mu} + \frac{1}{\mu(\mu-1)} = \frac{\mu^2-3\mu+3}{\mu(\mu-1)} > 0, \tag{4.66}$$

and for $m \geq 1$

$$\Phi^{(m)}(\mu) = \mathbf{1}_{m=1} + (-1)^{m-1}(m-1)!\left(\frac{1}{(\mu-1)^m} - \frac{3}{\mu^m}\right). \tag{4.67}$$

In particular $\Phi'(\mu) = 1 + \mathcal{O}(\mu^{-1})$*, and* $\mu^2 - 3\mu + 3 > 0$ *for all real* μ *since its discriminant is* -3*. Writing* $\varrho_\pm = \frac{1}{2}(3 \pm \mathrm{i}\sqrt{3})$ *for the roots of* $\mu^2 - 3\mu + 3$*,*

$$F'(\mu) = \kappa\,\mathrm{e}^{\mu}\,\frac{\mu^2 - 3\mu + 3}{\mu^4}, \qquad \frac{\mathrm{d}^m}{\mathrm{d}\mu^m}\log F'(\mu) = \mathbf{1}_{m=1} + (-1)^{m-1}(m-1)!\left(\frac{1}{(\mu-\varrho_+)^m} + \frac{1}{(\mu-\varrho_-)^m} - \frac{4}{\mu^m}\right). \tag{4.68}$$

Proof. Differentiate eq. (4.65) in the form $\mu - 3\log\mu + \log(\mu - 1)$ and clear denominators for the three displayed shapes of Φ'; eq. (4.67) is $m-1$ further differentiations of $-3\mu^{-1} + (\mu-1)^{-1}$. For eq. (4.68), $F = \kappa\mathrm{e}^{\mu}(\mu-1)\mu^{-3}$ gives $F' = F\bigl(1 + \frac{1}{\mu-1} - \frac{3}{\mu}\bigr) = F\,\Phi'(\mu)$, and $F\Phi' = \kappa\mathrm{e}^{\mu}\frac{\mu-1}{\mu^3}\cdot\frac{\mu^2-3\mu+3}{\mu(\mu-1)}$. Then $\log F' = \log\kappa + \mu + \log(\mu-\varrho_+) + \log(\mu-\varrho_-) - 4\log\mu$, and differentiate. □

Theorem 4.50 (The dominant-core correction; specialisation of theorem 1.25)**.** *Put* $z = X^{-1}$*. There is a unique* $U \in z\,\mathbb{Q}[[z]]$ *with*

$$U = 3\log(1 + zU) - \log(1 - z + zU), \tag{4.69}$$

and U *is analytic in a neighbourhood of* $z = 0$*. The dominant core is*

$$\mu_0(y) = X(y) + U\bigl(X(y)^{-1}\bigr), \qquad U(z) = \sum_{m\geq 1} c_m z^m. \tag{4.70}$$

Moreover eq. (4.69) *is exactly the fixed-point equation* $U = \Psi(z, U)$ *of theorem 1.25 at the dictionary values of table 4.1, so*

$$c_m = \sum_{q=1}^{m}\frac{1}{q}\,[z^m v^{q-1}]\,\Bigl(3\log(1+zv) - \log(1-z+zv)\Bigr)^q \tag{4.71}$$

by theorem 1.16, and equivalently

$$c_m = [z^m]\,\Bigl\{3\log\bigl(1 + zU_{<m}\bigr) - \log\bigl(1 - z + zU_{<m}\bigr)\Bigr\}, \qquad U_{<m}(z) = \sum_{j=1}^{m-1} c_j z^j, \tag{4.72}$$

a triangular recurrence over $\mathbb{Q}$*.*

Proof. Put $\mu = X + U$ and divide eq. (4.64) by the carrier identity $y = \kappa\mathrm{e}^X X^{-2}$:

$$\mathrm{e}^U(1+zU)^{-2}\left(1 - \frac{z}{1+zU}\right) = 1, \qquad z = \frac{1}{X},$$

since $\mu = X(1 + zU)$ and $1/\mu = z/(1+zU)$. As $1 - \frac{z}{1+zU} = \frac{1-z+zU}{1+zU}$, taking logarithms gives $U - 3\log(1+zU) + \log(1-z+zU) = 0$, i.e. eq. (4.69). Write $G(z, U)$ for the left side: $G(0,0) = 0$ and $\partial_U G(0,0) = 1$, so the analytic implicit function theorem gives a unique analytic solution near 0, necessarily with rational Taylor coefficients since G has rational ones.

For the identification with theorem 1.25: that theorem's fixed-point map at parameters a, b, Q is $\Psi(t,u) = -\frac{1}{a}\bigl[b\log(1+tu) + Q\bigl(\frac{t}{1+tu}\bigr)\bigr]$. At $a = 1$, $b = -2$, $Q(s) = \log(1-s)$ this is

$$\Psi(t,u) = 2\log(1+tu) - \log\left(1 - \frac{t}{1+tu}\right) = 2\log(1+tu) - \log(1-t+tu) + \log(1+tu) = 3\log(1+tu) - \log(1-t+tu),$$

which is the right side of eq. (4.69). Equation (4.71) is then theorem 1.16 verbatim. Equation (4.72) holds because in eq. (4.69) every occurrence of U on the right is multiplied by an explicit z, so $[z^m]$ of the right side involves only $c_1, \ldots, c_{m-1}$. □

Table 4.2: The centred-reversion coefficients c_m of eq. (4.70), recomputed here in exact rational arithmetic from eq. (4.72) (truncated power-series arithmetic over $\mathbb{Q}$, $M = 20$). All eighteen agree with the tables printed by the three sources, which call them b_m (S1), a_m (S2) and d_r (S3).

$c_1 = 1$	$c_2 = \frac{5}{2}$	$c_3 = \frac{13}{3}$	$c_4 = \frac{53}{12}$
$c_5 = -\frac{14}{5}$	$c_6 = -\frac{763}{30}$	$c_7 = -\frac{2179}{35}$	$c_8 = -\frac{42289}{630}$
$c_9 = \frac{132973}{1260}$	$c_{10} = \frac{3449711}{5040}$	$c_{11} = \frac{29813143}{18480}$	$c_{12} = \frac{496569589}{356400}$
$c_{13} = -\frac{4334124833}{926640}$	$c_{14} = -\frac{509723063851}{21621600}$	$c_{15} = -\frac{298798016297}{5896800}$	$c_{16} = -\frac{933244850273}{33633600}$
$c_{17} = \frac{4993013022640963}{23156733600}$	$c_{18} = \frac{208843210189406957}{231567336000}$		

Thus

$$\mu_0 = X + \frac{1}{X} + \frac{5}{2X^2} + \frac{13}{3X^3} + \frac{53}{12X^4} - \frac{14}{5X^5} - \frac{763}{30X^6} - \frac{2179}{35X^7} + \cdots . \tag{4.73}$$

Remark 4.51 (Implementation check). The cheapest correctness test for a table of c_m is the residual identity: with $U_M = \sum_{m \le M} c_m z^m$,

$$U_M - 3\log(1 + zU_M) + \log(1 - z + zU_M) = \mathcal{O}(z^{M+1}). \tag{4.74}$$

We verified eq. (4.74) at $M = 12$ symbolically: the residual is $\mathcal{O}(z^{13})$ exactly. This test is far more reliable than comparing printed digits, and it is the one we used to certify table 4.2.

4.8.4 The algebraic inverse index

Definition 4.52. $N_0(y) = \dfrac{1}{24} + \dfrac{3\,\mu_0(y)^2}{2\pi^2}$, the *algebraic* (phase-independent) inverse index; and $N_X(y) = \dfrac{1}{24} + \dfrac{3X(y)^2}{2\pi^2}$, the carrier index.

Corollary 4.53 (All algebraic coefficients of the inverse index)**.** *With c_m as in table 4.2, put*

$$\gamma_j = 3c_{j+1} + \frac{3}{2}\sum_{r=1}^{j-1} c_r c_{j-r} \qquad (j \ge 1). \tag{4.75}$$

Then

$$N_0(y) = \frac{3X^2}{2\pi^2} + \left(\frac{1}{24} + \frac{3}{\pi^2}\right) + \frac{1}{\pi^2}\sum_{j \ge 1} \frac{\gamma_j}{X^j}, \tag{4.76}$$

with

$$\gamma_1 = \tfrac{15}{2}, \quad \gamma_2 = \tfrac{29}{2}, \quad \gamma_3 = \tfrac{83}{4}, \quad \gamma_4 = \tfrac{559}{40}, \quad \gamma_5 = -\tfrac{611}{20}, \quad \gamma_6 = -\tfrac{112459}{840}, \tag{4.77}$$

$$\gamma_7 = -\tfrac{101329}{420}, \quad \gamma_8 = -\tfrac{228677}{3360}, \quad \gamma_9 = \tfrac{1709167}{1680}, \quad \gamma_{10} = \tfrac{325098619}{92400}. \tag{4.78}$$

Proof. Insert $\mu_0 = X + U$ into theorem 4.52. The cross term $\frac{3}{2\pi^2} \cdot 2XU = \frac{3}{\pi^2}\sum_{m \ge 1} c_m X^{1-m}$ contributes $3c_1/\pi^2 = 3/\pi^2$ to the constant and $3c_{j+1}/(\pi^2 X^j)$ at order X^{-j}; the square $\frac{3}{2\pi^2}U^2$ contributes $\frac{3}{2\pi^2}\left(\sum_{r=1}^{j-1} c_r c_{j-r}\right)X^{-j}$ and nothing at order X^0 or above. The listed values were recomputed from eq. (4.75); $\gamma_1, \dots, \gamma_8$ agree with the tables of all three sources, and γ_9, γ_{10} are new here. □

Corollary 4.54 (The carrier bias)**.** $N_0(y) - N_X(y) = \dfrac{3}{\pi^2} + \mathcal{O}(X^{-1})$*, and*

$$\frac{3}{\pi^2} = 0.30396355092701331433\ldots \tag{4.79}$$

Proof. Subtract $N_X = \frac{1}{24} + 3X^2/(2\pi^2)$ from eq. (4.76): the $3X^2/(2\pi^2)$ and $\frac{1}{24}$ cancel, leaving $3/\pi^2 + \pi^{-2}\sum_{j \ge 1}\gamma_j X^{-j}$. The constant was evaluated here to 20 digits. □

Remark 4.55 (Phase locking: do not expand the arithmetic phase about the carrier). This is the single most consequential structural point of the inverse problem, and only S2 states it explicitly.

The amplitudes $\widetilde{A}_k$ are *periodic on the index scale*, with period k. The core correction is small in the phase variable, $\mu_0 - X = \mathcal{O}(X^{-1})$; but $\mathrm{d}n/\mathrm{d}\mu = 3\mu/\pi^2 \asymp X$, so the induced change in the index is $\mathcal{O}(1)$ — exactly $3/\pi^2$ in the limit, by theorem 4.54. A shift of 0.304 in the index is *not* a small perturbation of $\cos(\pi\nu)$. Consequently:

- Taylor-expanding $\widetilde{A}_k(N(\mu_0))$ around N_X does not begin with a small correction, and produces the wrong phase already at the leading exponentially small order.
- Every arithmetic amplitude must be evaluated at the *full algebraic core* $N_0(y)$, not at $N_X(y)$ and not at any truncation of N_0 coarser than $o(1)$. By eq. (4.76), truncating the γ-series after M terms leaves an index error $\mathcal{O}(X^{-M})$, so $M \geq 1$ already suffices for $o(1)$; $M = 0$ (i.e. dropping the constant $3/\pi^2$) does not.
- Only the *subsequent* exponentially small displacement may be Taylor-expanded; that is what section 4.9 does.

The numerical signature of the effect is the first column of table 4.4, which tends to $-3/\pi^2$ and not to 0.

Remark 4.56 (Elementary-logarithm form). Substituting eq. (4.57) into eq. (4.76) gives the pure logarithmic block

$$N_0(y) = \frac{3R^2}{2\pi^2} + \frac{6R\ell}{\pi^2} + \frac{6\ell^2 + 12\ell + 3}{\pi^2} + \frac{1}{24} + \frac{3(8\ell^2 + 16\ell + 5)}{2\pi^2 R} + \mathcal{O}\left(\frac{\ell^3}{R^2}\right), \tag{4.80}$$

with $R = \log(y/\kappa)$ and $\ell = \log R$. We recomputed this block symbolically; it agrees with S2's display (which stops at the constant) and, after the change of variable $H = R + 2\log 2$, with S3's. For *numerical* work eq. (4.80) is markedly inferior to evaluating eq. (4.56) and eq. (4.73) directly, and it is recorded only to exhibit the transseries shape: N_0 is a polynomial of degree 2 in R with coefficients polynomial in ℓ, plus a descending series in R^{-1} with $\deg_\ell$ growing by one per order.

4.9 The full arithmetic inverse transseries

Everything so far is phase-independent: X, μ_0, N_0 are functions of y alone. The remaining correction is exponentially small and arithmetically oscillatory, and the oscillation must be frozen before "analytic inverse" has a meaning. The three sources do this in two different ways, both correct.

4.9.1 Two ways to freeze the phase

Definition 4.57 (Residue-frozen finite level; S1)**.** For $K \geq 1$ let $\Lambda_K = \mathrm{lcm}(1, 2, \dots, K)$ and fix a residue $r \in \mathbb{Z}/\Lambda_K\mathbb{Z}$. Since A_k has period $k \mid \Lambda_K$ for $k \leq K$, the numbers $A_k(r)$ are well defined, and

$$P_{K,r}(\mu) = \frac{\kappa}{\mu^2} \sum_{k=1}^{K} \frac{A_k(r)}{\sqrt{k}} \left[\mathrm{e}^{\mu/k}\left(1 - \frac{k}{\mu}\right) + \mathrm{e}^{-\mu/k}\left(1 + \frac{k}{\mu}\right) \right] \tag{4.81}$$

is a genuine analytic function of μ with constant coefficients. For large μ it is increasing (the $k = 1$ positive term dominates), and we write $\mu_{K,r}(y)$ for its large inverse branch.

Definition 4.58 (Interpolated level; S2, S3)**.** $\mathcal{P}_K(\mu)$ is eq. (4.81) with $A_k(r)$ replaced by $\widetilde{A}_k(n(\mu))$, $n(\mu) = \frac{1}{24} + 3\mu^2/(2\pi^2)$; its large inverse branch is written $\mu_K(y)$.

Proposition 4.59 (The two conventions agree below action one)**.** *Fix K, let y be large, and let $r \equiv \lfloor N_0(y) \rceil \pmod{\Lambda_K}$, where $\lfloor\cdot\rceil$ is nearest-integer rounding. Assume $|N_0(y) - r| \leq \frac{1}{2}$ modulo Λ_K has been resolved (which holds along the range $y = p(n)$ for large n by theorem 4.83). Then every inverse transmonomial of total action strictly less than 1 is the same for $\mu_{K,r}$ and μ_K; the two first differ at total action ≥ 1.*

Proof. Both inverses are constructed as $\mu_0 + v$ with v exponentially small, and in both the coefficient of the transmonomial $\mathrm{e}^{-\alpha_j\mu_0}$ at first order is $-r_j(\mu_0)/\Phi'(\mu_0)$ (theorem 4.69 below), where r_j differs between the two conventions only in the amplitude, $A_k(r)$ versus $\widetilde{A}_k(n(\mu_0)) = \widetilde{A}_k(N_0(y))$. Now $\widetilde{A}_k$ is entire and k-periodic with $\widetilde{A}_k(r) = A_k(r)$, so $|\widetilde{A}_k(N_0) - A_k(r)| \le \sup|\widetilde{A}_k'| \cdot |N_0 - r|$. Along the range, $|N_0(p(n)) - n| = \mathcal{O}(\mu^{5/2}\mathrm{e}^{-\mu/2})$ by theorem 4.79, so the amplitude discrepancy is $\mathcal{O}(k\mu^{5/2}\mathrm{e}^{-\mu/2})$ and its contribution to the sector carries the extra factor $\mathrm{e}^{-\mu/2}$, i.e. additional action $\frac{1}{2}$. Since the smallest primary action is $\frac{1}{2}$, the first affected total action is $\frac{1}{2} + \frac{1}{2} = 1$. Higher-order terms of the reversion have total action ≥ 1 already. □

Remark 4.60 (Which convention to prefer). The residue-frozen convention is intrinsic: it invents no interpolation, and at each finite level it reduces the problem to an ordinary analytic reversion with *constant* coefficients. Its cost is that the inverse is defined only after the residue class is known, which theorem 4.83 supplies. The interpolated convention defines a single function $\mathcal{N}$ for all real y and connects directly to the staircase (theorem 4.44); its cost is the interpolation dependence of theorem 4.11 and the appearance of the amplitude's own μ-derivatives in every higher coefficient. This chapter develops the reversion in a form that covers both: all formulas below are stated for a family of *blocks* $E_j(\mu)$, and the two conventions are two choices of block.

4.9.2 Blocks, multi-indices, and the reversion equation

Definition 4.61 (Sector blocks)**.** Fix $K \ge 1$ and put

$$J_K = \{(1,-1)\} \cup \{(k,\sigma) : 2 \le k \le K,\ \sigma = \pm 1\}. \tag{4.82}$$

For $j = (k,\sigma) \in J_K$ set $\alpha_j = 1 - \sigma/k$ as in eq. (4.21), and define the *block*

$$E_j(\mu) = \frac{\kappa}{\mu^2}\frac{a_k(\mu)}{\sqrt{k}}\left(1 - \frac{\sigma k}{\mu}\right)\mathrm{e}^{\sigma\mu/k} = F(\mu)\, r_j(\mu)\,\mathrm{e}^{-\alpha_j\mu}, \qquad r_j(\mu) = \frac{a_k(\mu)}{\sqrt{k}}\,\frac{1-\sigma k/\mu}{1-1/\mu}, \tag{4.83}$$

where $a_k(\mu)$ is either the constant $A_k(r)$ (residue-frozen convention, theorem 4.57) or $\widetilde{A}_k(n(\mu))$ (interpolated convention, theorem 4.58); $a_1 \equiv 1$ in both. A block may also be the whole paired tail $\mathcal{R}_{>K} \cdot F$ of eq. (4.25), or the zeta-condensation block of theorem 4.28, treated as one indivisible symbol.

Introduce independent formal variables ε_j, $j \in J_K$, and write $\boldsymbol{\varepsilon}^{\boldsymbol{m}} = \prod_j \varepsilon_j^{m_j}$, $|\boldsymbol{m}| = \sum_j m_j$, $\boldsymbol{m}! = \prod_j m_j!$ for multi-indices $\boldsymbol{m} \in \mathbb{N}^{J_K}$. The forward object at level K is

$$\mathcal{P}_K(\mu;\boldsymbol{\varepsilon}) = F(\mu) + \sum_{j\in J_K} \varepsilon_j E_j(\mu), \tag{4.84}$$

and the inverse problem is to solve $\mathcal{P}_K(\mu;\boldsymbol{\varepsilon}) = y$ for μ near $\mu_0 = F^{-1}(y)$, as a formal series in $\boldsymbol{\varepsilon}$ which is finally specialised at $\varepsilon_j = 1$. Two equivalent engines are available; both are finite at every prescribed multi-index.

4.9.3 Engine I: additive Lagrange–Bürmann

Theorem 4.62 (Additive reversion)**.** *Let F and E be analytic near μ_0 with $F'(\mu_0) \ne 0$ and $F(\mu_0) = y$. Write $\mathcal{D} = \frac{1}{F'(\mu)}\frac{\mathrm{d}}{\mathrm{d}\mu}$ evaluated along $\mu = \mu_0(y)$, i.e. $\mathcal{D} = \mathrm{d}/\mathrm{d}y$ in the coordinate y. Then the solution of $F(\mu) + \varepsilon E(\mu) = y$ near μ_0 is*

$$\mu(y;\varepsilon) = \mu_0(y) + \sum_{M\ge1}\frac{(-\varepsilon)^M}{M!}\,\mathcal{D}^{M-1}\left(\frac{E(\mu_0)^M}{F'(\mu_0)}\right). \tag{4.85}$$

Consequently, for the multivariate problem eq. (4.84)*, the coefficient of $\boldsymbol{\varepsilon}^{\boldsymbol{m}}$ with $M = |\boldsymbol{m}| \ge 1$ in $\mu - \mu_0$ is*

$$V_{\boldsymbol{m}}(\mu_0) = \frac{(-1)^M}{\boldsymbol{m}!}\,\mathcal{D}^{M-1}\left(\frac{E^{\boldsymbol{m}}(\mu_0)}{F'(\mu_0)}\right) = \frac{(-1)^M(M-1)!}{\boldsymbol{m}!}\,\operatorname*{Res}_{\mu=\mu_0}\frac{E^{\boldsymbol{m}}(\mu)}{(F(\mu)-F(\mu_0))^M}, \tag{4.86}$$

where $E^{\boldsymbol{m}} = \prod_j E_j^{m_j}$.

Proof. Both displays are theorem 1.18 in the same letters: eq. (4.85) is (1.34) and eq. (4.86) is (1.36).

What that theorem supplies is convergence for $|\varepsilon| < \varepsilon_0$, with ε_0 read off a small circle about μ_0; here the specialisation $\varepsilon_j = 1$ is wanted, and for the partition blocks it is legitimate. With $E = \sum_j E_j + \mathcal{R}_{>K} F$ we have $E/F = \mathcal{O}(\mu^C e^{-\mu/2})$ together with the same bound for finitely many derivatives, by theorem 4.23 and theorem 4.10; so on a fixed relative neighbourhood of a large μ_0 the Rouché comparison made in the proof of theorem 1.18 holds for all $|\varepsilon| \le 1 + \eta$ with some $\eta > 0$, and the Taylor series in ε converges at $\varepsilon = 1$. □

Remark 4.63 ($\mathcal{D}$ is iterated, not a plain derivative). In eq. (4.85) the operator $\mathcal{D}^{M-1}$ is the $(M-1)$st power of $\frac{1}{F'}\frac{\mathrm{d}}{\mathrm{d}\mu}$. Replacing it by $\frac{1}{(F')^{M-1}}\frac{\mathrm{d}^{M-1}}{\mathrm{d}\mu^{M-1}}$ after “pulling out” the powers of F' is wrong from $M = 3$ onwards, because F' is not constant. S3 flags this explicitly; it is the most common implementation bug in this formula.

Corollary 4.64 (Bell realisation). *Let $H_{\boldsymbol{m}} = E^{\boldsymbol{m}}/F'$ and $\Lambda_s^{(\boldsymbol{m})} = \frac{\mathrm{d}^s}{\mathrm{d}\mu^s}\log H_{\boldsymbol{m}}$, and let $\mu_0^{(s)} = \mathrm{d}^s\mu_0/\mathrm{d}y^s$ be generated by*

$$\mu_0^{(1)} = \frac{1}{F'(\mu_0)}, \qquad \mu_0^{(s)} = -\frac{1}{F'(\mu_0)}\sum_{q=2}^{s} F^{(q)}(\mu_0)\, B_{s,q}\big(\mu_0^{(1)}, \dots, \mu_0^{(s-q+1)}\big), \quad s \ge 2. \tag{4.87}$$

Then, with $M = |\boldsymbol{m}| \ge 1$,

$$V_{\boldsymbol{m}} = \frac{(-1)^M H_{\boldsymbol{m}}(\mu_0)}{\boldsymbol{m}!}\sum_{s=0}^{M-1} B_{M-1,s}\big(\mu_0^{(1)}, \mu_0^{(2)}, \dots\big)\, B_s\big(\Lambda_1^{(\boldsymbol{m})}, \dots, \Lambda_s^{(\boldsymbol{m})}\big), \tag{4.88}$$

with the conventions $B_{0,0} = B_0 = 1$.

Proof. Faà di Bruno (theorem 1.10) gives $\frac{\mathrm{d}^{M-1}}{\mathrm{d}y^{M-1}} H_{\boldsymbol{m}}(\mu_0(y)) = \sum_s H_{\boldsymbol{m}}^{(s)}(\mu_0) B_{M-1,s}(\mu_0^{(1)}, \dots)$, and writing $H_{\boldsymbol{m}} = \exp(\log H_{\boldsymbol{m}})$ gives $H_{\boldsymbol{m}}^{(s)} = H_{\boldsymbol{m}}\, B_s(\Lambda_1^{(\boldsymbol{m})}, \dots)$. Substitute in eq. (4.86). Equation (4.87) is obtained by differentiating $F(\mu_0(y)) = y$ s times and solving the Faà di Bruno identity for the term containing $\mu_0^{(s)}$. □

The point of eq. (4.88) is that every ingredient is elementary and closed-form for the partition blocks. Indeed, splitting each aggregate block into its primitive Fourier constituents

$$E_{k,h,\sigma,\epsilon}(\mu) = \frac{\kappa}{2\sqrt{k}}\,\frac{\mu - \sigma k}{\mu^3}\exp\left[\frac{\sigma\mu}{k} + \mathrm{i}\epsilon\Big(\pi s(h,k) - \frac{2\pi h}{k} n(\mu)\Big)\right], \qquad h \in U_k,\ \epsilon = \pm 1, \tag{4.89}$$

and using $n'(\mu) = 3\mu/\pi^2$, $n''(\mu) = 3/\pi^2$, $n^{(s)} = 0$ for $s \ge 3$, one gets the closed logarithmic jets

$$\frac{\mathrm{d}^s}{\mathrm{d}\mu^s}\log E_{k,h,\sigma,\epsilon} = (-1)^{s-1}(s-1)!\left(\frac{1}{(\mu - \sigma k)^s} - \frac{3}{\mu^s}\right) + \mathbf{1}_{s=1}\frac{\sigma}{k} - \mathrm{i}\epsilon\frac{2\pi h}{k}\, n^{(s)}(\mu), \tag{4.90}$$

while $\frac{\mathrm{d}^s}{\mathrm{d}\mu^s}\log F'$ is eq. (4.68). Hence $\Lambda_s^{(\boldsymbol{m})} = \sum_j m_j$ (eq. (4.90)) $-$ (eq. (4.68)) with no hidden differentiation anywhere. This is S3’s presentation and is the most explicit of the three.

4.9.4 Engine II: the triangular Newton–Hensel recurrence

The reversion may equally be organised in the phase coordinate, which is cheaper when many coefficients are wanted.

Theorem 4.65 (Triangular recurrence). *Put $\mu = \mu_0 + v$ and*

$$Q(v; \boldsymbol{\varepsilon}) = \log\Big(1 + \sum_{j \in J_K} \varepsilon_j B_j(v)\Big), \qquad B_j(v) = r_j(\mu_0 + v)\, e^{-\alpha_j v}, \tag{4.91}$$

the bookkeeping variable ε_j standing for the actual exponential $\mathrm{e}^{-\alpha_j\mu_0}$, which is substituted only at the very end; the residual v-dependence of $\mathrm{e}^{-\alpha_j(\mu_0+v)}$ is the factor $\mathrm{e}^{-\alpha_j v}$ inside B_j. Keeping the ε_j independent until the last step is what makes resonant exponent sums ($\alpha_j + \alpha_{j'} = \alpha_{j''}$, which does occur here) come out right. Then the reversion equation is

$$\Phi'(\mu_0)\, v + \sum_{s\geq 2} \frac{\Phi^{(s)}(\mu_0)}{s!} v^s + Q(v;\boldsymbol{\varepsilon}) = 0, \tag{4.92}$$

and its unique formal solution $v = \sum_{\boldsymbol{m}\neq\boldsymbol{0}} V_{\boldsymbol{m}}\boldsymbol{\varepsilon}^{\boldsymbol{m}}$ obeys

$$V_{\boldsymbol{m}} = -\frac{1}{\Phi'(\mu_0)}\,[\boldsymbol{\varepsilon}^{\boldsymbol{m}}]\left\{\sum_{s\geq 2}\frac{\Phi^{(s)}(\mu_0)}{s!}\,v_{<\boldsymbol{m}}^{s} + Q(v_{<\boldsymbol{m}};\boldsymbol{\varepsilon})\right\}, \tag{4.93}$$

where $v_{<\boldsymbol{m}}$ collects the already-computed coefficients. Each right-hand side is a finite expression in μ_0, $(\mu_0-1)^{-1}$, the actions α_j and the amplitudes a_k.

Proof. Taking logarithms in $\mathcal{P}_K(\mu;\boldsymbol{\varepsilon}) = y$ and subtracting $\Phi(\mu_0) = R$ gives $\Phi(\mu_0+v) - \Phi(\mu_0) + Q(v;\boldsymbol{\varepsilon}) = 0$; Taylor expansion of Φ at μ_0 is eq. (4.92), legitimate as a formal identity because v has no constant term in $\boldsymbol{\varepsilon}$. For triangularity: at a fixed $\boldsymbol{m}$, the unknown $V_{\boldsymbol{m}}$ occurs linearly only in the term $\Phi'(\mu_0)v$; each v^s with $s\geq 2$ needs a decomposition of $\boldsymbol{m}$ into at least two non-zero parts and so uses only strictly smaller multi-indices; and every monomial of Q carries at least one explicit ε_j, so an occurrence of $V_{\boldsymbol{m}}$ inside Q would be multiplied into a strictly larger multi-index. Solving the coefficient equation gives eq. (4.93). Finiteness holds because a fixed multi-index has finitely many additive decompositions. ☐

Corollary 4.66 (Multinomial expansion of Q)**.** *For $\boldsymbol{m}\neq\boldsymbol{0}$,*

$$[\boldsymbol{\varepsilon}^{\boldsymbol{m}}]\,Q(v;\boldsymbol{\varepsilon})\Big|_{v\text{ fixed}} = (-1)^{|\boldsymbol{m}|+1}\frac{(|\boldsymbol{m}|-1)!}{\boldsymbol{m}!}\prod_j B_j(v)^{m_j}. \tag{4.94}$$

Proof. Only $S^{|\boldsymbol{m}|}$ in $\log(1+S) = \sum_{d\geq 1}(-1)^{d+1}S^d/d$ contributes, with multinomial coefficient $|\boldsymbol{m}|!/\boldsymbol{m}!$. ☐

Corollary 4.67 (Closed multi-index Lagrange form in the phase)**.** *Let $G(v) = \big(\Phi(\mu_0+v) - \Phi(\mu_0)\big)/v$ for $v\neq 0$ and $G(0) = \Phi'(\mu_0)$. Then $v = -Q(v;\boldsymbol{\varepsilon})/G(v)$, and by theorem 1.14,*

$$V_{\boldsymbol{m}} = \sum_{q=1}^{|\boldsymbol{m}|}\frac{(-1)^q}{q}\,[v^{q-1}]\,\frac{1}{G(v)^q}\sum_{\substack{\boldsymbol{m}_1+\cdots+\boldsymbol{m}_q=\boldsymbol{m}\\ \boldsymbol{m}_i\neq\boldsymbol{0}}}\prod_{i=1}^{q}[\boldsymbol{\varepsilon}^{\boldsymbol{m}_i}]\,Q(v;\boldsymbol{\varepsilon}), \tag{4.95}$$

both sums being finite.

Proof. G is analytic near 0 with $G(0) = \Phi'(\mu_0)\neq 0$, and eq. (4.92) reads $vG(v) + Q(v;\boldsymbol{\varepsilon}) = 0$. Apply theorem 1.14 to $v = z\cdot\big(-Q(v;\boldsymbol{\varepsilon})/G(v)\big)$ and set $z=1$; since every monomial of Q has positive $\boldsymbol{\varepsilon}$-degree, only $q\leq|\boldsymbol{m}|$ contributes. Expanding Q^q and using eq. (4.94) gives eq. (4.95). ☐

Remark 4.68 (Three engines, one answer). Theorem 4.62 works in the ordinate y and is S3's; theorem 4.67 works in the phase μ and is S2's; theorem 4.65 is S1's. They compute the same coefficients: all three solve the same analytic equation with the same normalisation $V_{\boldsymbol{0}} = 0$, and the solution of a triangular system is unique. In practice eq. (4.93) is preferable when many multi-indices are wanted (it reuses everything), eq. (4.86) when a single high-order multi-index is wanted (it needs no lower ones), and eq. (4.88) when the answer is wanted symbolically in μ_0.

4.9.5 First-order sectors and the leading parity chirp

Theorem 4.69 (The first correction from each sector). *Let $\boldsymbol{e}_j$ be the unit multi-index at $j = (k, \sigma) \in J_K$. Then*

$$V_{\boldsymbol{e}_j}(\mu_0) = -\frac{r_j(\mu_0)}{\Phi'(\mu_0)} = -\frac{a_k(\mu_0)}{\sqrt{k}}\,\frac{\mu_0(\mu_0 - \sigma k)}{\mu_0^2 - 3\mu_0 + 3}, \tag{4.96}$$

so that the corresponding displacement of the inverse phase is $V_{\boldsymbol{e}_j}(\mu_0)\,\mathrm{e}^{-\alpha_j \mu_0}$, and the corresponding displacement of the inverse index is

$$\delta N^{(1)}_{k,\sigma}(y) = -\frac{3\,a_k(\mu_0)}{\pi^2\sqrt{k}}\,\frac{\mu_0^2(\mu_0 - \sigma k)}{\mu_0^2 - 3\mu_0 + 3}\,\mathrm{e}^{-(1-\sigma/k)\mu_0}. \tag{4.97}$$

Proof. Take $\boldsymbol{m} = \boldsymbol{e}_j$ in eq. (4.93): the $s \geq 2$ terms and all higher Q-monomials vanish, leaving $V_{\boldsymbol{e}_j} = -[\varepsilon_j]\,Q(0; \boldsymbol{\varepsilon})/\Phi'(\mu_0) = -B_j(0)/\Phi'(\mu_0) = -r_j(\mu_0)/\Phi'(\mu_0)$. Insert $r_j = \frac{a_k}{\sqrt{k}}\frac{1-\sigma k/\mu}{1-1/\mu} = \frac{a_k}{\sqrt{k}}\frac{\mu-\sigma k}{\mu-1}$ and eq. (4.66): the factors $\mu_0 - 1$ cancel, giving eq. (4.96). Finally $n(\mu) = \frac{1}{24} + \frac{3\mu^2}{2\pi^2}$ is exactly quadratic, so

$$n(\mu_0 + v) = N_0(y) + \frac{3\mu_0}{\pi^2}v + \frac{3}{2\pi^2}v^2, \tag{4.98}$$

and the linear term at first order gives eq. (4.97). □

Corollary 4.70 (Index coefficient of every transmonomial). *For every $\boldsymbol{m} \neq \boldsymbol{0}$, the coefficient of $\boldsymbol{\varepsilon}^{\boldsymbol{m}}$ in the inverse index is*

$$\mathfrak{N}_{\boldsymbol{m}}(\mu_0) = \frac{3\mu_0}{\pi^2}V_{\boldsymbol{m}} + \frac{3}{2\pi^2}\sum_{\substack{\boldsymbol{a}+\boldsymbol{b}=\boldsymbol{m}\\ \boldsymbol{a},\boldsymbol{b}\neq\boldsymbol{0}}} V_{\boldsymbol{a}}V_{\boldsymbol{b}}. \tag{4.99}$$

Proof. Expand eq. (4.98) with $v = \sum V_{\boldsymbol{m}}\boldsymbol{\varepsilon}^{\boldsymbol{m}}$ and compare coefficients. No higher powers occur because n is quadratic. □

The first two positive sectors, with $a_2 = \widetilde{A}_2(N_0) = \cos(\pi N_0)$ and $a_3 = \widetilde{A}_3(N_0) = 2\cos(\pi N_0)\cos(\frac{\pi}{18} + \frac{\pi N_0}{3})$ (interpolated convention) or $a_2 = (-1)^r$, $a_3 = 2\cos(\frac{\pi}{18} - \frac{2\pi r}{3})$ (residue-frozen), are

$$\delta N^{(1)}_{2,+} = -\frac{3\,a_2}{\pi^2\sqrt{2}}\,\frac{\mu_0^2(\mu_0 - 2)}{\mu_0^2 - 3\mu_0 + 3}\,\mathrm{e}^{-\mu_0/2}, \tag{4.100}$$

$$\delta N^{(1)}_{3,+} = -\frac{3\,a_3}{\pi^2\sqrt{3}}\,\frac{\mu_0^2(\mu_0 - 3)}{\mu_0^2 - 3\mu_0 + 3}\,\mathrm{e}^{-2\mu_0/3}. \tag{4.101}$$

Thus *parity of the index appears at relative scale* $\mathrm{e}^{-\mu_0/2}$, *the residue modulo* 3 *at* $\mathrm{e}^{-2\mu_0/3}$, *and the residue modulo k first appears in the growing kth sector at* $\mathrm{e}^{-(1-1/k)\mu_0}$.

4.9.6 The fractional-power staircase in the ordinate

The core equation converts exponentials in μ_0 into fractional powers of y exactly:

$$\mathrm{e}^{-\alpha\mu_0} = \left(\frac{\kappa(\mu_0 - 1)}{y\,\mu_0^3}\right)^{\alpha} \qquad \text{for every } \alpha, \tag{4.102}$$

directly from $y = \kappa\mathrm{e}^{\mu_0}(\mu_0 - 1)\mu_0^{-3}$.

Proposition 4.71 (Every first-order sector in closed y-form). *With $\alpha = \alpha_{k,\sigma}$ and $z = 1/\mu_0$,*

$$\delta N^{(1)}_{k,\sigma} = -\frac{3\kappa^{\alpha}}{\pi^2\sqrt{k}}\,a_k(\mu_0)\,y^{-\alpha}\,\mu_0^{1-2\alpha}\sum_{s\geq 0} g_s^{(k,\sigma)}\,\mu_0^{-s}, \tag{4.103}$$

where

$$\sum_{s\geq 0} g_s^{(k,\sigma)} z^s = \frac{(1-\sigma k z)(1-z)^{\alpha}}{1-3z+3z^2}, \qquad g_s^{(k,\sigma)} = \sum_{i=0}^{s} (-1)^i \binom{\alpha}{i} \big(\chi_{s-i} - \sigma k\, \chi_{s-i-1}\big), \tag{4.104}$$

with $\chi_{-1} = 0$ *and* $\chi_s = \dfrac{\varrho_+^{s+1} - \varrho_-^{s+1}}{\varrho_+ - \varrho_-}$ *the coefficients of* $(1-3z+3z^2)^{-1}$ ($\varrho_\pm = \frac{1}{2}(3 \pm \mathrm{i}\sqrt{3})$*;* $\chi_0 = 1, \chi_1 = 3, \chi_2 = 6, \chi_3 = 9, \chi_4 = 9, \chi_5 = 0, \ldots$*).*

Proof. Insert eq. (4.102) into eq. (4.97) and factor $\mu_0^{1-2\alpha}$ out of $\dfrac{\mu_0^2(\mu_0 - \sigma k)(\mu_0 - 1)^{\alpha}}{(\mu_0^2 - 3\mu_0 + 3)\mu_0^{3\alpha}}$; with $z = 1/\mu_0$ the remaining factor is $\dfrac{(1-\sigma k z)(1-z)^{\alpha}}{1-3z+3z^2}$. The coefficient formula is the Cauchy product of $(1-z)^{\alpha} = \sum_i (-1)^i \binom{\alpha}{i} z^i$ with $(1-\sigma k z)/(1-3z+3z^2)$, and χ_s is the standard solution of $\chi_s = 3\chi_{s-1} - 3\chi_{s-2}$, $\chi_0 = 1$, $\chi_1 = 3$, whose closed form is as stated since $\varrho_\pm$ are the roots of $\varrho^2 = 3\varrho - 3$. ☐

Corollary 4.72 (The inverse staircase of scales)**.** *Since* $\mu_0 \sim \log y$*, the primary positive sectors contribute, in order,*

$$y^{-1/2}, \qquad y^{-2/3}(\log y)^{-1/3}, \qquad y^{-3/4}(\log y)^{-1/2}, \qquad \ldots, \qquad y^{-(1-1/k)}(\log y)^{-1+2/k}, \tag{4.105}$$

accumulating at the scale y^{-1}*, where they meet the square of the first sector and the whole negative family.*

Proof. By eq. (4.103) the $(k,+)$ sector has size $y^{-\alpha}\mu_0^{1-2\alpha}(1+\mathcal{O}(\mu_0^{-1}))$ times a bounded amplitude, with $\alpha = 1-1/k$, so $1-2\alpha = -1+2/k$; and $\mu_0 \sim \log y$ because $\Phi(\mu_0) = \log(y/\kappa)$ with $\Phi(\mu) = \mu(1+o(1))$. Substituting $k = 2, 3, 4$ gives the first three displayed scales. Since $\alpha_{k,+} \uparrow 1$, the scales accumulate at y^{-1}; the square of the $(2,+)$ sector has action $\frac{1}{2} + \frac{1}{2} = 1$ and every negative action is ≥ 1. ☐

Theorem 4.73 (The leading parity chirp)**.** *In the interpolated convention,*

$$\mathcal{N}(y) = N_0(y) - \frac{3\sqrt{\kappa}}{\pi^2\sqrt{2}} \frac{\cos\big(\pi N_0(y)\big)}{\sqrt{y}} \left(1 + \mathcal{O}\left(\frac{1}{\log y}\right)\right) + \mathcal{O}\Big(y^{-2/3}(\log y)^{-1/3}\Big), \tag{4.106}$$

with the constant

$$\frac{3\sqrt{\kappa}}{\pi^2\sqrt{2}} = \frac{3^{1/4}}{2\pi} = 0.2094596846361762626265841\ldots \tag{4.107}$$

Since $N_0(y) = \frac{3}{2\pi^2}(\log y)^2(1+o(1))$*, the cosine is a* quadratic logarithmic chirp*, not a log-periodic oscillation.*

Proof. Take $k = 2$, $\sigma = +1$, $\alpha = \frac{1}{2}$ in eq. (4.103): the power $\mu_0^{1-2\alpha} = \mu_0^0$ is absent, $g_0^{(2,+)} = 1$, and $a_2 = \cos(\pi N_0)$ by theorem 4.10. The constant is $3\kappa^{1/2}/(\pi^2\sqrt{2})$, and $\sqrt{\kappa} = \pi/\sqrt{6\sqrt{3}}$ gives $\frac{3\pi}{\pi^2\sqrt{2}\sqrt{6\sqrt{3}}} = \frac{3}{\pi\sqrt{12\sqrt{3}}} = \frac{3}{2\pi\cdot 3^{3/4}} = \frac{3^{1/4}}{2\pi}$, which we evaluated to 25 digits. The relative $\mathcal{O}(1/\log y)$ is the $s \geq 1$ tail of eq. (4.103), and the next omitted contribution is the $k = 3$ sector at scale $y^{-2/3}(\log y)^{-1/3}$ by theorem 4.72; all products of two blocks have action ≥ 1. ☐

Remark 4.74 (Constant check). S2 prints this constant as 0.2094596846361762626; the closed form $3^{1/4}/(2\pi)$ is S3's. The two agree, and both agree with our independent 25-digit evaluation eq. (4.107). S1 does not state the constant, giving instead the equivalent phase-variable form eq. (4.100); the three are algebraically identical, as the cancellation of $\mu_0 - 1$ in the proof of theorem 4.69 shows.

4.10 The profinite arithmetic phase

The residue-frozen convention of theorem 4.57 organises itself into a projective system, which is the cleanest way to say what data an "arithmetically complete" inverse of p actually depends on.

Theorem 4.75 (Compatibility and the profinite parameter). *Let $\Lambda_K = \operatorname{lcm}(1, \ldots, K)$, so $\Lambda_K \mid \Lambda_{K+1}$. Suppose $r_K \in \mathbb{Z}/\Lambda_K\mathbb{Z}$ satisfy $r_{K+1} \equiv r_K \pmod{\Lambda_K}$, and let μ_{K,r_K} be the corresponding inverse branches of theorem 4.57. Then for every K, the transseries of $\mu_{K+1,r_{K+1}}$ obtained by deleting all monomials containing the variables of level $K+1$ coincides with the transseries of μ_{K,r_K}. Consequently the complete arithmetic inverse is a projective family indexed by the profinite integers $\widehat{r} \in \widehat{\mathbb{Z}} = \varprojlim \mathbb{Z}/m\mathbb{Z}$, and an actual partition index n supplies its own phase through the diagonal embedding $n \mapsto \widehat{n}$.*

Proof. Deleting the level-$(K+1)$ variables from the defining equation eq. (4.92) at level $K+1$ produces the defining equation at level K *provided* the retained amplitudes agree, i.e. provided $A_k(r_{K+1}) = A_k(r_K)$ for $k \leq K$. That is exactly the compatibility hypothesis, since A_k has period $k \mid \Lambda_K$ and $r_{K+1} \equiv r_K \pmod{\Lambda_K}$. Both truncated equations are triangular with the same normalisation $V_0 = 0$, and a triangular system has a unique solution, so the solutions agree coefficient by coefficient. The projective limit statement is the definition of $\widehat{\mathbb{Z}}$: compatible systems (r_K) are exactly the elements of $\varprojlim \mathbb{Z}/\Lambda_K\mathbb{Z} = \widehat{\mathbb{Z}}$, because $\{\Lambda_K\}$ is cofinal in the divisibility order on $\mathbb{N}$. □

Remark 4.76 (Why a profinite parameter and not a real one). At every algebraic order all indices share one expansion. Parity enters at relative order $\mathrm{e}^{-\mu/2}$; the residue mod 3 at $\mathrm{e}^{-2\mu/3}$; the residue mod k at $\mathrm{e}^{-(1-1/k)\mu}$. No finite real-valued smooth parameter records all of these congruence classes simultaneously, and no single real analytic function of n interpolates all A_k canonically (theorem 4.11). A profinite integer does so canonically and finitely at every exponential resolution: each finite action cutoff $\eta < 1$ sees only A_k for $k \leq (1-\eta)^{-1}$, i.e. only $\widehat{r}$ modulo $\Lambda_{\lfloor (1-\eta)^{-1} \rfloor}$. This is S1's observation and is absent from S2 and S3.

Remark 4.77 (A structure suggested, not proved). S1 proposes an "arithmetic transseries sheaf" with base $\widehat{\mathbb{Z}}$ and fibres the finite-level analytic transseries algebras, and calls theorem 4.75 its elementary case. We record this as a suggestion. Nothing beyond theorem 4.75 — namely, projective compatibility of the coefficient systems — is established here or in the sources: no topology on the fibres, no gluing statement, and no continuity of $\widehat{r} \mapsto \mu_{K,\widehat{r}}$ in any topology on $\widehat{\mathbb{Z}}$ has been proved. It is stated as theorem 4.96 in section 4.15, not as a theorem.

4.11 Validity, truncation control, and integer recovery

4.11.1 Remainder transport

Theorem 4.78 (Inverse error after K Rademacher levels). *Fix $K \geq 1$. Let $y = p(n)$, let $\mu = \lambda\sqrt{n - 1/24}$, and let μ_K denote either inverse branch of theorem 4.57 (with $r \equiv n$) or of theorem 4.58. Then*

$$\mu_K(y) - \mu = \mathcal{O}_K\Big(\mu^{3/2}\,\mathrm{e}^{-(1-\frac{1}{K+1})\mu}\Big), \qquad n_K(y) - n = \mathcal{O}_K\Big(\mu^{5/2}\,\mathrm{e}^{-(1-\frac{1}{K+1})\mu}\Big), \tag{4.108}$$

where $n_K = \frac{1}{24} + 3\mu_K^2/(2\pi^2)$. Explicitly, with the constant of theorem 4.23, for $\mu \geq \mu_*$ (any threshold at which $\Phi' \geq \frac{1}{2}$ and the omitted tail is $\leq \frac{1}{2}$ in absolute value),

$$|n_K(y) - n| \leq \frac{3\mu}{\pi^2} \cdot \frac{2}{1} \cdot \frac{8\mu^2}{3(1-\mu^{-1})\sqrt{K}} \exp\Big[-\Big(1 - \frac{1}{K+1}\Big)\mu\Big]\,\big(1 + o(1)\big). \tag{4.109}$$

Proof. This is theorem 1.44 with the forward remainder supplied by theorems 4.23 and 4.24. In detail: let $\mathcal{F}_K = \log \mathcal{P}_K$ and $\mathcal{F} = \log p$ regarded as functions of the phase. By theorem 4.17, $\mathcal{F}(\mu) - \mathcal{F}_K(\mu) = \log\big(1 + \mathcal{R}_{>K}/(1 + \mathcal{R}_{\leq K})\big)$, which by theorem 4.24 and $|\log(1+u)| \leq 2|u|$ for $|u| \leq \frac{1}{2}$ is $\mathcal{O}_K(\mu^{3/2}\mathrm{e}^{-(1-1/(K+1))\mu})$. Since $\mathcal{F}_K'(\mu) = \Phi'(\mu)(1 + o(1)) \to 1$, the mean value theorem converts this logarithmic residual into the phase error, which is the first half of eq. (4.108). Finally $\mathrm{d}n/\mathrm{d}\mu = 3\mu/\pi^2$ gives the index error. The explicit form eq. (4.109) replaces theorem 4.24 by the explicit theorem 4.23 (losing $\mu^{-1/2}$, gaining a constant) and $\mathcal{F}_K' \geq \frac{1}{2}$. □

Corollary 4.79 (Beyond-all-orders accuracy of the core inverse). *For $y = p(n)$,*

$$N_0(y) - n = \mathcal{O}\Big(\mu^{5/2}\mathrm{e}^{-\mu/2}\Big) = \mathcal{O}\Big(n^{5/4}\exp\Big(-\tfrac{1}{2}\pi\sqrt{2n/3}\Big)\Big), \tag{4.110}$$

so $N_0(p(n)) - n \to 0$ faster than every power of n^{-1}.

Proof. Theorem 4.78 at $K = 1$; note $N_0 = n_1$ because the $K = 1$ level of theorem 4.57 contains the sectors $(1, \pm)$, and the $(1, -)$ sector has action 2, hence contributes below the $K = 1$ error bound. □

Proposition 4.80 (A posteriori certificate; specialisation of theorem 1.42). *Let $\mathcal{F}_K(\mu) = \log \mathcal{P}_K(\mu)$ and let $\rho(\widetilde{\mu}) = \mathcal{F}_K(\widetilde{\mu}) - \log y$ be the residual of a candidate $\widetilde{\mu}$. If $\mathcal{F}_K' \geq m > 0$ on an interval I containing both $\widetilde{\mu}$ and the exact root $\mu_K(y)$, then*

$$|\widetilde{\mu} - \mu_K(y)| \leq \frac{|\rho(\widetilde{\mu})|}{m}, \qquad |\widetilde{n} - n_K(y)| \leq \frac{3}{2\pi^2}\big(|\widetilde{\mu}| + |\mu_K(y)|\big)\frac{|\rho(\widetilde{\mu})|}{m}. \tag{4.111}$$

Proof. Mean value theorem applied to $\mathcal{F}_K$, then the factorisation $\widetilde{\mu}^2 - \mu_K^2 = (\widetilde{\mu} - \mu_K)(\widetilde{\mu} + \mu_K)$. Since $\mathcal{F}_K' \to 1$, one may take $m = \frac{1}{2}$ beyond an effectively checkable threshold. □

Remark 4.81 (Why a residual certificate rather than term counting). Theorem 4.80 separates the *local inversion* error (how well $\widetilde{\mu}$ solves the level-K equation) from the *Rademacher tail* (how far the level-K equation is from the truth). Only the second needs theorem 4.23. This modularity is what makes it possible to certify a truncated symbolic transseries and a numerically refined Newton root by the same inequality.

4.11.2 Algebraic truncation

Proposition 4.82. *With $U_M(z) = \sum_{m \leq M} c_m z^m$ as in theorem 4.50,*

$$\mu_0 - \big(X + U_M(X^{-1})\big) = \mathcal{O}(X^{-M-1}), \qquad N_0 - \left[\frac{1}{24} + \frac{3}{2\pi^2}\big(X + U_M(X^{-1})\big)^2\right] = \mathcal{O}(X^{-M}), \tag{4.112}$$

the loss of one order coming from $\mathrm{d}n/\mathrm{d}\mu \asymp X$. In particular $M \geq 1$ is necessary *for the index to be correct to $o(1)$ and hence for the arithmetic phases of section 4.9 to be evaluated at the right point (theorem 4.55).*

Proof. U is analytic at 0 by theorem 4.50, so its Taylor remainder after M terms is $\mathcal{O}(z^{M+1})$; substitute $z = X^{-1}$ and use $N_0 - \widetilde{N} = \frac{3}{2\pi^2}(\mu_0 - \widetilde{\mu})(\mu_0 + \widetilde{\mu})$ with $\mu_0 + \widetilde{\mu} = \mathcal{O}(X)$. □

4.11.3 Integer recovery

Theorem 4.83 (Eventual exact recovery by rounding). *There is n_0 such that for every integer $n \geq n_0$,*

$$n = \big\lfloor N_0(p(n)) \big\rceil, \tag{4.113}$$

$\lfloor\cdot\rceil$ denoting nearest-integer rounding. Consequently the arithmetic phase $\widehat{n}$ of theorem 4.75 is recoverable from $y = p(n)$ before *any exponentially small sector is evaluated.*

Proof. By eq. (4.110), $|N_0(p(n)) - n| \to 0$; it is therefore $< \frac{1}{2}$ for all large n, at which point rounding is exact. The last sentence is the observation that $A_k(n)$ depends only on $n \bmod k$. □

Remark 4.84 (The threshold is very small in practice). Theorem 4.83 is asymptotic and claims no optimal n_0. We computed $N_0(p(n)) - n$ for all $1 \le n \le 59$ in 140-digit arithmetic; the largest absolute value is $0.1723\ldots$ at $n = 1$, and the sequence decreases in magnitude thereafter (through 2.8×10^{-2} at $n = 10$, 4.5×10^{-4} at $n = 50$, 1.5×10^{-5} at $n = 100$, 4.3×10^{-17} at $n = 1000$). So $n_0 = 1$ is consistent with the data, though we prove only the asymptotic statement; producing a certified n_0 requires effective constants in theorem 4.78, i.e. effective Rademacher remainder bounds, which is listed as theorem 4.95.

Remark 4.85 (Circularity, avoided). The arithmetic transseries needs the residue r of n, which appears to require knowing n. The order of operations resolves this: *(i)* compute X from eq. (4.56); *(ii)* compute μ_0 and N_0, which are phase-independent; *(iii)* round, obtaining n by theorem 4.83 for y on the range; *(iv)* now every $A_k(n)$ is known, and the exponentially small sectors refine an already identified integer. The arithmetic layer is thus a *refinement* of a known answer, not a device for guessing the phase. For y off the range, step (iii) yields $\lceil \mathcal{N}(y) \rfloor$ by theorem 4.44 instead, and the sectors describe $\mathcal{N}$ itself.

4.12 Numerical verification

Every number in this chapter was recomputed for this volume. Partition values were generated exactly by Euler's pentagonal recurrence

$$p(n) = \sum_{m \ge 1} (-1)^{m-1} \left[p\Big(n - \tfrac{m(3m-1)}{2}\Big) + p\Big(n - \tfrac{m(3m+1)}{2}\Big) \right], \qquad p(0) = 1,\ p(r) = 0\ (r < 0), \tag{4.114}$$

in exact integer arithmetic (checked against $p(10) = 42$, $p(50) = 204226$, $p(100) = 190569292$); Dedekind sums exactly as rationals from eq. (4.9); the transcendental factors in 140-digit floating point. Coefficient identities were verified in exact rational or exact symbolic arithmetic, never by floating-point spot checks. The tables check normalisation, signs, phases and the hierarchy of actions; they are used in no proof.

Table 4.3: Relative error $\big(S_K(n) - p(n)\big)/p(n)$ of the paired truncation $S_K = \sum_{k \le K}(P_{k,+} + P_{k,-})$, recomputed at 140 digits. Accuracy need not improve monotonically in K at small n: the amplitudes $A_k(n)$ change sign and can suppress or enhance an individual sector.

n	$p(n)$	$K=1$	$K=2$	$K=3$	$K=5$
10	42	$-8.8621 \cdot 10^{-3}$	$1.6606 \cdot 10^{-3}$	$3.7391 \cdot 10^{-4}$	$3.4722 \cdot 10^{-4}$
25	1 958	$1.0717 \cdot 10^{-3}$	$3.0045 \cdot 10^{-6}$	$-6.1015 \cdot 10^{-5}$	$-3.4866 \cdot 10^{-6}$
50	204 226	$-7.3074 \cdot 10^{-5}$	$3.9028 \cdot 10^{-6}$	$2.1050 \cdot 10^{-7}$	$-1.1350 \cdot 10^{-7}$
100	190 569 292	$-1.8220 \cdot 10^{-6}$	$8.6838 \cdot 10^{-9}$	$-4.9491 \cdot 10^{-9}$	$3.1454 \cdot 10^{-10}$
250	$2.3079 \cdot 10^{14}$	$-1.0760 \cdot 10^{-9}$	$7.2267 \cdot 10^{-13}$	$4.3021 \cdot 10^{-14}$	$2.1256 \cdot 10^{-15}$
500	$2.3002 \cdot 10^{21}$	$-2.4389 \cdot 10^{-13}$	$1.7537 \cdot 10^{-17}$	$-2.0017 \cdot 10^{-19}$	$-3.5933 \cdot 10^{-22}$
1000	$2.4061 \cdot 10^{31}$	$-1.6997 \cdot 10^{-18}$	$1.2574 \cdot 10^{-24}$	$-3.4744 \cdot 10^{-27}$	$3.9453 \cdot 10^{-30}$

Three features of table 4.4 confirm the theory point by point. *(i)* The first column tends to $-3/\pi^2 = -0.30396\ldots$, not to 0: this is theorem 4.54 and the numerical signature of theorem 4.55. (S1 and S2 tabulate $N_X = \frac{1}{24} + 3X^2/(2\pi^2)$ and report $\to -0.304$; S3 tabulates $3X^2/(2\pi^2)$ without the $\frac{1}{24}$ and reports $\to -0.346$. The columns differ by exactly $1/24$ and each is correct for its own definition; we keep the $\frac{1}{24}$ so that N_X and N_0 are directly comparable.) *(ii)* Restoring the amplitude factor $1 - 1/\mu$, i.e. passing from X to μ_0, removes that entire $\mathcal{O}(1)$ bias and leaves an exponentially small error. *(iii)* The residual after the core alternates with the parity of n: at $n = 49, 50, 51$ it is $-5.2643 \cdot 10^{-4}$, $+4.5102 \cdot 10^{-4}$, $-4.0920 \cdot 10^{-4}$, exactly as eq. (4.100) predicts through $\widetilde{A}_2 = \cos \pi N_0$. Adding the single first-order $k = 2$ sector reduces these to $-1.0890 \cdot 10^{-5}$, $-2.4087 \cdot 10^{-5}$, $+2.9029 \cdot 10^{-5}$; adding the first-order sectors through $k = 10$ reduces them to $-2.24 \cdot 10^{-7}$, $-9.95 \cdot 10^{-8}$, $-5.81 \cdot 10^{-8}$. This checks both the sign and the phase of the leading chirp eq. (4.106).

Table 4.4: Inverse errors at exact ordinates $y = p(n)$; entries are (approximation) $- n$. N_X is the bare Lambert carrier index, N_0 the dominant core, N_K the Newton inverse of the level-K interpolated sum (theorem 4.58). The residue-frozen inverses (theorem 4.57) reproduce the last three columns to all five digits shown for $n \geq 100$; at $n = 50$ they differ in the fifth digit of the $K = 3, 5$ entries ($-1.2992 \cdot 10^{-6}$, $7.0057 \cdot 10^{-7}$), at $n = 25$ in the fourth ($2.7926 \cdot 10^{-4}$, $1.5958 \cdot 10^{-5}$), at $n = 10$ in the third ($-5.3338 \cdot 10^{-3}$, $-1.2006 \cdot 10^{-3}$, $-1.1150 \cdot 10^{-3}$). That is exactly the pattern theorem 4.59 predicts.

n	$N_X - n$	$N_0 - n$	$N_2 - n$	$N_3 - n$	$N_5 - n$
10	$-3.9910 \cdot 10^{-1}$	$2.8447 \cdot 10^{-2}$	$-5.3290 \cdot 10^{-3}$	$-1.1863 \cdot 10^{-3}$	$-1.1015 \cdot 10^{-3}$
25	$-3.7877 \cdot 10^{-1}$	$-4.9052 \cdot 10^{-3}$	$-1.3751 \cdot 10^{-5}$	$2.7902 \cdot 10^{-4}$	$1.5941 \cdot 10^{-5}$
50	$-3.5044 \cdot 10^{-1}$	$4.5102 \cdot 10^{-4}$	$-2.4089 \cdot 10^{-5}$	$-1.2993 \cdot 10^{-6}$	$7.0059 \cdot 10^{-7}$
100	$-3.3600 \cdot 10^{-1}$	$1.5378 \cdot 10^{-5}$	$-7.3293 \cdot 10^{-8}$	$4.1772 \cdot 10^{-8}$	$-2.6548 \cdot 10^{-9}$
250	$-3.2364 \cdot 10^{-1}$	$1.3942 \cdot 10^{-8}$	$-9.3644 \cdot 10^{-12}$	$-5.5747 \cdot 10^{-13}$	$-2.7544 \cdot 10^{-14}$
500	$-3.1768 \cdot 10^{-1}$	$4.4042 \cdot 10^{-12}$	$-3.1668 \cdot 10^{-16}$	$3.6146 \cdot 10^{-18}$	$6.4888 \cdot 10^{-21}$
1000	$-3.1356 \cdot 10^{-1}$	$4.2960 \cdot 10^{-17}$	$-3.1780 \cdot 10^{-23}$	$8.7815 \cdot 10^{-26}$	$-9.9717 \cdot 10^{-29}$

Remark 4.86 (What was verified in exact arithmetic). (1) Equation (4.33) against the Taylor expansion of eq. (4.31), symbolically in β through order t^9: identical; the closed forms eqs. (4.37) and (4.38) were produced from eq. (4.33) and match S2 (through c_5) and S3 (through c_8). (2) $\omega_0, \ldots, \omega_4$ against both the S1 and the S2 grouping of the same numbers: identical (they look different only because S1 splits each into partial fractions in π). (3) $c_1, \ldots, c_{18}$ of table 4.2 from eq. (4.72) in exact rationals, plus the residual test eq. (4.74) at $M = 12$; all eighteen agree with the sources. (4) $\gamma_1, \ldots, \gamma_8$ against all three sources; γ_9, γ_{10} are new here. (5) $\Pi_1, \ldots, \Pi_8$ from the Lagrange form against the Stirling form and against each of the three printed shapes, confirming theorem 4.48; S1's printed P_6, P_7 are correct. (6) S3's u-expansion in H through H^{-4} and its N_0-expansion through H^{-3}, and S2's N_0-block in R through the constant term: correct as printed. (7) Equation (4.11) against direct evaluation of eq. (4.10) from exact Dedekind sums. (8) The chirp constant eq. (4.107) to 25 digits in both forms. No discrepancy with any source's arithmetic was found: every repair recorded in this chapter is structural, not numerical.

Remark 4.87 (Normalisation cross-checks). Because eq. (4.14) carries a factor $\sqrt{k}$ that some normalisations absorb into A_k, a missing or spurious power of k is the easiest error to make and the hardest to see. Three identifications of one quantity detect it: the kth derivative factor of eq. (4.14) equals $\frac{\mu}{4kN^{3/2}}[\mathrm{e}^{\mu/k}(1 - k/\mu) + \mathrm{e}^{-\mu/k}(1 + k/\mu)]$, and after multiplication by $\sqrt{k}/(\pi\sqrt{2})$ it may be written either as $\frac{1}{4\sqrt{3k}\,N}[\cdots]$ or as $\frac{\kappa}{\sqrt{k}\,\mu^2}[\cdots]$; at $k = 1$ the positive term is $\frac{\mathrm{e}^{\mu}}{4\sqrt{3}\,N}(1 - \frac{1}{\mu})$, which becomes eq. (4.2) after replacing N by n and dropping $1 - 1/\mu$. Any of these disagreeing with the others by a power of k localises the slip.

4.13 Algorithms

All formulas of this chapter are constructive and need only formal power-series arithmetic over $\mathbb{Q}(\pi)$ together with exact rational Dedekind sums.

Algorithm 4.88 (Forward sector coefficients)**.** Given k, σ, M: set $\beta = \sigma\pi/(6k)$ and evaluate the finite sum eq. (4.33) for $0 \leq m \leq M$ ($\mathcal{O}(m)$ field operations each, no series manipulation; keep β symbolic if sector-uniform output is wanted). If the whole array is wanted at once, evaluate $\ell_1, \ldots, \ell_M$ from eq. (4.39) and run the triangular recurrence eq. (4.41) instead. Assemble eq. (4.30), pairing the two signs before summing over large k (theorem 4.20).

Algorithm 4.89 (Phase-independent inverse)**.** Given a large y and an order M: (1) compute X from eq. (4.56) with a standard W_{-1} routine — do *not* substitute the logarithmic expansion eq. (4.57), which is numerically far inferior; (2) form $\mu_0 = X + \sum_{m \leq M} c_m X^{-m}$ from the universal rationals of table 4.2, optionally refined by one Newton step on $\Phi(\mu) = R$; (3) return $N_0 = \frac{1}{24} + 3\mu_0^2/(2\pi^2)$; (4) certify by evaluating $\rho = \mu_0 - 2\log\mu_0 + \log(1 - \mu_0^{-1}) - \log(y/\kappa)$ and applying theorem 4.80; (5) round (theorem 4.83) if y is known to lie on the range, else take $\lceil\cdot\rceil$ for the staircase (theorem 4.44).

Algorithm 4.90 (Arithmetic exponential correction)**.** Given y, a level K and a finite set of multi-indices: (1) compute μ_0, N_0 by theorem 4.89 and use N_0, never N_X, to evaluate amplitudes (theorem 4.55); (2) compute exact rational $s(h,k)$ for $k \le K$ and hence $A_k(r)$ (frozen, with r from theorem 4.83) or $\widetilde{A}_k(N_0)$ (interpolated); (3) build the actions and amplitudes of eq. (4.83); (4) traverse the multi-indices in a graded monomial order using eq. (4.93), or, for one isolated $\boldsymbol{m}$, use eq. (4.86) or the Bell form eq. (4.88) with the closed jets eq. (4.90); (5) substitute $\varepsilon_j = \mathrm{e}^{-\alpha_j \mu_0}$ and convert to the index by eq. (4.99); (6) certify by theorem 4.80 on $\mathcal{P}_K$, adding theorem 4.23 if the comparison is with the exact sequence.

Remark 4.91 (Modular structure of the computation). The layers are independent and should be cached separately: the carrier polynomials Π_j depend only on the exponent data $(a,b) = (1,-2)$; the rationals c_m only on the amplitude $1 - 1/\mu$; the inverse-derivative polynomials $\mu_0^{(s)}$ only on Φ; and the arithmetic enters solely through the numbers a_k in eq. (4.83). Universal tables are computed once and residue-dependent evaluation is then cheap.

4.14 A template for Rademacher-type sequences

Nothing above used the partition function except through four properties. S3 has the most careful version of this observation and S1 the most usable one; what follows merges them into a hypothesis list and a theorem, so that the template is a statement one can check rather than a slogan.

Definition 4.92 (Rademacher-type representation)**.** A sequence $(a(n))_{n \ge n_1}$ of positive reals is of *Rademacher type* if:

(T1) there is a real analytic, strictly increasing *phase* $\phi : (\nu_1, \infty) \to (0, \infty)$ with $\phi \to \infty$ and real analytic inverse, giving the large variable $\mu = \phi(n)$;

(T2) the *envelope* $F(\mu) = C\mathrm{e}^{a\mu^\alpha} \mu^b \exp Q(\mu^{-\delta})$ is an exponential–power model theorem 1.2 with $a > 0$, $\alpha \in (0, 1]$;

(T3) there are actions $\alpha_j > 0$ ($j \in J$) with $\inf_j \alpha_j > 0$, amplitudes $a_j(n)$ taking values in a fixed finite-dimensional space at each truncation level (e.g. periodic in n), and g_j analytic at ∞, with

$$a(n) = F(\mu)\Big(1 + \sum_{j \in J'} a_j(n) g_j(\mu) \mathrm{e}^{-\alpha_j \mu} + \mathcal{T}_{J'}(\mu; n)\Big) \quad \text{for every finite } J' \subset J; \tag{4.115}$$

(T4) $\{j : \alpha_j \le \eta\}$ is finite for every $\eta < \sup_j \alpha_j$, and either $\mathcal{T}_{J'} = \mathcal{O}(\mu^{C'} \mathrm{e}^{-\eta' \mu})$ for some $\eta' > \max_{j \in J'} \alpha_j$, or — if the actions accumulate — the grouped remainder is retained as one analytic block.

Theorem 4.93 (Template)**.** *Let* $(a(n))$ *be of Rademacher type. Then:*

1. (Reduction.) $\mu = \xi^{1/\alpha}$ *of theorem 1.4 brings* (T2) *to the linear-phase model with computable* $(C', 1, b', Q')$.
2. (Carrier.) $C' \mathrm{e}^\xi \xi^{b'} = y$ *is solved exactly by theorem 1.20; equivalently, for* $y = C\mathrm{e}^{a\mu^\alpha} \mu^{-\beta}$,

$$\mu = \left(-\frac{\beta}{a\alpha} W_{-1}\left(-\frac{a\alpha}{\beta}\left(\frac{C}{y}\right)^{\alpha/\beta}\right)\right)^{1/\alpha}. \tag{4.116}$$

3. (Amplitude.) *The envelope inverse is* $\xi_0 = X + U(X^{-1})$ *with* U *the unique solution of the fixed-point equation of theorem 1.25 at* $(1, b', Q')$*; coefficients by theorem 1.16 or by isolating the linear occurrence of* U.
4. (Phase locking.) *If some* a_j *is periodic on the index scale and* $\mathrm{d}n/\mathrm{d}\mu \to \infty$*, the amplitudes must be evaluated at* ξ_0*, not at* X*: the carrier bias* $n(\xi_0) - n(X)$ *is* $\mathcal{O}(1)$ *in general.*

5. (Sectors.) *With the phase frozen, every inverse transmonomial is given by eq.* (4.86) *(in y) or eq.* (4.95)*/eq.* (4.93) *(in μ), with first-order term*

$$\mu = \mu_0 - \frac{a_j g_j(\mu_0)}{(\log F)'(\mu_0)} \mathrm{e}^{-\alpha_j \mu_0} + \mathcal{O}\big(\mathrm{e}^{-2\alpha_{\min}\mu_0}\big). \tag{4.117}$$

6. (Transport.) *A relative forward remainder ϵ transports to a phase error $\epsilon/(\log F)'(\mu_0)$ and an index error $\epsilon\, n'(\mu_0)/(\log F)'(\mu_0)$, by theorem 1.44.*

7. (Staircase.) *If a is integer valued with gaps bounded below, the generalised inverse is the ceiling of the smooth one (theorem 1.39), with the order-one sawtooth of theorem 4.45 as an irreducible final sector.*

Proof. (i)–(iii) are theorems 1.4, 1.20 and 1.25 at the stated parameters. For the display in (ii): raising $y = C\mathrm{e}^{a\mu^\alpha}\mu^{-\beta}$ to the power α/β and rearranging gives $\big(\frac{a\alpha}{\beta}\mu^\alpha\big)\mathrm{e}^{-(a\alpha/\beta)\mu^\alpha} = \frac{a\alpha}{\beta}(C/y)^{\alpha/\beta}$, which is the Lambert equation with the stated argument, the branch -1 forced by $\mu \to \infty$. (iv) is the computation of theorem 4.55 with $3\mu/\pi^2$ replaced by $n'(\mu_0)$: what matters is only that $n' \to \infty$ while $\xi_0 - X \to 0$, so their product need not vanish. (v) is theorems 4.62, 4.65 and 4.67, whose proofs used no property of F or of the blocks beyond analyticity and $F' \neq 0$; the first-order display is eq. (4.96) in that generality. (vi) is the mean-value argument of theorem 4.78, and (vii) is theorem 4.44. ☐

Remark 4.94 (The partition case, and the scope of the template). Taking $\phi(n) = \pi\sqrt{2(n-1/24)/3}$, (C, a, α, b, Q) as in eq. (4.5), $J = \{(k,\sigma)\}$, $\alpha_{k,\sigma} = 1 - \sigma/k$ and $a_k = A_k$, hypotheses (T1)–(T3) hold by theorems 4.14 and 4.17, and (T4) holds in its second form: the actions *do* accumulate at 1, and the grouped block is the paired tail eq. (4.25) controlled by theorem 4.23. With $(\alpha, \beta) = (1, 2)$ in μ, eq. (4.116) reduces to eq. (4.56). The template applies verbatim to coefficient sequences of weakly holomorphic modular forms and eta-quotients with Rademacher-type expansions; the partition function is unusually clean because the Bessel index $3/2$ makes the kernel two exponentials with amplitudes linear in μ^{-1}, so step (iii) is the explicit rational recurrence eq. (4.72) rather than a general implicit solve. *No theorem about any other specific sequence is claimed here*: verifying (T1)–(T4) for a given family is exactly the work the template does not do.

4.15 Structural consequences and open questions

4.15.1 Convergent blocks inside an exponentially complete object

The exact sector sum eq. (4.19) converges; each sector's algebraic series in $n^{-1/2}$ converges on $|n^{-1/2}| < \sqrt{24}$ (theorem 4.30); and the reversion series U converges near 0 (theorem 4.50). Divergence is therefore *not* what creates the non-perturbative sectors here: they are supplied by modularity and the circle method, independently of the perturbative data. The useful trichotomy is that a *Poincaré expansion* records one dominant sector and forgets all exponentially small information; a *transseries completion* records the other sectors, whether or not the series in each diverges; and an *exact transseries representation*, as here, may itself converge. Compare Parts 2, 3 and 5, where the algebraic series do diverge and theorem 1.46 is the operative tool.

4.15.2 Arithmetic amplitudes are not Stokes constants

The $A_k(n)$ weight distinct exponential sectors and govern exponentially small corrections to the inverse, which is formally the role of Stokes constants. They are not Stokes constants: they are arithmetic, periodic in n rather than fixed, and no natural analytic continuation in the index produces them canonically (theorem 4.11). Theorem 4.75 says what replaces the Stokes data: a profinite integer, of which every finite exponential resolution sees only a finite quotient.

4.15.3 Open questions

Conjecture 4.95 (Effective rounding threshold)**.** *There is an explicit small n_0 — the evidence of theorem 4.84 suggests $n_0 = 1$ — with $n = \lfloor N_0(p(n)) \rceil$ for all $n \geq n_0$, provable by inserting effective Rademacher remainder bounds (Lehmer; Banerjee–Paule–Radu–Schneider) in place of $|A_k| \leq k$ in theorem 4.78.*

Conjecture 4.96 (Arithmetic transseries sheaf)**.** *The projective family of theorem 4.75 extends to a sheaf of transseries algebras over $\widehat{\mathbb{Z}}$, with $\widehat{r} \mapsto$ (level-K inverse transseries) continuous for the profinite topology and the finite-level algebras gluing over the standard clopen basis. Only the projective compatibility is proved here (theorem 4.77).*

Conjecture 4.97 (Uniform treatment at the accumulation action)**.** *The zeta condensation eq.* (4.29) *can serve as a single block in theorem 4.62, yielding a description of the inverse uniform across the accumulation action* 1*, with error bounds depending on the analytic behaviour of $\mathcal{Z}_n(s)$ rather than on a sector cutoff. This needs continuation and vertical growth information for the Kloosterman-type Dirichlet series $\mathcal{Z}_n$, which is not established here. S3 raises it as an optimisation problem rather than a gap, correctly: the exact Rademacher representation is always available as a fallback.*

Remark 4.98 (Classifying interpolations). Different strictly increasing smooth interpolations of p agree at every algebraic order of the inverse and on the integer range, but differ in the exponentially small off-lattice sectors: candidates include theorem 4.9, piecewise-logarithmic interpolation through $(n, \log p(n))$, Newton or cardinal series with prescribed growth, and continuations built from Poincaré series. Which part of the inverse transseries is interpolation-independent? Theorem 4.59 answers this for one pair of choices — they agree strictly below action 1 — and a general statement is open.

Remark 4.99 (Formalisation targets). Rademacher's theorem is now machine-checked (Eberl and Paulson, Archive of Formal Proofs, 2026). The natural next steps, in dependency order, are: (a) the normalisation theorem 4.14; (b) the generating function eq. (4.31) and the recurrence eq. (4.41); (c) the Catalan/residue proof of theorem 4.31; (d) the implicit construction of U and eq. (4.72); (e) the triangular multivariate inversion eq. (4.93); (f) residual-to-root transfer, theorem 4.80, under an explicit derivative lower bound. Steps (b), (d), (e) are Bell-polynomial infrastructure shared with Part 1.

4.16 Summary

1. In the shifted phase $\mu = \pi\sqrt{2(n - 1/24)/3}$, Rademacher's series is the exact sector sum eq. (4.17), each sector having a two-term amplitude linear in μ^{-1}. At $(k, \sigma) = (1, +1)$ it is the exponential–power model theorem 1.2 with $\alpha = 1/2$, reduced by theorem 1.4 to $\kappa e^{\mu}\mu^{-2}(1 - \mu^{-1})$ (theorem 4.1 and table 4.1).

2. Re-expansion in $n^{-1/2}$ is governed by one analytic function eq. (4.31); its coefficients have the finite closed form eq. (4.33) in *every* sector, which enters only through $\beta = \sigma\pi/(6k)$. Each such series converges while the total is only asymptotic — confusing the two is the characteristic error of the subject.

3. The actions $1 \mp 1/k$ accumulate at 1 from both sides, so the completed object is not a Hahn series and the two signs may not be separated (theorems 4.19 and 4.20); theorem 4.23 bounds the tail explicitly and theorem 4.28 packages the accumulation exactly.

4. The inverse carrier is $-2W_{-1}(-\frac{1}{2}\sqrt{\kappa/y})$ with Stirling flattening polynomials eq. (4.58); the amplitude is restored by theorem 1.25 at $\Psi(t, u) = 3\log(1 + tu) - \log(1 - t + tu)$, whose coefficients are table 4.2.

5. Since $dn/d\mu \asymp \mu$, that reversion produces an $\mathcal{O}(1)$ index shift $3/\pi^2$ which must be resummed *before* the periodic amplitudes are evaluated (theorem 4.55).

6. With the phase frozen — by a residue class mod $\operatorname{lcm}(1, \ldots, K)$ or by the entire interpolation theorem 4.9, which agree strictly below action 1 (theorem 4.59) — every inverse transmonomial is given by eq. (4.86), eq. (4.88), eq. (4.95) or eq. (4.93). The leading correction is the parity chirp eq. (4.106) with constant $3^{1/4}/(2\pi)$, and the primary sectors form the staircase eq. (4.105).

7. Along the range the core alone recovers n by rounding (theorem 4.83); for general y the staircase is the ceiling of the smooth inverse (theorem 4.44), whose order-one sawtooth dominates every arithmetic sector (theorem 4.45).

Remark 4.100 (Works referred to). Analytic input: Rademacher, Proc. Nat. Acad. Sci. USA **23** (1937), 78–84; Proc. London Math. Soc. (2) **43** (1938), 241–254; Ann. of Math. **44** (1943), 416–422; textbook treatment in Apostol, *Modular Functions and Dirichlet Series in Number Theory*, 2nd ed., Springer GTM 41, 1990; machine-checked in Eberl and Paulson, *Rademacher's series for the partition function*, Archive of Formal Proofs, 2026. Leading asymptotic: Hardy and Ramanujan, Proc. London Math. Soc. (2) **17** (1918), 75–115. Remainder estimates: Lehmer, J. London Math. Soc. **12** (1937), 171–176 and Trans. Amer. Math. Soc. **43** (1938), 271–295, **46** (1939), 362–373; Banerjee, Paule, Radu and Schneider, arXiv:2209.07887. The dominant-sector case of eq. (4.33) is O'Sullivan's, arXiv:2205.13468 and arXiv:2203.02868. Lambert W: Corless, Gonnet, Hare, Jeffrey and Knuth, Adv. Comput. Math. **5** (1996), 329–359; Jeffrey, Corless, Hare and Knuth, arXiv:math/9512230. The sequence is OEIS A000041.

Chapter 5

The swing factorial (A056040)

This chapter merges three independent treatments of the swinging factorial: *All-Orders Asymptotic Transseries for the Swing Factorial and Its Two Inverse Branches* (cited below as S1), *The Full Asymptotic Transseries of the Swing Factorial and Its Inverse* (S2), and *Parity-Resolved Transseries for the Swing Factorial and Its Inverse* (S3). The general apparatus of S2 — its universal coefficient calculus for exponential–power inversion and its Lambert-normalized reversion theorem — was promoted to Part 1; here it is only *cited* and *specialized.* S1 works in the half index m with the carrier constant $\log 4$; S2 and S3 work in the original index with $\log 2$ and differ only in the sign convention for the parity marker. The chapter adopts the S2/Part 1 normalization throughout and gives the translation dictionary in theorem 5.12. S1's genuinely distinct contribution — the centred (translation-symmetric) normal form — is kept, but recast so that it too is a single σ-uniform statement.
Every rational, polynomial and algebraic coefficient displayed in this chapter was recomputed from the stated recurrences in exact arithmetic (`sympy`, rational Bernoulli input, truncated power series over $\mathbb{Q}[\lambda, \lambda^{-1}, \sigma]/(\sigma^2 - 1)$); every decimal in the validation tables was recomputed at 50 significant digits. Where a recomputation reproduced a source table, that is said; where it did not, the discrepancy is recorded in a repairbox.

5.1 The sequence, the parity obstruction, and what "inverse" can mean

5.1.1 Definition and exact parity identities

Definition 5.1 (Swing factorial). For $n \in \mathbb{N}_0$ put

$$\mathrm{sw}(n) = \frac{n!}{\lfloor n/2 \rfloor!^{\,2}}. \tag{5.1}$$

This is OEIS A056040, the *swinging factorial.*

Splitting on the parity of n gives at once

$$\mathrm{sw}(2m) = \binom{2m}{m}, \qquad \mathrm{sw}(2m+1) = (2m+1)\binom{2m}{m}, \tag{5.2}$$

whence the one-step recurrence

$$\mathrm{sw}(0) = 1, \qquad \mathrm{sw}(n) = \begin{cases} n\,\mathrm{sw}(n-1), & n \text{ odd}, \\ \dfrac{4}{n}\,\mathrm{sw}(n-1), & n \text{ even}, \end{cases} \tag{5.3}$$

and the closed product form

$$\mathrm{sw}(n) = 2^{\,n-(n \bmod 2)} \prod_{k=1}^{n} k^{(-1)^{k+1}}. \tag{5.4}$$

The first values are

$$1,\,1,\,2,\,6,\,6,\,30,\,20,\,140,\,70,\,630,\,252,\,2772,\,924,\,12012,\,3432,\,51480,\,\ldots$$

Proposition 5.2 (Generating functions)**.**

$$\sum_{n\geq 0} \mathrm{sw}(n)\frac{z^n}{n!} = (1+z)\,I_0(2z), \tag{5.5}$$

$$\sum_{n\geq 0} \mathrm{sw}(n) z^n = \frac{1}{\sqrt{1-4z^2}} + \frac{z}{(1-4z^2)^{3/2}}, \tag{5.6}$$

where I_0 is the modified Bessel function of order zero.

Proof. By eq. (5.2) the even exponential part is $\sum_{m\geq 0} z^{2m}/(m!)^2 = I_0(2z)$ and the odd exponential part is $z\,I_0(2z)$, which is eq. (5.5). For the ordinary series put $F(v) = \sum_{m\geq 0}\binom{2m}{m}v^m = (1-4v)^{-1/2}$; then the even part is $F(z^2)$ and the odd part is $z\big(2vF'(v)+F(v)\big)\big|_{v=z^2} = z(1-4z^2)^{-3/2}$. □

Remark 5.3 (The generating function already shows the obstruction). The two singularities $z = \pm\frac{1}{2}$ of eq. (5.6) have equal modulus. Co-dominant singularities of equal modulus at $\pm\rho$ produce a period-two oscillation in the coefficient asymptotics, and no single non-oscillatory singular expansion can represent the whole sequence. Even/odd projection separates the two contributions exactly. This is the generating-function shadow of everything that follows.

5.1.2 The sequence is not eventually monotone

Proposition 5.4 (Exact adjacent ratios)**.** *For $m \geq 0$,*

$$\frac{\mathrm{sw}(2m+1)}{\mathrm{sw}(2m)} = 2m+1, \qquad \frac{\mathrm{sw}(2m+2)}{\mathrm{sw}(2m+1)} = \frac{2}{m+1}. \tag{5.7}$$

Consequently sw *rises strictly from every even index to the next odd index, and falls strictly from every odd index to the next even index as soon as $m \geq 2$. In particular* sw *is not eventually monotone and has no eventually defined single-valued inverse.*

Proof. Both ratios are eq. (5.3). The second is < 1 exactly when $m+1 > 2$. □

Warning 5.5 (There is no "the" inverse swing factorial). Any formula presented as *the* inverse of sw without naming a parity sheet or a generalized-inverse convention is incomplete: of the three inverse objects of theorem 1.37, none exists globally for sw and all three exist branchwise (section 5.10).

5.1.3 The two analytic branches

All three sources resolve the obstruction the same way: interpolate the two bisections separately by gamma quotients. We use the parity marker $\sigma \in \{-1,+1\}$ of S2, with

$$\sigma = -1 \text{ (even phase)}, \qquad \sigma = +1 \text{ (odd phase)}, \qquad \sigma_n := (-1)^{n+1}, \qquad \lambda := \log 2. \tag{5.8}$$

The coefficient ring of the whole chapter is the *parity algebra*

$$\mathscr{R} := \mathbb{Q}[\lambda, \lambda^{-1}, \sigma]/(\sigma^2-1), \tag{5.9}$$

in which every formula below is a single element carrying both sheets. This is not a notational economy only: σ_n is a discrete transmonomial of modulus one, and it must remain visible at every order of every expansion.

Definition 5.6 (Parity-resolved gamma branches). For $x > 0$ and $\sigma \in \{-1, +1\}$ set

$$\mathcal{S}_\sigma(x) := \frac{\Gamma(x+1)}{\Gamma\big(x/2 + (3-\sigma)/4\big)^2}. \tag{5.10}$$

Explicitly $\mathcal{S}_-(x) = \Gamma(x+1)/\Gamma(x/2+1)^2$ and $\mathcal{S}_+(x) = \Gamma(x+1)/\Gamma\big((x+1)/2\big)^2$, and

$$\mathcal{S}_-(2m) = \mathrm{sw}(2m), \qquad \mathcal{S}_+(2m+1) = \mathrm{sw}(2m+1) \qquad (m \in \mathbb{N}_0). \tag{5.11}$$

Proposition 5.7 (Duplication normal form and duality). *For $x > 0$,*

$$\mathcal{S}_\sigma(x) = \frac{2^x}{\sqrt{\pi}} \left(\frac{\Gamma(x/2+1)}{\Gamma(x/2+1/2)} \right)^\sigma, \tag{5.12}$$

and consequently

$$\mathcal{S}_+(x)\, \mathcal{S}_-(x) = \frac{4^x}{\pi}, \qquad \frac{\mathcal{S}_\sigma(x+2)}{\mathcal{S}_\sigma(x)} = \frac{16(x+1)(x+2)}{(2x+3-\sigma)^2} = \begin{cases} \dfrac{4(x+1)}{x+2}, & \sigma = -1, \\ \dfrac{4(x+2)}{x+1}, & \sigma = +1. \end{cases} \tag{5.13}$$

Proof. Legendre duplication in the form $\Gamma(x+1) = 2^x \pi^{-1/2} \Gamma\big((x+1)/2\big)\Gamma(x/2+1)$ turns eq. (5.10) into eq. (5.12); multiplying the two values of σ makes the gamma ratio cancel and leaves $4^x/\pi$. For the two-step ratio, $\Gamma(x+3)/\Gamma(x+1) = (x+1)(x+2)$ while the denominator contributes $\big(x/2 + (3-\sigma)/4\big)^{-2} = 16(2x+3-\sigma)^{-2}$. □

Proposition 5.8 (Strict monotonicity and existence of the real inverses). *For $x > 0$,*

$$\frac{\mathrm{d}}{\mathrm{d}x} \log \mathcal{S}_\sigma(x) = \lambda + \frac{\sigma}{2}\Big[\psi\big(\tfrac{x}{2}+1\big) - \psi\big(\tfrac{x}{2}+\tfrac{1}{2}\big)\Big], \tag{5.14}$$

where $\psi = \Gamma'/\Gamma$. The bracket decreases strictly from 2λ at $x = 0$ to 0 at $x = \infty$. Hence $\mathcal{S}_+$ has strictly positive derivative on $[0, \infty)$, $\mathcal{S}_-$ has derivative 0 at $x = 0$ and strictly positive derivative on $(0, \infty)$, and both are strictly increasing bijections of $[0, \infty)$ onto $[1, \infty)$. Write

$$\mathcal{I}_\sigma := \mathcal{S}_\sigma^{-1} : [1, \infty) \to [0, \infty). \tag{5.15}$$

Proof. Differentiating eq. (5.12) logarithmically gives eq. (5.14). Put $g(x) = \psi(x/2+1) - \psi(x/2+1/2)$. Then $g'(x) = \frac{1}{2}\big[\psi'(x/2+1) - \psi'(x/2+1/2)\big] < 0$ because the trigamma function $\psi'(t) = \sum_{k \geq 0}(t+k)^{-2}$ is strictly decreasing on $(0, \infty)$; and $g(x) \to 0$ as $x \to \infty$ by $\psi(t+\frac{1}{2}) - \psi(t) \to 0$. Finally $g(0) = \psi(1) - \psi(1/2) = 2\log 2 = 2\lambda$. Thus $0 < g(x) < 2\lambda$ for $x > 0$. For $\sigma = +1$, eq. (5.14) equals $\lambda + g/2 > 0$ on $[0, \infty)$; for $\sigma = -1$ it equals $\lambda - g/2$, which vanishes only at $x = 0$ and is positive afterwards. Both branches take the value 1 at $x = 0$ and tend to $+\infty$, so each is a continuous increasing bijection onto $[1, \infty)$. □

5.1.4 Why one cannot simply interpolate through both parities

Proposition 5.9 (Every smooth interpolation of the whole sequence oscillates). *With $R(x) = \Gamma(x/2+1)/\Gamma\big((x+1)/2\big)$, put*

$$\widehat{\mathcal{S}}(x) = \frac{2^x}{\sqrt{\pi}}\, R(x)^{-\cos \pi x}. \tag{5.16}$$

Then $\widehat{\mathcal{S}}(n) = \mathrm{sw}(n)$ for every $n \in \mathbb{N}_0$, yet $\widehat{\mathcal{S}}$ is not eventually monotone: its logarithmic derivative

$$\frac{\mathrm{d}}{\mathrm{d}x} \log \widehat{\mathcal{S}}(x) = \lambda + \pi \sin(\pi x) \log R(x) - \cos(\pi x)\, \frac{R'(x)}{R(x)} \tag{5.17}$$

has an oscillating term of amplitude $\sim \frac{\pi}{2}\log x$ against remaining terms that are $O(1)$, so it changes sign infinitely often. The same holds for the convex interpolation $\frac{1+\cos \pi x}{2}\mathcal{S}_- + \frac{1-\cos \pi x}{2}\mathcal{S}_+$. Consequently the phase-resolved *pair $(\mathcal{S}_-, \mathcal{S}_+)$, and not any single function, is the asymptotically stable continuation of* sw*; the corresponding inverse object is the pair $(\mathcal{I}_-, \mathcal{I}_+)$.*

Proof. At $x = n$ one has $-\cos \pi n = (-1)^{n+1} = \sigma_n$, so $\widehat{\mathcal{S}}(n) = (2^n/\sqrt{\pi})R(n)^{\sigma_n} = \mathcal{S}_{\sigma_n}(n) = \mathrm{sw}(n)$ by eq. (5.12) and eq. (5.11). Differentiating $\log \widehat{\mathcal{S}} = x\lambda - \frac{1}{2}\log \pi - \cos(\pi x)\log R(x)$ gives eq. (5.17); by theorem 5.15 below, $\log R(x) = \frac{1}{2}\log(x/2) + O(x^{-1})$ and $R'/R = O(x^{-1})$, so the oscillating term has amplitude $\frac{\pi}{2}\log x + O(1) \to \infty$ and dominates. The convex interpolation is treated the same way after writing it as $\mathcal{S}_-(x)$ times a factor whose logarithmic derivative oscillates with amplitude $\asymp \log x$. ☐

5.2 Exact normal form, Binet reduction, and Borel structure

5.2.1 The single scalar correction

Definition 5.10 (The gamma-ratio correction)**.** For $x > 0$ set

$$D(x) := \log \frac{\Gamma(x/2+1)}{\Gamma(x/2+1/2)} - \frac{1}{2}\log \frac{x}{2}. \tag{5.18}$$

Theorem 5.11 (Exact σ-uniform normal form)**.** *For $x > 0$ and $\sigma \in \{-1, +1\}$,*

$$\boxed{\mathcal{S}_\sigma(x) = C_\sigma\, \mathrm{e}^{\lambda x}\, x^{\sigma/2}\, \mathrm{e}^{\sigma D(x)}, \qquad C_\sigma = \frac{2^{-\sigma/2}}{\sqrt{\pi}}.} \tag{5.19}$$

At integer arguments this is the exact parity formula

$$\log \mathrm{sw}(n) = n\lambda - \tfrac{1}{2}\log \pi + \sigma_n\Big(\tfrac{1}{2}\log \tfrac{n}{2} + D(n)\Big), \qquad n \geq 1. \tag{5.20}$$

Proof. By eq. (5.18), $\Gamma(x/2+1)/\Gamma(x/2+1/2) = (x/2)^{1/2}\mathrm{e}^{D(x)}$. Substituting into eq. (5.12) gives $\mathcal{S}_\sigma = (2^x/\sqrt{\pi})(x/2)^{\sigma/2}\mathrm{e}^{\sigma D}$, and $(x/2)^{\sigma/2} = 2^{-\sigma/2}x^{\sigma/2}$. For eq. (5.20), take logarithms of eq. (5.19) at $x = n$, $\sigma = \sigma_n$, and use $\frac{1}{2}\log n - \frac{\lambda}{2} = \frac{1}{2}\log \frac{n}{2}$. ☐

Equation (5.19) is exactly theorem 1.2 with

$$\boxed{\alpha = 1, \qquad a = \lambda = \log 2, \qquad b = \frac{\sigma}{2}, \qquad C = C_\sigma = \frac{2^{-\sigma/2}}{\sqrt{\pi}}, \qquad q_j = \sigma d_j,} \tag{5.21}$$

where d_j are the coefficients of the asymptotic series of D (theorem 5.18). Because $\alpha = 1$ no monomial substitution is needed and theorem 1.4 is not invoked. We record the dictionary once and use it in every specialization below.

Remark 5.12 (Dictionary between the three sources). Let $\varepsilon_{\mathsf{S3}} \in \{+1, -1\}$ be the marker of S3 and $\epsilon_{\mathsf{S1}} \in \{0, 1\}$ that of S1. Then

$$\sigma = -\varepsilon_{\mathsf{S3}}, \qquad \sigma = 2\epsilon_{\mathsf{S1}} - 1, \qquad \mathsf{S3}:\ Q = -D, \quad q_r^{\mathsf{S3}} = -d_{2r-1}, \qquad \mathsf{S1}:\ \alpha_{\mathsf{S1}} = \log 4 = 2\lambda.$$

S1 parametrizes each sheet by the half index m with $S_\epsilon^{\mathsf{S1}}(m) = \mathrm{sw}(2m+\epsilon)$; as functions this is the reparametrization $x = 2m + \epsilon$ of theorem 5.6, so

$$S_\epsilon^{\mathsf{S1}}(m) = \mathcal{S}_{2\epsilon-1}(2m+\epsilon), \qquad \mathcal{I}_\sigma(y) = 2M_\epsilon^{\mathsf{S1}}(y) + \epsilon.$$

The reparametrization is invisible in the forward coefficients up to the substitution $x \mapsto 2m + \epsilon$, but it is *not* invisible in the inverse coefficients: the two normalizations invert different leading phases ($\lambda x + \frac{\sigma}{2}\log x$ against $2\lambda m + \frac{\sigma}{2}\log m$) and therefore have different Lambert cores. Only for the even sheet do the two cores agree exactly, through $X_-(y) = 2w_0^{\mathsf{S1}}(y)$; see theorem 5.37.

5.2.2 The Binet kernel collapses to a hyperbolic tangent

Theorem 5.13 (Exact Borel–Laplace representation). *For $\Re x > 0$,*

$$\boxed{D(x) = \int_0^\infty \mathrm{e}^{-xt}\,\widehat{D}(t)\,\mathrm{d}t, \qquad \widehat{D}(t) = \frac{\tanh(t/2)}{2t},} \tag{5.22}$$

the singularity of $\widehat{D}$ at $t = 0$ being removable with $\widehat{D}(0) = \frac{1}{4}$.

Proof. Put $z = x/2$ and $h(z) = \log\Gamma(z+1) - \log\Gamma(z+\frac{1}{2}) - \frac{1}{2}\log z$, so $h(x/2) = D(x)$. From the standard integral for a digamma difference,

$$\psi(z+1) - \psi(z+\tfrac{1}{2}) = \int_0^\infty \frac{\mathrm{e}^{-(z+1/2)s} - \mathrm{e}^{-(z+1)s}}{1-\mathrm{e}^{-s}}\,\mathrm{d}s = 2\int_0^\infty \frac{\mathrm{e}^{-2zv}}{\mathrm{e}^v+1}\,\mathrm{d}v,$$

the second equality by $s = 2v$ and $(\mathrm{e}^{-v} - \mathrm{e}^{-2v})/(1-\mathrm{e}^{-2v}) = 1/(\mathrm{e}^v+1)$. Since $1/(2z) = \int_0^\infty \mathrm{e}^{-2zv}\mathrm{d}v$,

$$h'(z) = \psi(z+1) - \psi(z+\tfrac{1}{2}) - \frac{1}{2z} = \int_0^\infty \mathrm{e}^{-2zv}\Big(\frac{2}{\mathrm{e}^v+1} - 1\Big)\mathrm{d}v = -\int_0^\infty \mathrm{e}^{-2zv}\tanh\frac{v}{2}\,\mathrm{d}v.$$

Both $h(z)$ and the right-hand side of eq. (5.22) tend to 0 as $\Re z \to +\infty$, so integrating from $+\infty$ down to z,

$$h(z) = \int_z^\infty\int_0^\infty \mathrm{e}^{-2\zeta v}\tanh\tfrac{v}{2}\,\mathrm{d}v\,\mathrm{d}\zeta = \int_0^\infty \mathrm{e}^{-2zv}\frac{\tanh(v/2)}{2v}\,\mathrm{d}v,$$

which is eq. (5.22) after $x = 2z$. Fubini applies because the double integral is absolutely convergent for $\Re z > 0$. ☐

S3 reaches eq. (5.22) from Binet's first formula, through the elegant identity $b(t) - 2b(2t) = -\frac{1}{2}\tanh(t/2)$ for $b(t) = (\mathrm{e}^t - 1)^{-1} - t^{-1} + \frac{1}{2}$; S2 reaches it from the digamma integral, as above. S1 keeps Binet's *second* formula (the $\arctan$ kernel) instead and never simplifies the difference of the two Binet integrals. The $\tanh$ collapse is the decisive simplification: it is what makes the Borel plane elementary, and it is adopted here.

Corollary 5.14 (Sign, monotonicity, and elementary bounds). *For $x > 0$,*

$$0 < D(x) < \frac{1}{4x}, \qquad 0 < -D'(x) < \frac{1}{4x^2}, \qquad (-1)^k D^{(k)}(x) > 0 \quad (k \geq 0). \tag{5.23}$$

In particular D is completely monotone and strictly decreasing.

Proof. $\widehat{D} > 0$ on $(0,\infty)$, so $(-1)^k D^{(k)}(x) = \int_0^\infty \mathrm{e}^{-xt}t^k\widehat{D}(t)\mathrm{d}t > 0$. From $0 < \tanh(t/2) < t/2$ we get $0 < \widehat{D}(t) < \frac{1}{4}$, and integrating $\mathrm{e}^{-xt}t^k$ gives the two displayed bounds. ☐

Corollary 5.15 (Asymptotics of the gamma ratio). *With $R(x) = \Gamma(x/2+1)/\Gamma((x+1)/2)$ as in eq. (5.16),*

$$\log R(x) = \tfrac{1}{2}\log(x/2) + D(x) = \tfrac{1}{2}\log(x/2) + O(x^{-1}), \qquad \frac{R'(x)}{R(x)} = \frac{1}{2x} + D'(x) = O(x^{-1}).$$

Proof. Immediate from eq. (5.18) and eq. (5.23). ☐

5.2.3 The Borel singularity lattice

Proposition 5.16 (Mittag–Leffler expansion and poles). *The Borel transform of eq.* (5.22) *satisfies*

$$\widehat{D}(t) = 2\sum_{k\geq 0} \frac{1}{t^2+\pi^2(2k+1)^2}, \tag{5.24}$$

is even and meromorphic on $\mathbb{C}$, and has exactly the simple poles

$$t = \pm\mathrm{i}\pi(2k+1), \qquad k\geq 0, \qquad \mathrm{Res}_{t=t_k}\widehat{D}(t) = \frac{1}{t_k}. \tag{5.25}$$

The positive real ray is free of singularities; the nearest singularities lie at distance π, so the formal series of D is Gevrey-1 with singulant modulus π.

Proof. $\tanh(w) = \sum_{k\geq 0} 2w/\big(w^2+\pi^2(k+\frac{1}{2})^2\big)$ is the classical Mittag–Leffler expansion; putting $w = t/2$ and dividing by $2t$ gives eq. (5.24), both sides being even, meromorphic, with the same poles and residues and the same value $\frac{1}{4}$ at $t=0$ (using $\sum_{k\geq 0}(2k+1)^{-2} = \pi^2/8$). At $t = t_k$ the function $\tanh(t/2)$ has residue 2, so $\widehat{D}$ has residue $2/(2t_k) = 1/t_k$. □

Remark 5.17 (What the exponentially small scale is, and is not). The quantity $\mathrm{e}^{-\pi x}$ that governs optimal truncation (section 5.9) is the envelope set by the distance π of the nearest *off-axis* Borel poles; it is not an ambiguity of the positive-axis sum, since the ray $\arg t = 0$ meets no singularity and eq. (5.22) is exact. Genuine Stokes jumps appear only when the Laplace direction is rotated onto the imaginary rays. S1, retaining an unsimplified Binet integral, makes the weaker but correct statement that no Stokes constant need be assigned on the real axis; the Borel picture explains why.

5.3 The logarithmic coefficients and a sharp enveloping remainder

5.3.1 All coefficients

We use the Bernoulli convention $z/(\mathrm{e}^z-1) = \sum_{n\geq 0} B_n z^n/n!$, so $B_1 = -\frac{1}{2}$, and the Bernoulli polynomials $t\mathrm{e}^{wt}/(\mathrm{e}^t-1) = \sum_n B_n(w)t^n/n!$.

Theorem 5.18 (All logarithmic coefficients). *As $x\to\infty$ in any closed subsector of $|\arg x| < \pi/2$,*

$$D(x) \sim \sum_{j\geq 1}\frac{d_j}{x^j}, \qquad d_{2r} = 0, \qquad d_{2r+1} = \frac{(2^{2r+2}-1)B_{2r+2}}{(2r+1)(2r+2)}. \tag{5.26}$$

Equivalently

$$d_{2r+1} = (-1)^r\,\frac{2\,(2r)!}{\pi^{2r+2}}\big(1-2^{-2r-2}\big)\zeta(2r+2) = (-1)^r\,\frac{2\,(2r)!}{\pi^{2r+2}}\sum_{k\geq 0}\frac{1}{(2k+1)^{2r+2}}, \tag{5.27}$$

and the formal Borel transform of the series is exactly the Taylor expansion of $\widehat{D}$:

$$\widehat{D}(t) = \sum_{r\geq 0} d_{2r+1}\frac{t^{2r}}{(2r)!} = \frac{\tanh(t/2)}{2t}. \tag{5.28}$$

Proof. Insert into eq. (5.24) the finite geometric identity

$$\frac{1}{t^2+a^2} = \sum_{r=0}^{N-1}\frac{(-1)^r t^{2r}}{a^{2r+2}} + \frac{(-1)^N t^{2N}}{a^{2N}(t^2+a^2)}, \qquad a = a_k = \pi(2k+1),$$

and integrate the finite sum against e^{-xt} using $\int_0^\infty \mathrm{e}^{-xt} t^{2r}\mathrm{d}t = (2r)!x^{-2r-1}$. This gives the second expression in eq. (5.27), hence the first by $\zeta(2r+2)(1-2^{-2r-2}) = \sum_{k\geq 0}(2k+1)^{-(2r+2)}$, and hence eq. (5.26) by Euler's evaluation $B_{2m} = (-1)^{m-1}2(2m)!\zeta(2m)/(2\pi)^{2m}$. The unexpanded last term is the remainder controlled in theorem 5.20, which in particular proves the asymptotic statement on $\Re x > 0$; the identity eq. (5.28) is the Taylor expansion of $\widehat{D}$ at the origin. □

Remark 5.19 (Sector of validity: a disagreement between the sources). S2 and S3 state eq. (5.26) on closed subsectors of $|\arg x| < \pi/2$, which is what the Laplace representation eq. (5.22) gives. S1 states the corresponding expansion on $|\arg m| \leq \pi - \delta$. Both are correct: the wider sector follows not from the Laplace integral but from the Tricomi–Erdélyi expansion of a gamma ratio,

$$\log\frac{\Gamma(z+a)}{\Gamma(z+b)} \sim (a-b)\log z + \sum_{n\geq 1}\frac{(-1)^{n+1}}{n(n+1)z^n}\big(B_{n+1}(a) - B_{n+1}(b)\big), \qquad |\arg z| \leq \pi - \delta, \tag{5.29}$$

applied with $a = 1$, $b = \frac{1}{2}$, $z = x/2$. The volume states the Laplace sector where an *exact* representation and a signed remainder are wanted, and the wider sector where only a Poincaré statement is wanted. The two routes give the same coefficients: with $B_m(\frac{1}{2}) = (2^{1-m}-1)B_m$ and $B_m(1) = B_m$ for $m \neq 1$, eq. (5.29) yields $d_n = 2^n\,\frac{(-1)^{n+1}}{n(n+1)}\big(2-2^{-n}\big)B_{n+1}$, which is eq. (5.26).

Table 5.1: The logarithmic coefficients d_{2r+1} of theorem 5.18, recomputed from eq. (5.26) in exact rational arithmetic and cross-checked against the Taylor expansion of $\tanh(t/2)/(2t)$.

j	1	3	5	7	9	11	13
d_j	$\frac{1}{4}$	$-\frac{1}{24}$	$\frac{1}{20}$	$-\frac{17}{112}$	$\frac{31}{36}$	$-\frac{691}{88}$	$\frac{5461}{52}$

Thus

$$D(x) \sim \frac{1}{4x} - \frac{1}{24x^3} + \frac{1}{20x^5} - \frac{17}{112x^7} + \frac{31}{36x^9} - \frac{691}{88x^{11}} + \frac{5461}{52x^{13}} - \cdots \tag{5.30}$$

5.3.2 An enveloping remainder, not merely a Poincaré one

Theorem 5.20 (Exact remainder and strict next-term envelope)**.** *For $x > 0$ and $N \geq 0$ write $D(x) = \sum_{r=0}^{N-1} d_{2r+1}x^{-(2r+1)} + R_N(x)$ (empty sum for $N = 0$). Then, with $a_k = \pi(2k+1)$,*

$$R_N(x) = 2(-1)^N\sum_{k\geq 0} a_k^{-2N}\int_0^\infty \frac{\mathrm{e}^{-xt}t^{2N}}{t^2+a_k^2}\,\mathrm{d}t, \tag{5.31}$$

and consequently

$$\boxed{0 < (-1)^N R_N(x) < \frac{|d_{2N+1}|}{x^{2N+1}}} \qquad (x > 0,\ N \geq 0). \tag{5.32}$$

Every truncation therefore brackets $D(x)$ together with its successor.

Proof. Substitute the finite geometric identity of the previous proof into eq. (5.24) and then into eq. (5.22); Tonelli's theorem justifies the interchange since all terms are positive. This gives eq. (5.31), in which every integral is positive. Replacing $(t^2+a_k^2)^{-1}$ by the strict upper bound a_k^{-2} and using $\int_0^\infty \mathrm{e}^{-xt}t^{2N}\mathrm{d}t = (2N)!x^{-2N-1}$ gives

$$0 < (-1)^N R_N(x) < 2(2N)!\,x^{-(2N+1)}\sum_{k\geq 0} a_k^{-(2N+2)} = \frac{|d_{2N+1}|}{x^{2N+1}},$$

by the last expression in eq. (5.27). □

Remark 5.21. Theorem 5.20 is strictly stronger than the Poincaré statement, and it is what makes every later analytic claim in this chapter effective: all remainder constants below are explicit rational multiples of the first omitted $|d_{2N+1}|$. S2 and S3 both prove it; S1 does not have it, because it never simplifies the Binet kernel. It was verified numerically at $x = 5$ and $x = 10$ for $0 \le N \le 4$: the bracketing eq. (5.32) holds with the two sides agreeing to within a factor 1.33 at $x = 5$ and 1.09 at $x = 10$.

5.3.3 Large order and least term

Proposition 5.22 (Least term and its envelope). *As $r \to \infty$, $d_{2r+1} \sim (-1)^r 2(2r)!\pi^{-2r-2}$, so the ratio of consecutive nonzero term magnitudes in eq.* (5.30) *is* $\sim (2r+2)(2r+1)/(\pi^2 x^2)$. *The least term therefore occurs at $r \approx \pi x/2$ and has magnitude*

$$\frac{2\sqrt{2}}{\pi\sqrt{x}}\,\mathrm{e}^{-\pi x}\big(1 + O(x^{-1})\big). \tag{5.33}$$

By theorem 5.20, truncation at the least term attains an error of exactly this scale, with no hidden constant.

Proof. The asymptotic form of d_{2r+1} is eq. (5.27) with $\zeta(2r+2) \to 1$. The ratio statement is immediate. Setting $2r = \pi x$ and applying Stirling's formula to $(2r)!$ gives $2(2r)!\pi^{-2r-2}x^{-2r-1} \sim 2\sqrt{2\pi^2 x}\,\mathrm{e}^{-\pi x}/(\pi^2 x) = 2\sqrt{2}\,\mathrm{e}^{-\pi x}/(\pi\sqrt{x})$. □

Remark 5.23 (Numerical confirmation). Recomputed: at $x = 10$ the least nonzero term of eq. (5.30) occurs at $r = 15$ (against $\pi x/2 = 15.708$) with magnitude 6.5448×10^{-15}, while eq. (5.33) predicts 6.4659×10^{-15}; at $x = 20$ the least term occurs at $r = 31$ (against 31.416) with magnitude 1.03859×10^{-28} against a predicted 1.03837×10^{-28}.

5.4 The forward transseries

5.4.1 Exponentiation

Theorem 5.24 (All forward coefficients). *Define $A_0 = 1$ and, for $n \ge 1$,*

$$A_n = [t^n]\,\exp\Big(\sum_{j\ge1} d_j t^j\Big) = \frac{1}{n!}\,B_n\big(1!\,d_1, 2!\,d_2, \dots, n!\,d_n\big) = \sum_{\substack{k_1,k_2,\dots\ge0\\ \sum_j jk_j=n}}\;\prod_{j\ge1}\frac{d_j^{k_j}}{k_j!}, \tag{5.34}$$

with B_n the complete exponential Bell polynomial of theorem 1.10. Equivalently and most cheaply,

$$A_0 = 1, \qquad nA_n = \sum_{j=1}^{n} j d_j A_{n-j} = \sum_{\substack{1\le j\le n\\ j \text{ odd}}} j d_j A_{n-j}. \tag{5.35}$$

Then, for every fixed $N \ge 1$ and uniformly on closed subsectors of $\Re x > 0$,

$$\boxed{\mathcal{S}_\sigma(x) = C_\sigma\, 2^x x^{\sigma/2}\Big(\sum_{n=0}^{N-1}\frac{\sigma^n A_n}{x^n} + O_N(x^{-N})\Big).} \tag{5.36}$$

Proof. The three expressions in eq. (5.34) are the definition of B_n, its partition form, and the expansion of the exponential as a product; eq. (5.35) follows from $A'(t) = A(t)\sum_j j d_j t^{j-1}$ for $A(t) = \exp\sum_j d_j t^j$, together with $d_{2r} = 0$. The parity of the σ-dependence is exact: since $\sum_j d_j t^j$ is an odd formal series,

$$\exp\Big(-\sum_j d_j t^j\Big) = \exp\Big(\sum_j d_j(-t)^j\Big) = \sum_{n\ge0}(-1)^n A_n t^n,$$

so $\exp(\sigma D)$ has coefficients $\sigma^n A_n$ in either sheet. For the remainder, theorem 5.25 below gives an explicit bound. □

Proposition 5.25 (Effective forward remainder). *Fix $N \geq 1$ and let $x \geq x_N := \max\{1, 2N\}$. Then*

$$\Big|\mathrm{e}^{\sigma D(x)} - \sum_{n=0}^{N-1} \frac{\sigma^n A_n}{x^n}\Big| \leq \frac{K_N}{x^N}, \qquad K_N := \mathrm{e}^{1/4}\Big(|d_{2\lceil N/2\rceil+1}| + \sum_{n\geq N} \frac{|A_n|}{x_N^{n-N}}\Big) < \infty, \tag{5.37}$$

where the tail sum converges because $A_n = O(C^n n!^{1/2})$ for some C.

Proof. Write $D_{<M}(x) = \sum_{r<M} d_{2r+1}x^{-(2r+1)}$ with $M = \lceil N/2\rceil$, so that by theorem 5.20 $|D(x) - D_{<M}(x)| \leq |d_{2M+1}|x^{-(2M+1)} \leq |d_{2M+1}|x^{-N}$ for $2M+1 \geq N$. Since $|D(x)| \leq \frac{1}{4}$ and $|D_{<M}(x)| \leq \frac{1}{4}$ for $x \geq 1$ by eq. (5.32) with $N = 0$, the mean value theorem gives $|\mathrm{e}^{\sigma D} - \mathrm{e}^{\sigma D_{<M}}| \leq \mathrm{e}^{1/4}|D - D_{<M}|$. Finally $\mathrm{e}^{\sigma D_{<M}(x)}$ is an absolutely convergent power series in $1/x$ for $x \geq x_N$ whose coefficients agree with $\sigma^n A_n$ for $n < N$ up to terms already accounted for, and whose tail is bounded by $\sum_{n\geq N} |A_n|x^{-n}$. Collecting the two bounds gives eq. (5.37). The growth bound on A_n follows from $d_{2r+1} = O((2r)!\pi^{-2r})$ and the Bell-polynomial partition form. □

Source claim. S1 obtains its multiplicative remainder by writing "$e^{R_N(m)} = 1 + O(m^{-N})$, so formal exponentiation also gives the claimed analytic remainder". **Defect.** That sentence exponentiates a *formal* truncation error and silently assumes that the tail of the exponentiated series is of the same order; no bound on the coefficients A_n enters, although the series $\sum A_n x^{-n}$ diverges. **Repair.** Theorem 5.25 supplies the missing ingredients (a bound on $D - D_{<M}$ from the enveloping theorem, a uniform bound on the exponential's Lipschitz constant, and an explicit tail bound), so the constant in eq. (5.36) is effective.

5.4.2 Coefficient table and the two sectors

Table 5.2: The universal forward coefficients A_n of eq. (5.34), recomputed by eq. (5.35) in exact rational arithmetic. The table reproduces the corresponding tables of all three sources (S3 tabulates $C_m = (-1)^n A_n$; S1 tabulates $A_n/2^n$ in its half index).

n	A_n	n	A_n
0	1	8	$-334477/8388608$
1	$1/4$	9	$28717403/33554432$
2	$1/32$	10	$59697183/268435456$
3	$-5/128$	11	$-8400372435/1073741824$
4	$-21/2048$	12	$-34429291905/17179869184$
5	$399/8192$	13	$7199255611995/68719476736$
6	$869/65536$	14	$14631594576045/549755813888$
7	$-39325/262144$	15	$-4251206967062925/2199023255552$

Explicitly, with $C_- = \sqrt{2/\pi}$ and $C_+ = 1/\sqrt{2\pi}$,

$$\mathcal{S}_-(x) \sim \sqrt{\frac{2}{\pi}}\,\frac{2^x}{\sqrt{x}}\Big(1 - \frac{1}{4x} + \frac{1}{32x^2} + \frac{5}{128x^3} - \frac{21}{2048x^4} - \frac{399}{8192x^5} + \frac{869}{65536x^6} + \cdots\Big), \tag{5.38}$$

$$\mathcal{S}_+(x) \sim \frac{2^x\sqrt{x}}{\sqrt{2\pi}}\Big(1 + \frac{1}{4x} + \frac{1}{32x^2} - \frac{5}{128x^3} - \frac{21}{2048x^4} + \frac{399}{8192x^5} + \frac{869}{65536x^6} + \cdots\Big), \tag{5.39}$$

and for the integer sequence itself,

$$\boxed{\mathrm{sw}(n) \sim \frac{2^n}{\sqrt{\pi}}\, 2^{-\sigma_n/2}\, n^{\sigma_n/2} \sum_{k\geq 0} \frac{\sigma_n^k A_k}{n^k}, \qquad \sigma_n = (-1)^{n+1}.} \tag{5.40}$$

Remark 5.26 (Period-two, not oscillatory error). Equation (5.40) is a two-phase (period-two) transseries: the discrete transmonomial σ_n has modulus one, so it is never absorbed into a remainder, and it changes the algebraic scale from $2^n n^{-1/2}$ to $2^n n^{+1/2}$. Equivalently, with the idempotents $\chi_\sigma(n) = (1+\sigma\sigma_n)/2$,

$$\mathrm{sw}(n) \sim \sum_{\sigma=\pm 1} \chi_\sigma(n)\, C_\sigma 2^n n^{\sigma/2} \sum_{k\geq 0} \sigma^k A_k n^{-k}.$$

Corollary 5.27 (Central binomial coefficient and the reciprocal)**.** *Putting* $x = 2m$ *in eq.* (5.38),

$$\binom{2m}{m} \sim \frac{4^m}{\sqrt{\pi m}}\Big(1 - \frac{1}{8m} + \frac{1}{128m^2} + \frac{5}{1024m^3} - \frac{21}{32768m^4} - \frac{399}{262144m^5} + \cdots\Big), \tag{5.41}$$

that is, the half-index coefficients of S1 are $A_n/2^n$ *on the even sheet. Moreover*

$$\frac{1}{\mathcal{S}_\sigma(x)} \sim \frac{1}{C_\sigma}\, 2^{-x} x^{-\sigma/2} \sum_{n\geq 0} \frac{(-\sigma)^n A_n}{x^n}. \tag{5.42}$$

Proof. For eq. (5.41) substitute $x = 2m$ and $C_- 2^{2m}(2m)^{-1/2} = 4^m/\sqrt{\pi m}$. For eq. (5.42) note that the reciprocal of eq. (5.19) carries $\mathrm{e}^{-\sigma D}$, whose coefficients are $(-\sigma)^n A_n$ by the oddness argument in the proof of theorem 5.24. □

5.5 The centred normal form

This section is S1's distinctive contribution: a translation of the index that makes the correction series *sparse in a second way* and equalizes the two leading constants. S1 states it as two parallel statements in its half index; it is recast here as a single σ-uniform identity in the original index, and the coefficients are recomputed in that normalization.

Theorem 5.28 (Centred exact normal form)**.** *Put*

$$u := \frac{x}{2} + \frac{1}{4}. \tag{5.43}$$

Then, exactly and for both parities,

$$\boxed{\mathcal{S}_\sigma(x) = \frac{4^u}{\sqrt{2\pi}}\left(\frac{\Gamma(u+3/4)}{\Gamma(u+1/4)}\right)^{\sigma},} \tag{5.44}$$

and, as $u \to \infty$ *in* $|\arg u| \leq \pi - \delta$,

$$\mathcal{S}_\sigma(x) \sim \frac{4^u}{\sqrt{2\pi}}\, u^{\sigma/2} \exp\Big(\sigma \sum_{q\geq 1} \frac{\nu_{2q}}{u^{2q}}\Big), \qquad \nu_{2q} = \frac{B_{2q+1}(1/4)}{q(2q+1)}. \tag{5.45}$$

In particular every odd inverse power is absent *from the centred correction, and the leading constant* $1/\sqrt{2\pi}$ *is the same on both sheets.*

Proof. From $u = x/2 + 1/4$ we get $x/2 + 1 = u + 3/4$ and $x/2 + 1/2 = u + 1/4$, while $2^x = 2^{2u-1/2} = 4^u/\sqrt{2}$; substituting in eq. (5.12) gives eq. (5.44). For eq. (5.45) apply eq. (5.29) with $a = \frac{3}{4}$, $b = \frac{1}{4}$, $z = u$: the logarithmic term is $(a-b)\log u = \frac{1}{2}\log u$, and the n-th coefficient is $\frac{(-1)^{n+1}}{n(n+1)}\big(B_{n+1}(\frac{3}{4}) - B_{n+1}(\frac{1}{4})\big)$. Since $a+b=1$, the reflection formula $B_m(1-w) = (-1)^m B_m(w)$ gives $B_{n+1}(\frac{1}{4}) = (-1)^{n+1} B_{n+1}(\frac{3}{4})$, so the difference vanishes for odd n and equals $2B_{n+1}(\frac{3}{4}) = -2B_{n+1}(\frac{1}{4})$ for even $n = 2q$. This yields $\nu_{2q} = B_{2q+1}(1/4)/(q(2q+1))$. □

Table 5.3: Centred logarithmic coefficients ν_{2q} of eq. (5.45), recomputed from Bernoulli polynomials at $1/4$. They agree with S1's $\widehat{\lambda}_{2q}^{(0)} = -\nu_{2q}$ after the marker translation of theorem 5.12.

$2q$	2	4	6	8	10
ν_{2q}	$\frac{1}{64}$	$-\frac{5}{2048}$	$\frac{61}{49152}$	$-\frac{1385}{1048576}$	$\frac{50521}{20971520}$

Remark 5.29 (Two different sparsities). The uncentred correction $\sigma D(x)$ contains only *odd* powers of $1/x$; the centred correction contains only *even* powers of $1/u$. There is no contradiction: the two statements concern different remainder functions, because the $\frac{\sigma}{2}\log$ term is attached to x in one and to u in the other. Both are instances of one rule, recorded in theorem 5.30 below, and the swing shifts are precisely the optimally centred ones.

Lemma 5.30 (Optimal centring of a gamma ratio). *Let $a, b \in \mathbb{R}$ with $\beta := a - b \neq 0$, and set $s = (a+b-1)/2$ and $z = w + s$. Then $\Gamma(w+a)/\Gamma(w+b) = \Gamma(z + \frac{1+\beta}{2})/\Gamma(z + \frac{1-\beta}{2})$, the two shifts sum to 1, and*

$$\log \frac{\Gamma(z + \frac{1+\beta}{2})}{\Gamma(z + \frac{1-\beta}{2})} \sim \beta \log z - \sum_{q \geq 1} \frac{B_{2q+1}\left(\frac{1-\beta}{2}\right)}{q(2q+1)}\, z^{-2q}, \qquad |\arg z| \leq \pi - \delta, \tag{5.46}$$

with no odd inverse powers.

Proof. The shift identity is immediate. Apply eq. (5.29) and the argument of theorem 5.28: complementary shifts make the Bernoulli difference vanish in odd degree by $B_m(1-w) = (-1)^m B_m(w)$. ☐

For the swing factorial $a = 1$, $b = \frac{1}{2}$, $w = x/2$, so $s = \frac{1}{4}$ and $z = u$, and $\beta = \frac{1}{2}$: eq. (5.45) is exactly eq. (5.46) raised to the power σ. The first centred expansions are

$$\mathcal{S}_-(x) \sim \frac{4^u}{\sqrt{2\pi u}}\left(1 - \frac{1}{64u^2} + \frac{21}{8192u^4} - \frac{671}{524288u^6} + \cdots\right), \tag{5.47}$$

$$\mathcal{S}_+(x) \sim \frac{4^u\sqrt{u}}{\sqrt{2\pi}}\left(1 + \frac{1}{64u^2} - \frac{19}{8192u^4} + \frac{631}{524288u^6} + \cdots\right), \tag{5.48}$$

whose coefficients were recomputed from $\exp\left(\sigma \sum_q \nu_{2q} z^q\right)$ and agree with S1.

5.6 The two Lambert cores, and why the branch matters

5.6.1 Specialization of the core theorem

Write

$$L_\sigma(y) := \log \frac{y}{C_\sigma} = \log y + \tfrac{1}{2}\log \pi + \tfrac{\sigma}{2}\lambda. \tag{5.49}$$

Dropping the correction in eq. (5.19) leaves the *core equation*

$$L_\sigma(y) = \lambda X + \frac{\sigma}{2}\log X, \qquad \text{equivalently} \qquad C_\sigma\, 2^X X^{\sigma/2} = y. \tag{5.50}$$

Theorem 5.31 (Explicit Lambert cores). *Theorem 1.20 with the dictionary eq.* (5.21) *gives*

$$X_{\sigma,k}(y) = \frac{\sigma}{2\lambda} W_k\left(2\sigma\lambda\, e^{2\sigma L_\sigma(y)}\right), \tag{5.51}$$

and the unique large positive real solutions are

$$\boxed{X_+(y) = \frac{W_0\left(4\pi\lambda y^2\right)}{2\lambda}, \qquad X_-(y) = -\frac{W_{-1}\left(-4\lambda/(\pi y^2)\right)}{2\lambda}.} \tag{5.52}$$

X_+ is defined for all $y > 0$; X_- is defined exactly for

$$y \geq y_\star := 2\sqrt{\frac{\mathrm{e}\lambda}{\pi}} = 1.548870223880\ldots, \tag{5.53}$$

and satisfies $X_-(y) \geq 1/(2\lambda) = 0.72134752\ldots$ there.

Proof. With $a = \lambda$, $b = \sigma/2$, theorem 1.20 reads $X = (b/a)W_k\big((a/b)\mathrm{e}^{L/b}\big)$, and $1/\sigma = \sigma$ gives eq. (5.51). For $\sigma = +1$, $C_+ = 1/\sqrt{2\pi}$ so $\mathrm{e}^{2L_+} = 2\pi y^2$ and the argument is $4\pi\lambda y^2 > 0$, on which W_0 is the only real branch. For $\sigma = -1$, $C_- = \sqrt{2/\pi}$ so $\mathrm{e}^{-2L_-} = 2/(\pi y^2)$ and the argument is $-4\lambda/(\pi y^2) < 0$; it lies in $[-1/\mathrm{e}, 0)$ exactly when $y^2 \geq 4\lambda\mathrm{e}/\pi$, i.e. $y \geq y_\star$. On that interval both W_0 and W_{-1} are real; $W_{-1} \leq -1$ gives $X_- \geq 1/(2\lambda)$, and $W_{-1}(z) \to -\infty$ as $z \to 0^-$ gives $X_-(y) \to \infty$. Theorem 5.32 identifies the two roots and shows that W_0 gives the wrong one. □

5.6.2 Why W_0 is the wrong root on the even sheet

Proposition 5.32 (The even carrier is not injective). *Let $g_-(X) = C_- 2^X X^{-1/2}$ be the even carrier of eq.* (5.50)*. Then g_- is strictly decreasing on* $(0, 1/(2\lambda))$*, strictly increasing on* $(1/(2\lambda), \infty)$*, and*

$$\min_{X>0} g_-(X) = g_-\Big(\frac{1}{2\lambda}\Big) = 2\sqrt{\frac{\mathrm{e}\lambda}{\pi}} = y_\star. \tag{5.54}$$

Hence for $y > y_\star$ the equation $g_-(X) = y$ has exactly two positive roots $X^{\text{small}}(y) < 1/(2\lambda) < X^{\text{large}}(y)$; they coalesce at $y = y_\star$. In terms of the Lambert branches,

$$X^{\text{large}}(y) = -\frac{W_{-1}\big(-4\lambda/(\pi y^2)\big)}{2\lambda} = X_-(y) \xrightarrow[y\to\infty]{} \infty, \qquad X^{\text{small}}(y) = -\frac{W_0\big(-4\lambda/(\pi y^2)\big)}{2\lambda} = \frac{2}{\pi y^2}\big(1+O(y^{-2})\big). \tag{5.55}$$

The principal branch therefore returns a root that tends to zero*; it is not a large-index inverse at any order, and no amount of asymptotic correction can repair the choice. On the odd sheet the carrier $g_+(X) = C_+ 2^X X^{1/2}$ is a strictly increasing bijection of $(0,\infty)$ onto $(0,\infty)$, its Lambert argument is positive, and W_0 is the only real branch.*

Proof. $(\log g_-)'(X) = \lambda - 1/(2X)$ vanishes only at $X = 1/(2\lambda)$ and changes sign there from $-$ to $+$; the minimum value is $C_-\mathrm{e}^{\lambda/(2\lambda)}(2\lambda)^{1/2} = \sqrt{2/\pi}\cdot\sqrt{\mathrm{e}}\cdot\sqrt{2\lambda} = 2\sqrt{\mathrm{e}\lambda/\pi}$, which is eq. (5.54), and equals $y_\star$ of eq. (5.53). Strict monotonicity on each side gives exactly two roots for $y > y_\star$. Both are given by eq. (5.51) with $\sigma = -1$ and a real branch of W; since $W_{-1} \leq -1 \leq W_0 \leq 0$ on $[-1/\mathrm{e}, 0)$, the branch W_{-1} gives the root $\geq 1/(2\lambda)$ and W_0 the root $\leq 1/(2\lambda)$. Finally $W_0(z) = z + O(z^2)$ as $z \to 0$, so $X^{\text{small}} = -\big(-4\lambda/(\pi y^2) + O(y^{-4})\big)/(2\lambda) = 2/(\pi y^2) + O(y^{-4})$. □

Remark 5.33 (Numerical confirmation and the branch point). Recomputed at 50 digits: for $y = 10^2, 10^3, 10^5$ the principal-branch root is 6.366759642×10^{-5}, 6.366203342×10^{-7}, $6.366197724\times10^{-11}$, against $2/(\pi y^2) = 6.366197724\times10^{-5}$, 6.366197724×10^{-7}, $6.366197724\times10^{-11}$. The threshold $y_\star$ where the two roots coalesce is exactly the value at which the Lambert argument reaches the branch point $-1/\mathrm{e}$; the agreement of these two descriptions is a useful consistency check on the whole normalization. Note also $\mathcal{S}_-\big(1/(2\lambda)\big) = 1.15213\ldots < y_\star$, a fact used in theorem 5.35.

Remark 5.34 (The branch rule is forced by $\operatorname{sign} b$). All three sources agree on the assignment: W_{-1} on the even sheet, W_0 on the odd sheet. In the model theorem 1.2 the rule is $b < 0 \Rightarrow W_{-1}$, $b > 0 \Rightarrow W_0$ for the large real root, and $b = \sigma/2$; so the branch distinction is forced by the sign of the power of x in the forward asymptotics — that is, by the fact that the even bisection carries $n^{-1/2}$ and the odd bisection $n^{+1/2}$. It is a consequence of the parity split, not an independent convention.

5.6.3 Two exact inequalities

The sources deduce the *sign* of the first inverse correction from $D > 0$. The following sharpening is exact at all y, not only to first order, and costs nothing.

Proposition 5.35 (The core brackets the true inverse)**.** *For every $y > 1$,*

$$\mathcal{I}_+(y) < X_+(y), \tag{5.56}$$

and for every $y > y_\star$,

$$\mathcal{I}_-(y) > X_-(y). \tag{5.57}$$

Proof. By eq. (5.19) and theorem 5.14, $\mathcal{S}_+(x) = g_+(x)\mathrm{e}^{D(x)} > g_+(x)$ for all $x > 0$, where g_+ is the odd carrier. Both $\mathcal{S}_+$ and g_+ are strictly increasing. Put $x_1 = \mathcal{I}_+(y)$; then $g_+(x_1) < \mathcal{S}_+(x_1) = y = g_+(X_+(y))$, and g_+ increasing gives $x_1 < X_+(y)$.

For $\sigma = -1$, $\mathcal{S}_-(x) = g_-(x)\mathrm{e}^{-D(x)} < g_-(x)$. Let $x_0 = \mathcal{I}_-(y)$ with $y > y_\star$. Since $\mathcal{S}_-$ is increasing and $\mathcal{S}_-(1/(2\lambda)) = 1.15213\ldots < y_\star < y$, we have $x_0 > 1/(2\lambda)$; also $X_-(y) \geq 1/(2\lambda)$ by theorem 5.31. On $[1/(2\lambda), \infty)$ the carrier g_- is strictly increasing (theorem 5.32), and $g_-(x_0) > \mathcal{S}_-(x_0) = y = g_-(X_-(y))$, whence $x_0 > X_-(y)$. □

Remark 5.36. This explains without computation the signs $c_1^{(+1)} = -1/(4\lambda) < 0$ and $c_1^{(-1)} = +1/(4\lambda) > 0$ found in section 5.7, and supplies a rigorous one-sided bracket at every y, used as a check in theorem 5.63.

5.6.4 A scaled form, and the half-index cores

Putting $w_\sigma = 2\lambda X_\sigma$ turns eq. (5.50) into the single scalar equation

$$w + \sigma \log w = \Lambda_\sigma, \qquad \Lambda_\sigma := \log(\pi y^2) + \sigma \log(4\lambda), \tag{5.58}$$

so that $w_+ = W_0(\mathrm{e}^{\Lambda_+})$ and $w_- = -W_{-1}(-\mathrm{e}^{-\Lambda_-})$. This is S3's normalization, with its $\varepsilon = -\sigma$; it is convenient in section 5.8.

Remark 5.37 (Half-index cores). S1's carriers, in the half index and with $\alpha = \log 4$, are $w_0^{\mathsf{S1}} = -\frac{1}{2\alpha}W_{-1}(-2\alpha/(\pi y^2))$ and $w_1^{\mathsf{S1}} = \frac{1}{2\alpha}W_0(\pi\alpha y^2/2)$. On the even sheet $n = 2m$ exactly, and one checks directly that $X_-(y) = 2w_0^{\mathsf{S1}}(y)$; the two normalizations then agree, and their inverse coefficients are related by $c_n^{(-1)} = 2^{n+1}d_n^{\mathsf{S1},(0)}$, which we verified symbolically for $1 \leq n \leq 8$. On the odd sheet $n = 2m+1$, so the two carriers solve *different* equations ($C_+2^X X^{1/2} = y$ against $K_1 4^w w^{1/2} = y$) and their coefficient lists are *not* termwise comparable, although both expansions converge asymptotically to the same function $\mathcal{I}_+$. Failing to notice this is the easiest way to "discover" a contradiction between S1 and S2/S3 where there is none.

5.7 All-orders reversion around the core

5.7.1 Specialization of the reversion theorem

Theorem 5.38 (Swing specialization of theorem 1.25)**.** *Let $X = X_\sigma(y)$ solve eq.* (5.50)*, let $t = X^{-1}$, and put*

$$\Psi_\sigma(t,u) = -\frac{1}{\lambda}\left[\frac{\sigma}{2}\log(1+tu) + \sigma\sum_{j\geq 1} d_j\Big(\frac{t}{1+tu}\Big)^j\right]. \tag{5.59}$$

Then the unique $U \in t\mathscr{R}[[t]]$ with $U = \Psi_\sigma(t,U)$ has coefficients

$$c_n^{(\sigma)} = [t^n]\,U = \sum_{m=1}^{n}\frac{1}{m}\,[t^n u^{m-1}]\,\Psi_\sigma(t,u)^m \tag{5.60}$$

by theorem 1.16, and equivalently, with $\gamma_1 = 0$ and $\gamma_r = c^{(\sigma)}_{r-1}$ for $r \geq 2$,

$$c^{(\sigma)}_n = -\frac{1}{\lambda}\left[\frac{\sigma}{2}\sum_{k=1}^{n}\frac{(-1)^{k+1}}{k}\widehat{B}_{n,k}(\gamma) + \sigma\sum_{\substack{1\leq j\leq n\\ j\text{ odd}}} d_j\sum_{k=0}^{n-j}(-1)^k\binom{j+k-1}{k}\widehat{B}_{n-j,k}(\gamma)\right], \tag{5.61}$$

with $\widehat{B}$ the ordinary partial Bell polynomials of theorem 1.9 and the expansions of theorem 1.12. Only $c^{(\sigma)}_1, \ldots, c^{(\sigma)}_{n-1}$ occur on the right. The coefficients lie in the parity algebra $\mathscr{R}$ of eq. (5.9)*; no logarithm of L_σ occurs.*

Proof. This is theorem 1.25 with the dictionary eq. (5.21): $a = \lambda$, $b = \sigma/2$, $q_j = \sigma d_j$, which turns the general Ψ into eq. (5.59). The restriction of the inner sum to odd j is $d_{2r} = 0$. ☐

5.7.2 Recomputed coefficients

The coefficients below were recomputed from eq. (5.59) by truncated power-series arithmetic over $\mathbb{Q}[\lambda, \lambda^{-1}]$, separately for $\sigma = +1$ and $\sigma = -1$, and then recombined into $\mathscr{R}$ by $c = \frac{1}{2}(c^+ + c^-) + \sigma \cdot \frac{1}{2}(c^+ - c^-)$. They agree with S2 for $1 \leq n \leq 10$ and with S3 for $1 \leq n \leq 8$ after the marker translation $\sigma = -\varepsilon_{\mathrm{S3}}$.

$$\begin{aligned}
c^{(\sigma)}_1 &= -\frac{\sigma}{4\lambda}, \qquad c^{(\sigma)}_2 = \frac{1}{8\lambda^2}, && (5.62)\\
c^{(\sigma)}_3 &= \frac{2\sigma\lambda^2 - 3\lambda - 3\sigma}{48\lambda^3}, \qquad c^{(\sigma)}_4 = \frac{-4\lambda^2 + 15\sigma\lambda + 6}{192\lambda^4}, && (5.63)\\
c^{(\sigma)}_5 &= -\frac{96\sigma\lambda^4 - 80\lambda^3 + 40\sigma\lambda^2 + 135\lambda + 30\sigma}{1920\lambda^5}, && (5.64)\\
c^{(\sigma)}_6 &= \frac{96\lambda^4 - 180\sigma\lambda^3 + 215\lambda^2 + 210\sigma\lambda + 30}{3840\lambda^6}, && (5.65)\\
c^{(\sigma)}_7 &= \frac{1}{53760\lambda^7}\big(8160\sigma\lambda^6 - 4312\lambda^5 + 1428\sigma\lambda^4 + 1050\lambda^3\\
&\qquad - 4025\sigma\lambda^2 - 2100\lambda - 210\sigma\big), && (5.66)\\
c^{(\sigma)}_8 &= \frac{1}{645120\lambda^8}\big(-48960\lambda^6 + 56056\sigma\lambda^5 - 40908\lambda^4 + 15015\sigma\lambda^3\\
&\qquad + 50610\lambda^2 + 17010\sigma\lambda + 1260\big), && (5.67)\\
c^{(\sigma)}_9 &= -\frac{1}{1290240\lambda^9}\big(1111040\sigma\lambda^8 - 413184\lambda^7 + 79616\sigma\lambda^6 + 43512\lambda^5\\
&\qquad - 83748\sigma\lambda^4 + 83475\lambda^3 + 92610\sigma\lambda^2 + 22050\lambda + 1260\sigma\big), && (5.68)\\
c^{(\sigma)}_{10} &= \frac{1}{5160960\lambda^{10}}\big(2222080\lambda^8 - 1756032\sigma\lambda^7 + 779152\lambda^6 - 203280\sigma\lambda^5\\
&\qquad - 162582\lambda^4 + 487305\sigma\lambda^3 + 309960\lambda^2 + 55440\sigma\lambda + 2520\big). && (5.69)
\end{aligned}$$

Setting $\sigma = \pm 1$ and $\lambda = \log 2$,

$$\mathcal{I}_+(y) \sim X_+ - \frac{1}{4\lambda X_+} + \frac{1}{8\lambda^2 X_+^2} + \frac{2\lambda^2 - 3\lambda - 3}{48\lambda^3 X_+^3} + \frac{-4\lambda^2 + 15\lambda + 6}{192\lambda^4 X_+^4} + \cdots, \tag{5.70}$$

$$\mathcal{I}_-(y) \sim X_- + \frac{1}{4\lambda X_-} + \frac{1}{8\lambda^2 X_-^2} + \frac{-2\lambda^2 - 3\lambda + 3}{48\lambda^3 X_-^3} + \frac{-4\lambda^2 - 15\lambda + 6}{192\lambda^4 X_-^4} + \cdots. \tag{5.71}$$

The signs of the first corrections are those forced by theorem 5.35.

5.7.3 A differential form of the same coefficients

All three sources also give an operator form of the reversion, which is the convenient one for the Stokes analysis of section 5.9. We state it as a chapter-local lemma; it is a candidate for promotion, being independent of the swing factorial.

Lemma 5.39 (Perturbed inverse in phase coordinates)**.** *Let F_0 be invertible near the point considered with $F_0' \neq 0$, let g be a formal correction, and let X satisfy $F_0(X) = z$. Introduce*

$$\mathscr{D} := \frac{1}{F_0'(X)} \frac{\mathrm{d}}{\mathrm{d}X}. \tag{5.72}$$

Then the solution x of $F_0(x) + \eta\, g(x) = z$ with $x|_{\eta=0} = X$ is, as a formal power series in the marker η,

$$x = X + \sum_{k\geq 1} \frac{(-\eta)^k}{k!} \mathscr{D}^{k-1}\left(\frac{g(X)^k}{F_0'(X)}\right). \tag{5.73}$$

Proof. This is theorem 1.18 under the dictionary $F \mapsto F_0$, $E \mapsto g$, $\varepsilon \mapsto \eta$, $\mu_0 \mapsto X$, $y \mapsto z$ and $\mathcal{D} \mapsto \mathscr{D}$, which carries (1.32) to eq. (5.72) and (1.34) to eq. (5.73). Since g is here a formal and in general divergent series, no contour is available and the statement invoked is the coefficientwise one, theorem 1.19, whose only hypothesis is that $F_0'(X)$ be invertible. □

Proposition 5.40 (Differential formula for $c_n^{(\sigma)}$)**.** *Let $\mathscr{Q}_\sigma(t) = \sigma \sum_{j\geq 1} d_j t^j$ and*

$$\mathscr{L}_\sigma := -\frac{t^2}{\lambda + \sigma t/2} \frac{\mathrm{d}}{\mathrm{d}t} \qquad (t = X^{-1}), \tag{5.74}$$

which is eq. (5.72) *for $F_{0,\sigma}(X) = \lambda X + \frac{\sigma}{2} \log X$ expressed in the variable t. Then, for every $n \geq 1$,*

$$\boxed{c_n^{(\sigma)} = [t^n] \sum_{k=1}^{\lfloor (n+1)/2 \rfloor} \frac{(-1)^k}{k!} \mathscr{L}_\sigma^{k-1}\left(\frac{\mathscr{Q}_\sigma(t)^k}{\lambda + \sigma t/2}\right),} \tag{5.75}$$

a finite *sum requiring only $d_1, \ldots, d_n$.*

Proof. Apply theorem 5.39 with $F_0 = F_{0,\sigma}$, $g = \sigma D$ and $\eta = 1$; the substitution $t = X^{-1}$ turns $F_{0,\sigma}'(X) = \lambda + \sigma/(2X)$ into $\lambda + \sigma t/2$ and $\mathrm{d}/\mathrm{d}X$ into $-t^2\mathrm{d}/\mathrm{d}t$. Setting the marker η to 1 is legitimate coefficientwise because $\mathscr{Q}_\sigma(t)^k = O(t^k)$ while each application of $\mathscr{L}_\sigma$ raises the order in t by one, so the k-th summand begins at t^{2k-1} and can contribute to $[t^n]$ only when $2k - 1 \leq n$. □

Source claim. S2 writes its operator expansion as an *equality*, $\mathcal{I}_\sigma(y) = X + \sum_{m\geq 1} \frac{(-\sigma)^m}{m!} \mathscr{D}^{m-1}\big[D(X)^m / F_0'(X)\big]$, after setting its coupling parameter to 1. **Defect.** With the marker removed there is no small parameter, the series over m does not converge, and the left-hand side is a number while the right-hand side is a divergent series; the statement as printed is false. **Repair.** The correct statement is the asymptotic one: the m-th summand is $O(X^{-(2m-1)})$, so for each fixed M the partial sum through $m = M$ differs from $\mathcal{I}_\sigma(y)$ by $O(X^{-(2M+1)})$, and regrouping by powers of X^{-1} reproduces eq. (5.75). S1 and S3 state the same formula correctly, as a coefficient identity.

5.7.4 The analytic statement, with a proof

Source claim. All three sources assert the Poincaré statement "$\mathcal{I}_\sigma(y) = X_\sigma(y) + \sum_{n<M} c_n^{(\sigma)} X_\sigma(y)^{-n} + O(X_\sigma(y)^{-M})$". **Defect.** S3 gives no proof at all ("Likewise, for each fixed M,

…"); S1 gives a two-line sketch that appeals to the mean value theorem but never bounds the residual of the truncated ansatz, and in particular never uses any control on the forward remainder at the *perturbed* point; S2 proves the ingredients (a residual-to-index bound and derivative lower bounds) but does not assemble them. This is the one place in the three sources where a formal computation is upgraded to an analytic conclusion without an argument. **Repair.** Theorem 5.41 below supplies the proof, and makes the constant effective by using the enveloping remainder theorem 5.20.

Theorem 5.41 (Inverse transseries, analytically). *Fix $M \geq 1$ and $\sigma \in \{-1, +1\}$. Then, as $y \to +\infty$,*

$$\boxed{\mathcal{I}_\sigma(y) = X_\sigma(y) + \sum_{n=1}^{M-1} \frac{c_n^{(\sigma)}}{X_\sigma(y)^n} + O\big(X_\sigma(y)^{-M}\big)\,,} \tag{5.76}$$

with an effective implied constant depending only on M and σ. Since $X_\sigma(y) = \big(\log y\big)/\lambda\,(1 + o(1))$, the expansion is one in inverse powers of $\log y$ once the dominant power–logarithmic phase has been inverted exactly.

Proof. Write $X = X_\sigma(y)$ and $x_M := X + \sum_{n=1}^{M-1} c_n^{(\sigma)} X^{-n}$; for y large, $x_M = X(1 + O(X^{-2}))$, so $x_M > 1$ and $x_M \asymp X$. Consider the phase residual

$$\rho := \log \mathcal{S}_\sigma(x_M) - \log y = \lambda(x_M - X) + \frac{\sigma}{2} \log \frac{x_M}{X} + \sigma D(x_M),$$

where we used eq. (5.19) and $\log y = \log C_\sigma + \lambda X + \frac{\sigma}{2} \log X$. Put $t = X^{-1}$ and $U_{M-1}(t) = \sum_{n<M} c_n^{(\sigma)} t^n$, so that $x_M = X(1 + tU_{M-1})$ and

$$\rho = \lambda\, X \underbrace{\Big[tU_{M-1} + \frac{\sigma t}{2\lambda} \log(1 + tU_{M-1}) + \frac{\sigma t}{\lambda} D_{<M}(x_M) \Big]}_{=: t\,\mathcal{E}(t)} + \sigma\big(D(x_M) - D_{<M}(x_M)\big),$$

where $D_{<M}$ denotes the truncation of the series eq. (5.26) after the terms of order $x^{-(M-1)}$. Two estimates finish the proof.

(i) The algebraic residual. Substituting $x_M = X(1{+}tU_{M-1})$ and $D_{<M}(x_M) = \mathcal{Q}_\sigma\big(t/(1{+}tU_{M-1})\big)/\sigma$ truncated, the bracket $\mathcal{E}(t)$ is exactly $U_{M-1}(t) - \Psi_\sigma\big(t, U_{M-1}(t)\big)$ modulo terms of order t^M coming from the truncation of $\mathcal{Q}_\sigma$. By theorem 5.38 the coefficients of $t^1, \ldots, t^{M-1}$ in $U - \Psi_\sigma(t, U)$ vanish when U is replaced by U_{M-1} — this is the triangularity of the recurrence — so $\mathcal{E}(t) = O(t^M)$ with an explicit rational-in-λ constant obtained from $c_1^{(\sigma)}, \ldots, c_{M-1}^{(\sigma)}$ and $d_1, \ldots, d_M$. Hence $\lambda X \cdot t\,\mathcal{E}(t) = O(X^{-M+1} \cdot X^{-1}) \cdot X = O(X^{-M+1})$; more carefully, $X\,t\,\mathcal{E}(t) = \mathcal{E}(t) = O(X^{-M})$.

(ii) The transcendental residual. By theorem 5.20 with $N = \lceil M/2 \rceil$, $|D(x_M) - D_{<M}(x_M)| \leq |d_{2N+1}|\, x_M^{-(2N+1)} \leq c_M\, X^{-M}$ for an explicit c_M, since $2N + 1 \geq M$ and $x_M \asymp X$.

Therefore $|\rho| \leq C_M X^{-M}$ with C_M explicit. Finally, $\big(\log \mathcal{S}_\sigma\big)'(x) = \lambda + \frac{\sigma}{2x} + \sigma D'(x) \geq \lambda - \frac{1}{2x} - \frac{1}{4x^2}$ by theorem 5.14, which exceeds $\lambda/2$ once $x \geq 2/\lambda$. Theorem 1.42 (residual-to-root conversion) then gives

$$|x_M - \mathcal{I}_\sigma(y)| \leq \frac{|\rho|}{\lambda/2} \leq \frac{2C_M}{\lambda}\, X^{-M},$$

which is eq. (5.76). The final sentence follows from $\lambda X + \frac{\sigma}{2} \log X = L_\sigma(y)$ and $L_\sigma(y) = \log y + O(1)$. □

Remark 5.42 (An exact algebraic certificate). Step (i) of the proof is also the recommended *implementation* check. Given a candidate truncation U_{M-1}, substitute it into $U - \Psi_\sigma(t, U)$ and expand: the coefficients of $t^1, \ldots, t^{M-1}$ must vanish identically in $\mathbb{Q}[\lambda, \lambda^{-1}, \sigma]/(\sigma^2 - 1)$. This is an exact algebraic certificate for every table of $c_n^{(\sigma)}$, and it is how the tables in eq. (5.62)–eq. (5.69) were verified here, independently of the recurrence that produced them.

5.7.5 The centred reversion

Applying theorem 1.25 to the centred normal form eq. (5.45) — that is, with $a = 2\lambda$, $b = \sigma/2$, $C = 1/\sqrt{2\pi}$ and $q_j = \sigma\nu_j$ ($\nu_j = 0$ for odd j) — gives the centred cores

$$\mathcal{U}_+(y) = \frac{W_0\big(8\pi\lambda y^2\big)}{4\lambda}, \qquad \mathcal{U}_-(y) = -\frac{W_{-1}\big(-2\lambda/(\pi y^2)\big)}{4\lambda}, \tag{5.77}$$

and the centred inverse expansion

$$\boxed{\mathcal{I}_\sigma(y) \sim 2\,\mathcal{U}_\sigma(y) - \frac{1}{2} + 2\sum_{n\ge 2}\frac{\widehat{c}_n^{(\sigma)}}{\mathcal{U}_\sigma(y)^n}, \qquad \widehat{c}_1^{(\sigma)} = 0.} \tag{5.78}$$

The vanishing of $\widehat{c}_1^{(\sigma)}$ is immediate from $\nu_1 = 0$. Recomputed in the present normalization,

$$\widehat{c}_2^{(\sigma)} = -\frac{\sigma}{128\lambda}, \qquad \widehat{c}_3^{(\sigma)} = \frac{1}{512\lambda^2}, \qquad \widehat{c}_4^{(\sigma)} = \frac{\sigma(5\lambda^2-2)}{4096\lambda^3}, \tag{5.79}$$

$$\widehat{c}_5^{(\sigma)} = -\frac{7\lambda^2-2}{16384\lambda^4}, \qquad \widehat{c}_6^{(\sigma)} = -\frac{\sigma\big(244\lambda^4-57\lambda^2+12\big)}{393216\lambda^5}, \qquad \widehat{c}_7^{(\sigma)} = \frac{334\lambda^4-75\lambda^2+12}{1572864\lambda^6}, \tag{5.80}$$

and

$$\widehat{c}_8^{(\sigma)} = \frac{\sigma\big(4155\lambda^6-460\lambda^4+96\lambda^2-12\big)}{6291456\lambda^7}. \tag{5.81}$$

These agree with S1's centred table after the substitution $\alpha = 2\lambda$ and the marker translation of theorem 5.12; observe that $\widehat{c}_n$ is σ-even for odd n and σ-odd for even n, a reflection symmetry of eq. (5.44) that the uncentred coefficients do not have.

Source claim. S1 concludes that "the centred one is often the preferable computational normal form". **Defect.** Its own table already contradicts the unqualified form of this claim: at $m = 200$ the centred core beats the uncentred core by 10^3 (2.8×10^{-7} against 4.5×10^{-4}), but at $R = 4$ the *uncentred* expansion is slightly better (7.3×10^{-16} against 1.1×10^{-15}). **Repair.** The correct statement, confirmed by the recomputation in table 5.5 and table 5.6, is: centring removes the order-X^{-1} term entirely, so it improves the *bare core* by a factor $\asymp X$ and is the right choice when no corrections are to be carried; once several corrections are carried, the two normalizations are comparable, with neither dominating uniformly.

5.8 Flattening the core into a polynomial–logarithmic transseries

5.8.1 The specialization

Set

$$H_\sigma(y) := \log\frac{L_\sigma(y)}{\lambda}, \tag{5.82}$$

and recall from theorem 1.33 that the inverse of the model theorem 1.2 has the unique power–logarithmic form $x \sim a^{-1}\big[L - bH + \sum_{n\ge1} P_n(H)L^{-n}\big]$ with $\deg P_n \le n$ and leading term $b^{n+1}H^n/n$. With the dictionary eq. (5.21),

$$\boxed{\mathcal{I}_\sigma(y) \sim \frac{1}{\lambda}\left[L_\sigma - \frac{\sigma}{2}H_\sigma + \sum_{n\ge1}\frac{P_n^{(\sigma)}(H_\sigma)}{L_\sigma^n}\right],} \tag{5.83}$$

where the $P_n^{(\sigma)}$ are generated by theorem 1.33 from $b = \sigma/2$, $a = \lambda$, $q_j = \sigma d_j$; the fixed-order meaning of eq. (5.83) is the analytic clause of theorem 1.33, namely an error $O\big((1+H_\sigma)^{N+1}L_\sigma^{-(N+1)}\big)$ after N

correction terms, whose hypotheses hold here by theorem 5.41. The recomputed polynomials are

$$P_1^{(\sigma)}(H) = \frac{H - \sigma\lambda}{4}, \tag{5.84}$$

$$P_2^{(\sigma)}(H) = \frac{\sigma H^2 - 2(\lambda+\sigma)H + 2\lambda}{16}, \tag{5.85}$$

$$P_3^{(\sigma)}(H) = \frac{1}{96}\Big[2H^3 - (6\sigma\lambda+9)H^2 + (18\sigma\lambda+6)H + 4\sigma\lambda^3 - 6\lambda^2 - 6\sigma\lambda\Big], \tag{5.86}$$

$$\begin{aligned} P_4^{(\sigma)}(H) = \frac{1}{384}\Big[&3\sigma H^4 - (12\lambda+22\sigma)H^3 + (66\lambda+36\sigma)H^2 \\ &+ (24\lambda^3 - 36\sigma\lambda^2 - 72\lambda - 12\sigma)H - 8\lambda^3 + 30\sigma\lambda^2 + 12\lambda\Big], \end{aligned} \tag{5.87}$$

$$\begin{aligned} P_5^{(\sigma)}(H) = \frac{1}{3840}\Big[&12H^5 - (60\sigma\lambda+125)H^4 + (500\sigma\lambda+350)H^3 \\ &+ (240\sigma\lambda^3 - 360\lambda^2 - 1050\sigma\lambda - 300)H^2 \\ &+ (-280\sigma\lambda^3 + 780\lambda^2 + 600\sigma\lambda + 60)H \\ &- 192\sigma\lambda^5 + 160\lambda^4 - 80\sigma\lambda^3 - 270\lambda^2 - 60\sigma\lambda\Big]. \end{aligned} \tag{5.88}$$

These reproduce S2 exactly for $1 \le n \le 5$; they were recomputed for $\sigma = \pm 1$ separately and recombined in $\mathscr{R}$, and independently re-derived through $n = 6$. They also satisfy the cross-normalization identity $P_n^{(\sigma)}(0) = \lambda^{n+1}c_n^{(\sigma)}$ of theorem 1.33: for instance $P_1^{(\sigma)}(0) = -\sigma\lambda/4 = \lambda^2 c_1^{(\sigma)}$ and $P_2^{(\sigma)}(0) = \lambda/8 = \lambda^3 c_2^{(\sigma)}$, which ties eq. (5.84)–eq. (5.88) to eq. (5.62)–eq. (5.69) and catches any transcription error in either list.

5.8.2 The pure Lambert block in closed form

The pure block — the case $Q \equiv 0$, in which the whole correction series is suppressed and only the Lambert core survives — is theorem 1.34, proved there in the shortest of the three forms the sources use. Specializing its data to the swing branches, $a = \lambda$ and $b = \sigma/2$, gives

$$P_n^{(0)}(H) = \left(\frac{\sigma}{2}\right)^{n+1} \sum_{m=1}^{n} (-1)^{m+1} \begin{bmatrix} n \\ m \end{bmatrix} \frac{H^{n-m+1}}{(n-m+1)!}, \tag{5.89}$$

with $\left[{n \atop m}\right]$ the unsigned Stirling numbers of the first kind. Reducing with $\sigma^2 = 1$, the two parities differ only by the overall sign σ^{n+1}: the even and odd pure blocks coincide for odd n and are opposite for even n.

Writing $b = \sigma/2$, the first few are

$$P_1^{(0)} = b^2 H, \quad P_2^{(0)} = \tfrac{b^3}{2}(H^2 - 2H), \quad P_3^{(0)} = \tfrac{b^4}{6}(2H^3 - 9H^2 + 6H), \quad P_4^{(0)} = \tfrac{b^5}{12}(3H^4 - 22H^3 + 36H^2 - 12H),$$

and eq. (5.89) was recomputed here against the recurrence of theorem 1.33 for $1 \le n \le 8$, in exact arithmetic with b symbolic. In particular the leading coefficient is b^{n+1}/n, which is theorem 1.33's degree statement. Theorem 1.34 is the precise bridge between Lambert asymptotics and the first-kind Stirling calculus, and it is stated there rather than here because all four chapters use it.

5.8.3 The core series and its convergence

The flattening can also be organized as "expand the core, then compose", which is S3's route and which is constructive at every order.

Proposition 5.43 (All coefficients of the scaled core)**.** *Let w solve $w + \sigma \log w = \Lambda$ as in eq.* (5.58)*, and put $\ell = \log \Lambda$. Define*

$$A_k(\ell) := \frac{1}{k+1}[z^k]\left(\ell + \log(1+z)\right)^{k+1}, \qquad k \ge 0. \tag{5.90}$$

Then

$$w \;=\; \Lambda + \sum_{k\geq 0} \frac{(-\sigma)^{k+1} A_k(\ell)}{\Lambda^k}, \tag{5.91}$$

and, with $s(k,j)$ the signed Stirling numbers of the first kind,

$$A_0(\ell) = \ell, \qquad A_k(\ell) = \frac{1}{(k+1)\,k!} \sum_{j=1}^{k} \binom{k+1}{j} j!\, s(k,j)\, \ell^{\,k+1-j} \quad (k \geq 1). \tag{5.92}$$

The first polynomials are

$$A_0 = A_1 = \ell, \quad A_2 = \tfrac{\ell(2-\ell)}{2}, \quad A_3 = \tfrac{\ell(2\ell^2-9\ell+6)}{6}, \quad A_4 = -\tfrac{\ell(3\ell^3-22\ell^2+36\ell-12)}{12},$$

$$A_5 = \tfrac{\ell(12\ell^4-125\ell^3+350\ell^2-300\ell+60)}{60}, \quad A_6 = -\tfrac{\ell(20\ell^5-274\ell^4+1125\ell^3-1700\ell^2+900\ell-120)}{120}.$$

Proof. Write $w = \Lambda(1+v)$ and $z = \Lambda^{-1}$. The equation becomes $\Lambda v = -\sigma\big(\ell + \log(1+v)\big)$, i.e. $v = -\sigma z\big(\ell + \log(1+v)\big)$. Lagrange inversion gives $v = \sum_{k\geq 0} \frac{(-\sigma z)^{k+1}}{k+1} [v^k]\,(\ell + \log(1+v))^{k+1}$; multiplying by $\Lambda = z^{-1}$ yields eq. (5.91) with A_k as in eq. (5.90). Expanding the $(k+1)$-st power and using $\big(\log(1+z)\big)^j = j! \sum_{k\geq j} s(k,j) z^k/k!$ gives eq. (5.92). ☐

The sign in eq. (5.91) matches S3 after $\varepsilon = -\sigma$; explicitly $w_- = \Lambda_- + \ell + \ell/\Lambda_- + \cdots$ on the even sheet (W_{-1}) and $w_+ = \Lambda_+ - \ell + \ell/\Lambda_+ - \cdots$ on the odd sheet (W_0).

Proposition 5.44 (An explicit convergence criterion). *Fix $\rho \in (0,1)$ and put $M_\rho(\ell) = |\ell| + \log \frac{1}{1-\rho}$. Then*

$$|A_k(\ell)| \leq \frac{M_\rho(\ell)^{\,k+1}}{(k+1)\rho^k}, \tag{5.93}$$

so the series in eq. (5.91) *converges absolutely whenever $M_\rho(\ell) < \rho|\Lambda|$ for some $\rho \in (0,1)$, and its sum is then the solution branch $w \sim \Lambda$. In particular, for real $\Lambda \geq 10$ the choice $\rho = \frac{1}{2}$ already gives convergence.*

Proof. On $|z| = \rho$ one has $|\log(1+z)| \leq \sum_{k\geq 1} \rho^k/k = \log \frac{1}{1-\rho}$, so $|\ell + \log(1+z)| \leq M_\rho(\ell)$ there; Cauchy's estimate applied to eq. (5.90) gives eq. (5.93). Summing the geometric-type majorant gives absolute convergence for $M_\rho(\ell) < \rho|\Lambda|$. The sum solves $w + \sigma \log w = \Lambda$ because Lagrange's theorem identifies the series with the unique analytic solution $v(z)$ of $v = -\sigma z(\ell + \log(1+v))$ vanishing at $z = 0$; the branch with $w \sim \Lambda$ is the one selected. For $\Lambda = 10$, $\ell = \log 10 = 2.3026$ and $M_{1/2} = 2.3026 + 0.6931 = 2.9957 < 5$. ☐

Source claim. S3 asserts that "the pure Lambert logarithmic expansion has a nonzero convergence domain for sufficiently large argument, as in the classical analysis of Lambert W", citing Corless et al. but giving no criterion. **Defect.** The cited classical result concerns the doubly-indexed series in $(\log z)^{-1}$ and $\log\log z$; the single series eq. (5.91), whose coefficients are polynomials in the slowly varying ℓ, needs its own statement, since convergence depends on the ratio ℓ/Λ and fails for ℓ comparable to Λ. **Repair.** Theorem 5.44 supplies an explicit, proved criterion, which also delimits the regime where eq. (5.91) may be summed rather than truncated.

5.8.4 Composition, and the equivalence of the two normalizations

Theorem 5.45 (Constructive equivalence of the two inverse normalizations). *With $v_j = (-\sigma)^j A_{j-1}(\ell)$ and*

$$R_r^{(m)}(\ell) = [z^r] \Big(1 + \sum_{j\geq 1} v_j z^j\Big)^{-m} = \sum_{k=0}^{r} (-1)^k \binom{m+k-1}{k} \widehat{B}_{r,k}(v), \qquad R_0^{(m)} = 1, \tag{5.94}$$

the Lambert-normalized expansion eq. (5.76) *and the flattened expansion eq.* (5.83) *are term-by-term equivalent: in the scaled variables of eq.* (5.58)*,*

$$\mathcal{I}_\sigma(y) \sim \frac{\Lambda_\sigma}{2\lambda} - \frac{\sigma\,\ell_\sigma}{2\lambda} + \sum_{N\ge 1} \frac{\Pi_N^{(\sigma)}(\ell_\sigma)}{\Lambda_\sigma^N}, \qquad \Pi_N^{(\sigma)}(\ell) = \frac{(-\sigma)^{N+1} A_N(\ell)}{2\lambda} + \sum_{m=1}^{N} c_m^{(\sigma)} (2\lambda)^m R_{N-m}^{(m)}(\ell), \tag{5.95}$$

with $\deg \Pi_N^{(\sigma)} = N$ *and leading coefficient* $(-1)^{N-1}(-\sigma)^{N+1}/(2\lambda N)$.

Proof. Insert $X_\sigma = w_\sigma/(2\lambda)$ into eq. (5.76), write $w = \Lambda(1+V)$ with $V = \sum_j v_j z^j$ from eq. (5.91), and collect the coefficient of Λ^{-N}: the core contributes the first term of eq. (5.95) and the m-th correction contributes $c_m^{(\sigma)}(2\lambda)^m [z^{N-m}]\,(1+V)^{-m}$. The Bell form of $R_r^{(m)}$ is theorem 1.12. Degrees: A_N has degree N with leading coefficient $(-1)^{N-1}/N$ by eq. (5.92), while every correction with $m \ge 1$ contributes degree at most $N-m$. □

Source claim. S1 asserts the equivalence of its two inverse normalizations in prose: "expanding the Lambert carrier in the scale $X^{-1}\mathbb{R}[\ell]$ and then substituting that expansion … reproduces the recurrence coefficient by coefficient." **Defect.** No proof and no formula; the assertion is exactly the sort of "successive matching" that the coefficient calculus is supposed to eliminate. **Repair.** Theorem 5.45, which follows S3, makes the equivalence a finite explicit composition with a proof, and identifies the degrees and leading coefficients.

Remark 5.46 (Three flattening normalizations, and which one the volume uses). The three sources flatten in three different variables: S2 in $L_\sigma = \log(y/C_\sigma)$ with $H_\sigma = \log(L_\sigma/\lambda)$; S3 in $\Lambda_\sigma = \log(\pi y^2) + \sigma\log(4\lambda)$ with $\ell_\sigma = \log\Lambda_\sigma$; S1 in $X_\epsilon = \log(y\alpha^{\beta_\epsilon}/K_\epsilon)$ in the half index. Since $\Lambda_\sigma = 2L_\sigma + \sigma(\log(4\lambda) - \lambda) \ne 2L_\sigma$, these are genuinely different variables, the three coefficient lists are not comparable termwise, and none is in error — all were recomputed and are correct in their own normalization. The volume adopts S2's variables in eq. (5.83) (they are those of theorem 1.33) and records S3's in eq. (5.95), where the composition proof is cleanest.

Corollary 5.47 (Leading elementary forms). *As* $y \to \infty$*,*

$$\mathcal{I}_\sigma(y) = \frac{1}{\lambda}\Big(L_\sigma(y) - \frac{\sigma}{2}\log\frac{L_\sigma(y)}{\lambda} + O\Big(\frac{\log L_\sigma(y)}{L_\sigma(y)}\Big)\Big). \tag{5.96}$$

The opposite signs of the $\frac{1}{2}\log L$ *term on the two sheets are the inverse-side signature of the forward powers* $n^{-1/2}$ *and* $n^{+1/2}$.

5.9 Asymptotic meaning, optimal truncation, and Stokes structure

5.9.1 What each expansion means at fixed order

For each fixed M the forward statement is eq. (5.36) with the effective constant of theorem 5.25, and the inverse statement is eq. (5.76). Both are Poincaré statements about divergent series: the coefficients d_j grow factorially (theorem 5.22), hence so do the A_n and, through the reversion, the $c_n^{(\sigma)}$. The divergence is not a defect but a datum: the Borel transform and the exact sum of the logarithmic series are known in closed form (theorem 5.13, theorem 5.16). By contrast the pure core series eq. (5.91) *converges* in the regime delimited by theorem 5.44; the full inverse is asymptotic only because of the gamma-ratio correction.

5.9.2 Optimal truncation and its image in the inverse variable

Proposition 5.48 (Least-term truncation, forward and inverse). *Theorem 1.46 applied to eq.* (5.30) *with the singulant modulus* π *of theorem 5.16 gives the least-term index* $r \approx \pi x/2$ *and the beyond-all-orders floor*

$$\big|R_N(x)\big|_{N\approx \pi x/2} = \frac{2\sqrt{2}}{\pi\sqrt{x}}\,\mathrm{e}^{-\pi x}\big(1+O(x^{-1})\big), \tag{5.97}$$

which by theorem 5.20 is an equality up to the stated factor and not merely an upper bound. Transporting it through the inverse by theorem 1.44 and $\big(\log \mathcal{S}_\sigma\big)' = \lambda + O(x^{-1})$ *gives the inverse floor*

$$O\Big(X_\sigma^{-1/2}\mathrm{e}^{-\pi X_\sigma}\Big) = O\Big(y^{-\pi/\lambda}\,(\log y)^{-1/2+\sigma\pi/(2\lambda)}\Big), \qquad \frac{\pi}{\lambda} = 4.5323601418\ldots \tag{5.98}$$

Proof. Equation (5.97) is theorem 5.22. For the image, use the core equation in the form $\mathrm{e}^{\lambda X} = (y/C_\sigma)X^{-\sigma/2}$, so that $\mathrm{e}^{-\pi X} = (y/C_\sigma)^{-\pi/\lambda}X^{\sigma\pi/(2\lambda)}$; multiply by $X^{-1/2}$ and use $X \asymp \log y$. □

Remark 5.49. Exponential smallness in the index becomes *algebraic* smallness in the level: an $\mathrm{e}^{-\pi x}$ forward sector maps to a $y^{-\pi/\lambda}$ inverse sector. This is why hyperasymptotic corrections to the inverse live on a strictly finer grid than the algebraic powers of $1/\log y$ produced by eq. (5.83) — they are invisible to that scale at every order.

5.9.3 Lateral sums and the aggregate Stokes multipliers

For a direction θ let $\mathcal{S}_{\theta\pm} D(x) = \int_0^{\mathrm{e}^{\mathrm{i}(\theta\pm 0)}\infty} \mathrm{e}^{-xt}\widehat{D}(t)\,\mathrm{d}t$ whenever the rotated Laplace integral converges, and let Disc_θ denote the lower-minus-upper lateral difference.

Proposition 5.50 (Bridge equations). *With the poles and residues of eq.* (5.25),

$$\mathrm{Disc}_{\pi/2} D(x) = 2\sum_{k\ge 0}\frac{\mathrm{e}^{-\mathrm{i}\pi(2k+1)x}}{2k+1} = 2\,\mathrm{artanh}\big(\mathrm{e}^{-\mathrm{i}\pi x}\big) \qquad (\Im x < 0), \tag{5.99}$$

$$\mathrm{Disc}_{-\pi/2} D(x) = -2\sum_{k\ge 0}\frac{\mathrm{e}^{\mathrm{i}\pi(2k+1)x}}{2k+1} = -2\,\mathrm{artanh}\big(\mathrm{e}^{\mathrm{i}\pi x}\big) \qquad (\Im x > 0), \tag{5.100}$$

the closed forms giving the analytic continuations beyond those half-planes. Consequently $\mathrm{Disc}_\theta \log \mathcal{S}_\sigma = \sigma\,\mathrm{Disc}_\theta D$*, and the multiplicative Stokes factor of the forward branch across the ray* $\theta = \pm\pi/2$ *is* $\exp\big(\sigma\,\mathrm{Disc}_{\pm\pi/2} D(x)\big)$*, that is*

$$\mathfrak{M}_+^{(\sigma)}(x) = \left(\frac{1+\mathrm{e}^{-\mathrm{i}\pi x}}{1-\mathrm{e}^{-\mathrm{i}\pi x}}\right)^{\sigma}, \qquad \mathfrak{M}_-^{(\sigma)}(x) = \left(\frac{1-\mathrm{e}^{\mathrm{i}\pi x}}{1+\mathrm{e}^{\mathrm{i}\pi x}}\right)^{\sigma}. \tag{5.101}$$

On the positive real axis no lateral choice arises, because $\arg t = 0$ *is a regular direction.*

Proof. Rotating the Laplace contour past the ray through $t_k = \mathrm{i}\pi(2k+1)$ picks up $2\pi\mathrm{i}$ times the residue of $\mathrm{e}^{-xt}\widehat{D}(t)$ there, namely $2\pi\mathrm{i}\,\mathrm{e}^{-t_k x}/t_k = 2\mathrm{e}^{-\mathrm{i}\pi(2k+1)x}/(2k+1)$; summing over k and using $\mathrm{artanh}\,z = \sum_{k\ge 0} z^{2k+1}/(2k+1)$ gives eq. (5.99). The series converges for $|\mathrm{e}^{-\mathrm{i}\pi x}| = \mathrm{e}^{\pi\Im x} < 1$, that is $\Im x < 0$. The lower ray is the conjugate computation. The last two displays follow from $\log \mathcal{S}_\sigma = \log C_\sigma + \lambda x + \frac{\sigma}{2}\log x + \sigma D$, in which only D is discontinuous, together with $2\,\mathrm{artanh}\,z = \log\frac{1+z}{1-z}$. □

Remark 5.51 (Convention dependence). S2 defines the discontinuity as lower minus upper lateral sum; S3 instead adopts the residue convention in which crossing a singular ray contributes $+2\pi\mathrm{i}$ times the enclosed residues, and works with $Q = -D$. The two sets of formulae differ by an overall sign, hence by a reciprocal in the multipliers, and both are internally consistent. A Stokes generator is only defined relative to a stated convention, so the volume states the convention explicitly in theorem 5.50 and does not declare either source's signs canonical.

5.9.4 Transport of an exponential sector through inversion

Proposition 5.52 (Sector transport). *Let F be one lateral branch of $\log \mathcal{S}_\sigma$ and let the other be $F + \Delta$. As a formal expansion in the sector Δ,*

$$x_{\mathrm{new}} = x + \sum_{k\geq 1} \frac{(-1)^k}{k!} \left(\frac{1}{F'(x)} \frac{\mathrm{d}}{\mathrm{d}x} \right)^{k-1} \left(\frac{\Delta(x)^k}{F'(x)} \right), \tag{5.102}$$

with $\Delta = \sigma \operatorname{Disc}_\theta D$ and $F'(x) = \lambda + \frac{\sigma}{2x} + \sigma D'(x)$. Quantitatively, and with no formal series at all: if $|\Delta| \leq \delta$ on the interval joining x and x_{new} and $F' \geq \mu > 0$ there, then

$$|x_{\mathrm{new}} - x| \leq \frac{\delta}{\mu}, \qquad x_{\mathrm{new}} - x = -\frac{\Delta(x)}{F'(x)} + O(\delta^2), \tag{5.103}$$

the implied constant depending only on $\sup |(F^{-1})''|$ *on that interval.*

Proof. Equation (5.102) is theorem 5.39 with $F_0 = F$, $g = \Delta$ and $\eta = 1$, read as an identity of formal series in the sector marker. The first inequality in eq. (5.103) is theorem 1.42; the second is Taylor's theorem for F^{-1} at the point $F(x)$. □

Source claim. S1 states its transport rule as a *theorem* ("Transseries-sector transport through inversion") whose hypothesis is that "E is smaller than the algebraic scale retained in Φ_{alg}", whose conclusion is prefaced by the word "formally", and whose displayed first terms carry an unjustified error symbol $O(E^3, E^2E', E(E')^2)$. **Defect.** The hypothesis is never used; the conclusion is a formal-series identity, not an analytic one; and the error term is asserted rather than proved. A statement that holds only formally must not stand in a `theorem` environment. **Repair.** Theorem 5.52 separates the two claims: the all-orders identity is formal (and is exactly theorem 5.39), while the quantitative first-order statement is proved by the mean value theorem under stated hypotheses. S2 and S3 already state their versions correctly, as formal expansions.

Source claim. S1's section on complex inverse sheets asserts that for every Lambert branch with $|w_{\epsilon,k}| \to \infty$ in a suitable sector, "the same coefficients $d_r^{(\epsilon)}$ give $M_{\epsilon,k}(y) \sim w_{\epsilon,k}(y) + \sum_r d_r^{(\epsilon)} w_{\epsilon,k}(y)^{-r}$". **Defect.** For $k \neq 0, -1$ nothing has been said, let alone proved, about $M_{\epsilon,k}$ being an inverse of anything: the existence of an analytic branch of $\mathcal{S}_\sigma^{-1}$ along the corresponding path is a hypothesis, not a consequence, and it fails when the path meets a pole of the gamma quotient or a Stokes direction. **Repair.** The statement is demoted to theorem 5.53, with the two missing hypotheses displayed. S2 states the same thing correctly, as a *formal* complex branch whose analytic validity is a separate requirement.

Remark 5.53 (Complex branches, formally). For $k \in \mathbb{Z}$ let $X_{\sigma,k}$ be as in eq. (5.51) after fixing branches of the powers and logarithms in L_σ. Suppose that along a path with $|X_{\sigma,k}(y)| \to \infty$ *(i)* the forward expansion eq. (5.36) is valid, which requires the path to stay in a sector free of the poles of $\Gamma(x/2 + (3-\sigma)/4)^{-2}$ and away from the Stokes directions of theorem 5.50, and *(ii)* an analytic branch $\mathcal{I}_{\sigma,k}$ of the inverse exists along it. Then that branch satisfies $\mathcal{I}_{\sigma,k}(y) \sim X_{\sigma,k}(y) + \sum_{n\geq 1} c_n^{(\sigma)} X_{\sigma,k}(y)^{-n}$ with the *same* coefficients, by the proof of theorem 5.41. The coefficient sequence does not depend on k: all branch dependence sits in W_k and in the chosen logarithm. Without hypotheses (i) and (ii) the display is a formal statement only.

5.10 Discrete inversion and index recovery

5.10.1 The three inverse objects, parity by parity

Theorem 1.37 attaches three objects to a discrete sequence: the functional inverse of a chosen interpolation, the integer staircase, and the round-to-nearest generalized inverse.

Warning 5.54 (None of the three exists for the unsplit sequence). By theorem 5.4 the set $\{n : \mathrm{sw}(n) \le y\}$ is not an initial segment of $\mathbb{N}_0$ — for instance $\mathrm{sw}(5) = 30 > 20 = \mathrm{sw}(6)$ — so the staircase of sw is not monotone and does not invert it; the round-to-nearest inverse is ambiguous whenever two indices of opposite parity are nearly equidistant in value; and by theorem 5.9 no eventually monotone interpolation exists at all. All three objects exist, are unique, and obey theorem 1.39 *on each parity class separately.*

Proposition 5.55 (Exact branchwise staircases). *For $y \ge 1$ put*

$$N_-^{\le}(y) = \max\{n \textit{ even} : \mathrm{sw}(n) \le y\}, \qquad N_+^{\le}(y) = \max\{n \textit{ odd} : \mathrm{sw}(n) \le y\}, \tag{5.104}$$

both sets being non-empty because $\mathrm{sw}(0) = \mathrm{sw}(1) = 1$. *Then*

$$\boxed{N_-^{\le}(y) = 2\left\lfloor \frac{\mathcal{I}_-(y)}{2} \right\rfloor, \qquad N_+^{\le}(y) = 2\left\lfloor \frac{\mathcal{I}_+(y) - 1}{2} \right\rfloor + 1,} \tag{5.105}$$

and the least index of each parity that reaches the level y is

$$N_-^{\ge}(y) = 2\left\lceil \frac{\mathcal{I}_-(y)}{2} \right\rceil, \qquad N_+^{\ge}(y) = 2\left\lceil \frac{\mathcal{I}_+(y) - 1}{2} \right\rceil + 1. \tag{5.106}$$

These identities are exact, *not asymptotic. The quantity eq.* (5.106) *is the staircase N_* of theorem 1.37, and the displayed formula for it is theorem 1.39's exact-rounding identity $N_* = \lceil \nu \rceil$, applied to the bisection* $(\mathrm{sw}(2m + \epsilon))_{m \ge 0}$ *with its admissible interpolation $m \mapsto \mathcal{S}_\sigma(2m + \epsilon)$.*

Proof. By eq. (5.11) and theorem 5.8, $\mathcal{S}_-$ is a strictly increasing bijection agreeing with sw at the even integers, so for even n we have $\mathrm{sw}(n) \le y \iff \mathcal{S}_-(n) \le y \iff n \le \mathcal{I}_-(y)$. The largest such even n is $2\lfloor \mathcal{I}_-(y)/2 \rfloor$ and the least even $n \ge \mathcal{I}_-(y)$ is $2\lceil \mathcal{I}_-(y)/2 \rceil$. The odd case is identical after the shift $n \mapsto n - 1$. The boundary case in which $\mathcal{I}_\sigma(y)$ is an integer of the relevant parity is covered, because then y is exactly a sequence value and floor and ceiling return it. ☐

The round-to-nearest inverses on the two parity classes are

$$n_-^{\mathrm{near}}(y) = 2\left\lfloor \frac{\mathcal{I}_-(y)}{2} + \frac{1}{2} \right\rfloor, \qquad n_+^{\mathrm{near}}(y) = 2\left\lfloor \frac{\mathcal{I}_+(y) - 1}{2} + \frac{1}{2} \right\rfloor + 1. \tag{5.107}$$

Remark 5.56 (The rounding must be certified, not trusted). Equation (5.105)–eq. (5.107) are exact statements about the real numbers $\mathcal{I}_\sigma(y)$, whereas a truncated transseries supplies only an approximation to them. When y lies exponentially close to a sequence value the floor can flip. The remedy is cheap: use the transseries to produce a candidate integer, then decide the inequality $\mathrm{sw}(n) \le y$ exactly, either in integer arithmetic or by comparing $\log \mathcal{S}_\sigma(n)$ with $\log y$ at the candidate and its same-parity neighbour, working through $\log \Gamma$ to stay in range. Theorem 1.39 is the general separation condition and theorem 1.42 converts the logarithmic residual into a certified bracket for $\mathcal{I}_\sigma(y)$.

5.10.2 Which parity wins, and by how much

Theorem 5.57 (The sheet gap). *Put $Y = \log(y\sqrt{\pi})$ and $h = \log(Y/\lambda)$, so that $L_\sigma = Y + \sigma\lambda/2$. Then, as $y \to \infty$,*

$$\boxed{\mathcal{I}_-(y) - \mathcal{I}_+(y) = \frac{h}{\lambda} - 1 + \frac{1}{2Y} + O\left(\frac{h^2}{Y^2}\right) = \log_2\left(\frac{\log(y\sqrt{\pi})}{\log 2}\right) - 1 + o(1).} \tag{5.108}$$

In particular the gap tends to $+\infty$: every sufficiently large level is reached first on the odd *sheet, about* $\log_2 \log y$ *indices before the even sheet catches up.*

Proof. Subtract the two instances of eq. (5.83) truncated after $n = 1$. The leading terms contribute $(L_- - L_+)/\lambda = -1$. Next, $H_\sigma = \log\big((Y + \sigma\lambda/2)/\lambda\big) = h + \sigma\lambda/(2Y) + O(Y^{-2})$, and the two contributions $-(\sigma/2\lambda)H_\sigma$ combine to $(H_- + H_+)/(2\lambda) = h/\lambda + O(Y^{-2})$, the σ-odd parts cancelling. Finally, with $P_1^{(\sigma)} = (H - \sigma\lambda)/4$ from eq. (5.84),

$$\frac{1}{\lambda}\left[\frac{P_1^{(-1)}(H_-)}{L_-} - \frac{P_1^{(+1)}(H_+)}{L_+}\right] = \frac{1}{4\lambda}\left[\frac{H_- + \lambda}{L_-} - \frac{H_+ - \lambda}{L_+}\right] = \frac{1}{4\lambda}\cdot\frac{2\lambda}{Y} + O\left(\frac{h}{Y^2}\right) = \frac{1}{2Y} + O\left(\frac{h}{Y^2}\right).$$

Collecting the three displays gives eq. (5.108); the second form follows from $h/\lambda = \log_2(Y/\lambda)$ and $1/(2Y) \to 0$. The error term $O(h^2/Y^2)$ absorbs the $n = 2$ terms of eq. (5.83), whose polynomials have degree 2 in H_σ. □

Remark 5.58 (Numerical confirmation, and agreement with S1). Recomputed with eight Lambert-normalized corrections on each sheet: for $\log y = 10, 30, 100, 400$ the gap $\mathcal{I}_-(y) - \mathcal{I}_+(y)$ equals 2.97806, 4.47863, 6.18565, 8.17591, against the prediction $h/\lambda - 1 + 1/(2Y)$ equal to 2.97829, 4.47928, 6.18583, 8.17593. S1 states the same result in its half index as $N_0 - N_1 = \frac{2}{\alpha}\log\log y + \frac{2}{\alpha}\log\frac{2}{\alpha} - 1 + o(1)$ with $\alpha = 2\lambda$; since $\frac{2}{\alpha} = \frac{1}{\lambda}$ and $\log\frac{2}{\alpha} = -\log\lambda$, this reads $\frac{1}{\lambda}(\log\log y - \log\lambda) - 1 + o(1)$, which is eq. (5.108). Directly from the sequence, the least index reaching $y = 10, 10^3, 10^6, 10^{12}$ is $5, 11, 21, 39$ on the odd sheet against $6, 14, 24, 44$ on the even sheet: gaps $1, 3, 3, 5$, consistent with the very slow $\log_2\log y$ growth.

Corollary 5.59 (Counting function)**.** *Let* $N(y) := \#\{n \in \mathbb{N}_0 : \mathrm{sw}(n) \le y\}$. *Then*

$$N(y) = \frac{\mathcal{I}_-(y) + \mathcal{I}_+(y)}{2} + O(1) = \log_2\big(y\sqrt{\pi}\big) + O(1). \tag{5.109}$$

Proof. By theorem 5.55 the even indices counted are $0, 2, \ldots, 2\lfloor\mathcal{I}_-(y)/2\rfloor$, that is $\mathcal{I}_-(y)/2 + O(1)$ of them, and similarly there are $\mathcal{I}_+(y)/2 + O(1)$ odd ones; this is the first equality. For the second, average the two instances of eq. (5.83): as in the proof of theorem 5.57 the σ-odd term $-(\sigma/2\lambda)H_\sigma$ cancels in the average, leaving $\frac{1}{2\lambda}(L_- + L_+) + O(\log\log y/\log y) = Y/\lambda + O(1)$, and $Y/\lambda = \log_2(y\sqrt{\pi})$. □

5.11 Algorithms, certification, and validation

5.11.1 Generating the coefficients

Algorithm 5.60 (Exact coefficient generation)**.** Every coefficient family of this chapter is produced by a finite exact recurrence over $\mathbb{Q}[\lambda, \lambda^{-1}, \sigma]/(\sigma^2 - 1)$:

1. d_{2r+1} from eq. (5.26) (rational Bernoulli input; $d_{2r} = 0$);
2. A_n from the triangular recurrence eq. (5.35);
3. $c_n^{(\sigma)}$ from eq. (5.61), or by solving $U = \Psi_\sigma(t, U)$ coefficientwise, or from the differential formula eq. (5.75);
4. $P_n^{(\sigma)}(H)$ from theorem 1.33 with the dictionary eq. (5.21), or from eq. (5.95) after theorem 5.43;
5. the exact certificate of theorem 5.42 after every step.

Truncating every intermediate series at order n is exact for the n-th coefficient, because all three recurrences are triangular; at order N the cost is $\widetilde{O}(N^2)$ coefficient operations. No numerical fitting and no order-specific ansatz occurs anywhere.

Remark 5.61 (How this chapter's tables were produced). The three sources supply three implementations of the above. For this chapter every coefficient was regenerated by a fourth, independent one: truncated power-series arithmetic over exact rationals with λ symbolic, run separately at $\sigma = \pm 1$ and recombined in $\mathscr{R}$. Every displayed coefficient agreed with the source that printed it; the corrections recorded in this chapter concern statements, never numbers.

5.11.2 Certified numerical inversion

Proposition 5.62 (Derivative bounds for certification). *For $x \geq 1$,*

$$\big(\log \mathcal{S}_+\big)'(x) > \lambda + \frac{1}{2x} - \frac{1}{4x^2} > \lambda, \qquad \big(\log \mathcal{S}_-\big)'(x) > \lambda - \frac{1}{2x}, \tag{5.110}$$

the exact value being eq. (5.14). *Hence if $r_* = \log \mathcal{S}_\sigma(x_*) - \log y$ and $\big(\log \mathcal{S}_\sigma\big)' \geq \mu > 0$ on the interval joining x_* and $\mathcal{I}_\sigma(y)$, then $|x_* - \mathcal{I}_\sigma(y)| \leq |r_*|/\mu$ by theorem 1.42.*

Proof. $\big(\log \mathcal{S}_\sigma\big)'(x) = \lambda + \frac{\sigma}{2x} + \sigma D'(x)$ by eq. (5.19); apply $0 < -D'(x) < 1/(4x^2)$ from theorem 5.14, and $1/(2x) - 1/(4x^2) > 0$ for $x > 1/2$. □

Algorithm 5.63 (Branchwise inversion of a level). Given a large y and a parity σ:

1. Evaluate the Lambert core: $X_+ = W_0(4\pi\lambda y^2)/(2\lambda)$ or $X_- = -W_{-1}(-4\lambda/(\pi y^2))/(2\lambda)$. *The branch choice is not optional* (theorem 5.32).
2. Add corrections $\sum_n c_n^{(\sigma)} X^{-n}$ by Horner nesting in $1/X$, stopping at or before the least term rather than summing indefinitely; alternatively use the centred form eq. (5.78), whose bare core is already accurate to $O(\mathcal{U}^{-2})$.
3. Certify: compute $r_* = \log \mathcal{S}_\sigma(x_*) - \log y$ through $\log \Gamma$ and bound the error by theorem 5.62. The inequalities eq. (5.56) and eq. (5.57) give an independent one-sided check that costs nothing.
4. For higher accuracy run Newton's method on the logarithmic equation,
$$x_{j+1} = x_j - \frac{\log \mathcal{S}_\sigma(x_j) - \log y}{\lambda + \frac{\sigma}{2}\big[\psi(x_j/2+1) - \psi(x_j/2+1/2)\big]},$$
which is stable even when $\mathcal{S}_\sigma(x)$ is far outside floating-point range. Theorem 5.8 guarantees a positive denominator, and the transseries seed lies in the basin for large y.
5. For a discrete answer, round by eq. (5.105) and verify the final inequality exactly (theorem 5.56).

5.11.3 Validation

The tables below were recomputed at 50 significant digits against exact gamma-quotient reference values, with y generated as $\mathcal{S}_\sigma(x_0)$ so that the exact inverse is x_0. They reproduce the corresponding tables of S2 and S3 to every printed digit, and, after the reparametrization of theorem 5.12, those of S1 as well.

Table 5.4: Relative error of the forward expansion eq. (5.36) after the term $\sigma^M A_M x^{-M}$. Recomputed.

x	σ	$M=0$	$M=2$	$M=4$	$M=6$	$M=8$
10	-1	2.5273×10^{-2}	3.8527×10^{-5}	4.7159×10^{-7}	1.4186×10^{-8}	7.8542×10^{-10}
10	$+1$	2.4650×10^{-2}	3.8626×10^{-5}	4.7375×10^{-7}	1.4233×10^{-8}	7.8709×10^{-10}
20	-1	1.2573×10^{-2}	4.8642×10^{-6}	1.5087×10^{-8}	1.1546×10^{-10}	1.6336×10^{-12}
20	$+1$	1.2417×10^{-2}	4.8704×10^{-6}	1.5121×10^{-8}	1.1565×10^{-10}	1.6353×10^{-12}
50	-1	5.0122×10^{-3}	3.1226×10^{-7}	1.5560×10^{-10}	1.9152×10^{-13}	4.3651×10^{-16}
50	$+1$	4.9872×10^{-3}	3.1242×10^{-7}	1.5573×10^{-10}	1.9164×10^{-13}	4.3668×10^{-16}

Warning 5.64 (These tables are not convergence). The decreasing entries at fixed x_0 do not indicate convergence as $K \to \infty$. Both the forward and the inverse series are divergent, with the beyond-all-orders floor eq. (5.98); the tables stop far short of the least term (at $x = 20$ the least term of eq. (5.30) sits at $r = 31$, of size 10^{-28}). The non-monotone entries in the odd column at $x_0 = 10$ are ordinary behaviour of an asymptotic series at small argument, not an error. Beyond the least term one should use the exact Laplace representation eq. (5.22), or a Newton step against $\log \Gamma$, or an exponentially improved representation transported by theorem 5.52.

Table 5.5: Absolute error $\left|\mathcal{I}_\sigma^{[K]}(\mathcal{S}_\sigma(x_0)) - x_0\right|$ of the Lambert-normalized inverse eq. (5.76) with K corrections retained; $K = 0$ is the bare core. Recomputed.

x_0	σ	$K = 0$	$K = 2$	$K = 4$	$K = 6$	$K = 8$
10	-1	3.8813×10^{-2}	1.6744×10^{-5}	2.9057×10^{-7}	2.4840×10^{-8}	1.7249×10^{-9}
10	$+1$	3.3589×10^{-2}	2.2677×10^{-4}	3.9282×10^{-6}	5.5615×10^{-8}	2.0051×10^{-9}
20	-1	1.8701×10^{-2}	1.2039×10^{-6}	6.9374×10^{-9}	1.8267×10^{-10}	3.2855×10^{-12}
20	$+1$	1.7399×10^{-2}	3.0220×10^{-5}	1.3176×10^{-7}	4.7142×10^{-10}	4.3075×10^{-12}
50	-1	7.3186×10^{-3}	4.2970×10^{-8}	5.6559×10^{-11}	2.8529×10^{-13}	8.3851×10^{-16}
50	$+1$	7.1104×10^{-3}	2.0095×10^{-6}	1.4043×10^{-9}	8.1104×10^{-13}	1.1816×10^{-15}
100	-1	3.6329×10^{-3}	3.9487×10^{-9}	1.6091×10^{-12}	2.1882×10^{-15}	1.6191×10^{-18}
100	$+1$	3.5808×10^{-3}	2.5440×10^{-7}	4.4455×10^{-11}	6.4401×10^{-15}	2.3379×10^{-18}

Table 5.6: The same errors for the centred inverse eq. (5.78). The bare core is better by a factor $\asymp X$, because $\widehat{c}_1^{(\sigma)} = 0$; once corrections are retained the two normalizations are comparable. Recomputed.

x_0	σ	$K = 0$	$K = 2$	$K = 4$	$K = 6$	$K = 8$
10	-1	8.7330×10^{-4}	5.5308×10^{-5}	1.1593×10^{-7}	2.7056×10^{-9}	4.4842×10^{-11}
10	$+1$	7.6102×10^{-4}	5.6715×10^{-5}	2.3489×10^{-7}	7.9121×10^{-9}	4.0347×10^{-10}
20	-1	2.2206×10^{-4}	7.4914×10^{-6}	5.2119×10^{-9}	3.7661×10^{-11}	3.3070×10^{-13}
20	$+1$	2.0696×10^{-4}	7.5956×10^{-6}	7.4297×10^{-9}	6.3564×10^{-11}	8.0880×10^{-13}
50	-1	3.5860×10^{-5}	5.0352×10^{-7}	6.5130×10^{-11}	8.3102×10^{-14}	1.4333×10^{-16}
50	$+1$	3.4850×10^{-5}	5.0641×10^{-7}	7.5153×10^{-11}	1.0250×10^{-13}	2.0275×10^{-16}
100	-1	8.9913×10^{-6}	6.3982×10^{-8}	2.1684×10^{-12}	7.1217×10^{-16}	3.2358×10^{-19}
100	$+1$	8.8632×10^{-6}	6.4167×10^{-8}	2.3300×10^{-12}	7.9120×10^{-16}	3.8477×10^{-19}

5.12 Summary of the coefficient hierarchy

The whole calculation is a chain of constructive coefficient transformations, each with a closed extraction formula and a triangular recurrence:

$$\boxed{B_{2r+2} \xrightarrow{\text{gamma ratio}} d_{2r+1} \xrightarrow{\exp,\,B} A_n \xrightarrow{\text{Lagrange},\,\widehat{B}} c_n^{(\sigma)} \xrightarrow{\text{unfold}\,W} P_n^{(\sigma)}(H),} \tag{5.111}$$

while the Borel kernel $\tanh(t/2)/(2t)$ supplies, in one stroke, the exact sum of the first stage, the sign and size of every remainder (theorem 5.20), the singulant π, the least-term scale, and the complete Stokes data. Two facts are specific to the swing factorial and cannot be relegated to the general theory of Part 1:

1. *Parity is leading-order data.* The forward object is a period-two transseries eq. (5.40) whose discrete transmonomial σ_n switches the algebraic scale between $2^n n^{-1/2}$ and $2^n n^{+1/2}$; the inverse object is a *pair* of monotone branches, and theorem 5.9 shows that this is forced, not chosen.

2. *The two sheets need different Lambert branches.* W_{-1} on the even sheet, W_0 on the odd sheet, the assignment being forced by $\operatorname{sign} b = \sigma$ in theorem 1.20. The principal branch on the even sheet does not merely lose accuracy: by theorem 5.32 it returns the root $2/(\pi y^2)\big(1 + O(y^{-2})\big) \to 0$, so that choice is fatal before any correction is computed.

Chapter 6

Synthesis: what the four subjects share, and where they part

This chapter belongs to the consolidation rather than to any source. It compares the four inversions carried out in Parts 2 to 5, states precisely how much of each is an instance of Part 1, and records what the twelve articles left open. Nothing here is asserted beyond what the earlier chapters prove.

6.1 One inversion, four dictionaries

The four subjects of this volume were written about independently, and their *forward* analyses have almost nothing in common: a singularity analysis of a Pólya functional equation, a pair of gamma-quotient branches, a convergent Rademacher series over root-of-unity sectors, and a second, quite different pair of gamma quotients. Their *inverse* analyses are, to a degree that the individual articles could not see, the same computation.

The reason is theorem 1.8. In each subject one first isolates a *dominant core* Φ_0 — an increasing function whose inverse is available in closed form — and writes the exact equation for the index x in terms of the observed value y as

$$\Phi_0(x) + \text{(a correction that is small relative to } \Phi_0'\text{)} = L(y).$$

The core is then inverted exactly by the Lambert W function, and the correction is removed to all orders by one triangular recurrence, theorem 1.29 (in the exponential–power case, theorem 1.25). The recurrence does not know which sequence it came from: it sees only the core data and the coefficients q_j of the correction.

Table 1.1 records the resulting dictionary. Three points in it deserve emphasis, because each was invisible from inside a single subject.

The partition numbers are the same problem after one substitution. Of the four, the partition numbers look least like the others: the Hardy–Ramanujan phase is $\pi\sqrt{2N/3}$, a *square root* of the variable, where the other three carry a phase linear in the index or in $x \log x$. The three partition articles accordingly build their inversion by hand around the equation they write as $\rho - 2\log\rho = L$. By theorem 1.4 that equation *is* the linear-phase core $a\xi + b\log\xi$ in the variable $\xi = \sqrt{N}$, with $b/\alpha = -2$; the substitution costs nothing and removes the apparent exception. What the substitution does *not* do is commute with the arithmetic shift, and theorem 1.6 shows why the order matters: re-expanding $N = n - \frac{1}{24}$ in powers of n after the reduction manufactures an entire second tower of exponents $n^{\alpha-k}$, which is exactly the infinite algebraic array that one source reports and attributes, correctly, to the shift alone.

The two "factorials" are not the same kind of object. It is tempting — and this volume's own brief made the mistake — to expect the double factorial and the swing factorial to sit side by side. They do not. The swing factorial $n!/\lfloor n/2\rfloor!^2$ has its factorials cancel to leave growth $2^n n^{\pm 1/2}$, so it is an exponential–power model with $a = \log 2$ (Part 5). The double factorial does not: its logarithm is $\frac{s}{2}(\log s - 1) + \mathcal{O}(\log s)$ with $s = n + 1$, a phase of factorial type, and since $s \log s/s^{\alpha} \to \infty$ for every $\alpha \leq 1$, no choice of model data reproduces it (Part 3). What survives the difference is exactly the axiomatized core: both $aX + b\log X$ and $X(\kappa \log X + d)$ are admissible in the sense of theorem 1.8, both are inverted exactly by W (theorems 1.20 and 1.23), and the reversion above them is one theorem, theorem 1.29. The right statement is therefore not "four instances of one model" but "four instances of one *core calculus*, of which three share a model".

The Lambert branch is determined by a sign. Each of the twelve articles verifies by hand, for its own case, which branch of W produces the large root; several remark that the appearance of W_{-1} is essential and surprising. With the signed convention of theorem 1.1 it is neither: by theorem 1.21 the branch is W_0 when $b > 0$ and W_{-1} when $b < 0$, with the fold at $X_* = |b|/a$ and the threshold $L_c = |b|(1 - \log(|b|/a))$. Reading table 1.1 down the b column now predicts the branch in every chapter, including the two branches of Part 5, which differ only in the sign of σ.

6.2 Where the common frame stops

The shared apparatus covers the passage from an exact forward description to an all-orders inverse. It covers neither end of that passage, and it is worth being explicit about which parts of the four chapters are irreducibly their own.

(a) *Getting the forward description at all.* The four forward analyses share no method. Part 2 needs a singularity analysis of $T(z) = z \exp \sum_{k\geq 1} T(z^k)/k$, the location and nature of the Otter singularity, and a transfer theorem; Parts 3 and 5 need Binet's representation and Stirling's series for gamma quotients; Part 4 needs the circle method, or rather its exact endpoint, Rademacher's convergent series. Nothing in Part 1 produces any of these, and nothing in one of them helps with another.

(b) *The arithmetic layer.* The sectors of Part 4 are indexed by a denominator k and carry Kloosterman-type amplitudes built from Dedekind sums. This structure has no analogue in the other three subjects, whose exponentially small corrections come from a competing singularity (Part 2) or from Borel geometry (Parts 3 and 5). The frame transports each sector through the inversion once the sector is given; it does not explain where the sectors come from.

(c) *Parity.* Two of the four sequences are not the restriction of any single smooth interpolation, and in both cases this is forced, not accidental: the even and odd subsequences have different exponents ($n^{\pm 1/2}$ for the swing factorial) or different gamma representations (the double factorial). Theorem 1.37 and theorem 1.39(4) handle the consequence uniformly — the integer inverse is a minimum over residue classes — but the *fact* of the split is a property of the sequence.

(d) *Which interpolation.* For a discrete sequence the asymptotic inverse is well defined only after an interpolation is chosen, and the choice is not innocent. Theorem 1.38 separates the three inverse objects; Part 3 shows what changes when the OEIS periodic interpolation is used instead of the branchwise gamma continuation, namely a log-periodic oscillation in the inverse that is absent from either branch. One partition source records the sharper version of the same phenomenon: the algebraic part of the inverse is interpolation-independent, the nonperturbative part is not.

6.3 The three inverse objects, subject by subject

Theorem 1.37 distinguishes the staircase inverse N_*, the inverse of a chosen interpolation, and the asymptotic inverse. The four chapters do not stand in the same relation to them, and the differences are not cosmetic.

Chapter	interpolation available	parity split	exact staircase recovery
2 rooted trees	from the singular expansion	no	along the range
3 double factorial	two branches, or OEIS	yes	per residue class
4 partitions	from Rademacher	no	along the range
5 swing factorial	two branches	yes	per residue class

Two cautions apply to the whole table. First, by theorem 1.39(5) the staircase N_* is *not* uniformly approximable by any continuous function: $\sup_{y\geq Y}|N_*(y)-g(y)|\geq\frac{1}{2}$ for every continuous g and every Y. Statements of the form "$N_*(y)=\nu(y)+o(1)$" are therefore false as uniform statements, however natural they look, and the volume asserts only the exact ceiling identity, recovery along the range $y=A_n$, and the residue-class refinement. Second, the error floor of the asymptotic inverse is not a defect of the reversion but is inherited: by theorem 1.46 the forward series has a least-term envelope, and theorem 1.42 converts it into a floor for the inverse. No amount of further reversion crosses it.

6.4 What this consolidation establishes

Several statements in this volume are not in any of its twelve sources. They are listed here because a reader who knows the sources should be able to see quickly what has changed; the corrections themselves are boxed where they occur and collected in Part B.

(1) *A correct proof of the Lagrange fixed-point formula.* Five of the twelve articles derive $[t^n]\,U=\sum_{m\leq n}\frac{1}{m}[t^nu^{m-1}]\,\Psi^m$ by introducing a marker z and applying Lagrange–Bürmann in z. That application is not available: Lagrange–Bürmann needs $\varphi(0)$ invertible, and here $\varphi(0)=\Psi(t,0)\in tR[[t]]$ is a non-unit. The identity is nonetheless true, and theorem 1.15 proves it by universality — it is a polynomial identity over $\mathbb{Q}$ in the coefficients of φ, valid in a localization into which the relevant domain injects. Every reversion in this volume rests on this lemma, so the repair is not cosmetic.

(2) *The branch rule* theorem 1.21, with the fold point and threshold, replacing four case analyses.

(3) *The normalization constant of the flattened blocks.* The sources state $P_n(0)=D_n$ in a case where $a=1$; the correct general statement is $P_n(0)=a^{n+1}c_n$ (theorem 1.33(3)), and since $a\neq 1$ in every chapter of this volume the factor is needed throughout.

(4) *A quantitative obstruction for the staircase*, theorem 1.39(5), in place of an $o(1)$ claim that is false uniformly.

(5) *Backward error as a certificate.* Theorem 1.42 upgrades the sources' conditional bound — which presupposes that a root exists — to an existence statement, by a sign change and the intermediate value theorem.

(6) *One closed form where the sources had three.* Theorem 1.34 is proved equal to all three published forms of the pure Lambert block, and the equality was verified in exact rational arithmetic for $1\leq n\leq 8$.

(7) *A non-recursive formula for the factorial-core corrections.* Theorem 1.17 extends the fixed-point formula to $\Psi=h+f$ with $h\in u^2R[[u]]$, which is what Part 3 needs; the source reaches the same coefficients only through an ad hoc compositional inverse.

6.5 Open questions

(i) *A Stokes-transport theorem.* The mechanism by which an exponentially small forward sector becomes an exponentially small inverse sector is uniform — it is the perturbation-operator form theorem 1.31 — but a theorem needs each subject's Borel geometry as a hypothesis. The four chapters supply four separate instances; the general statement is not proved here.

(ii) *The counting function.* One source states that the number of indices with $A_n \leq y$ equals the average of the branch inverses plus $\mathcal{O}(1)$. The statement is plausible for every subject with a parity split, but it needs a per-subject monotonicity hypothesis and is not proved in this volume.

(iii) *Rademacher-type sequences in general.* Part 4 gives a template for inverting a sequence presented as a convergent sum over arithmetic sectors. Whether the template applies to the other classical Rademacher-type expansions is untested here.

(iv) *Multifactorials.* One double-factorial source sketches the m-fold generalization $n!^{(m)}$. The core is again of factorial type, so theorem 1.29 applies verbatim; what is not done is the residue-class analysis of the staircase for general m, which theorem 1.39(4) reduces to a finite computation but does not carry out.

(v) *Relation to the formalized Lambert layer of this repository.* The surrounding development contains a Lean-verified treatment of both real branches of W, a lower-branch bracket with an explicit remainder rate, and the saddle expansions of the Fabius function. The Lambert core these four chapters are built on is therefore already formalized to a degree none of the sources assumes. Which of the reversion machinery above is within reach of that layer — theorem 1.20 and theorem 1.21 look closest — is an open and, in this repository, a natural question.

Appendix A

Provenance

This volume consolidates twelve independently written articles, three on each of its four subjects. All twelve were filed in this repository on 3 September 2026 as a single quick-intake batch, at `drafts/series-and-transse` the directories below are relative to that path. Nothing in the twelve was edited in place: this volume is a new document, and the sources remain available in git history.

Source directory	Source	PDF	Absorbed into
Chapter 2: Butcher–Pólya rooted trees			
`Butcher_Tree_Transseries`	1 674 lines	21 pp, A4	Chapter 2
`Butcher_Tree_Transseries-2`	2 853 lines	62 pp, A4	Chapter 2
`Butcher_Tree_Transseries-3`	2 382 lines	35 pp, A4	Chapter 2
Chapter 3: the double factorial			
`Double_Factorial_Transseries`	1 552 lines	18 pp, A4	Chapter 3
`Double_Factorial_Transseries-2`	2 387 lines	28 pp, A4	Chapter 3
`double_factorial_transseries-1`	2 238 lines	44 pp, A4	Chapter 3
Chapter 4: the partition numbers			
`Partition_Number_Transseries_and_Asymptotic_Inverse`	1 901 lines	32 pp, A4	Chapter 4
`Partition_Number_Transseries_and_Inverse`	2 020 lines	27 pp, A4	Chapter 4
`Partition_Numbers_Transseries_and_Inverse`	1 905 lines	33 pp, Letter	Chapter 4
Chapter 5: the swing factorial			
`Swing_Factorial_Transseries`	1 885 lines	28 pp, A4	Chapter 5
`Swing_Factorial_Transseries_Article`	1 824 lines	25 pp, A4	Chapters 1 and 5
`Swing_Factorial_Transseries_and_Inverse`	1 389 lines	22 pp, A4	Chapter 5

The shared apparatus of Chapter 1 is drawn from all twelve. Its organizing normal form, and the Lagrange fixed-point route to the reversion coefficients, follow the presentation in `Swing_Factorial_Transseries_Article` most closely; the α-reduction that brings the partition numbers into the same frame is this volume's, since no source needed it.

Appendix B

Repairs to the sources

Every correction this volume makes to one of its twelve sources is boxed in the text where it occurs and listed here. A repair is recorded when a source states something false, states an asymptotic conclusion that its argument does not support, or asserts a named result without proof. Differences of convention are not repairs and are recorded in the text as remarks instead.

B.1 Chapter 1: the common apparatus

A proof that does not apply (five sources). Swing_Factorial_Transseries_Article, Partition_Number_Transseries_and_ (twice), Partition_Numbers_Transseries_and_Inverse, Partition_Number_Transseries_and_Asymptotic_Inverse and double_factorial_transseries-2 all prove the Lagrange fixed-point formula $[t^n]\,U = \sum_{m\le n} \frac{1}{m}[t^n u^{m-1}]\,\Psi^m$ by introducing an auxiliary marker z, solving $U = z\Psi(t,U)$, and applying Lagrange–Bürmann in z. Lagrange–Bürmann requires the relevant $\varphi(0)$ to be invertible; here $\varphi(0) = \Psi(t,0) \in tR[[t]]$ is a non-unit, and the residue proof genuinely breaks, since $\psi = w/\varphi(w)$ is then not an element of $w\,R[[t]][[w]]$. The identity is true; theorem 1.15 proves it by universality instead. Every reversion in this volume depends on the lemma, so the repair is structural rather than cosmetic.

An asymptotic claim that is false uniformly. Several sources state, or leave the reader to infer, that the integer inverse satisfies $N_*(y) = \nu(y) + o(1)$. No continuous function approximates a staircase uniformly: theorem 1.39(5) shows $\sup_{y\ge Y}|N_*(y) - g(y)| \ge \frac{1}{2}$ for every continuous g and every Y. The volume asserts only the exact ceiling identity, nearest-integer recovery *along the range*, and the residue-class refinement.

A normalization constant. The sources record $P_n(0) = D_n$ for the flattened blocks, in a case where $a = 1$. The general statement is $P_n(0) = a^{n+1}c_n$ (theorem 1.33(3)); since $a \ne 1$ in every chapter of this volume the factor is needed throughout.

Two truncation conventions in one article. Swing_Factorial_Transseries_Article uses the exclusive convention in its forward error table and the inclusive one in its inverse table, so those two tables are not on a common scale. The volume is inclusive throughout (theorem 1.1); tables inherited from that article's forward section are re-indexed.

A claim of this volume's own, corrected. The brief under which Part 1 was written asserted that the case $\alpha = 1$ of theorem 1.2 covers the double factorial. It does not: $\log\bigl((x+1)!!\bigr) = \frac{s}{2}(\log s - 1) + \mathcal{O}(\log s)$ with $s = x+1$, and $s\log s/s^{\alpha} \to \infty$ for every $\alpha \le 1$. The sources never claimed membership. Theorem 1.8 axiomatizes the dominant core instead, and theorem 1.23 inverts the factorial phase exactly.

Corrupt source text, not reproduced. Swing_Factorial_Transseries_Article equation `eq:inverse-Stokes-leadin` and Butcher_Tree_Transseries-2 lines 2482–2483 contain corrupt control sequences. The affected statements are re-derived here rather than copied.

B.2 Chapter 3: the double factorial

Convergence asserted without a threshold. The first source states that the periodic carrier "has the convergent local Lagrange–Bürmann expansion" for sufficiently large Λ, with no threshold and no proof of convergence as opposed to formal solvability; the second proves only a contraction estimate, from which convergence of the Lagrange series does not follow. Theorem 3.46 separates the two: existence and uniqueness of the real root by contraction for $\Lambda > 2\pi a$, and convergence of the Lagrange series under the explicit sufficient condition $\mu = 2a/\Lambda \le 1/5$.

Two assertions in place of proofs. The deepest-pole coefficient ("only the quadratic kernel and the first Bernoulli coefficient can contribute") is exactly the step needing the degree bookkeeping, and is supplied here by induction on the Λ^{-1}-degree. The coefficientwise sign law is argued in one source with "use a positive rooted-tree contribution …; such a tree exists", exhibiting no such object; theorem 3.37 keeps the correct positivity reduction and replaces the strictness step with an explicit attainability induction, verified symbolically for $n \le 7$.

A formal-to-analytic slide. The third source presents an all-orders phase reversion as a theorem with an asymptotic conclusion, introduced by "setting $\tau = 1$ as an asymptotic expansion". The identity is formal; making it asymptotic needs estimates, uniform in the oscillatory phase, that are neither stated nor proved. It is demoted here to the formal theorem 3.50, with a proved remainder supplied separately.

A silent three-way inconsistency. One source centres the periodic inverse at the even-branch constant, the other two at the parity mean; the leading displacements consequently differ by $2a/\Lambda$. Both are correct for their own centring, but no source says so, and a reader comparing them would conclude that one is wrong. The volume adopts the mean centring and states the relation.

A notation collision across three sources. The letter C denotes the multiplicative amplitude, the logarithmic constant, and the branch prefactor in the three siblings respectively, and α, β carry two incompatible meanings. A crosswalk table fixes the volume's usage; the core parameters are never abbreviated, to avoid a fourth collision with κ in theorem 1.8.

A qualitative warning made quantitative. All three sources warn that a different smooth interpolation "can change this sector" without exhibiting one. Theorem 3.8 exhibits the family $\mathcal{D}(x)\mathrm{e}^{\lambda \sin \pi x}$, which interpolates $n!!$ at every non-negative integer and changes the leading oscillatory amplitude from $2a/\Lambda$ to $2\sqrt{a^2 + \lambda^2}/\Lambda$ — a change larger than the entire Bernoulli grid. This is what makes the chapter's organizing claim a theorem rather than a caution.

B.3 Chapter 2: the rooted-tree numbers

Digits printed beyond the accuracy of the computation. One source prints the constants of the inherited $-\sqrt{\rho}$ sector to forty decimal digits, but only about twenty-five of them are correct. Two independent recomputations here — including a from-scratch direct summation at eighty digits — give

$$r_0^- = 0.46863662792484554808720262193181 9314\ldots,$$

and the volume prints only digits it has verified. This is the one place in the twelve articles where a printed numeral is wrong; it is a precision failure rather than a mistake in a formula.

An audit appendix that contradicts its own main text. The same source's audit appendix reports $D_3 = -3.582177326832123\ldots$, disagreeing with the value its main text derives. The main text is correct; the appendix entry is not, and the volume uses the recomputed value.

A symbol used for two different constants. One source writes α for the growth constant where the other two write λ, and the volume's Chapter 1 uses α for the phase exponent of theorem 1.2. The chapter fixes λ for the growth constant throughout and never uses α for it.

An asymptotic statement with a formal proof. A source states its inverse-coefficient theorem asymptotically but proves it only formally, and separately displays a global transseries with no hypothesis under which it could hold. The first is proved here in its asymptotic form; the second is replaced by a conditional theorem with a finite radius and an explicit hypothesis.

A circular derivation of the singularity bounds. The elementary bounds $\frac{1}{4} \leq \rho \leq \mathrm{e}^{-1}$ are used in the sources in a way that presupposes what they are used to prove. They are established here independently, from $x \leq T\mathrm{e}^{-T} \leq \mathrm{e}^{-1}$ and the Catalan bound, and the critical characterization is then obtained non-circularly.

B.4 Chapter 5: the swing factorial

A divergent series written as an equality. One source sets its bookkeeping coupling to 1 and prints the resulting operator expansion as an equality. There is then no small parameter, the sum diverges, and a number is equated to a divergent series; as printed the statement is false. The m-th summand is $\mathcal{O}(X^{-(2m-1)})$, so the correct statement is asymptotic, and regrouping gives the finite differential form used here. The other two sources state the same expansion correctly.

A theorem whose hypothesis is never used. A sector-transport result is labelled a theorem, its hypothesis is not used in the argument, and its conclusion is prefaced "formally". It is split here into the formal all-orders identity, which is what the proof gives, and a quantitative first-order statement proved by the mean value theorem under stated hypotheses.

An overclaim about complex branches. One source asserts that every Lambert branch with $|w| \to \infty$ yields an inverse branch with the same coefficients. For $k \neq 0, -1$ the existence of an analytic branch of the inverse along the path is a hypothesis, not a consequence, and it fails at gamma poles and Stokes directions. The claim is demoted to a remark with the two hypotheses displayed.

The analytic inverse statement, unproved in all three sources. One source writes "likewise, for each fixed $M, \ldots$" with no proof; the second sketches an argument that never bounds the residual of the truncated ansatz nor uses forward control at the perturbed point; the third proves the ingredients but never assembles them. This is the one formal-to-analytic slide common to all three. Theorem 5.41 proves the statement in full, with an effective constant.

A recommendation contradicted by its own table. A remark asserts that the centred normalization is "often the preferable computational normal form". The same article's table shows the centred core better by a factor $\approx 10^3$ at one end and the uncentred form slightly better at the other. The corrected statement is that centring kills the X^{-1} term and so improves the *bare core* by a factor $\asymp X$, while with several corrections retained the two are comparable and neither dominates uniformly.

Two convergence claims without criteria. An equivalence of the Lambert and logarithmic normalizations is asserted in prose with no formula and no proof; it is proved here as a finite explicit composition. A convergence claim appeals to "the classical analysis of Lambert W", but the classical result concerns a doubly indexed series, whereas the series at hand has coefficients polynomial in ℓ and converges only when ℓ/L is small; an explicit criterion is proved instead.

B.5 Chapter 4: the partition numbers

A definition that contradicts its own use. The source abbreviated S2 in Part 4 defines the real Rademacher amplitude by a sum over $1 \leq h \leq k$ with $\gcd(h,k) = 1$. For $k \geq 2$ that is the intended index set, but at $k = 1$ it returns $\cos(2\pi\nu)$ rather than the constant 1, while the same article uses $A_1 \equiv 1$ throughout. With the printed definition the interpolant would not interpolate p: its dominant term would oscillate off-lattice at full relative strength, and the article's own eventual-monotonicity statement would be false. The volume takes $h \in U_k$ with $U_1 = \{0\}$; every later formula of that source is then correct as printed.

An understated radius. S1 states only that its universal sector-coefficient series "converges for all sufficiently small $|t|$". The radius is exactly $\sqrt{24}$ in $t = n^{-1/2}$, the nearest singularities being the branch points of $\sqrt{1 - t^2/24}$; convergence therefore holds for every real $n > 1/24$.

A convergent series and an asymptotic total, conflated. All three partition sources are loose about the distinction. The algebraic series $\sum_m \omega_m n^{-m/2}$ converges, but its sum is the leading Rademacher block, *not* $p(n)$; the Poincaré statement about $p(n)$ carries an error that is the sum of a convergent algebraic tail and an exponentially small Rademacher remainder. Summing the algebraic series to convergence does not compute $p(n)$: the residual stalls at the $\mathrm{e}^{-\mu/2}$ floor.

A false intermediate bound. S1's kernel estimate asserts $g(z) \leq z^2/3$ on $[0,1]$, which fails at $z = 1$ where $g(1) = \mathrm{e}^{-1} = 0.3679\ldots > 1/3$. The chapter proves $g(z) \leq (z^2/3)\cosh z$ instead, which is what the tail bound actually needs.

An overclaim in this volume's own Chapter 1. Theorem 1.3 originally said that the correction series Q is divergent of Gevrey type "in all four chapters". It is not: for the partition numbers $Q(t) = \log(1-t)$ has radius of convergence 1. The remark now distinguishes the three divergent chapters from Part 4, where theorem 1.46 has no content and the error floor is set by the Rademacher remainder instead.